WESTERN ILLINOIS UNIVERSITY – QUAD CITIES
School of Engineering

Final Design Review

Team #1

Project Rock-E

Team Members: Reilly Bacorn, Ethan Call, Aaron Flemming, Jacob Kruse, Kanika Palecanda,

Justin Scott, Harsh Shah, Brendan Webster, Caiyun Wurslin

May 1, 2024

Executive Summary

Autonomous vehicles (AV's) are currently being integrated into society and will certainly remain relevant to engineering careers for years to come. Building and programming an AV for the 2024 Intelligent Ground Vehicle Competition (IGVC) ensures WIU engineering students have valuable problem-solving skills for industry.

Team 1's (appendix A) contribution to progress in this industry is an AV named Rock-E (appendix G, Figure 1); it is designed to compete in the 2024 IGVC. Rock-E is specifically crafted to abide by the rules of the competition (restricting size, propulsion, speed, and safety features) and remain under WIU's budget of $5,000 + donations (appendix S, C).

The design, inspired by various benchmarks from industry and previous competitions (appendix E, F, Q), aspired to remain simple and functional in its systems such as chassis, propulsion, and electrical (appendix E, Q, R, T) to facilitate integration with the navigational subsystem that was in parallel development. Ideas about chassis configuration and propulsion were conceived by each team member (appendix F) before a decision was made (appendix G, M). Following this, subsystems were designed: electronics and safety features (appendix K, L), wheels (appendix E, G), and motors (appendix Q).

The electrical layout of Rock-E is a combination of schematics of purchased components and a master wiring diagram (appendix K, L). Of peak importance to the design is safety; wiring best practices are followed (appendix T) and the design eliminates electrical hazards and possibility of a runaway vehicle through proper configuration and component choice.

The propulsion system is reliant on the correct selection of motors, so necessary torques were calculated preceding motor procurement and physical verification testing took place (appendix O, P, Q).

Further verification of design choices was obtained by performing simulations. Static, thermal, and air flow (appendix H, I, J) analyses were conducted, verifying component choice, structural configuration, as well as airflow and cooling of electrical components.

Navigational systems were first developed as standalone subsystems (appendices F-M) before being integrated into the complex network of sensors, processors, and logic that compose the Rock-E navstack (appendix V).

The project charter and work breakdown structure are found in appendices B and D. The lessons learned from the low- and mid-fidelity prototypes are found in appendix N. The standards, sources referenced, and ethics involved in the project are located in appendices U, and V, and navP.

With Rock-E's design having been sufficiently validated, autonomous navigation achieved, and a team prepared to compete, Team 1 is requesting to represent WIU at the 2024 Intelligent Ground Vehicle Competition.

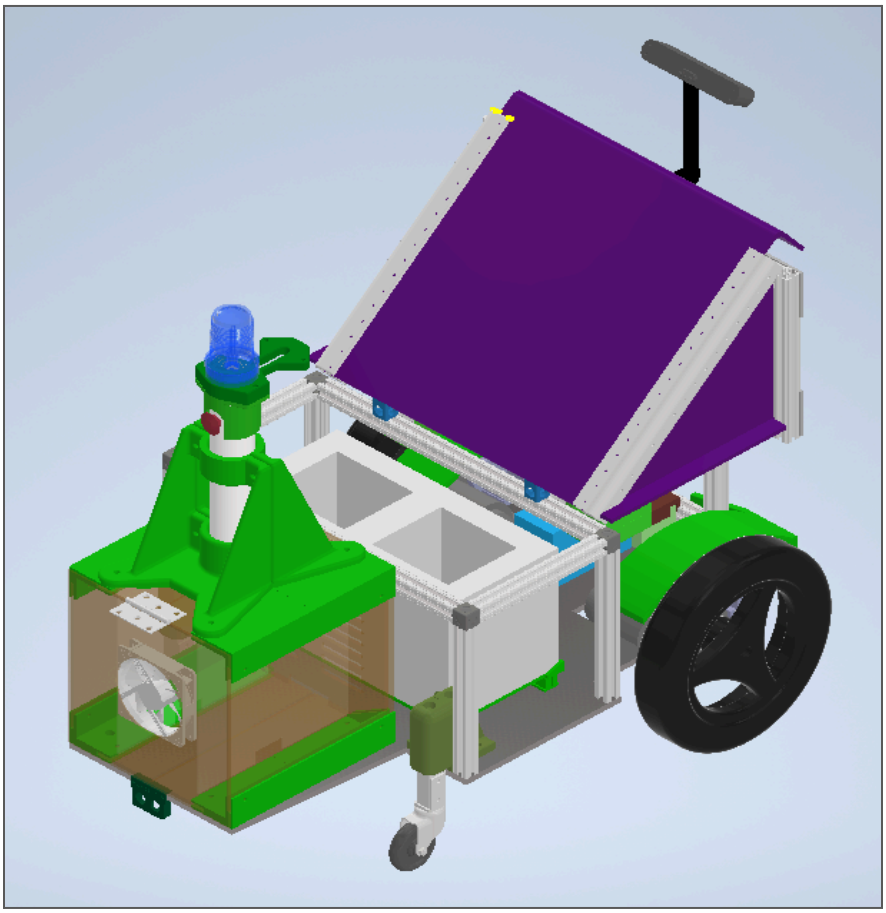

Figure 1: Final FDR Design

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Appendix A: Team Members and Organization Structure

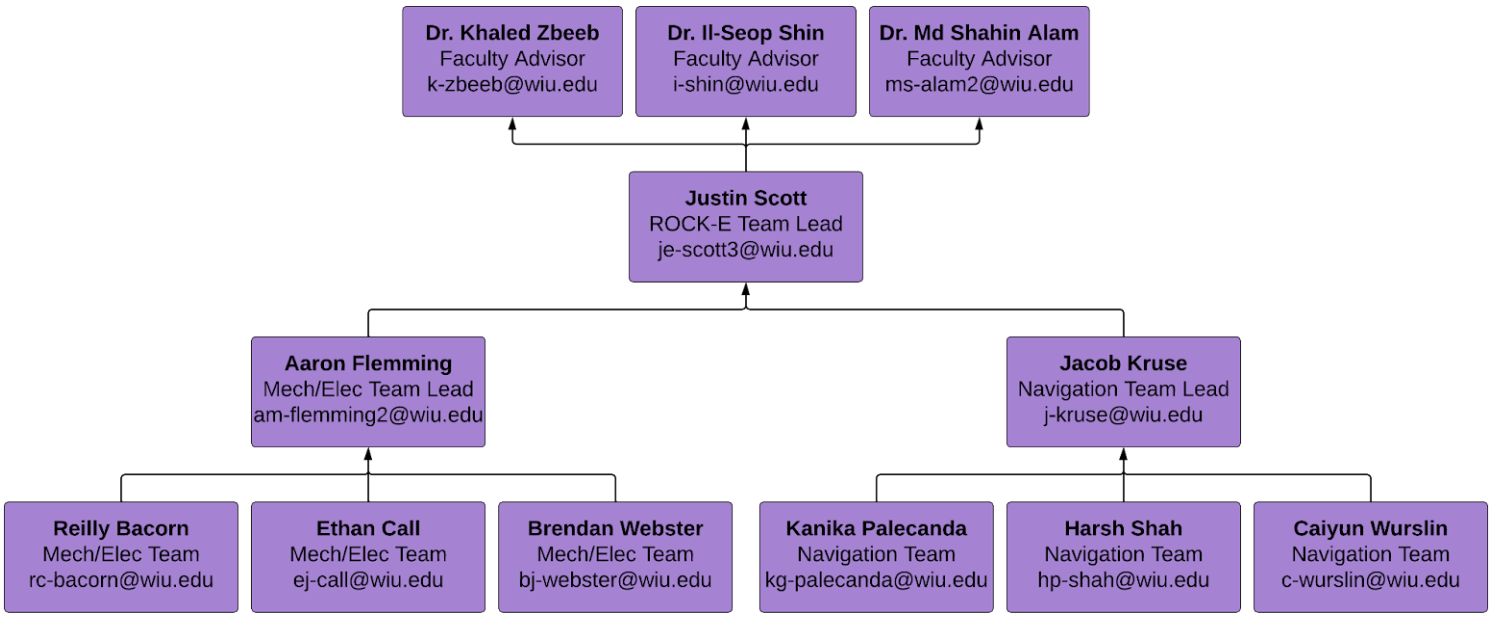


Figure A-1: Team Rock-E Organization

Table A-2: Team Rock-E Responsibilities

Name	Contact	Major	Responsibility
Reilly Bacorn	rc-bacorn@wiu.edu	Mechanical Engineering / Engineering Physics	Mobility
Ethan Call	ej-call@wiu.edu	Mechanical Engineering	Chassis
Aaron Flemming	am-flemming2@wiu.edu	Mechanical Engineering	Chassis
Jacob Kruse	j-kruse@wiu.edu	Electrical Engineering	GPS
Kanika Palecanda	kg-palecanda@wiu.edu	Mechanical Engineering	LiDAR
Justin Scott	je-scott3@wiu.edu	Mechanical Engineering	Safety / Power
Harsh Shah	hp-shah@wiu.edu	Mechanical Engineering	Camera
Brendan Webster	bj-webster@wiu.edu	Mechanical Engineering	Power / Cooling
Caiyun Wurslin	c-wurslin@wiu.edu	Electrical Engineering	IMU

Appendix B: Project Charter

Project Charter			
Project Title	Autonomous Navigation Vehicle	Team	ROCK-E Navigation Team
Start Date	8/26/2023	End Date	5/1/2024

Vision Statement
To expand our education and training into the world of engineering through the creation and design of an intelligent autonomous ground vehicle .

Project Scope	Deliverables
Program a functioning, autonomous vehicle using various sensors over the course of two semesters with the goal of competing in the IGVC	Preliminary Design Report/Presentation Critical Design Report/Presentation Final Design Report/Presentation
Risks	**Assumptions/Dependencies**
• parts not in stock • going over budget • missing deadlines	• Budget is secure • mentors provide advice/feedback

Financials
The budget to complete the project is $5,000; this does not include items currently in stock at WIU or donated to the project.

Milestones	Target completion	Delivery date
Project proposal	9/4/2023	9/4/2023
Low-Fidelity Prototype	10/18/2023	10/18/2023
Preliminary Design Report	10/18/2023	10/18/2023
Preliminary Design Presentation	10/18/2023	10/18/2023
Critical Design Presentation	12/6/2023	12/06/23
Critical Design Report	12/6/2023	12/06/23
Mid-Fidelity Prototype	12/6/2023	12/06/23
Final Prototype	5/1/2024	5/1/2024
Final Design Presentation	5/1/2024	5/1/2024

Project Team		Review Committee	
Team Member	Jacob Kruse	Professor	Dr. Khaled Zbeeb
Team Member	Kanika Palecanda	Mentor	Dr. Il-Seop Shin
Team Member	Harsh Shah	Mentor	Dr. Md Shahin Alam
Team Member	Caiyun Wurslin	Mentor	Rafael Obregon
Team Member	Justin Scott		
Team Member	Aaron Flemming		
Team Member	Ethan Call		
Team Member	Reilly Bacorn		
Team Member	Brendan Webster		

Appendix C: Business Case and Project Budget

Business Case

The proposal for an autonomous vehicle project at Western Illinois University (WIU) is a visionary endeavor designed to offer mechanical engineering students a comprehensive and hands-on learning experience. As the world of autonomous transportation rapidly expands and integrates itself into various industries, there is a growing need to equip students with practical knowledge in designing, building, and programming autonomous vehicles. By engaging students in this project, universities aim to bolster their employability in a field with ever-increasing demand for skilled professionals. Not only does this initiative provide a valuable opportunity for students to acquire the practical skills necessary for the industry, but it also holds the potential to position WIU as a notable contributor to the advancement of autonomous vehicle technology for engineering students. The project's multifaceted approach encompasses not only technical skills development but also collaboration, interdisciplinary research, and community engagement. Consequently, this endeavor has the potential to yield a multitude of benefits for both the students and the university as a whole, thereby solidifying WIU's reputation as a leader in mechanical engineering education and innovation.

Project Budget

Western Illinois University Quad Cities has generously provided each team with a $5000 budget for their autonomous vehicle along with permission to fundraise and secure grants as desired. With this budget, each team must carefully manage their money to purchase the needed equipment for their vehicle. Each team must also be careful not to go over budget. This may result in an incomplete vehicle and ultimately failure during the competition. Each team must also not go exceedingly under budget as this may result in a cheap or incapable vehicle.

Appendix D: Work Breakdown Structure / Project Schedule

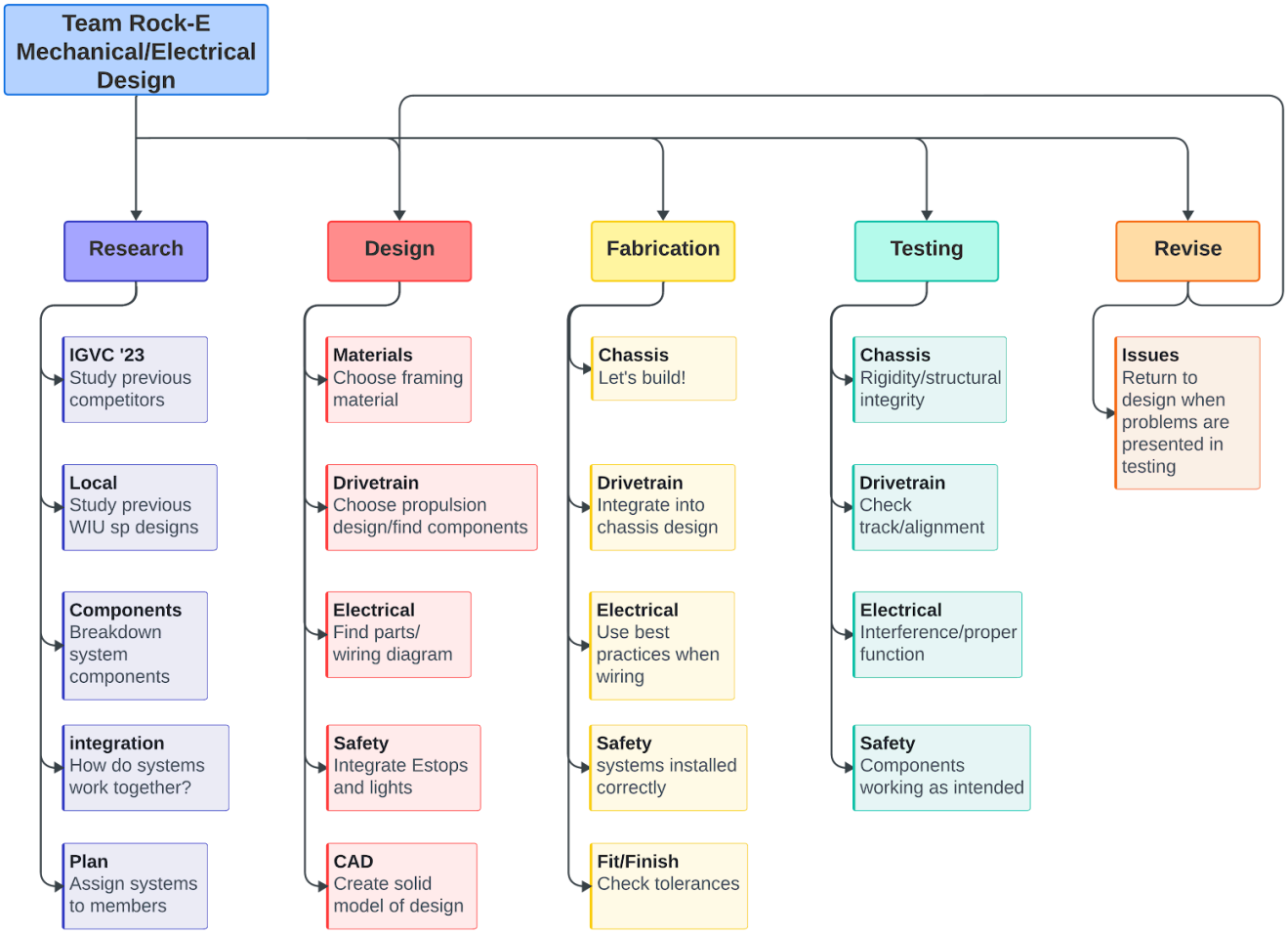


Figure D-1: Mechanical/Electrical Work Breakdown Structure

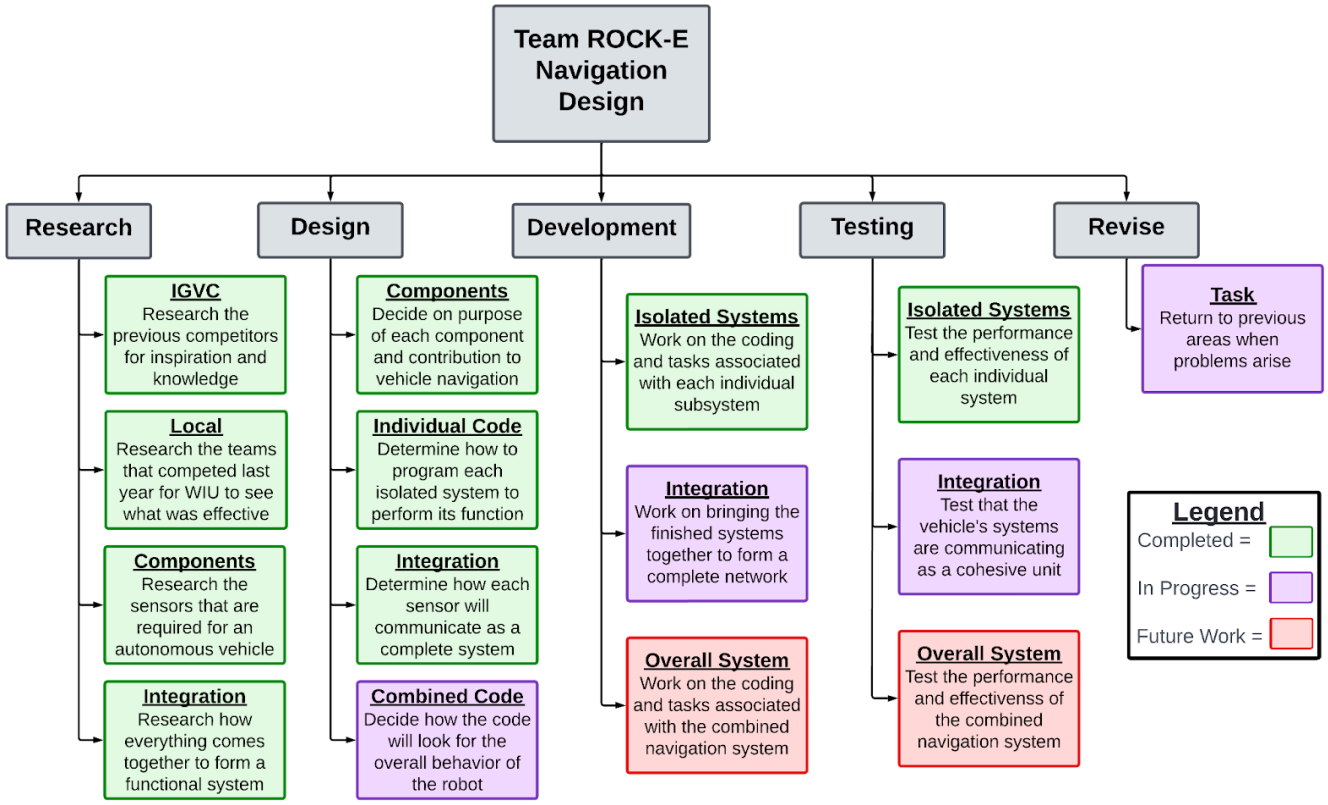
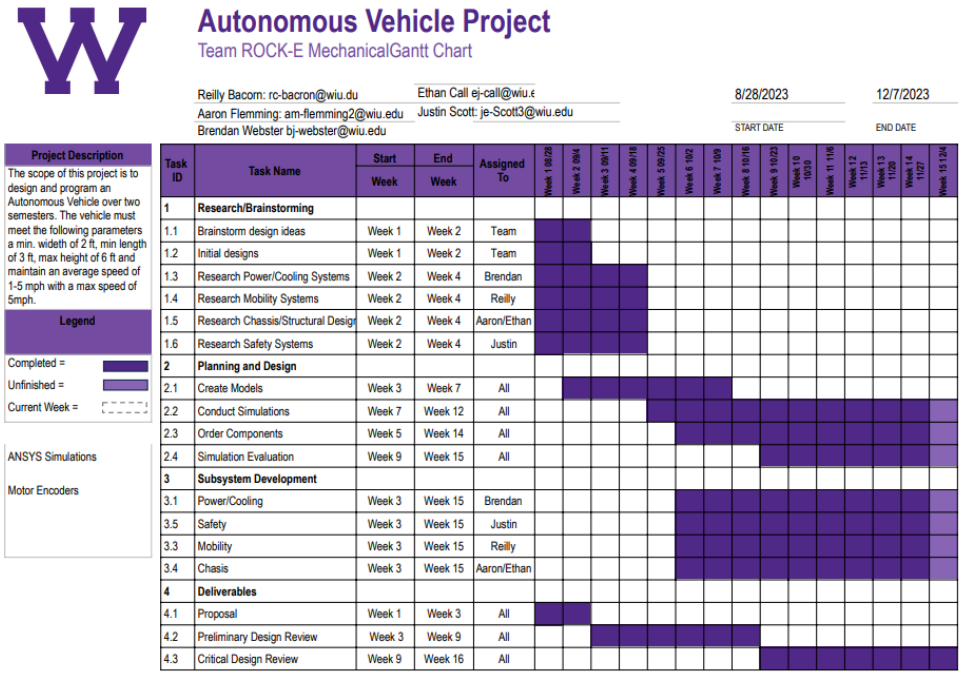


Figure D-2: Navigation Work Breakdown Structure

The Mechanical/Electrical and Navigation plans for the project can be seen in **Figures D-1 and D-2**. The project schedules or Gantt Charts for the fall and spring semesters can be seen in **Figures D-3, D-4, and D-5**. The first half of the first semester was dedicated to brainstorming ideas for Rock-E's design, researching components, and creating initial designs. The second half of the first semester was spent finishing the design work and CAD modeling, most of the simulations, and most of the testing of the components, while also having ordered the majority of parts to enable the move to fabrication. Some errors were encountered when attempting the simulations which put them slightly behind schedule. The initial plan was to begin fabricating the chassis by the end of this semester; however, due to unforeseen complications with incorporating the motor encoders this was also delayed. During the break between semesters, fabrication began starting with the base and frame. The second semester was spent finishing the fabrication of the vehicle, testing the vehicle and revising any parts.



Autonomous Vehicle Project

Team ROCK-E MechanicalGantt Chart

Reilly Bacorn: rc-bacron@wiu.du | Ethan Call ej-call@wiu.e | 8/28/2023 (START DATE) | 12/7/2023 (END DATE)
Aaron Flemming: am-flemming2@wiu.edu | Justin Scott: je-Scott3@wiu.edu
Brendan Webster bj-webster@wiu.edu

Project Description
The scope of this project is to design and program an Autonomous Vehicle over two semesters. The vehicle must meet the following parameters a min. wideth of 2 ft, min length of 3 ft, max height of 6 ft and maintain an average speed of 1-5 mph with a max speed of 5mph.

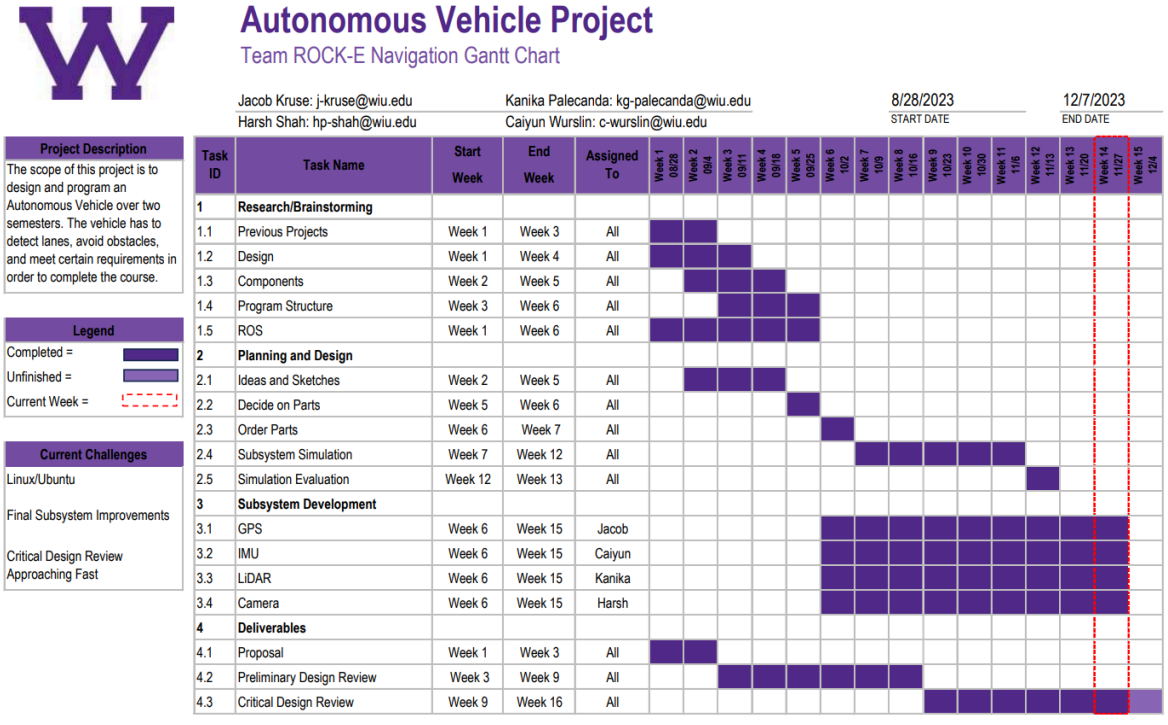
Legend
Completed =
Unfinished =
Current Week =

ANSYS Simulations

Motor Encoders

Task ID	Task Name	Start Week	End Week	Assigned To
1	**Research/Brainstorming**			
1.1	Brainstorm design ideas	Week 1	Week 2	Team
1.2	Initial designs	Week 1	Week 2	Team
1.3	Research Power/Cooling Systems	Week 2	Week 4	Brendan
1.4	Research Mobility Systems	Week 2	Week 4	Reilly
1.5	Research Chassis/Structural Desig	Week 2	Week 4	Aaron/Ethan
1.6	Research Safety Systems	Week 2	Week 4	Justin
2	**Planning and Design**			
2.1	Create Models	Week 3	Week 7	All
2.2	Conduct Simulations	Week 7	Week 12	All
2.3	Order Components	Week 5	Week 14	All
2.4	Simulation Evaluation	Week 9	Week 15	All
3	**Subsystem Development**			
3.1	Power/Cooling	Week 3	Week 15	Brendan
3.5	Safety	Week 3	Week 15	Justin
3.3	Mobility	Week 3	Week 15	Reilly
3.4	Chasis	Week 3	Week 15	Aaron/Ethan
4	**Deliverables**			
4.1	Proposal	Week 1	Week 3	All
4.2	Preliminary Design Review	Week 3	Week 9	All
4.3	Critical Design Review	Week 9	Week 16	All

Figure D-3: Mechanical/Electrical Fall Semester Gantt Chart



Autonomous Vehicle Project

Team ROCK-E Navigation Gantt Chart

Jacob Kruse: j-kruse@wiu.edu | Kanika Palecanda: kg-palecanda@wiu.edu | 8/28/2023 (START DATE) | 12/7/2023 (END DATE)
Harsh Shah: hp-shah@wiu.edu | Caiyun Wurslin: c-wurslin@wiu.edu

Project Description
The scope of this project is to design and program an Autonomous Vehicle over two semesters. The vehicle has to detect lanes, avoid obstacles, and meet certain requirements in order to complete the course.

Legend
Completed =
Unfinished =
Current Week =

Current Challenges
Linux/Ubuntu

Final Subsystem Improvements

Critical Design Review Approaching Fast

Task ID	Task Name	Start Week	End Week	Assigned To
1	**Research/Brainstorming**			
1.1	Previous Projects	Week 1	Week 3	All
1.2	Design	Week 1	Week 4	All
1.3	Components	Week 2	Week 5	All
1.4	Program Structure	Week 3	Week 6	All
1.5	ROS	Week 1	Week 6	All
2	**Planning and Design**			
2.1	Ideas and Sketches	Week 2	Week 5	All
2.2	Decide on Parts	Week 5	Week 6	All
2.3	Order Parts	Week 6	Week 7	All
2.4	Subsystem Simulation	Week 7	Week 12	All
2.5	Simulation Evaluation	Week 12	Week 13	All
3	**Subsystem Development**			
3.1	GPS	Week 6	Week 15	Jacob
3.2	IMU	Week 6	Week 15	Caiyun
3.3	LiDAR	Week 6	Week 15	Kanika
3.4	Camera	Week 6	Week 15	Harsh
4	**Deliverables**			
4.1	Proposal	Week 1	Week 3	All
4.2	Preliminary Design Review	Week 3	Week 9	All
4.3	Critical Design Review	Week 9	Week 16	All

Figure D-4: Navigation Fall Semester Gantt Chart

Autonomous Vehicle Project

Team Rock-E Gantt Chart

Western Illinois Quad Cities - 3300 River Dr Moline, Illinois

ENGR 490: Senior Project

1/22/2024 START DATE

5/3/2024 END DATE

Project Description

The scope of this project is to design and program an Autonomous Vehicle over two semesters. The vehicle has to detect lanes, avoid obstacles, and meet certain requirements in order to complete the course.

Group Members

Reilly Bacorn
Ethan Call
Aaron Flemming
Jacob Kruse
Kanika Palecanda
Justin Scott
Harsh Shah
Brendan Webster
Caiyun Wurslin

Legend

Completed =
Unfinished =
Current Week =

Current Challenges

Writing the FDR

Optimization of Navigation Program

Finishing touches on Robot

Task ID	Task Name	Start Week	End Week	Assigned To	Week 1 01/22	Week 2 01/29	Week 3 02/05	Week 4 02/12	Week 5 02/19	Week 6 02/26	Week 7 03/04	Week 8 03/11	Week 9 03/18	Week 10 03/25	Week 11 04/01	Week 12 04/08	Week 13 04/15	Week 14 04/22	Week 15 04/29
1	**Fabrication**																		
1.1	Mobility	Week 1	Week 4	Mechanical	■	■	■	■											
1.2	Safety	Week 1	Week 5	Mechanical	■	■	■	■	■										
1.3	Power	Week 1	Week 5	Mechanical	■	■	■	■	■										
1.4	Chassis	Week 1	Week 10	Mechanical	■	■	■	■	■	■	■	■	■	■					
1.5	Cooling	Week 1	Week 11	Mechanical	■	■	■	■	■	■	■	■	■	■	■				
2	**Integration**																		
2.1	LiDAR and Camera (Costmaps)	Week 1	Week 4	Harsh and Kanika	■	■	■	■											
2.2	GPS, IMU, Encoders (Filter)	Week 1	Week 4	Jacob	■	■	■	■											
2.3	Motors and Motor Drivers	Week1	Week 4	Caiyun	■	■	■	■											
2.4	Vehicle Remote Control	Week 3	Week 6	Navigation			■	■	■	■									
2.5	Path Planning Algorithm	Week 4	Week 7	Navigation				■	■	■	■								
2.6	Combination of Everything	Week 5	Week 8	Navigation					■	■	■	■							
3	**Testing**																		
3.1	Obstacle Avoidance	Week 6	Week 11	Harsh and Kanika						■	■	■	■	■	■				
3.2	Waypoint Navigation / Localization	Week 6	Week 11	Jacob						■	■	■	■	■	■				
3.3	Motor Control	Week 6	Week 11	Caiyun						■	■	■	■	■	■				
3.4	Mobility	Week 6	Week 11	Mechanical						■	■	■	■	■	■				
3.5	Safety	Week 6	Week 11	Mechanical						■	■	■	■	■	■				
3.6	System Simulation	Week 7	Week 12	Navigation							■	■	■	■	■	■			
3.7	Chassis	Week 7	Week 12	Mechanical							■	■	■	■	■	■			
3.8	Path Planning Algorithm	Week 8	Week 13	Navigation								■	■	■	■	■	■		
3.9	Power	Week 8	Week 13	Mechanical								■	■	■	■	■	■		
3.10	Total Navigation Program	Week 9	Week 14	Navigation									■	■	■	■	■	■	
3.11	Cooling	Week 9	Week 14	Mechanical									■	■	■	■	■	■	
4	**Deliverables**																		
4.1	Final Design Review Presentation	Week 1	Week 15	All	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■
4.2	Final Design Review Report	Week 1	Week 15	All	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■

Figure D-5: Combined Spring Semester Gantt Chart

Appendix E: Risk Mitigation

Risk mitigation is inherently factored into the design in all subsystems. The primary choice in this respect is that of a conventional configuration: a front-wheel drive, two motor, two caster configuration that is appropriate for the application at hand. Such configurations are common in previous years of the IGVC competition and have seen success in commercial industries, such as the MuL Technologies MARC (**Fig. E-1**) [1].

Figure E-1: MuL Technologies MARC [1]

Other, more complex configurations for Rock-E's design were considered, such as mecanum wheels with skid-steering, tracks, legs, or hovercraft. These designs are high-risk and do not yield benefits for this use case on a paved surface, so they were avoided almost immediately in discussion. Though it was not immediately apparent, a six-wheel design was ruled out only after a scale model was constructed and found to behave unsatisfactorily when turning (**Fig. E-2**). This was due to unanticipated skid-steering, which is unacceptable for use on pavement.

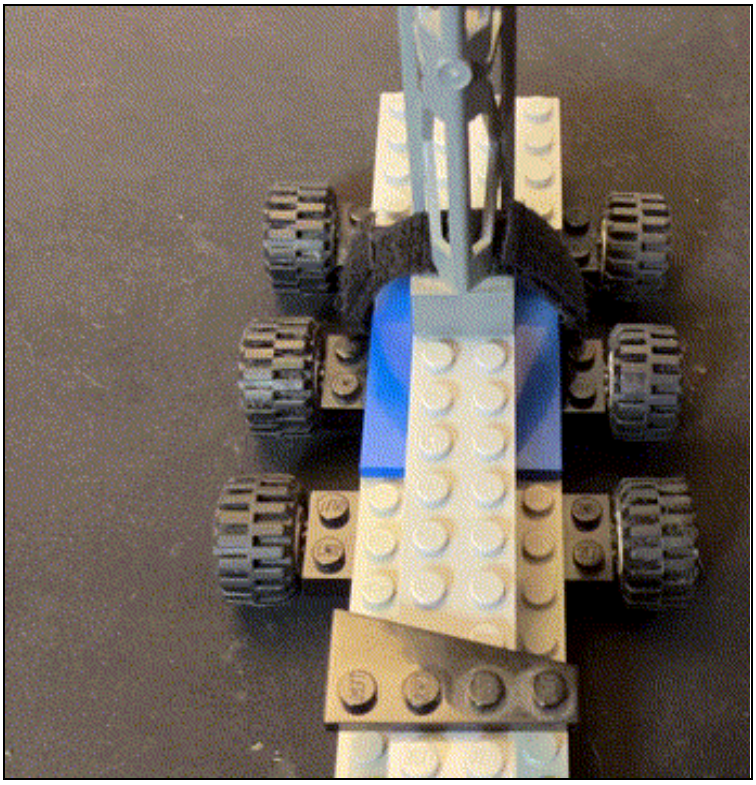


Figure E-2: Scale Model of Six-Wheel Design, Showing Skid-Steering

Further precautions have been taken in the design by allowing variability. Having considered the high likelihood of needing to modify elements of the platform in the future, it was sought to avoid making anything permanent that does not absolutely need to be. Mounting systems for both navigational and other components have been left variable. Casters have been considered and multiple options remain available in case there is need to pivot to a different approach. Internal designs at this early stage have been more along the lines of compartments or "black boxes" instead of discrete components and exact placement. Due to the high likelihood of change needed in the future, the team is more prepared to adjust accordingly than if it had attempted to define exact placement, meaning less cost to time and budget when the inevitable need to change happens.

Another aspect of risk mitigation has been the early procurement of components. For example, once the drive motors were chosen, it was sought to order them as soon as possible. Unfortunately, the supplier's (SuperDroid Robotics) website of the custom motors that was

thought to be set to order was inaccurate, and the motors were no longer being offered for sale. To recover from this, the team researched, located, and ordered separately the components that the supplier was incorporating into what they were calling their PCS-250 motor package with the intent to manufacture the same design in-house [2]. If a conscious effort was not made to contact them early, it may not have been found out until much later that they were unable to supply the chosen motors, which would have been a major setback. Having these motors in hand is good protection in case the supply dries up and it is not possible to order more (global issues are out of local control and can ruin plans without warning). This also allows the team to test and verify specifications of the purchased devices to ensure they meet needs and are behaving as advertised.

Risks to the components themselves are present when dealing with the forces of gravity and electricity, as well as fatigue in general. Rock-E's electrical system follows a cohesive strategy to minimize risks to man and machine, through devices such as grounding, fuzes, and thermal runaway protection, as well as general safety practices. The system is reasonably weatherproofed to prevent the influence of moisture, dust, and other undesirable media on the equipment, most of which could threaten delicate electronics.

To minimize communication faults during autonomous use, accommodations such as a quality E-Stop, wire insulation, and ferrite beads are implemented. To prevent physical damage, lift and pick points were chosen so that there are structurally-approved locations to grab when lifting and transporting Rock-E. The use of a suspension and other forms of insulation help prevent the components from becoming damaged via vibration during movement of the robot.

To minimize risks to the project time table and expensive materials, simulations were used to determine and check the validity of component choices and configuration. A thermal simulation allowed the team to check the cooling system before the build physically begins thus limiting the potential to overheat components. Static and dynamic structural analyses were used to check the structural design and component choice: namely of concern here is the thickness of the baseplate, suspension components, and structural members. In general, it would be far less costly to time and budget if it can be preliminarily assured the structure will not fail during use or transportation, perhaps itself leading to the damage of costly components.

Appendix F: Sketches

The design of the Rock-E robot has seen many ideas about its configuration throughout the early phases of the project. Each team member has contributed sketches to best collect thoughts during the development of the design. These sketches were used in meetings and discussions on how best to proceed with aspects of the design and are found below in this appendix.

Having already decided on a wheels-and-casters approach, the first set of sketches were submitted by each mechanical team member where it was discussed what was liked and disliked about each configuration. All members had looked into designs that had been successful at the IGVC in 2022, so most discussed were similar in form to some of the contenders [1]. After weighing the merits of each, the team decided that Ethan's configuration (**Figures F-3 and F-4**), inspired by a zero-turn lawn mower, would be the basis of the design moving forward.

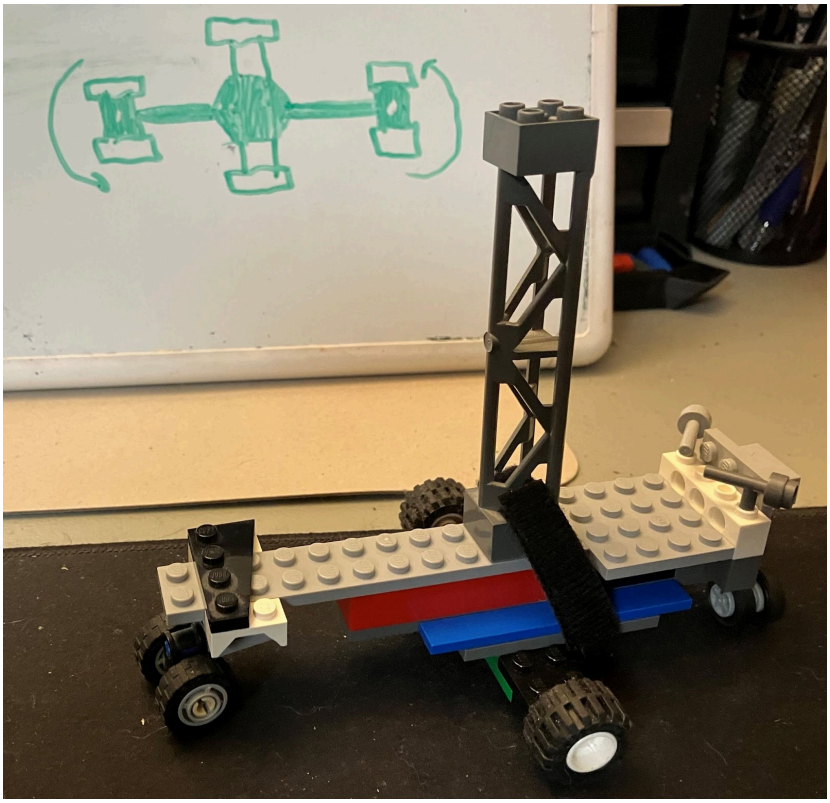


Figure F-1: Scale Model and Week 1 Configuration Sketch

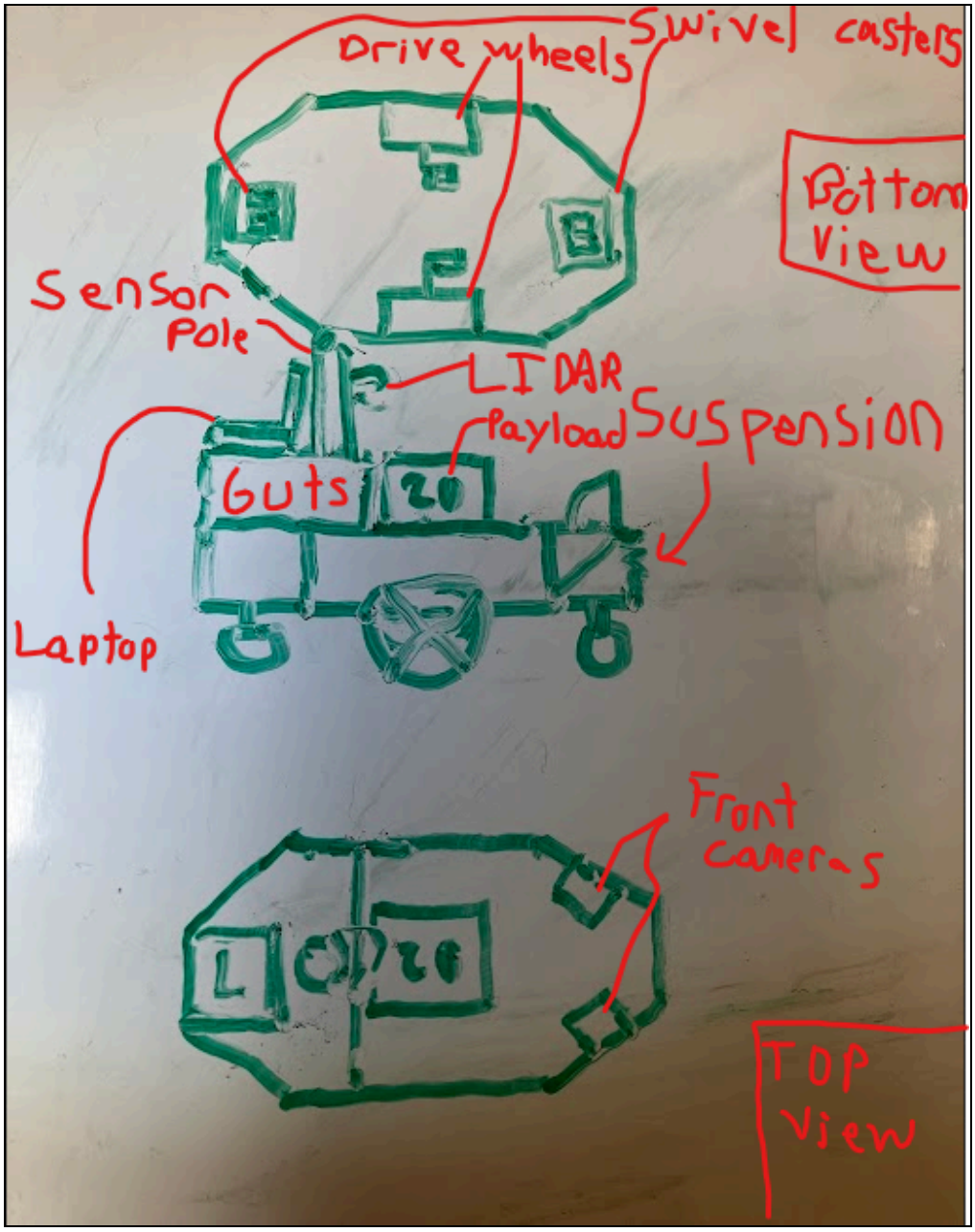


Figure F-2: Week 1 Configuration Sketch

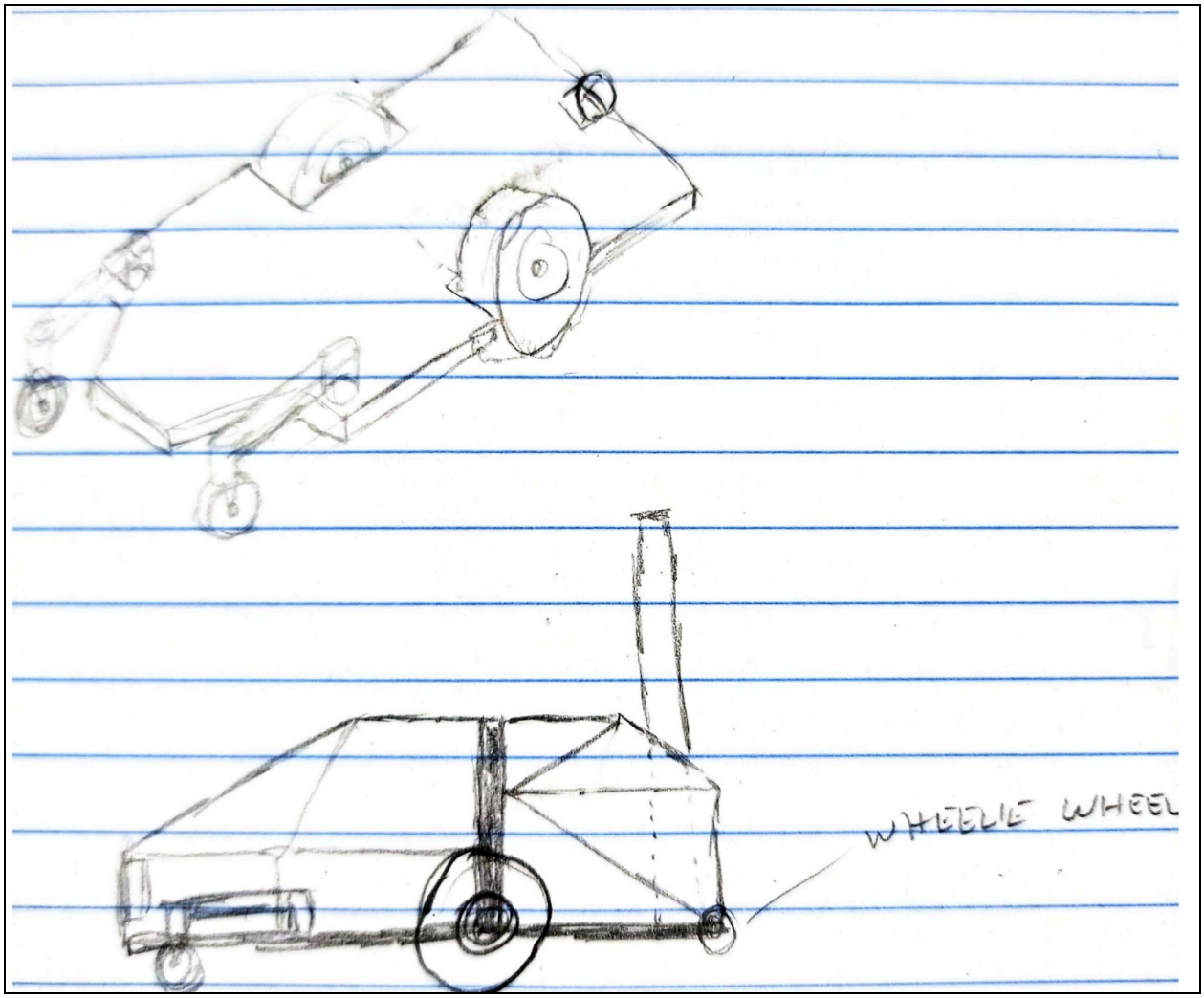


Figure F-3: Week 1 Configuration Sketch

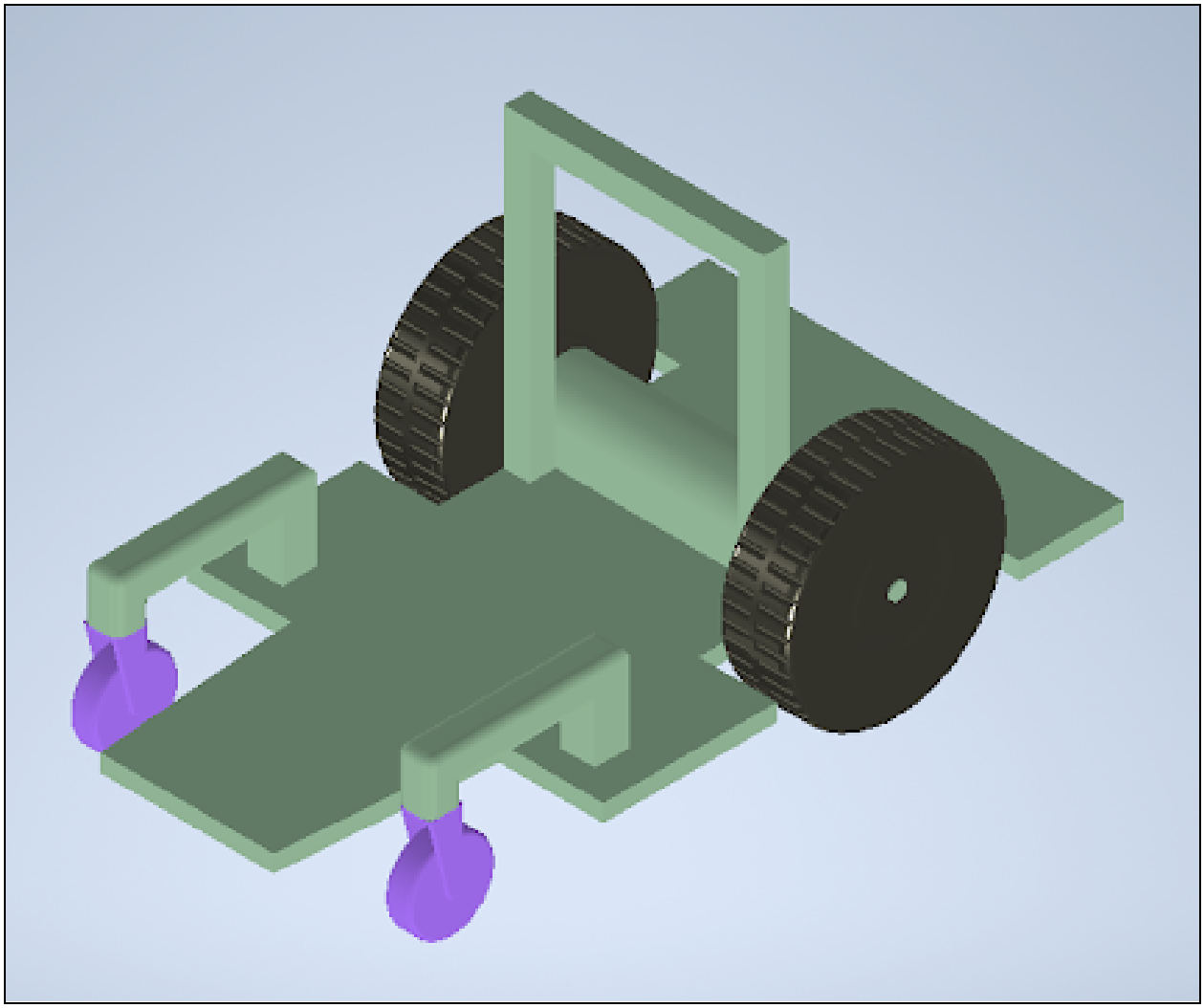


Figure F-4: Week 1 3D Design

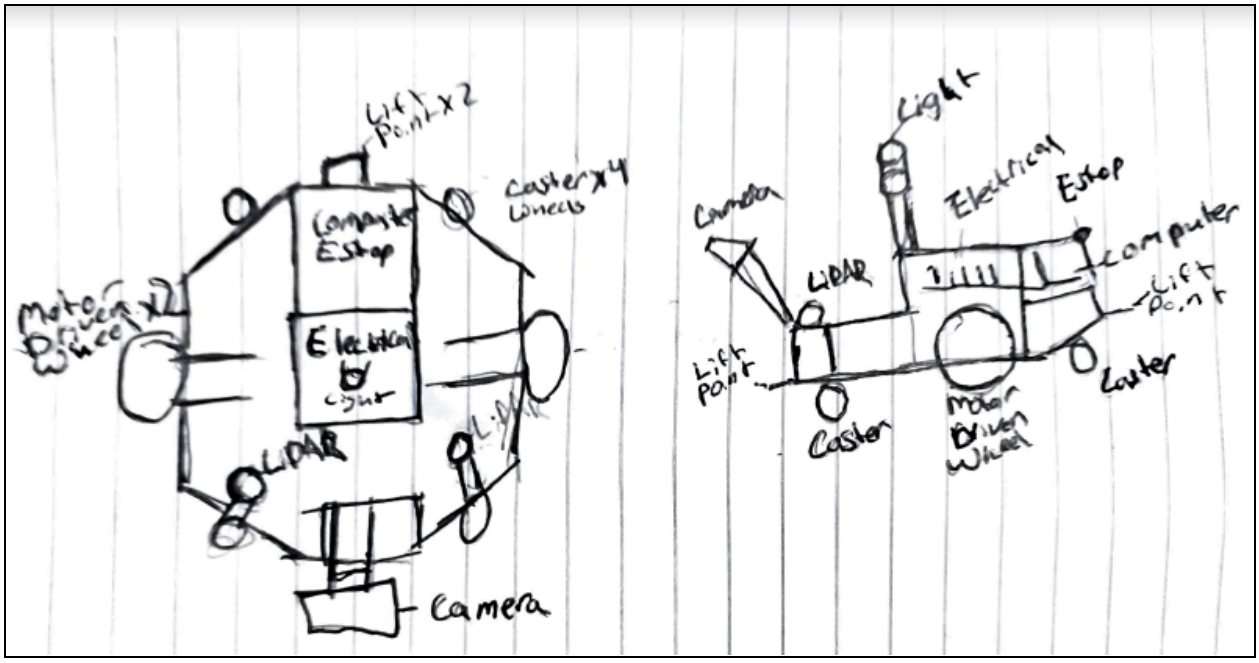


Figure F-5: Week 1 Configuration Sketch

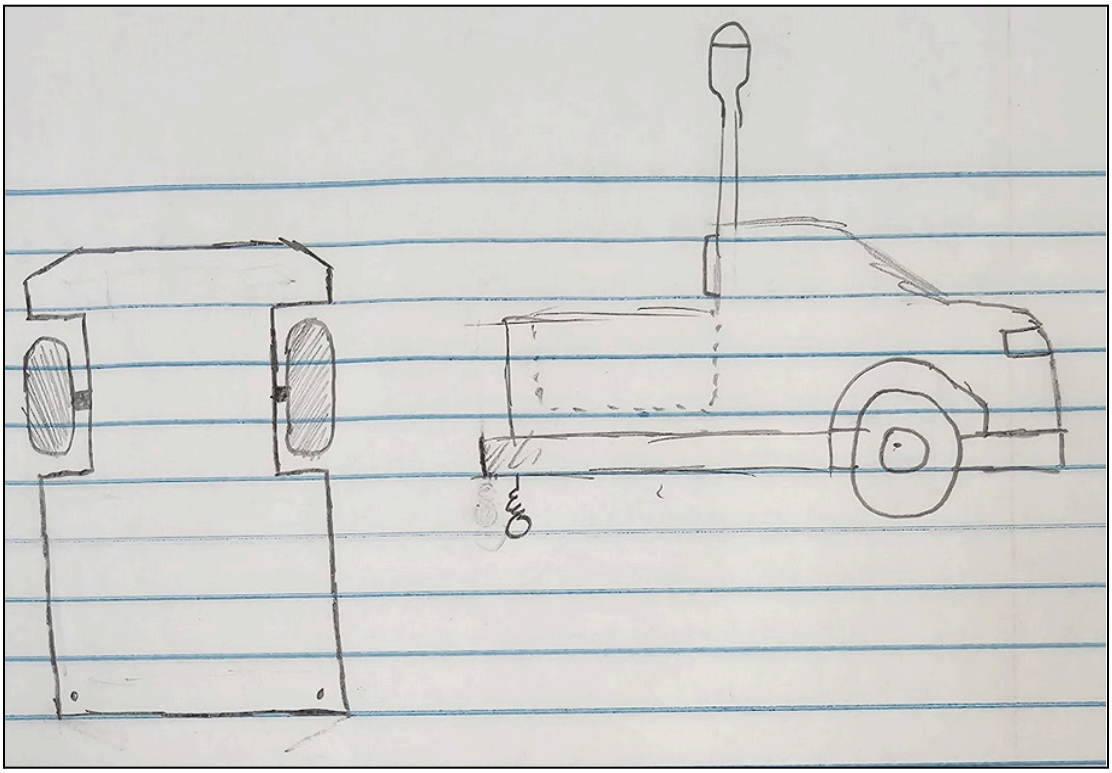


Figure F-6: Week 1 Configuration Sketch

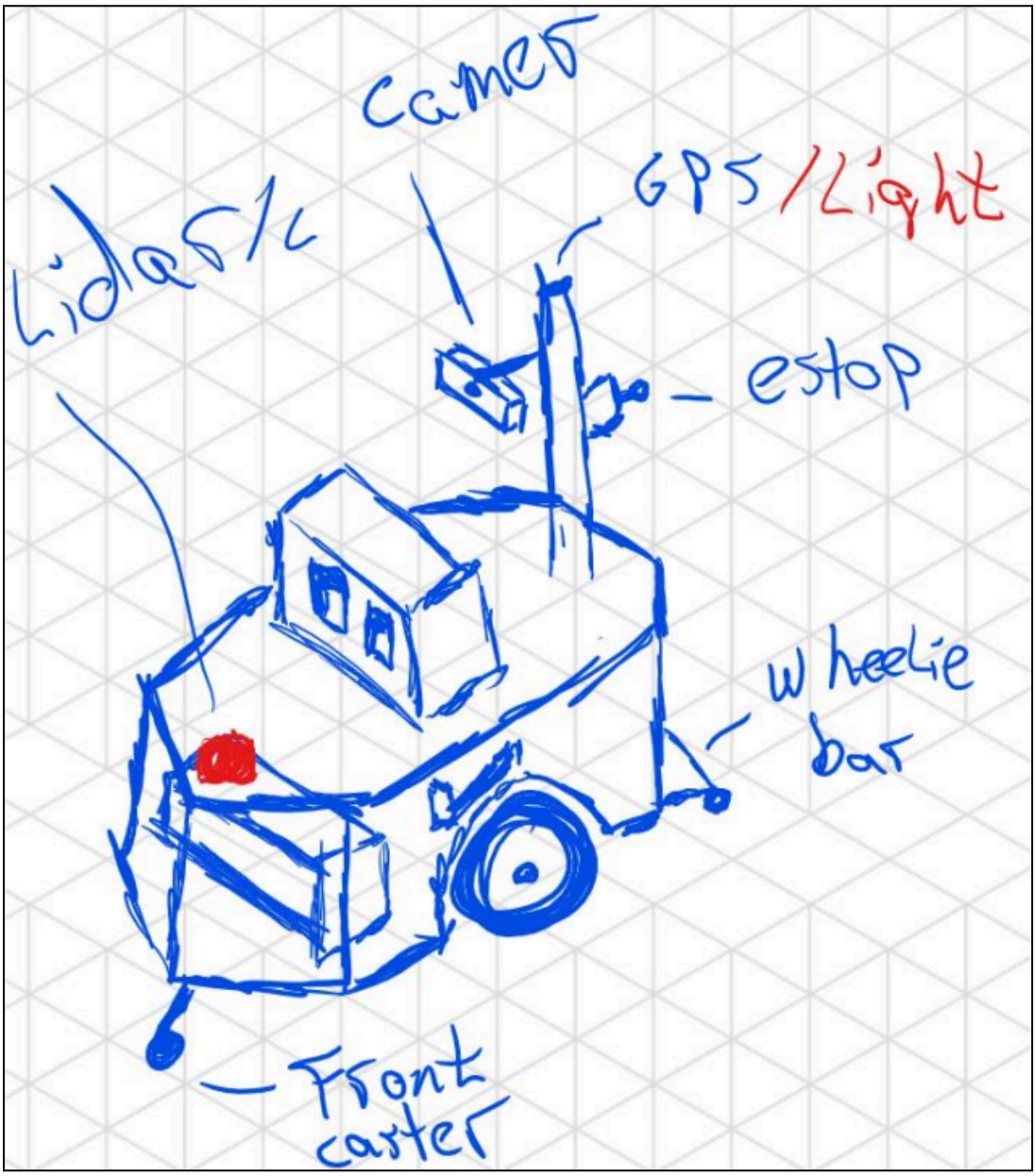


Figure F-7: Week 1 Configuration Sketch

Having decided upon a general chassis design, the team continued to iterate and discuss various design and subsystem concepts. Week two featured Brendan's use of sketches to conceptualize different options regarding component arrangement and cooling systems within the robot (**Figures F-8, F-9**).

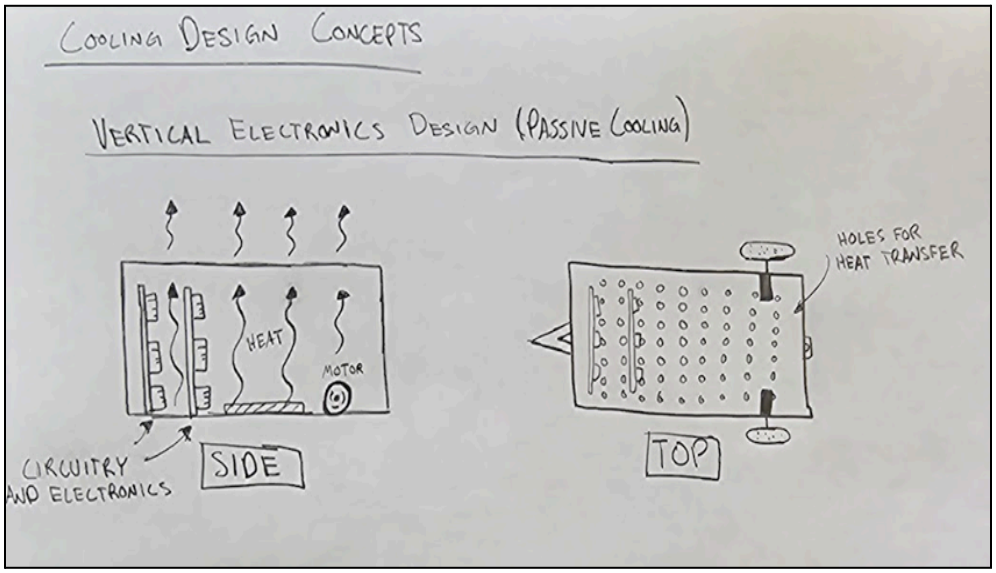


Figure F-8: Week 2 Cooling Sketch

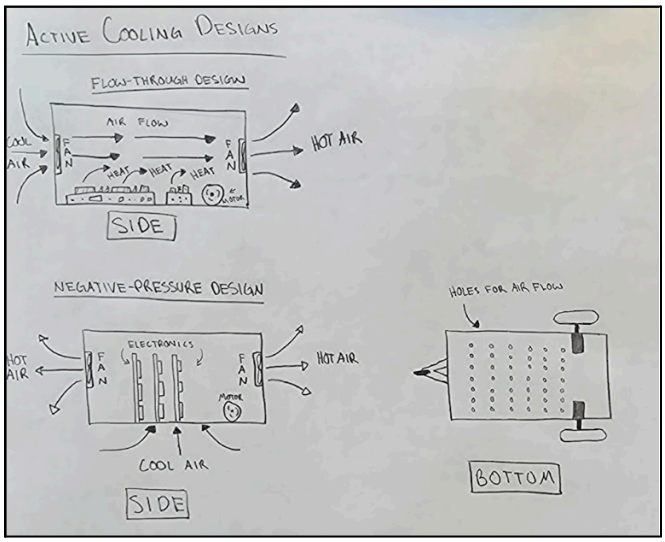


Figure F-9: Week 2 Cooling Sketch

Week three saw the further development of ideas in the suspension and chassis by Ethan in the form of sketches (**Figures F-10 and F-11**) and CAD drawings (see Appendix G), as well as the beginning of some electrical block diagrams by Justin (see Appendix L).

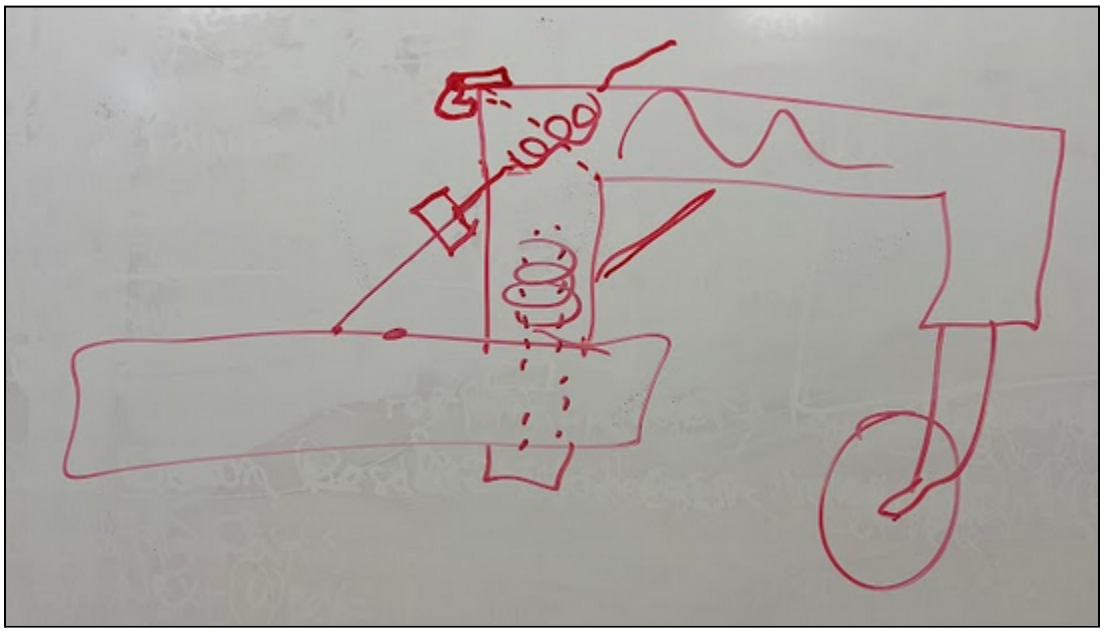


Figure F-10: Week 3 Suspension Sketch

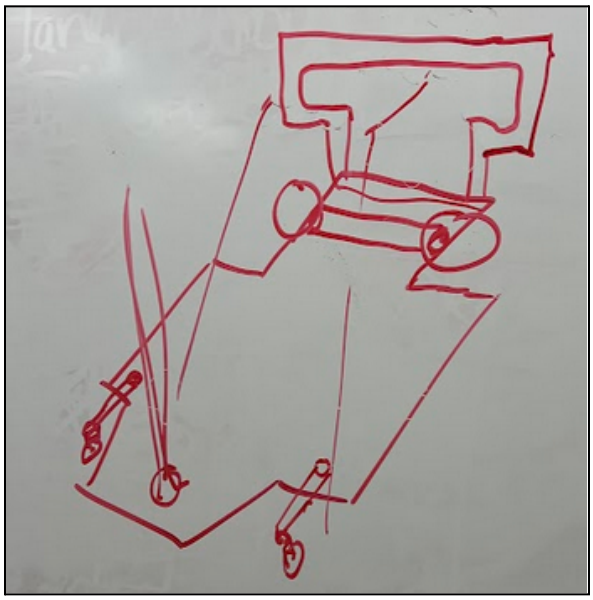


Fig. F-11: Week 3 Chassis Sketch

Weeks four and five featured sketches in use for an electrical diagram by Brendan and Justin (which was refined, see Appendix L). As most of the ideation phase was passed, many contributions hereafter were not demonstrable via sketches.

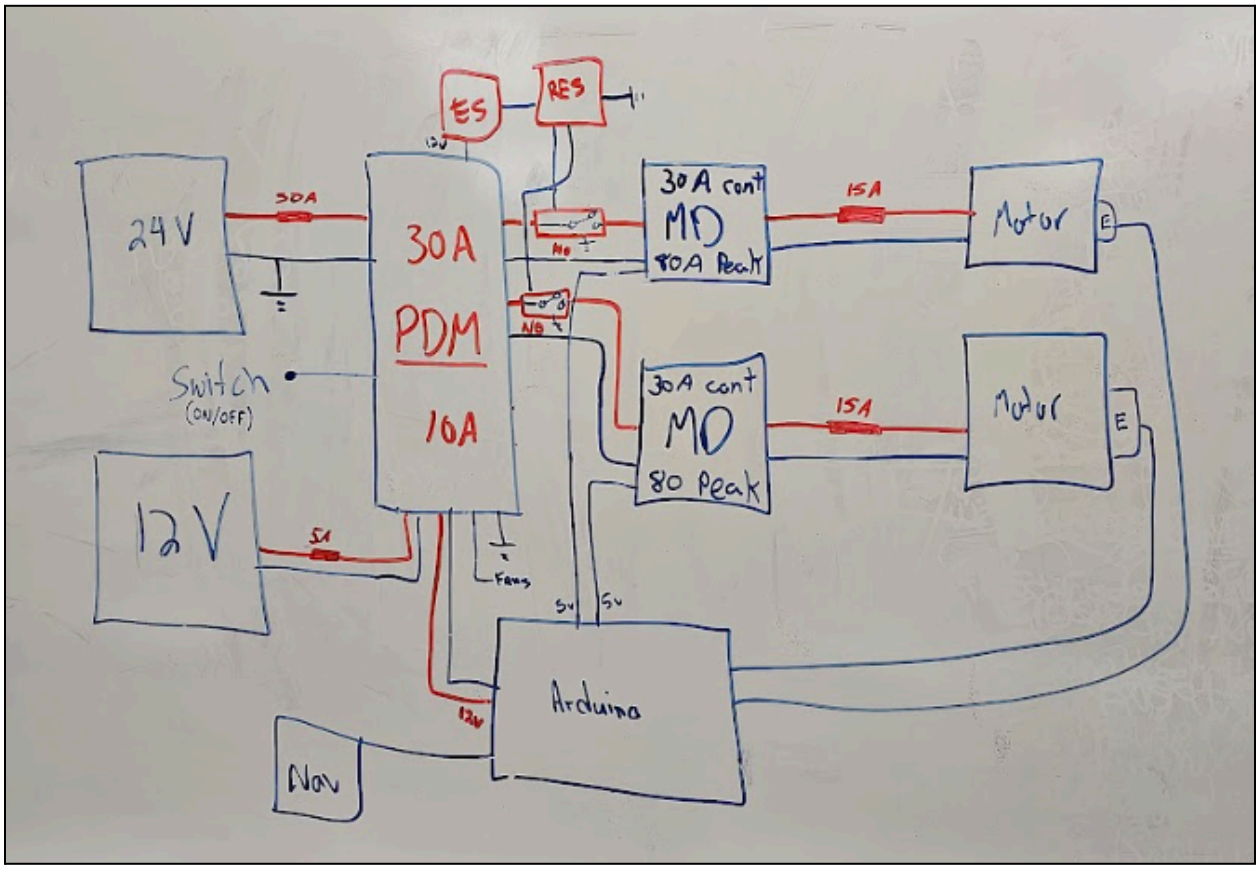


Figure F-12: Week 4/5 Electrical Sketch

Appendix G: Mechanical CAD

The design that was chosen for the final design was a product of several iterations. In the initial brainstorming phase all submitted ideas on the form Rock-E should take and considered several types of propulsion and forms of the vehicle (This is discussed in further depth in Appendix F). After all team members submitted their ideas a vote was conducted on which design to use as a basis for the CAD design moving forward; that design can be seen below (**Figure G-1**).

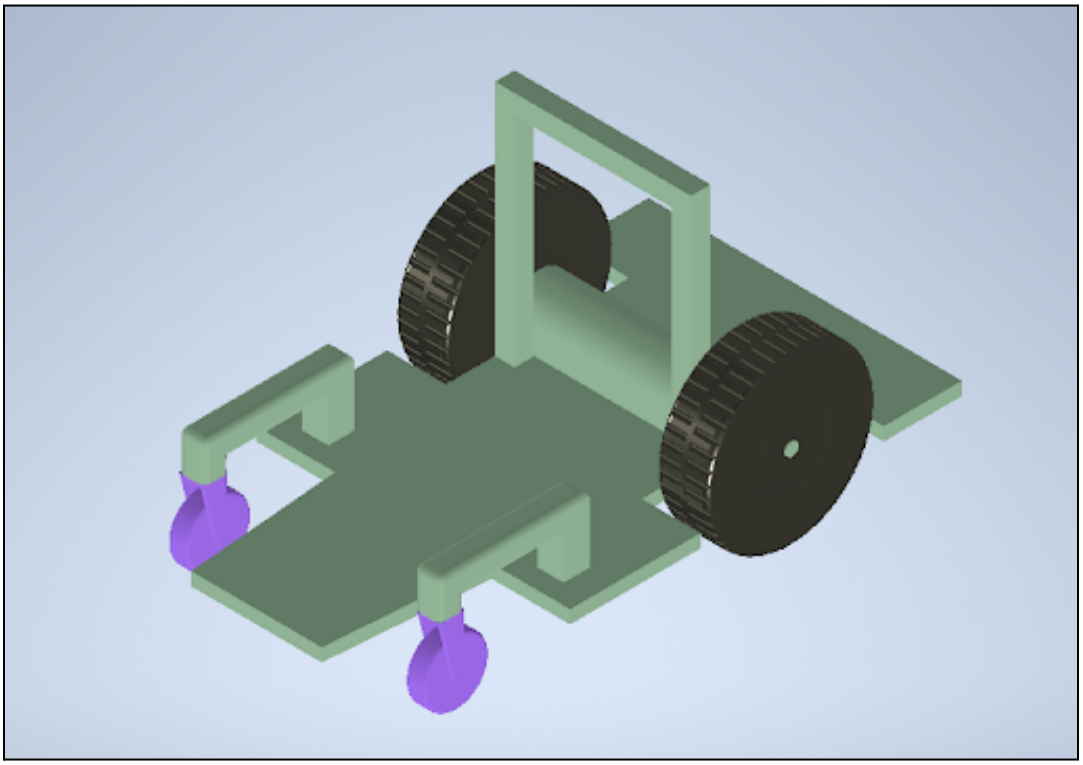


Figure G-1: Initial CAD Design

The design uses two main drive wheels that are each driven by separate wheelchair-type motors and supported by two caster wheels for a nice, stable quad-wheel design. The inspiration was taken greatly from the drive system of a zero turn lawn-mower with the drive wheels located at the rear of the vehicle and the casters at the front. However, in the planning phase it was decided to flip the design and have the drive wheels at the front of the vehicle rather than the rear. This change was because all of the sensory gear was planned to be mounted to the front

region of the vehicle and it was desired to eliminate the need to incorporate the distance between the sensor devices and the drive wheels in any programming, avoiding the sort of lagging such an arrangement would require adjusting to.

Over the course of the design process it was decided to use a ¼" aluminum plate as the base of the vehicle and the main source of the vehicle's rigidity. The needed framing on the vehicle would then be provided by the T-slot extruded aluminum and would be covered in a combination of polycarbonate and 3D-printed materials. It was also decided to mount the casters on a suspension system that would be fabricated out of mainly 3D-printed materials. The initial suspension assembly design can be seen below (**Figure G-2**).

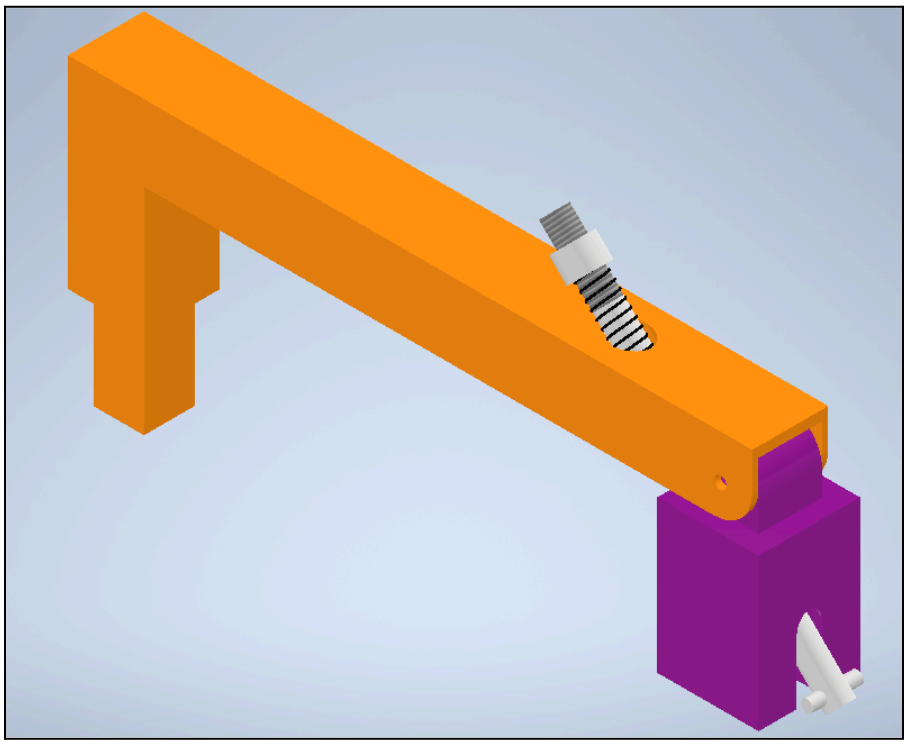


Figure G-2: Caster Suspension System Genesis

After synthesizing the design ideas the next phase of the design was created as seen below (**Figure G-3**). The design was two feet by three feet (the minimum size allowed by the IGVC), and had a ¼" aluminum plate as the base. The forward frame is to provide a mounting point for the LIDAR and the rear pole is to mount the light and E-stop. Also, a design for a

central point for mounting the camera was created. The hinged top is able to lift open and thereby access the electrical box at the center and the computer which will sit on the front shelf. The cinder block was located at the rear.

Figure G-3: Proposed Final Design

After reviewing the proposed design a few adjustments to the layout of the vehicle were suggested by the team members (These modifications are discussed in further depth in Appendix N). The final PDR design swapped the locations of the cinder block cargo and the electronics box, removing the central camera mount in favor of a mount on the forward frame, and a few small adjustments were made to the shelf for the laptop and its support frame (**Figure G-4**).

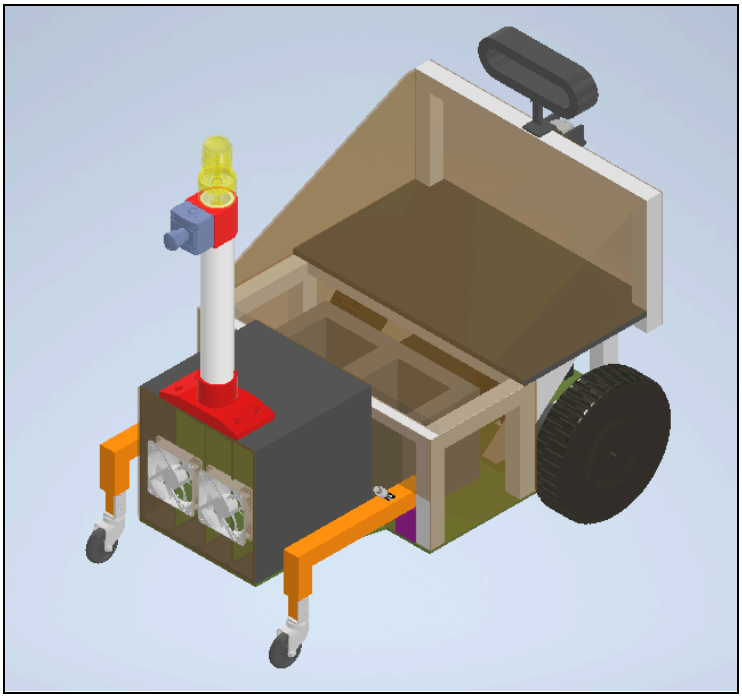


Figure G-4: Final PDR Design

Moving into the CDR report, several small changes were made to the PDR design. The overall shape and idea of the CAD design remained the same but some components were replaced and others were added.

One of the first changes to the CDR mid-fidelity prototype was the switch to the metric system; this was done in order to transition from an unwieldy mix of the Imperial system and a few metric pieces. Unfortunately for the timetable, this effort required a complete rebuild of all the parts. Though somewhat time consuming this decision was made to standardize the measuring system across the project, use a more practical measuring system for the task at hand, and would likely have to have been completed later regardless.

During the rebuild some new design plans were made and greater detail was added by now including all the joints and hardware, taken from imported CAD models from McMaster-Carr. This helped to determine just how the framework would be put together and add necessary refinement required from the mid-fidelity prototype.

Another aspect that was changed was the simplification of the suspension system. The original multi-part design was determined to be far too flimsy at the joint with a large amount of lateral deflection when tested. Instead, a solid arm design was made (**Figure G-5**)

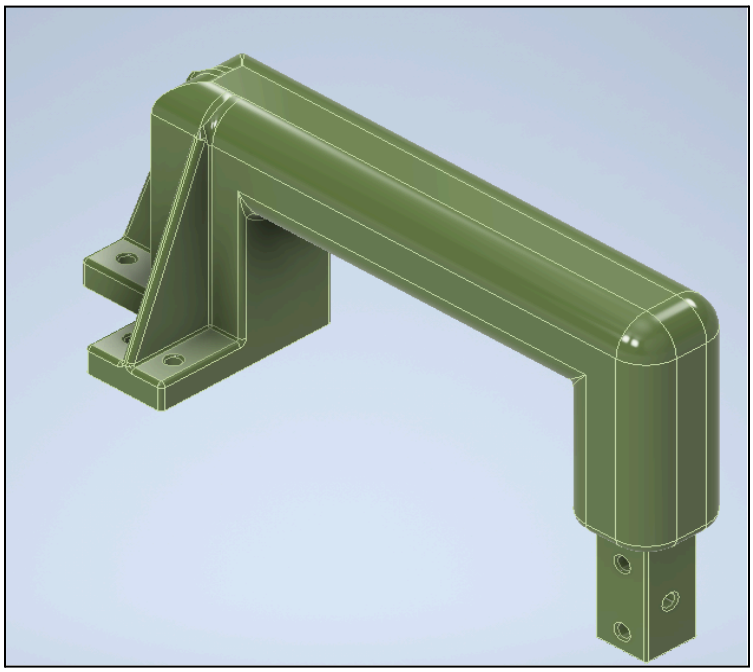


Figure G-5: Solid Suspension Design

This new design had much less deflection and was considered adequate enough for use as a final design. However, it was decided that the solid arm may have insufficient vibration mitigation and a new variant was dreamt up. This new model retains the solid arm design as the majority but uses another piece as a mount to the actual caster. This additional part would be inserted vertically into the main arm tube along with a stack of additively-manufactured springs which will provide an adjustable level of shock absorption. This new and improved suspension design can be seen in **Figure G-6**.

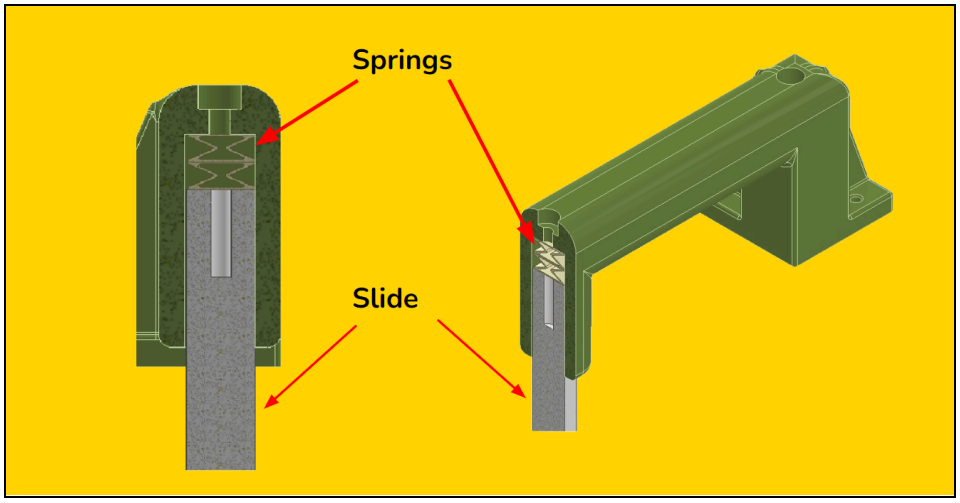


Figure G-6: Solid Suspension Arm with Slide and 3D-Printed Springs

When the suspension arm design in figure G-6 was tested, the spring setup was working well but the arm itself was still deflecting too much. This would cause Rock-E to lose position in the rear of the vehicle and need to be corrected by the central processor. To overcome this issue the suspension arm was redesigned to be much shorter than the original (**Figure G-7**). With this design change the suspension is now performing to required specifications.

Figure G-7: Final Suspension Arm Design

Another change that was made was the addition of an encoder mount to the motor mounts. The new external encoders that were chosen utilize a belt-driven system; therefore another mounting system needed to be added to the motor mounts that could mount the encoder close enough to the motor's drive shaft to allow a belt to be connected across a gear mounted to

the shaft of each. A face-mounted fitting will be attached to the encoder itself and the fitting then will be attached to the motor mount by an adjustable slot system (**Figure G-7**).

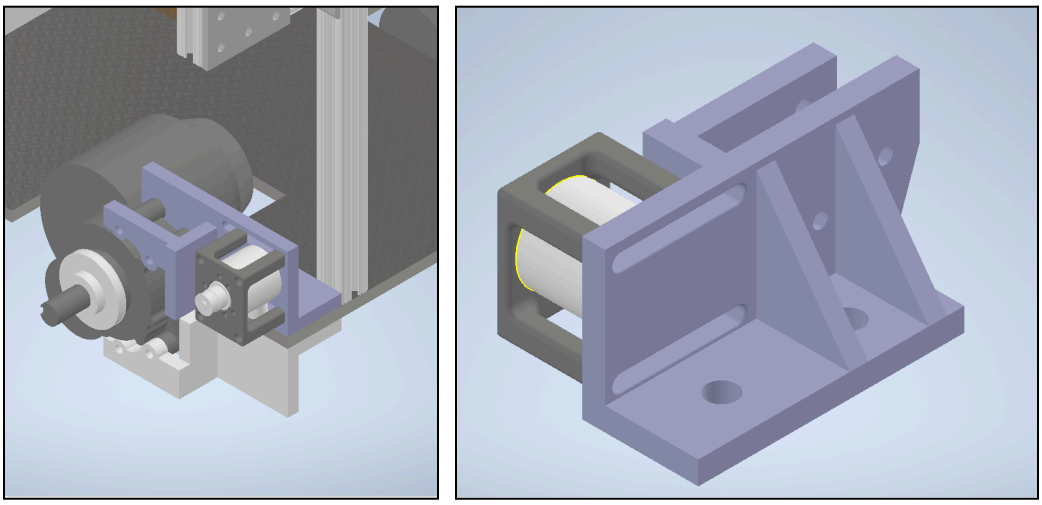


Figure G-7: Encoder Mounting System

Other small changes were made to the design at this stage, such as the removal of several plexiglass panels, notably around the cargo area. It was decided that since the only cargo that was being transported was a cinder block, there was no need for it to be protected from the elements. The laptop hood was also shrunk for the same reason. These omissions simplify the build process and cut back on overall materials needed therefore reducing overall cost and weight. The camera mount was modified from the original design since it was found to be too large and was raised up to over a meter above the ground per a request from the navigation team to assist visibility. Other small changes were also made but they were merely design choices, not providing any notable advantages. The final design of the CDR, the mid-fidelity prototype, can be seen in **Figure G-8**.

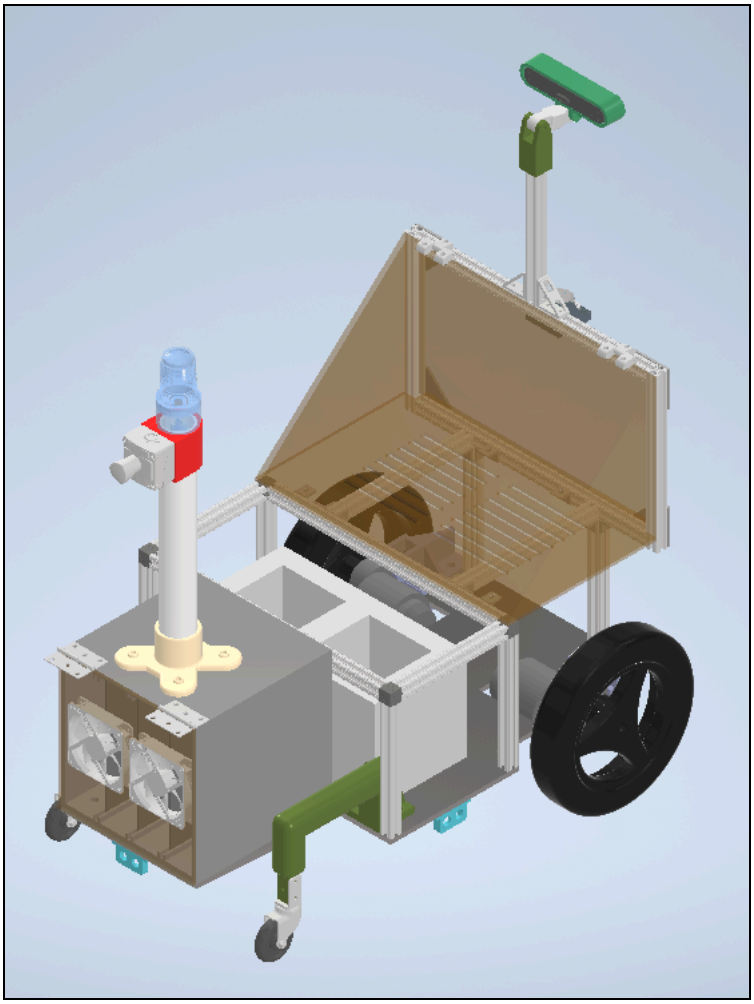


Figure G-8: CDR Final Design

The sensory equipment used by the navigation team needs to be securely mounted to the chassis so it maintains its calibration. Another factor to consider in the design of the mounting methods is that the camera's angle will need to be adjusted as the navigation team fine-tunes the lane and pothole detection needed to navigate the course, though it should be resilient enough to remain calibrated. The mount was designed so that the angle in the vertical direction can be adjusted (**Figure G-9**). The camera is secured directly into the mount using two M3 screws which are designated as the mounting screws to use in the datasheet from the manufacturer.

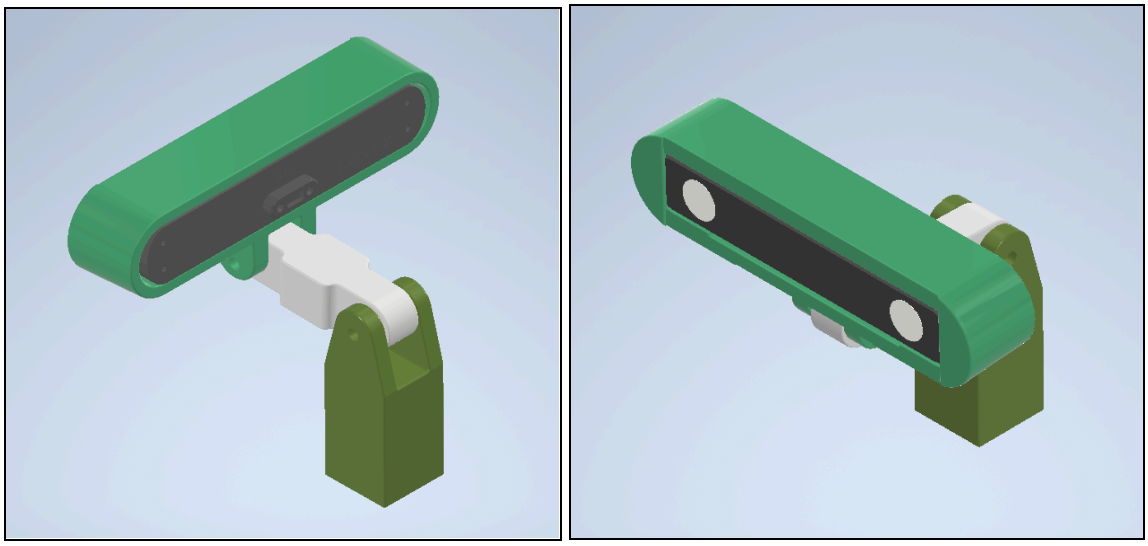


Figure G-9: Camera Mount

The ultrasonic sensors intended for use on the sides and rear of Rock-E also needed a proper method of mounting (**Figure G-10**). These sensors are exposed to the elements and lack proper weatherproofing so protection from light rain was also needed in the design. This mount anchors the ultrasonic sensors directly to the mounting bracket and the bracket is mounted to the chassis frame via double-sided tape and zip ties. While looking for specifications for the ultrasonic sensors, some dimensions were found to not be in agreement on different sources, so some of the dimensions were measured by hand using a caliper.

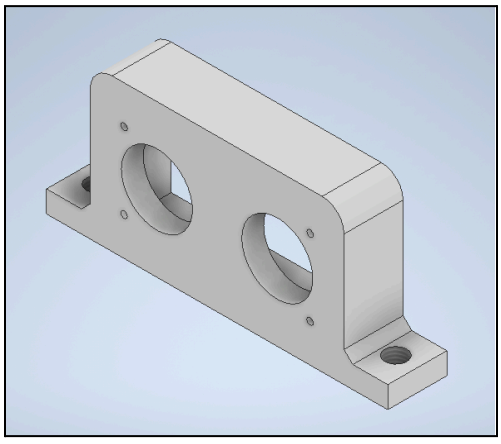


Figure G-10: Ultrasonic Sensor Mount

As development progressed into actual construction of the vehicle, the final form of the robot slowly came together. Multiple changes, large and small were made to improve the vehicle as well as its manufacturability. Some of those changes included the electronics box which was broken into multiple pieces in order to enable its ability to be printed on the 3D Printer.

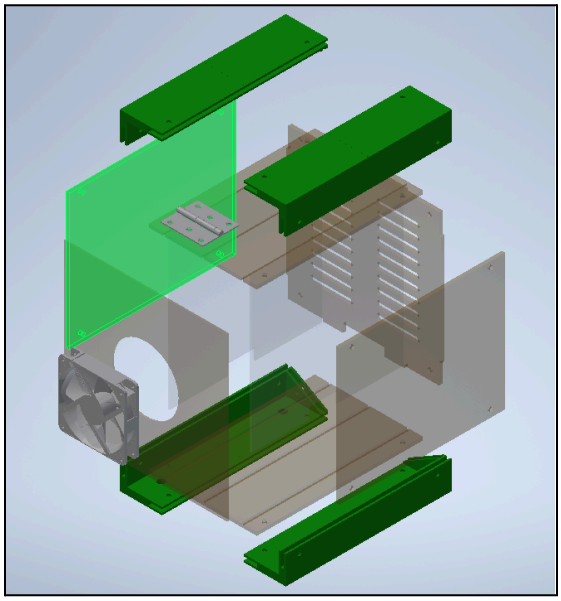


Figure G-11: Exploded View of Final Electronics Box Design

A new design for the Light Pole and its base was also developed that shortened the overall pole and added a new light mount and GPS mount.

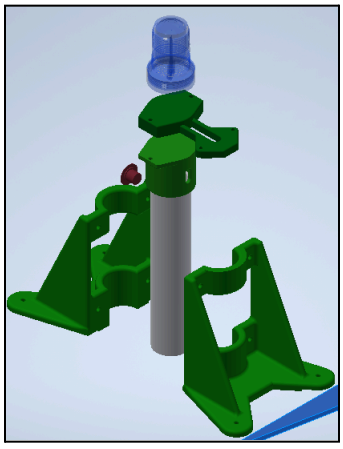


Figure G-12: Exploded View of Light Pole Design

These changes along with several others; including added fenders, motor driver cases, changes to the laptop cover, etc., resulted in the final form of the robot shown below.

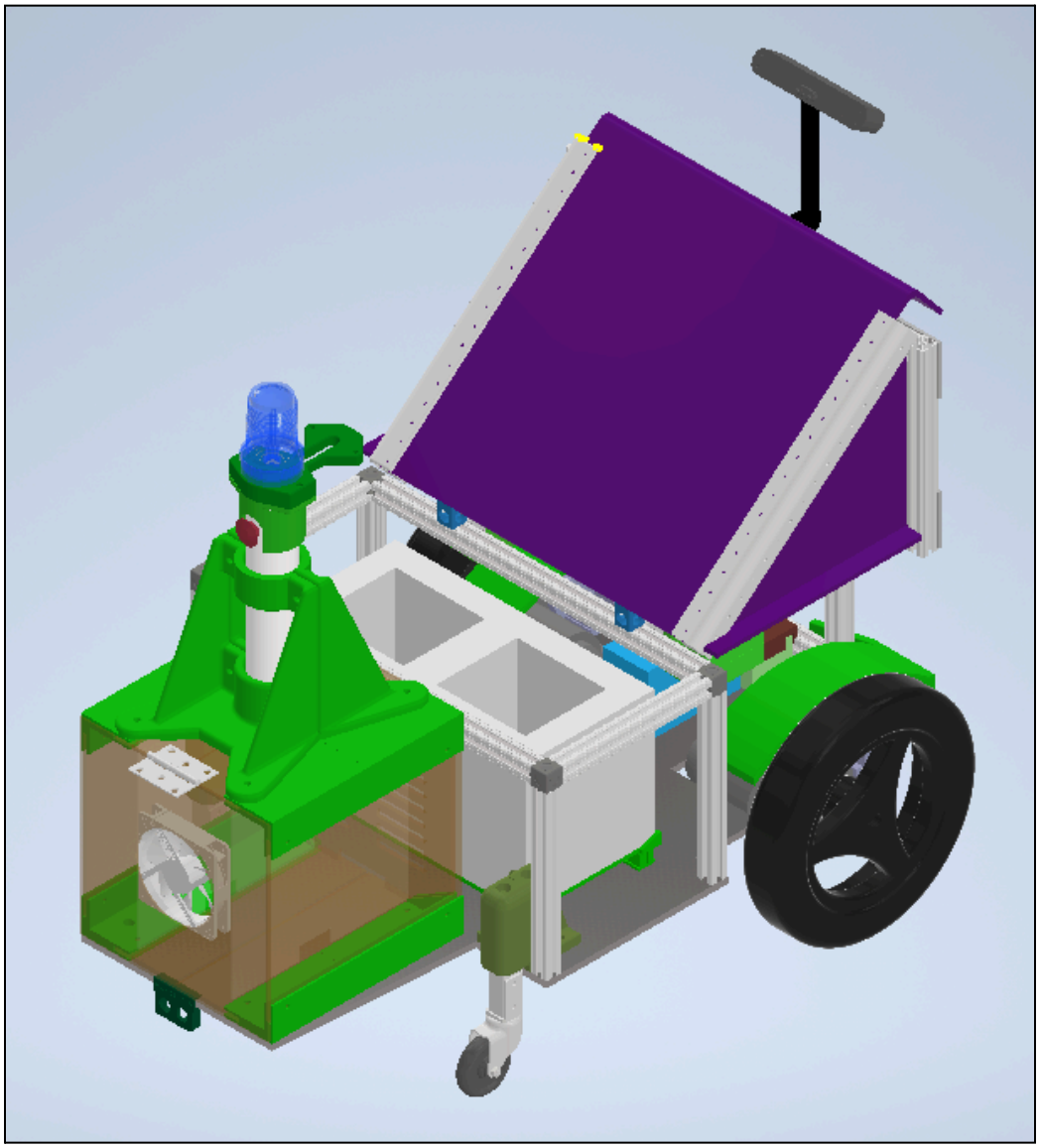


Figure G-13: Final CAD Design

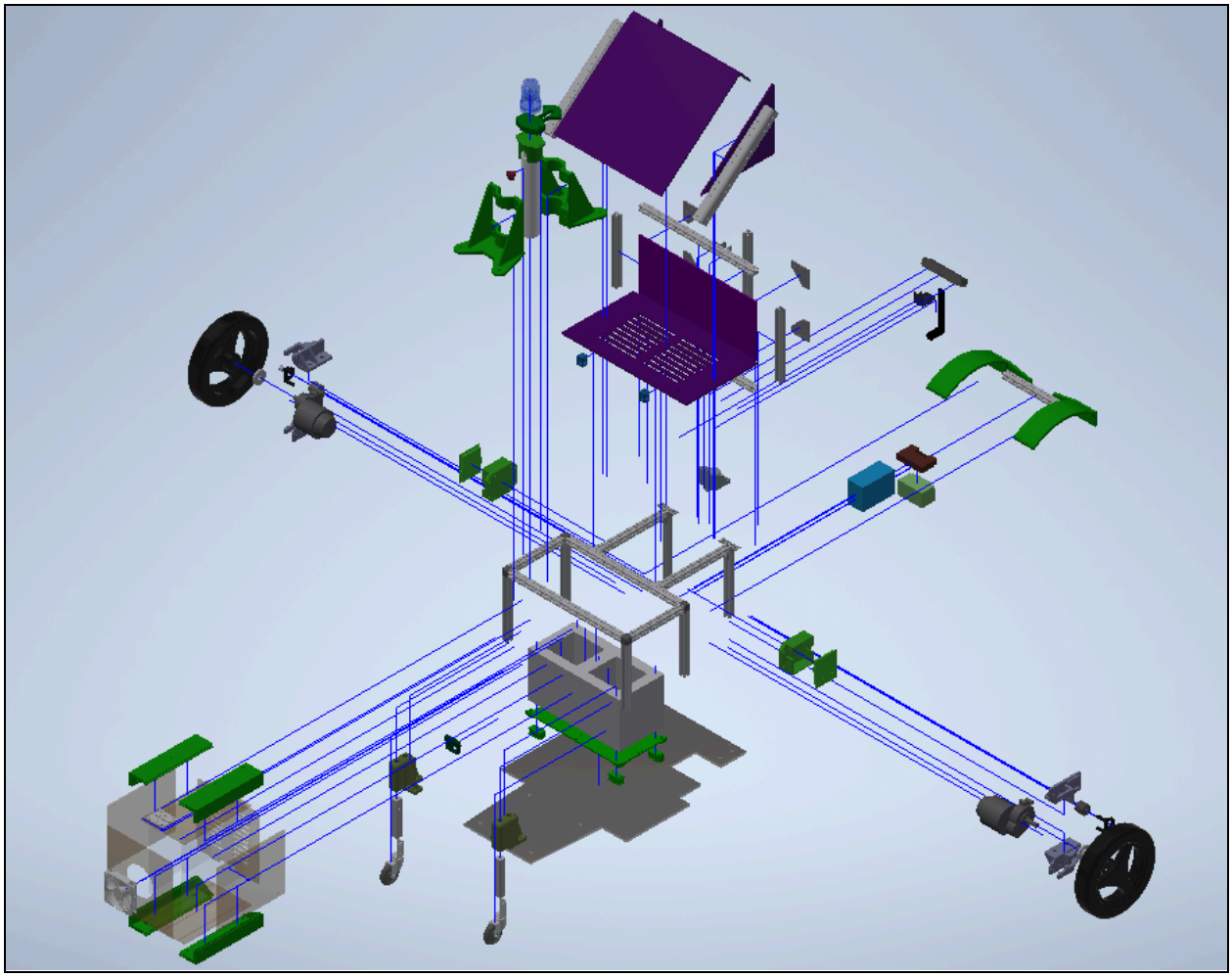


Figure G-13: Exploded View of Final CAD Design

Appendix H: Structural Analysis Simulations

To confirm the frame would be able to handle the load, Autodesk Inventor was used to simulate the forces that the frame would be subjected to. In order to ensure simulations were completed in a reasonable time frame some changes were made to the model in order to simplify the geometry, (though the model's accuracy is not sacrificed). The t-slotted aluminum was replaced with solid bars of aluminum and the model was reduced to just the frame. The bolt holes where the wheels attach at the motor mount and suspension arm were used as the fixed points. The forces applied were a 444.8 N load to the base and a 66.7 N force to the laptop shelf. The actual force applied by the laptop was 13.7 N which is well under the assumed load. This was to assume some components would be on the heavier side of possible options and to give a minimum factor of safety of 2. Under these conditions, the maximum displacement found was 0.9253 mm and the maximum stress was 9.881 MPa (**Figure H-1**) . The frame will be able to withstand the loading it will be subjected to as the base is rated for 240 MPa and the displacement is negligible under a far greater load than it will be subjected to in real applications.

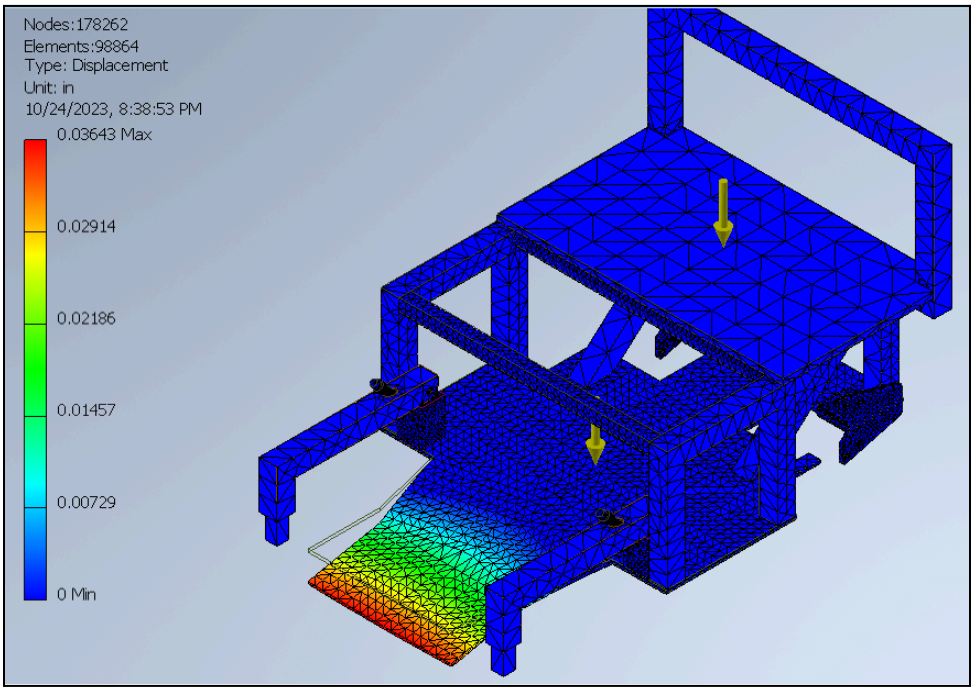


Figure H-1: Displacement in Static Structural Analysis

Dynamic Simulation

A dynamic was conducted to verify the strength of the suspension. Two versions of the simulation were run. One with the material properties of PETG and one with the material properties of carbon fiber reinforced Nylon. First a sine wave track (**Figure H-2**) was constructed to simulate rough terrain. Next it was necessary to properly constrain and join all the materials on ROCK-E together. It was important to establish 3D contact points between all the wheels and the track so the vehicle and the track would interact properly. After setting up the simulations the motion was established giving the vehicle a velocity of 4.8 kilometers per hour. Those results were then exported to FEA within Autodesk Inventor which can be seen in the following figures. With the PETG a maximum displacement of 14 mm (**Figure H3 & H4**) was recorded while the carbon fiber reinforced nylon only saw a maximum displacement of 0.58 mm (**Figure H5 & H6**). Both underwent the same amount of stress with the main areas of concern being where the suspension thinned around where the casters would attach (**Figure H-7 & H-8**). This area underwent a stress of around 127 MPa. Ultimately the length of the suspension arm would be decreased and the point where the casters attach would be made into a separate part to reduce this stress.

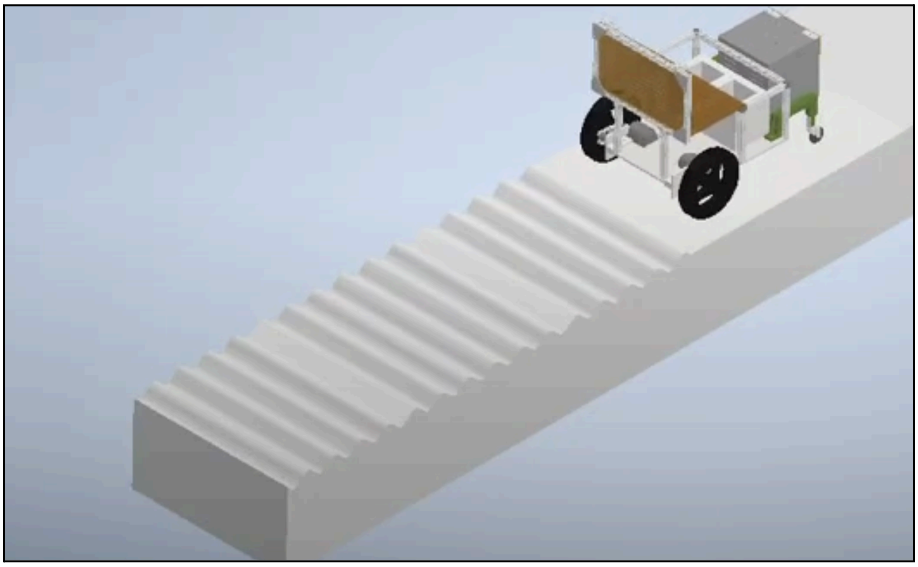


Figure H-2: Dynamic Simulation Track

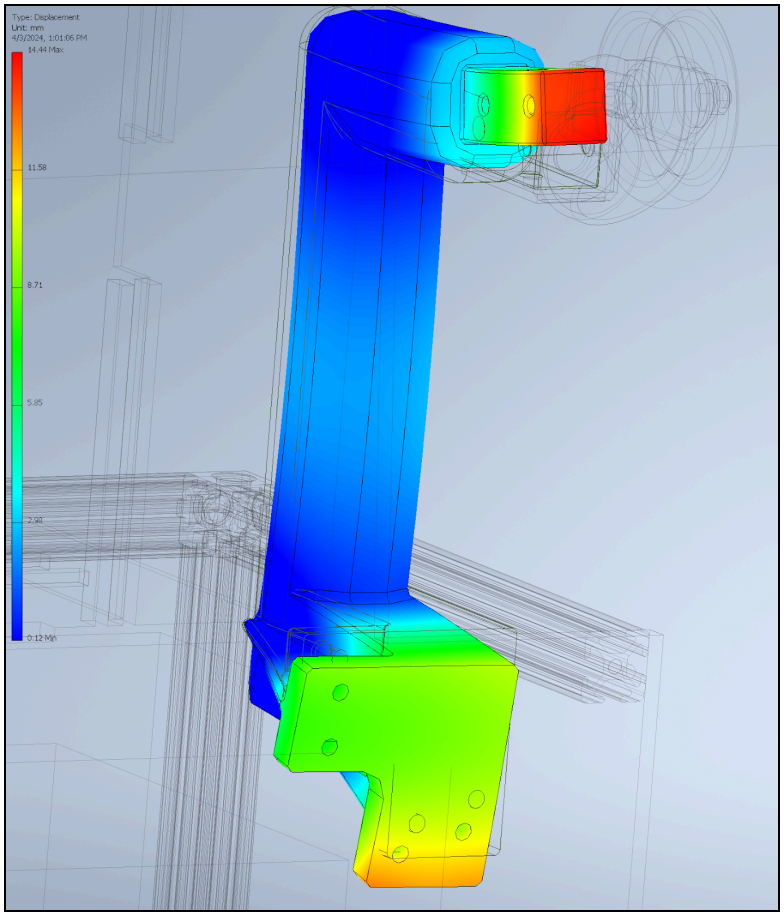


Figure H-3: Dynamic Sim PETG Displacement Bottom View

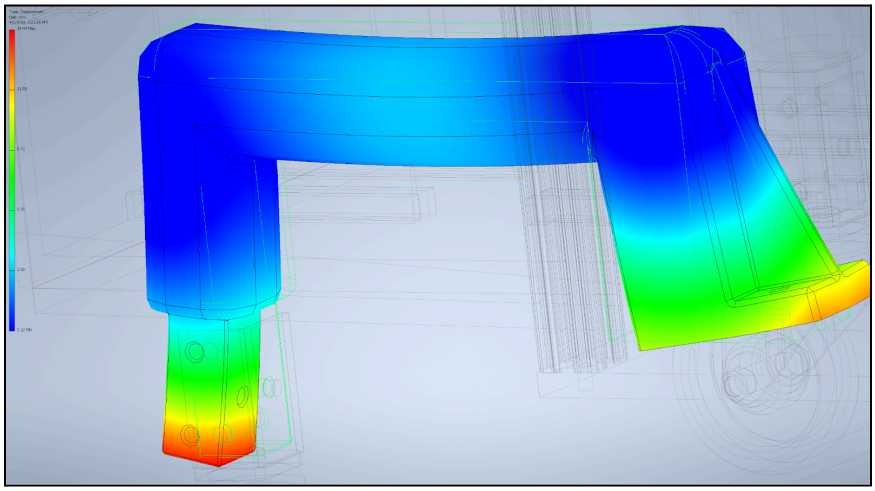


Figure H-4: Dynamic Sim PETG Displacement Side View

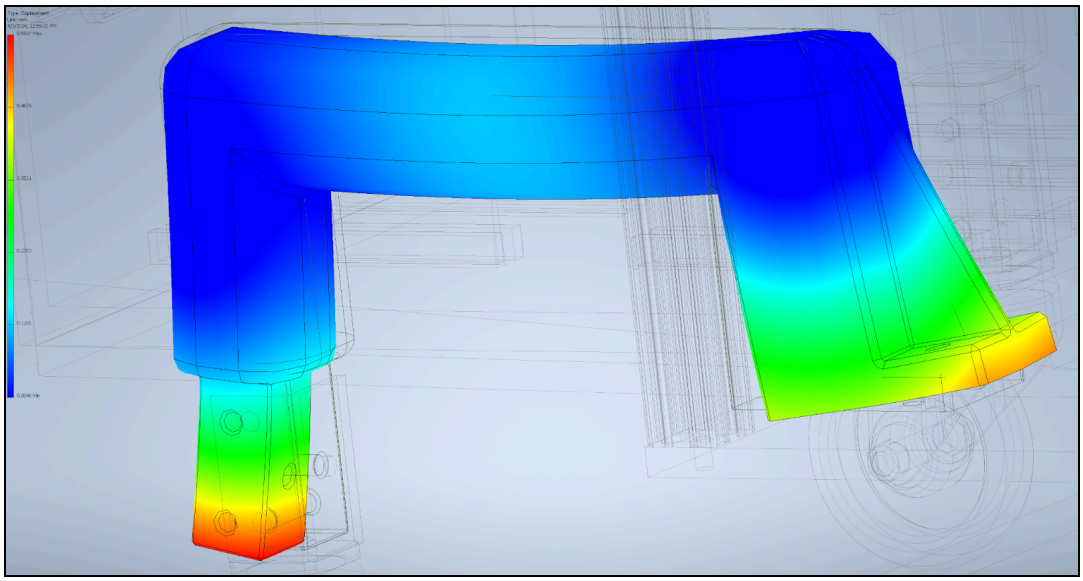


Figure H-5: Dynamic Sim CF reinforced Nylon Displacement Side View

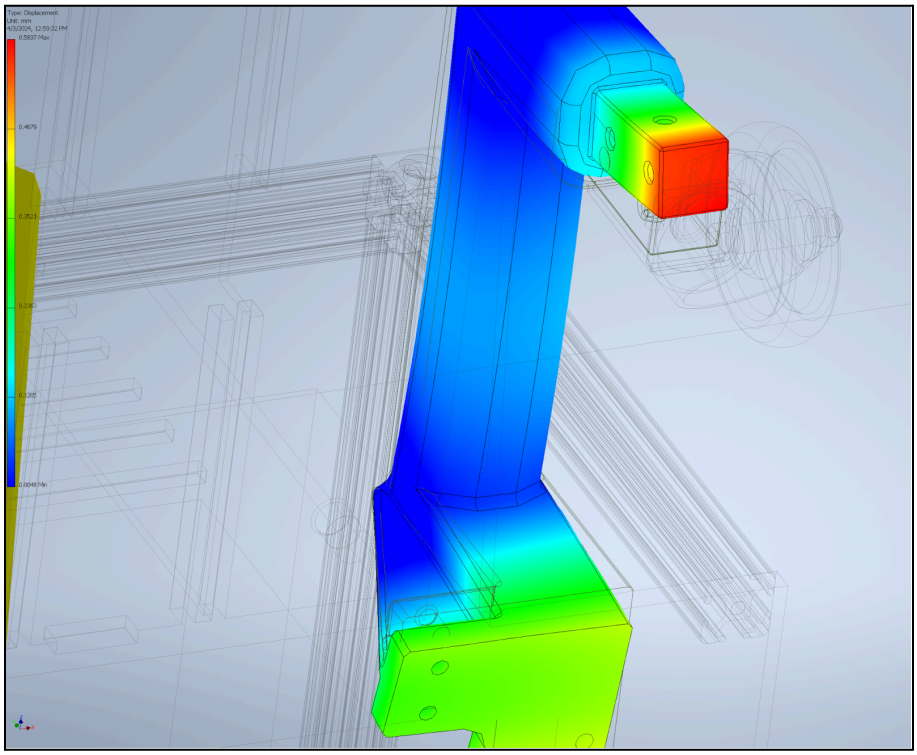


Figure H-6: Dynamic Sim CF reinforced Nylon Displacement Bottom View

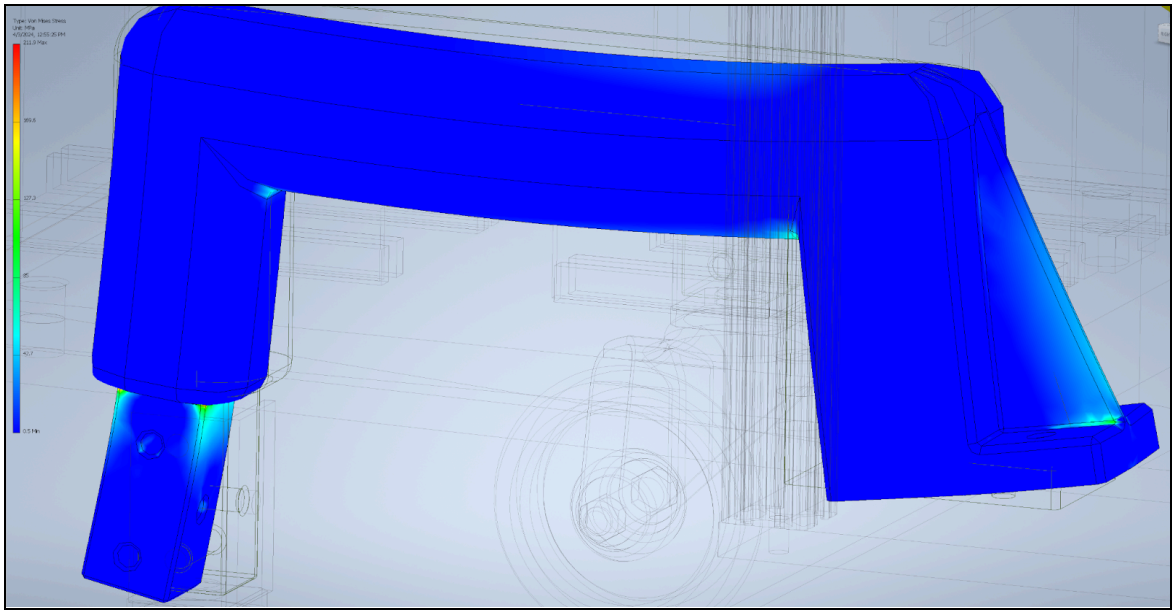


Figure H-7: Dynamic Sim CF reinforced Nylon Stress Side View

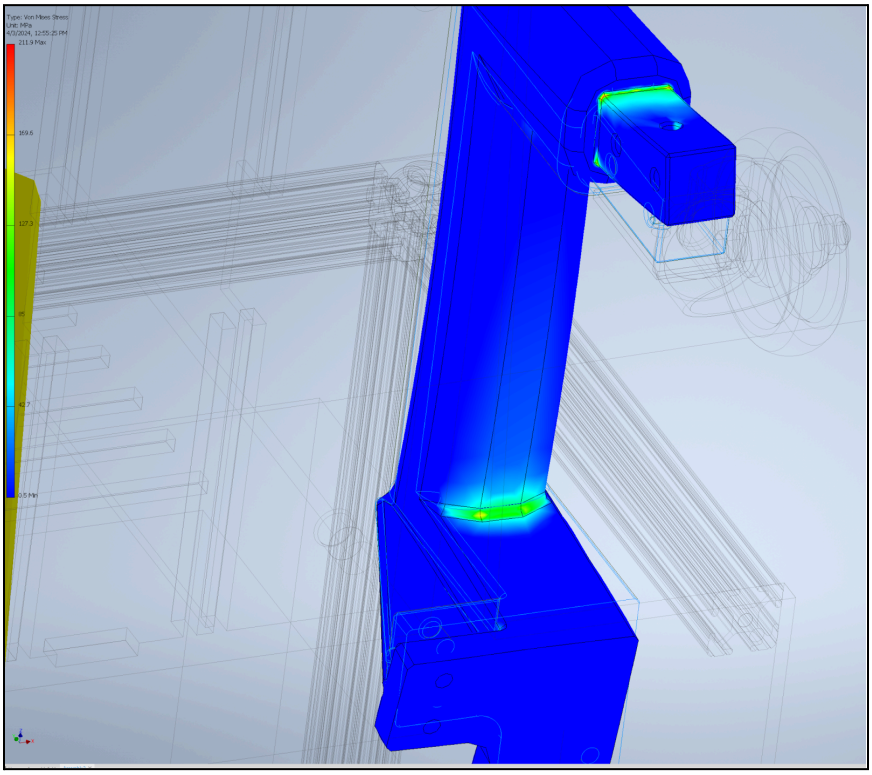


Figure H-8: Dynamic Sim CF reinforced Nylon Bottom View

Appendix I: Heat Simulation

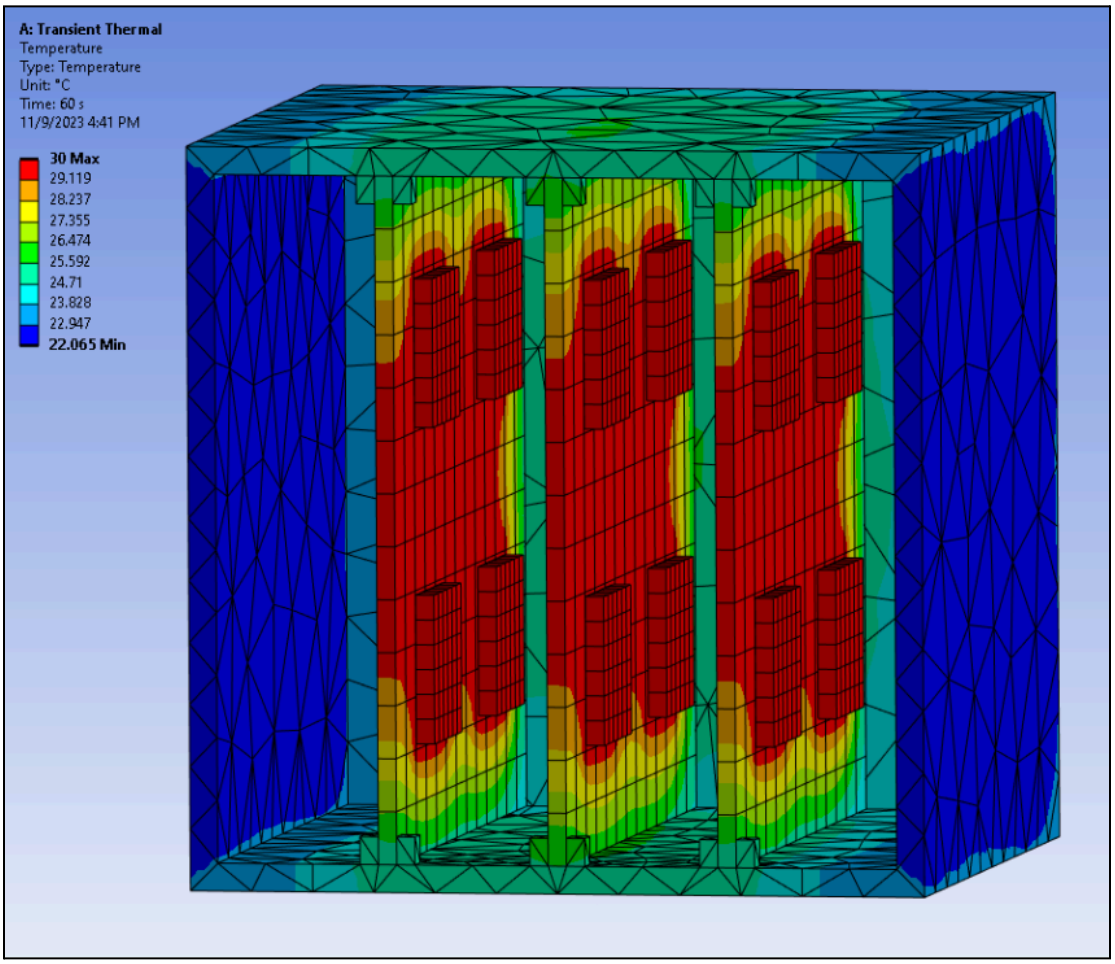


Figure I-1: Electronics Box Heat Simulation

Shown in **Figure I-1** is the heat simulation for the electronics box. This simulation was performed with Ansys software. The simulation was conducted to understand how heat would be distributed throughout the box. The idea behind the electronics box is that there will be vertical mounted plates that hold the circuit boards; this acts as a heat sink. There will then be fans mounted on the front cover of the electronics box that will draw hot air out of the box. To set up the simulation in Ansys, a model was created and assembled in a CAD software then converted to a STP file to successfully import into Ansys. In Ansys Workbench, a transient thermal node must be created with the geometry that was created previously. Once that is done, the environment will be created and thermal properties can be applied. This involves setting up the

material properties for the model. For this model in particular, the electronics box and the vertical plates are presumed to be aluminum. However the plates were later changed to acrylic and the circuit boards are set as PCB. The ambient temperature must be set up as well, which for this simulation was set to 22°C (71.6°F). The operating temperature for the circuit boards must also be assigned. To do this, the small squares that represent the circuit boards on the vertical plates are selected and their temperatures are set to 30°C (86°F). This value is based off of a value that was observed during motor testing. After the parameters are set, the simulation can then be run. The results of the simulation are shown in **Figure I-1**. By observing the figure, it is worth noting that there is not much heat transfer to the box itself, and all the heat is contained within the plates, which will then be cooled by the fans. This shows that this concept will work well and there will be no overheating issues when it comes to the electronics box.

Appendix J: Air Flow Simulation

To ensure that the electrical components will not overheat, Rock-E's electronics box was put through heat and airflow simulations. As seen in the heat simulation in **Figure I-1**, the heat tended to be isolated to where the components were and did not spread too much throughout the box. This works well with the airflow design as most of the volume of air moves through the area of the components. As seen in **Figures J-1 and J-2**, cool air comes in through the front of the box and gets pulled towards the rear with the exhaust fans and the hot air is expelled out the rear of the vehicle. There is a small amount of air that has the potential to be trapped in the corners of the box, but this is not a major concern as Rock-E's electronics will be centrally located on the boards in the middle of the box. The conditions of this experiment are assuming the inlet air will be from the vehicle moving at the slowest possible average speed of 0.447 m/s (1 mph) and the fans will be operating at 1000 rpm.

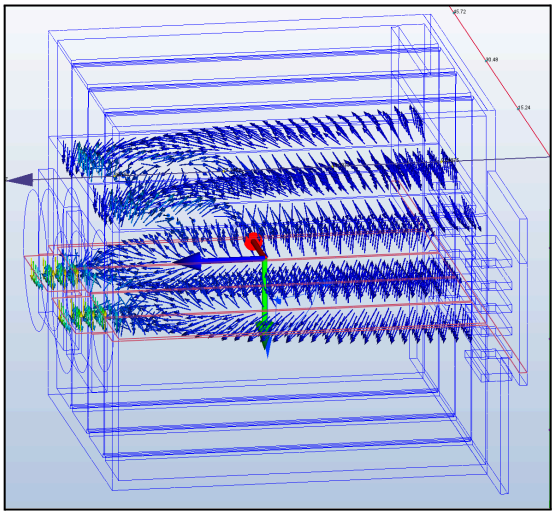


Figure J-1: Airflow in the X-plane

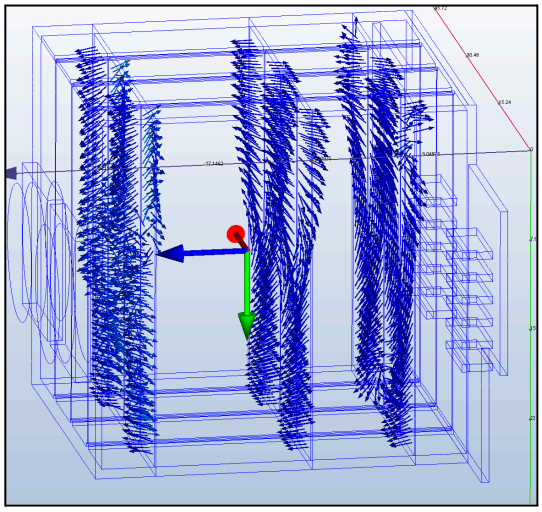


Figure J-2: Airflow in the Z-plane

Appendix K: Electronic Schematics

Power Distribution Module

Rock-E has incorporated a power distribution module (PDM) [1] in the electronics system (**Figure K-1 & K-2**). This module will give Rock-E a centralized distribution center for the discrete electronics on the robot as well as an on/off switching center for all electronics connected to the motor drive system. It includes two 24-volt high-amperage (30 amp) connections for the drive motors and one 24-volt low-amperage connection for discrete electronics (**Figure K-3**) along with six 12-volt low-amperage connections for discrete electronics. The PDM will also allow charging of the 12-volt and 24-volt batteries remotely since the board has connectors for this particular purpose

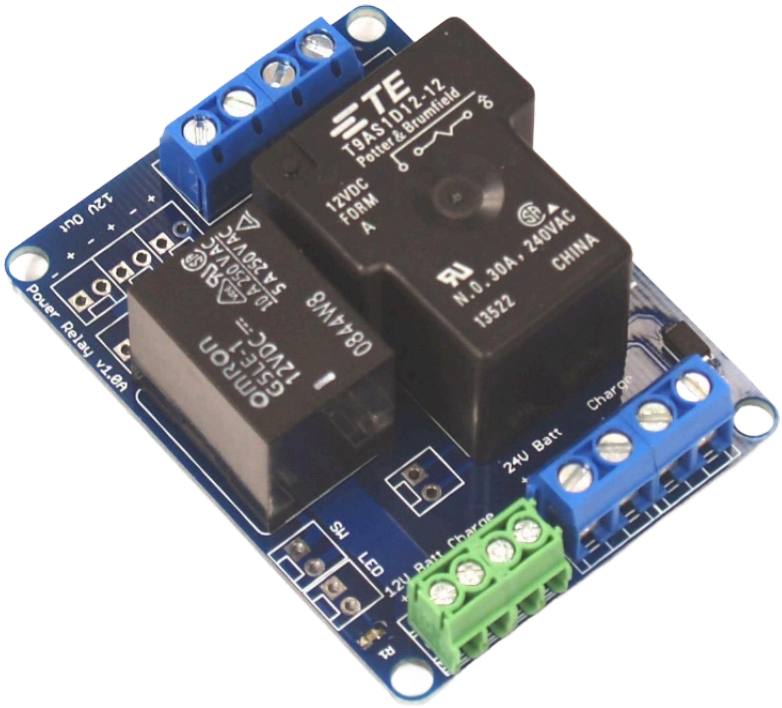


Figure K-1: Power Distribution Module

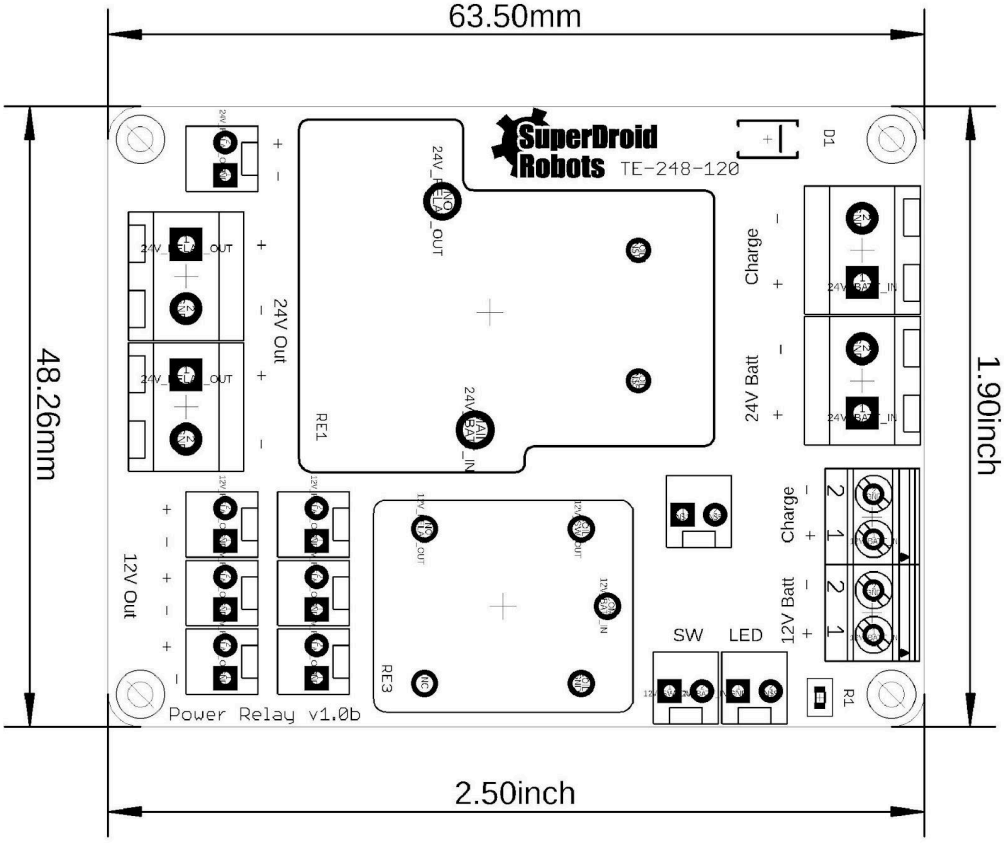


Figure K-2: Power Distribution Module

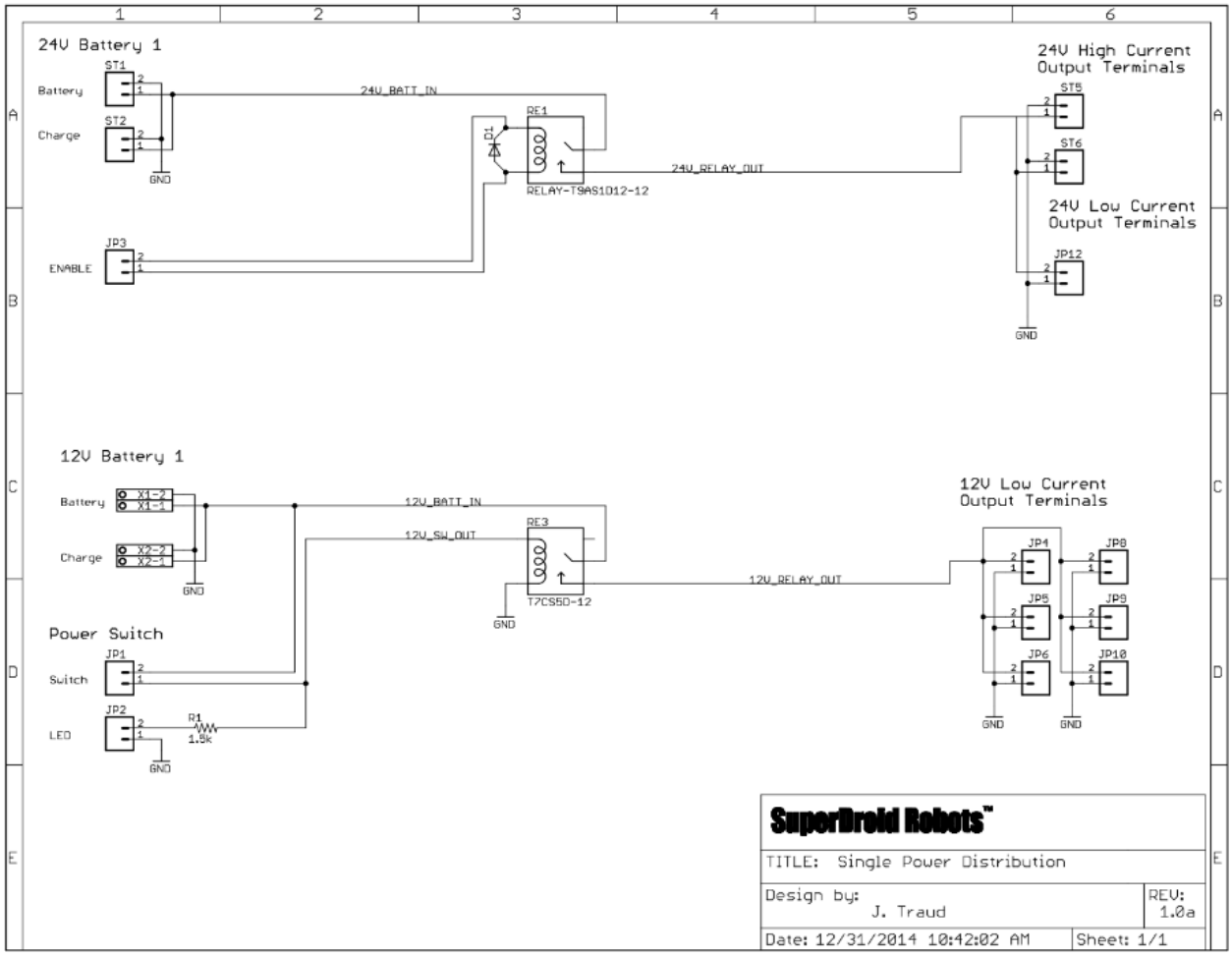


Figure K-3: Power Distribution Module Schematic

5-Pin Relays

Rock-E has incorporated 5-pin 24-volt automotive style relays (**Fig. K-4**) into the electronics. Their primary use is switching the high-amperage load for the motor drivers with a low amperage signal through the emergency stop system. This style of relay was chosen because it is cheaper than alternatives, can handle vibration, is weatherproof, and is easy to acquire. A schematic (**Fig. K-5a**) has been included.

Figure K-4: 5-pin Automotive Relay

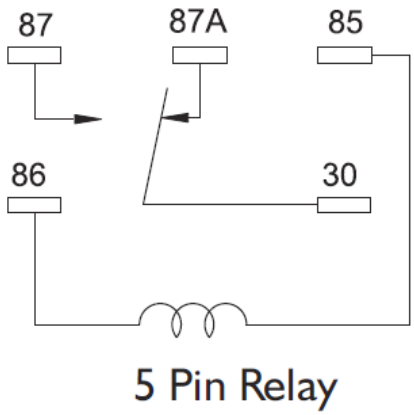


Figure K-5a: 5-pin Automotive Relay Pin Setup [2]

30 - common (connects to load) (+)

85 - trigger coil ground (-)

86 - trigger coil source (+)

87 - Normally open relay switch (NO) (-)

87A - Normally Closed Relay Switch (NC) (-)

Figure K-5a 5-pin Automotive Relay Pin Setup [2]

Motor Drivers

Rock-E will require DC motor drivers to control the two drive motors. In order to choose the appropriate motor driver listed features were aggregated:

- Must be able to handle 30 amps per motor continuously (double the motor stall current)
- Input voltage of 24 volts
- Pulse-width-modulation input for control
- Compatibility with Arduino microcontrollers
- Reasonable cost

Using the features listed above and staying within the approved vendors distributions, four different units were compared:

Table K-1: Motor Driver Options

Model (includes link)	Continuous Current	Operating Voltage	Motor Count	PWM Control?	Cost
Cytron MD30C	30 A	5-30V	1	Yes	$35.00
Cytron MDDS30	30 A	7-35V	2	Yes	$79.00
Sabertooth Dual 32A	32 A	6-30V	2	No	$170.40
RoboClaw Solo 30A	30 A	6-34V	1	Yes	$119.00

Rock-E will use the Cytron MD30C (**Figure K-5b**) since this unit will meet all needed features, is reliable, and is easy to acquire through Amazon. This motor driver also uses a Full NMOS H-Bridge, so a heat sink is not required. A special feature found in the MD30C is polarity protection. This feature will only allow the motor driver to function if the supply power is connected correctly. The final feature choice was not of the highest importance, but it is

worthwhile to mention; price. The MD30C is the cheapest option of the selected units. It can be noted, though, that the MD30C only controls one motor, whereas the MDDS30 can control two and is around twice the price. Either unit would have been sufficient, but a common thought is that if one of the MD30C motor controllers were to break, the robot would only need one replacement. If Rock-E were to use the MDDS30 and it damages one of the motor drivers, the entire unit will need to be replaced at twice the cost.

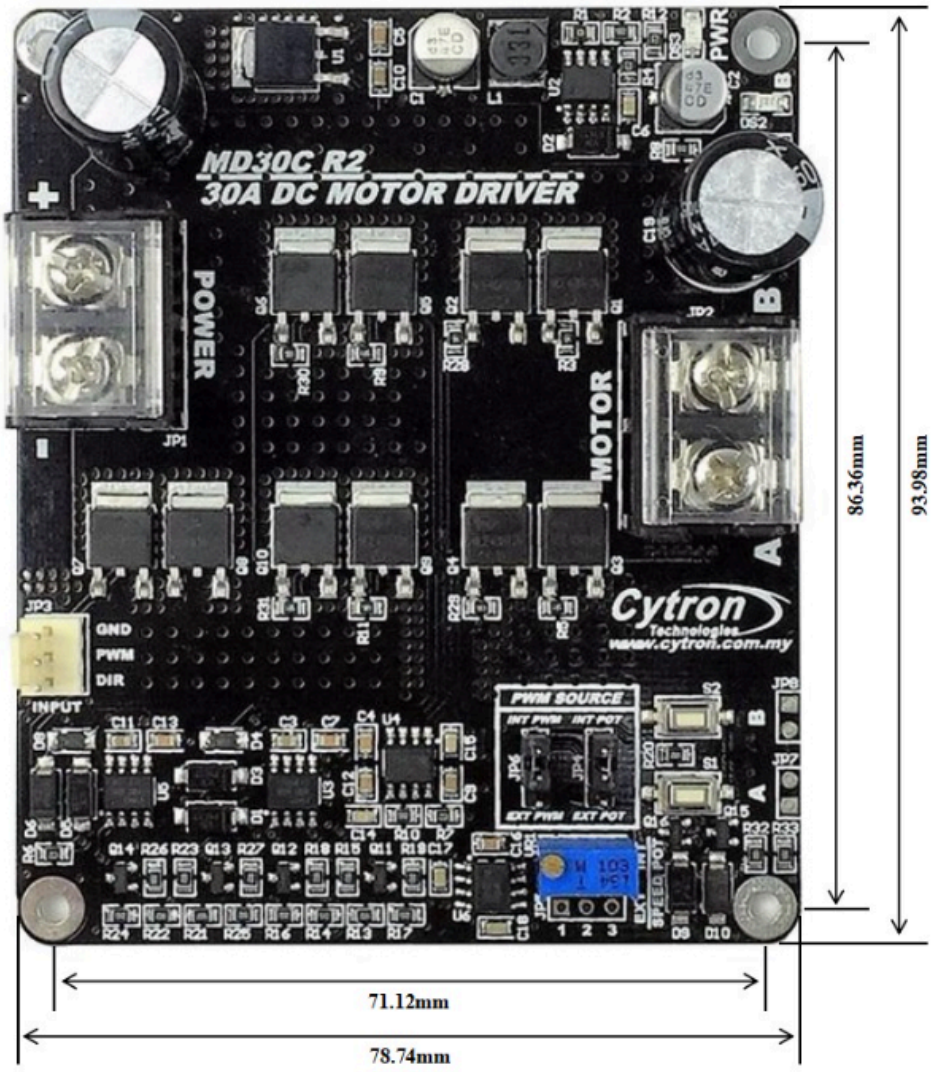


Figure K-5b: MD30C Motor Controller

Kar-Tech Wireless Emergency Stop System

IGVC competition rules require the autonomous robot to have a mechanical e-stop and a remote e-stop [3-p7]. A number of the robots in the competition are using cheap remote switches as a stand-in for a true wireless e-stop system [4]. After researching, it has been concluded that the reasoning behind using inferior simple remote switches instead of actual remote e-stop systems is cost. Remote switches can be purchased on Amazon (link) [4] for less than $50 while a remote e-stop system can run into the hundreds of dollars.

Rock-E is using a true wireless e-stop system for safety. The unit is the Kar-Tech Wireless Emergency Stop System (**Fig. K-6 & K-7**) [5]. The system consists of a transmitter and a receiver and transmits a two-way signal at 2.4 Ghz between the units. If the signal between the units is interrupted either by an e-stop condition or the units are out of range the receiving unit will go into an e-stop condition. Resetting the transmitter or bringing the units back into range will reset the e-stop condition. A wiring diagram for the unit has been provided (**Fig. K-8**)

Figure K-6: Wireless E-stop Receiver

Figure K-7: Wireless E-stop Transmitter

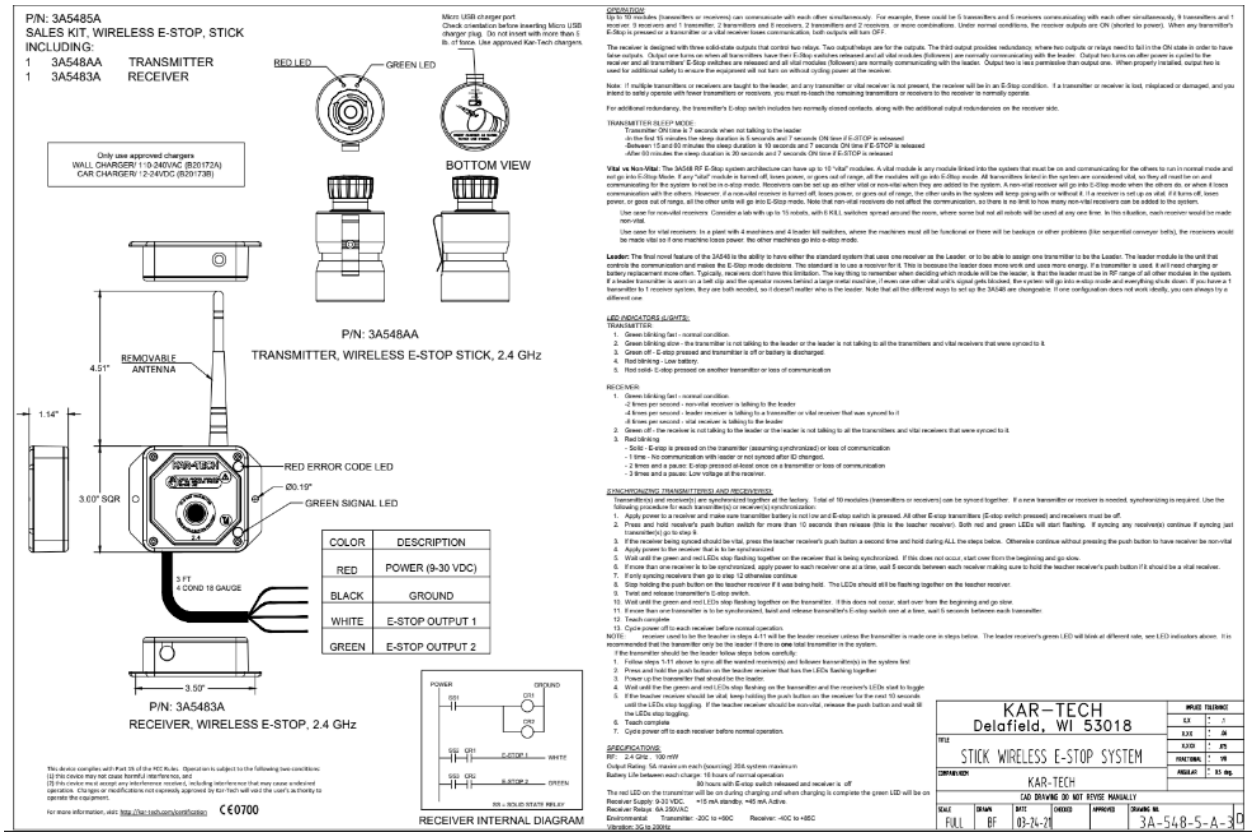


Figure K-8: Wireless E-stop Schematic

Mechanical E-Stop Switch

The mechanical e-stop (**Fig. K-9 & K-9B**) [6] for Rock-E is a true e-stop and has two switches incorporated into the design, one that is normally closed and one that is normally open. This is to ensure that if one switch fails the e-stop condition will still perform.

Figure K-9: Mechanical E-Stop with two switches

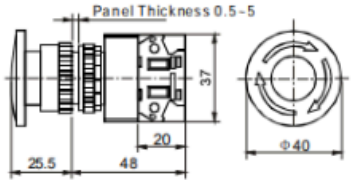
Shape	Type	Type Number	Dimension	Remarks
Emergency Pushbutton (can be illuminated)	Push-lock Turn-reset	Y090-□TS/△/▲		R G Y ◎ The dimension of head is: Φ30mm, Φ40mm

Figure K-9B: Mechanical E-Stop Information

Batteries

Rock-E will use two separate batteries to control electronics; A 12-volt lithium-ion battery (**Fig K-10**) [8] and a 24-volt lithium ion battery (**Fig. K-11**) [9]. The two batteries will be connected to electronics through the PDM (**Fig. K-1**). A two-battery system is being used since research has shown that interference is an issue between the motors, motor drivers, and discrete electronics in direct current powered systems [7] . The 24-volt battery will be used through the PDM to control the motor drivers, motors, E-stops, and powering of the PDM. The 12-volt battery will be used to power the Arduino Due, lights, and motor encoders.

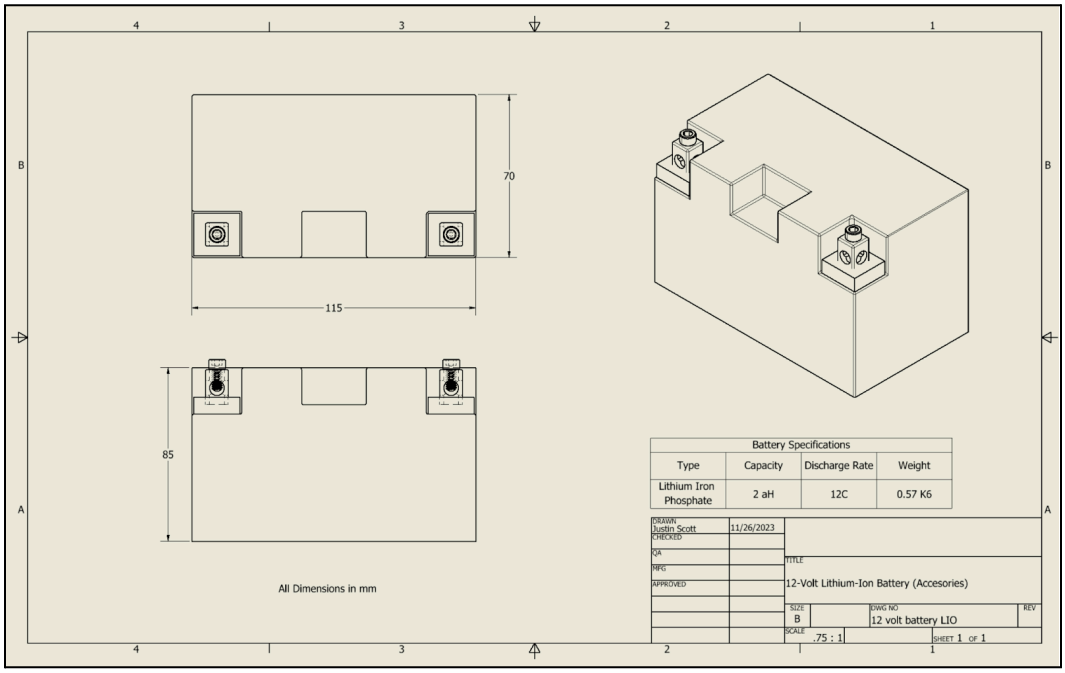


Figure K-10: 12-volt Lithium-ion Battery

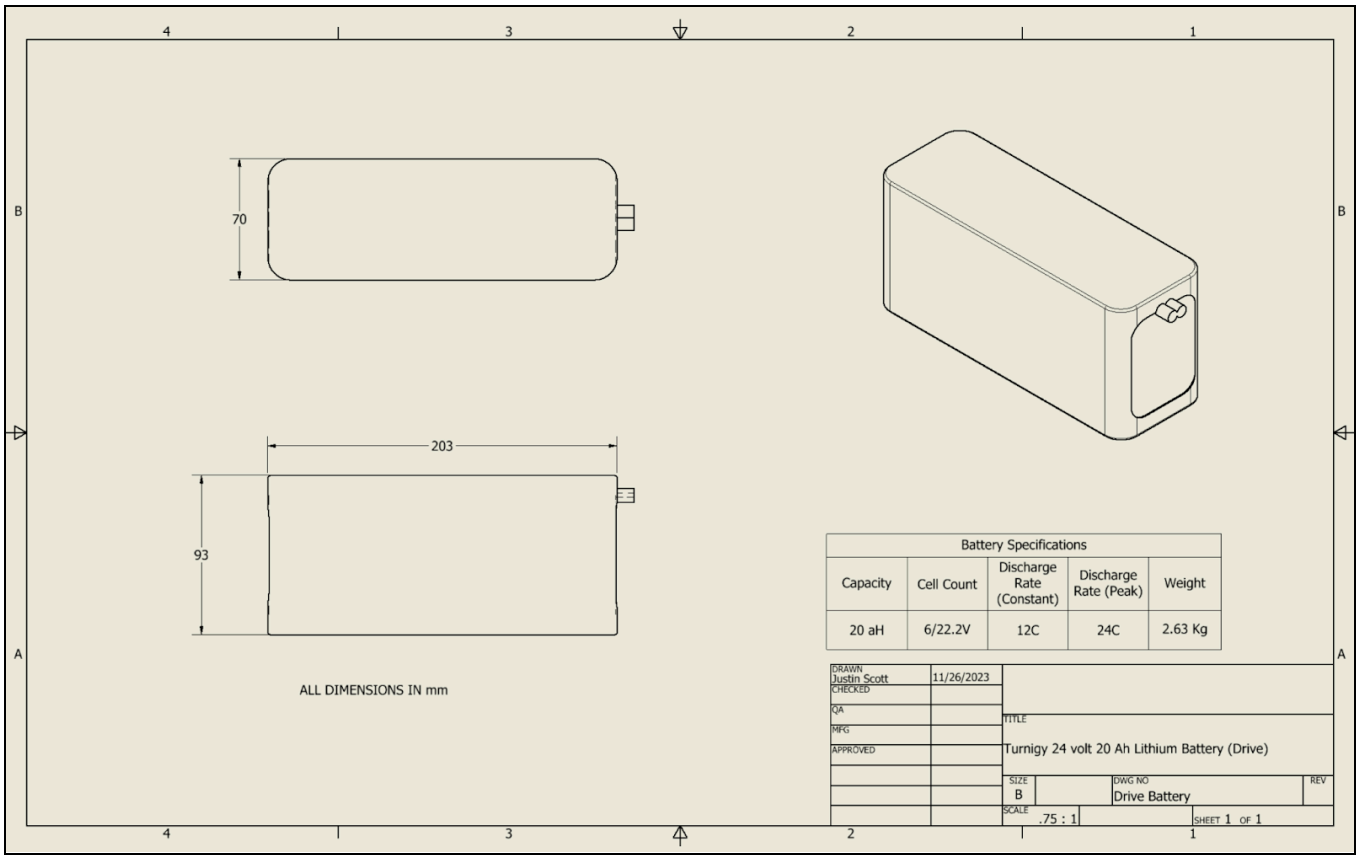


Figure K-11: 24-volt Lithium-ion Battery

Motor Encoders & Timing Gears

Rock-E will use two Taiss E38S6-360-24G rotary encoders (**Fig. K-12 & K-14**). These encoders are placed in a remote location and attached to each motor output shaft by a timing belt system (**Fig. K-13**). They will be connected to an Arduino Due to relay motor rotation to the ROS2 software. Each encoder has the ability to relay 360 pulses per revolution. When connected to the timing belt system (reduction ratio 4:1) each turn of a drive wheel will account for 1,440 pulses per revolution at each encoder.

Figure K-12: Taiss E38S6-360-24G Rotary Encoder [10]

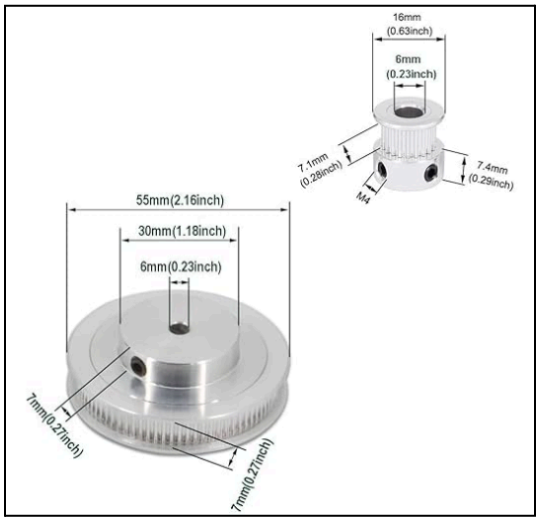


Figure K-13: Timing Belt Reduction Gears (4:1)

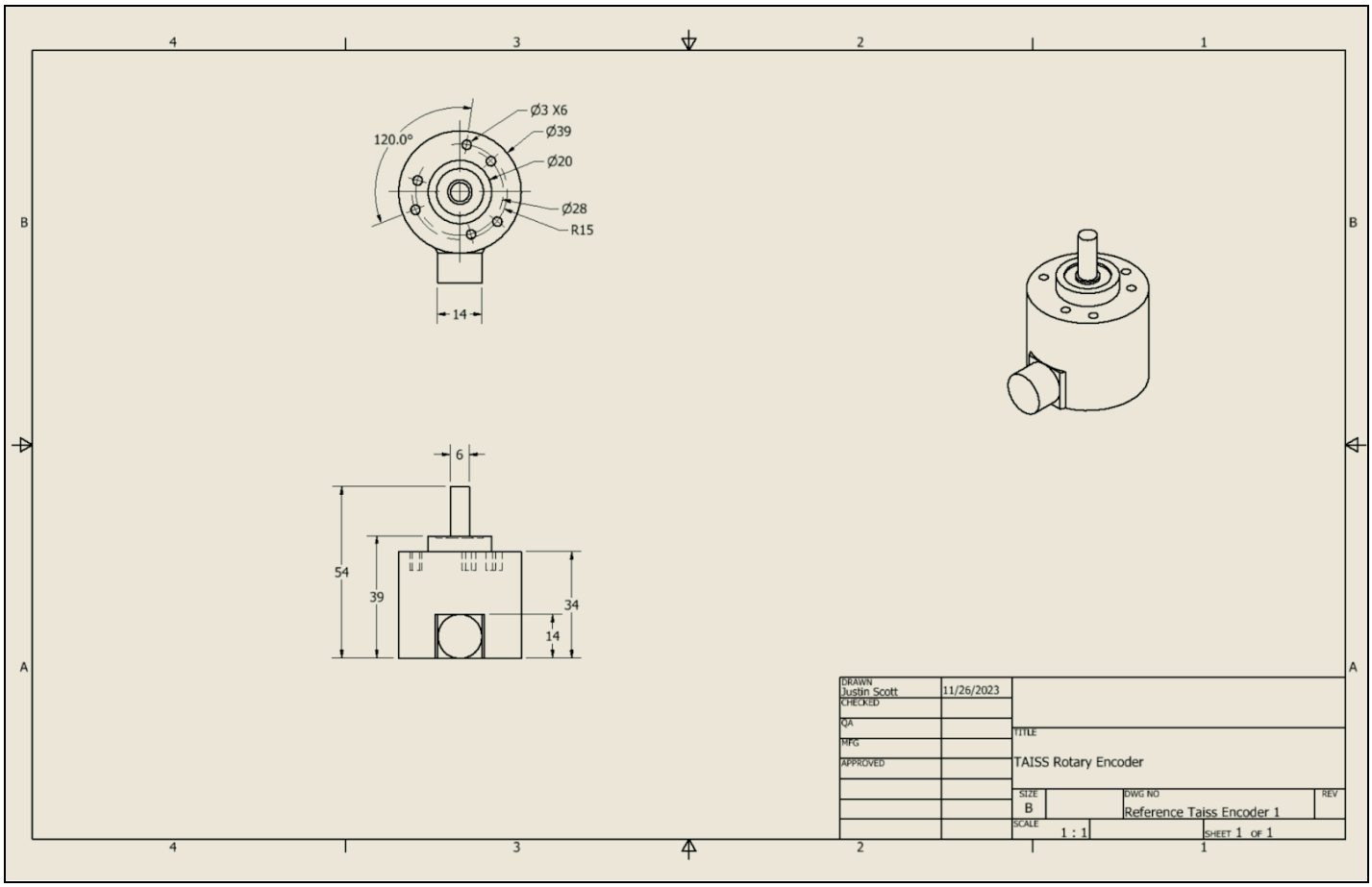


Figure K-14: Rotary Encoder Drawing

Buck Converters

Rock-E will be utilizing a Stemedu LM2596 DC to DC adjustable buck converter (**Fig. K-15**) to make the ultrasonic sensors function. The ultrasonic sensors (Appendix Z) require 5V in order to function properly, and since there was not a reliable source anywhere, the team had to step down 12V from the PDM. The buck converters have a voltage IN and voltage OUT. 12V goes into the buck converter and it will output whatever voltage it is set to. There is a small potentiometer on the converter that can be used to change the output voltage. This is how Rock-E will step down the voltage for the ultrasonic sensors to work correctly.

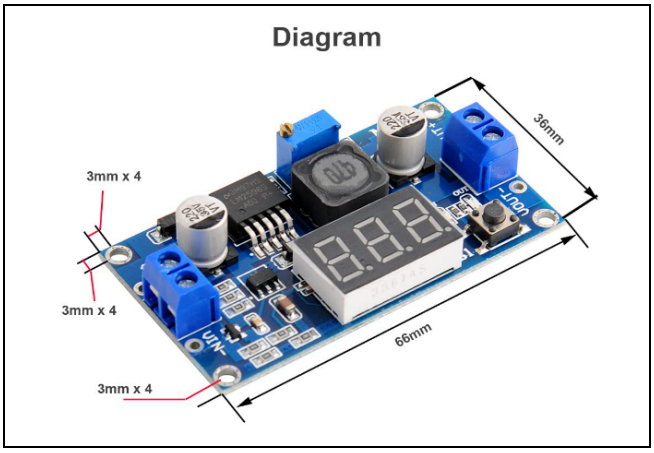


Figure K-15: DC to DC Buck Converter

Mosfet Module

For Rock-E's safety light a Onyehn MOS Driver Module for Arduinos will be used. This module works similar to a relay. With the help of the PDM and the Arduino, Rock-E will boost the signal from the arduino into a 12V signal to turn the light on and off. The mosfet enables this. The team actually had to remove the mosfet that it came with because the threshold voltage was too high to enable the light. The IRF520 mosfet was removed and instead the team soldered on a K2231 mosfet, which has a lower threshold voltage. The IRF520 has a threshold voltage of 4V, which is not enough for the Arduinos 3.3V signal. When switching to the K2231 mosfet, which has a 0.8-2.0V threshold, the arduino signal was then able to trigger the circuit and turn the light on.

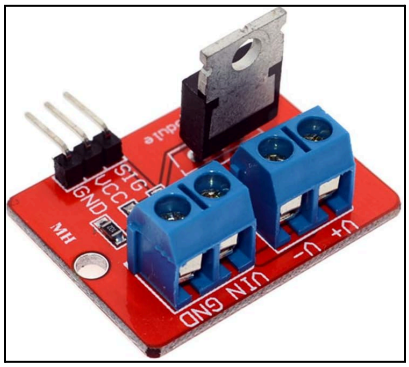


Figure K-16: Mosfet Module

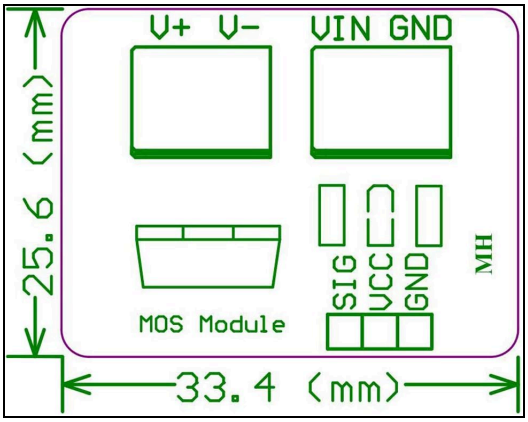


Figure K-17: Mosfet Module

Arduino Due

*Please see **appendix AE** for any and all information regarding this device.*

Lidar

*Please see **appendix AA** for any and all information regarding this device.*

Camera
*Please see **appendix AD** for any and all information regarding this device.*

GPS
*Please see **appendix AB** for any and all information regarding this device.*

IMU
*Please see **appendix AC** for any and all information regarding this device.*

Appendix L: Wiring Diagram

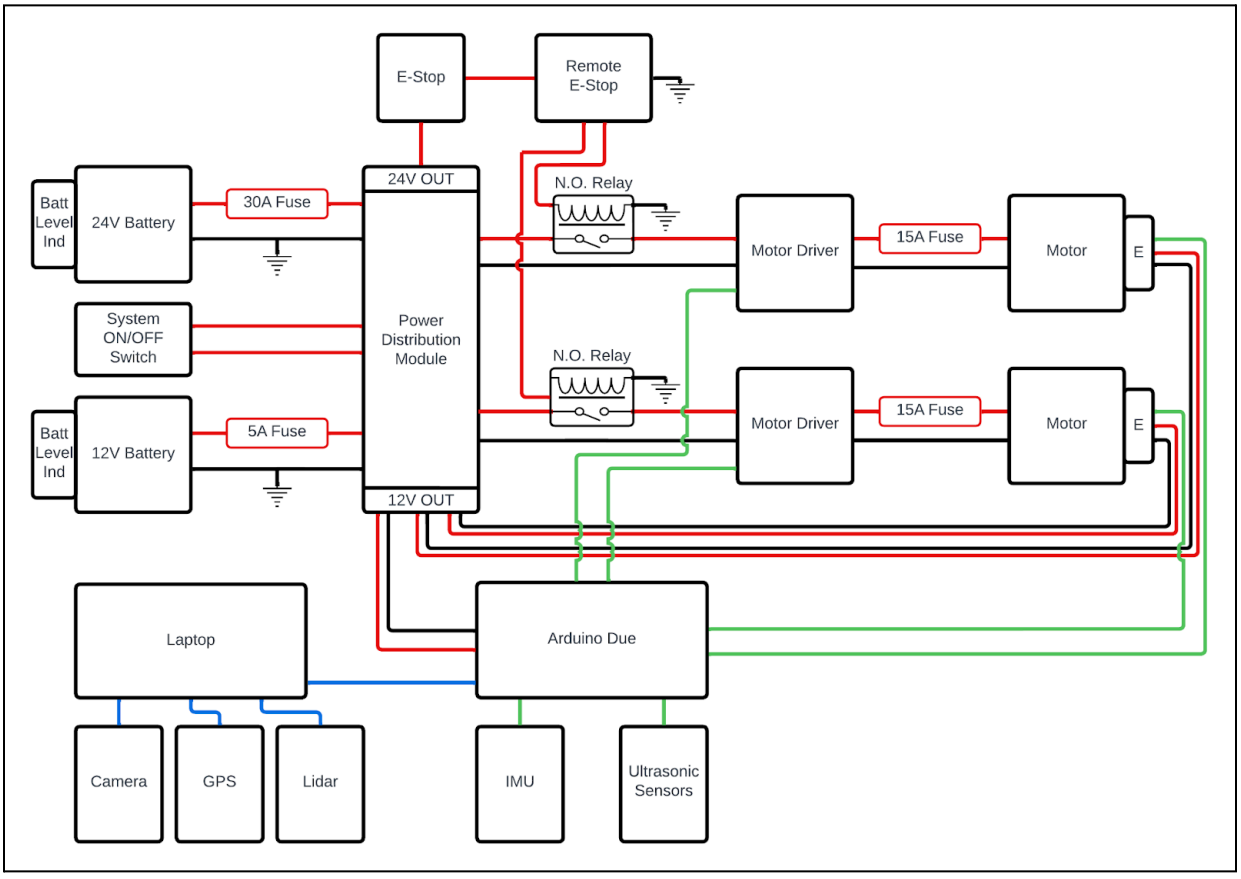


Figure L-1: CDR Wiring Diagram

(NOTE: This diagram and description is only accurate as of the CDR)

(See next page for FDR diagram and description)

The wiring diagram (**Figure L-1**) consists of many modules. There is a 24V battery that will power the motors. A separate 12V battery will power the Power Distribution Module (PDM) which allows it to power several other systems like the E-Stop circuit, cooling fans, and Arduino(s). Each battery has a battery level indicator that allows it to be monitored when needed. The PDM can be switched on and off via the connected System ON/OFF Switch. The PDM splits the 24V current 2 ways for each motor circuit. The motor drivers allow control over the

polarity and amount of power that goes to the motors via an Arduino. Between the PDM and the motor drivers, there are normally open relays. These relays do not allow current to pass through the motor drivers unless the E-stop is released. When the E-stop is depressed, it will open the relay and cut power to the motor circuit, stopping the vehicle. This principle also applies to the Remote E-Stop. Attached to the motors are the motor encoders. The encoders tell the Arduino how fast the motors are spinning. This is important for controlling how much power goes to the motors. Also connected to the arduino is IMU and the ultrasonic sensors. The IMU, which stands for Inertial Measurement Unit, is an electronic device that measures and reports acceleration, orientation, angular rates, and gravitational forces. The ultrasonic sensors measure the distance of objects in front of them. Everything that is connected to the Arduino is then transmitted back to the laptop as data via a USB Connection shown by the blue wire. Also connected to the laptop is the camera, GPS, and the LiDAR camera. The camera allows the vehicle to see where it is, the GPS allows the vehicle to know where it is, and the LiDAR, which stands for Light Detection and Ranging, is a remote sensing method that uses light in the form of a pulsed laser to measure ranges to obstacles. This creates a map for the vehicle. The CPU then processes that information and makes the necessary adjustments to the system. In the wiring diagram, there are fuses placed where high current is occurring. This is for safety reasons to prevent the system from melting down in the event that there is too much current draw. Red wires in the circuit represent positive connections. Black wires represent negative connections and grounds. Green wires represent 5V connections. Blue wires represent USB connections.

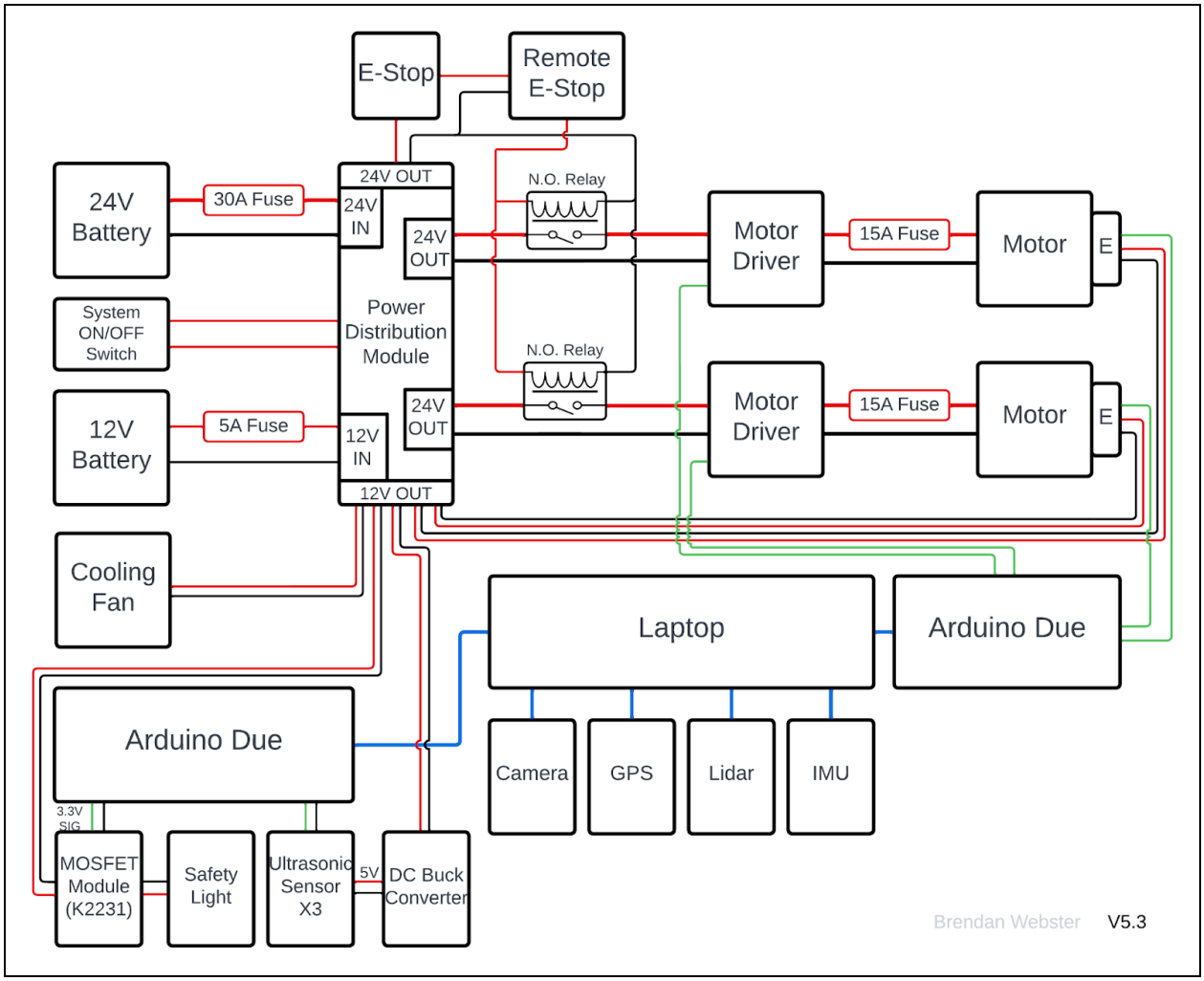


Figure L-2: FDR Wiring Diagram

The FDR wiring diagram (**Figure L-2**) consists of many modules. There is a 24V battery that will power the motors. A separate 12V battery will power the Power Distribution Module (PDM) which allows it to power several other systems like the E-Stop circuit, cooling fan, and Arduino(s). The PDM can be switched on and off via the connected System ON/OFF Switch. The PDM will split the 24V current 2 ways for each motor circuit. The motor drivers allow control over the polarity and amount of power that goes to the motors via an Arduino. Between the PDM and the motor drivers, there are normally open relays. These relays will not allow current to pass through the motor drivers unless the E-stop is released. When the E-stop is depressed, it will open the relay and cut power to the motor circuit, stopping the vehicle. This

principle also applies to the Remote E-Stop. Attached to the motors are the motor encoders. Encoders tell the Arduino how fast the motors are spinning. This is important for controlling how much power goes to the motors. Connected to the second Arduino are the ultrasonic sensors. The ultrasonic sensors measure the distance of objects in front of them. The ultrasonic sensors require a 5V connection which is made possible by the DC to DC Buck Converter, which turns 12V into 5V. Also connected to the second Arduino is the safety light. The safety light requires 12V to operate. By utilizing a Mosfet Module, the safety light operates by getting a 3.3V signal from the Arduino, which is then boosted to a 12V signal via the Mosfet Module. Everything that is connected to the Arduinos is then transmitted back to the laptop as data via a USB Connection shown by the blue wire. Also connected to the laptop is the IMU, Camera, GPS, and the Lidar camera. The IMU, which stands for Inertial Measurement Unit, is an electronic device that measures and reports acceleration, orientation, angular rates, and gravitational forces. The camera allows the vehicle to see where it is, the GPS allows the vehicle to know where it is, and the Lidar, which stands for Light Detection and Ranging, is a remote sensing method that uses light in the form of a pulsed laser to measure ranges to obstacles. This creates a map for the vehicle. The laptop then processes that information and makes the necessary adjustments to the system. In the wiring diagram, there are fuses placed where high current is occurring. This is for safety reasons to prevent the system from melting down in the event that there is too much current draw. Red wires in the circuit represent positive connections. Black wires represent negative connections and grounds. Green wires represent signal connections. Blue wires represent USB connections.

Appendix M: Manufacturing Drawings

A compilation of all the engineering drawings for the Rock-E Senior Design Project. Some were created by the members of the team and others were pulled from the site the materials were sourced from.

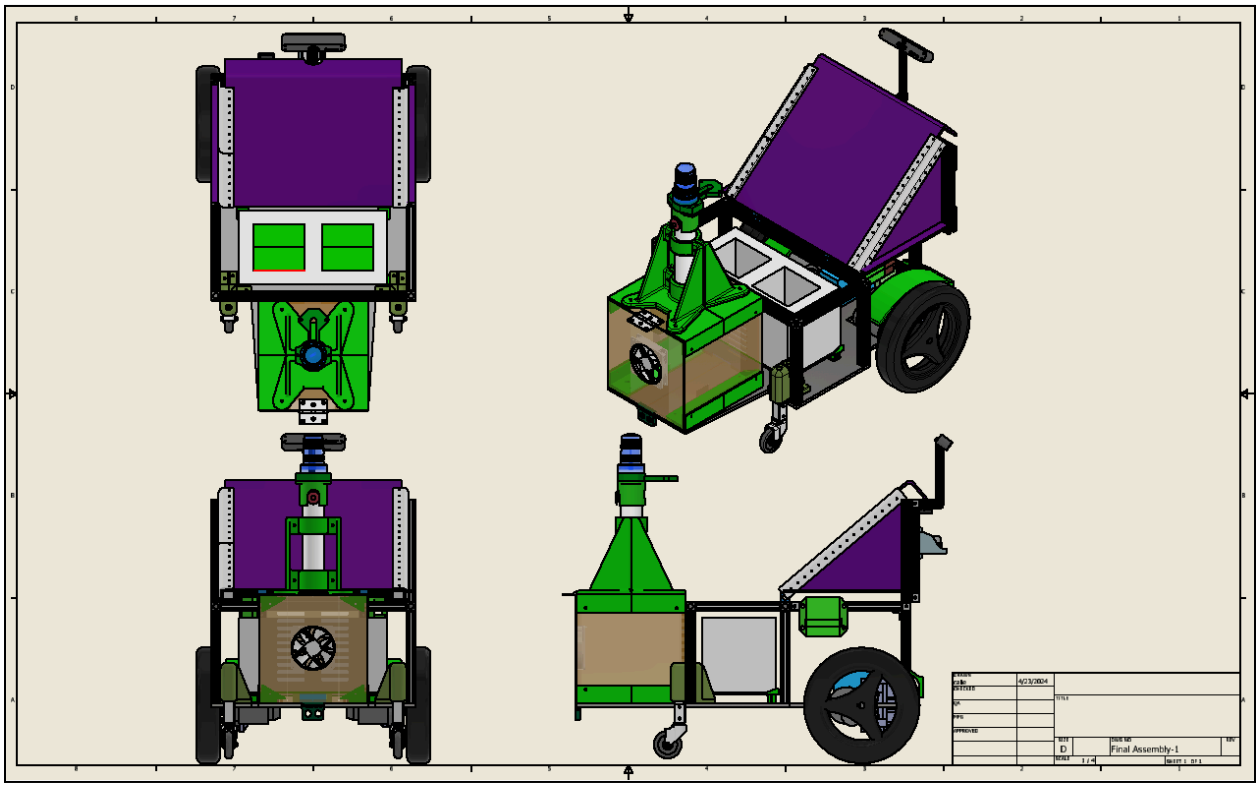


Figure M-1: FDR Complete Model

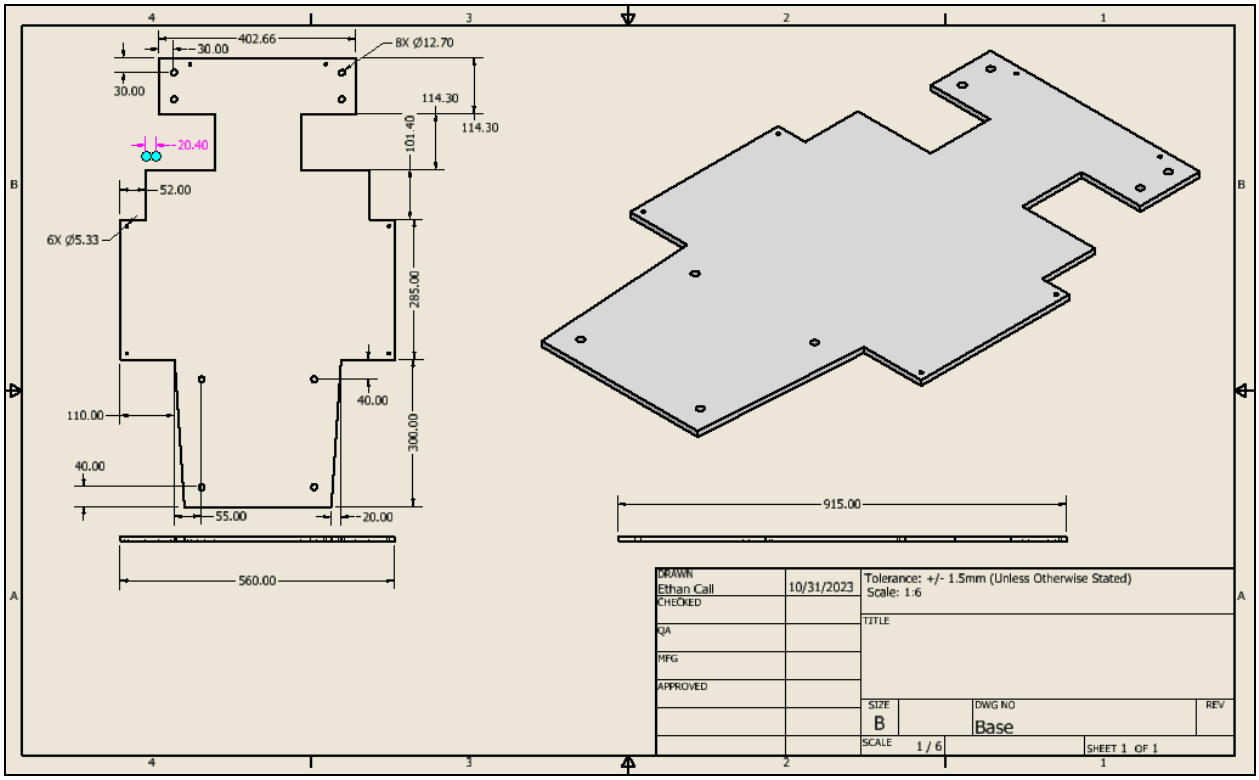


Figure M-2: Base Plate

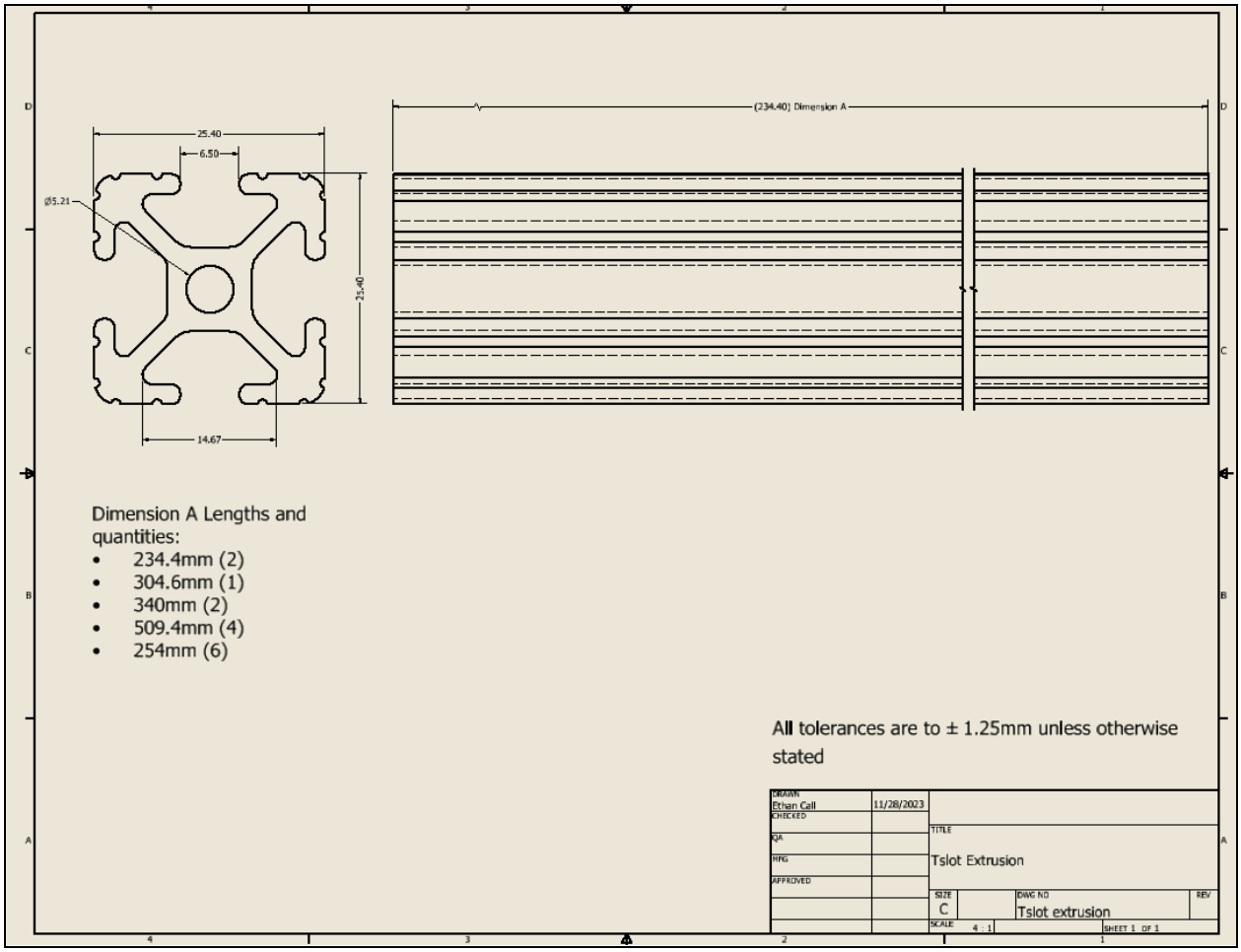


Figure M-3: T-slot Aluminum Extrusion

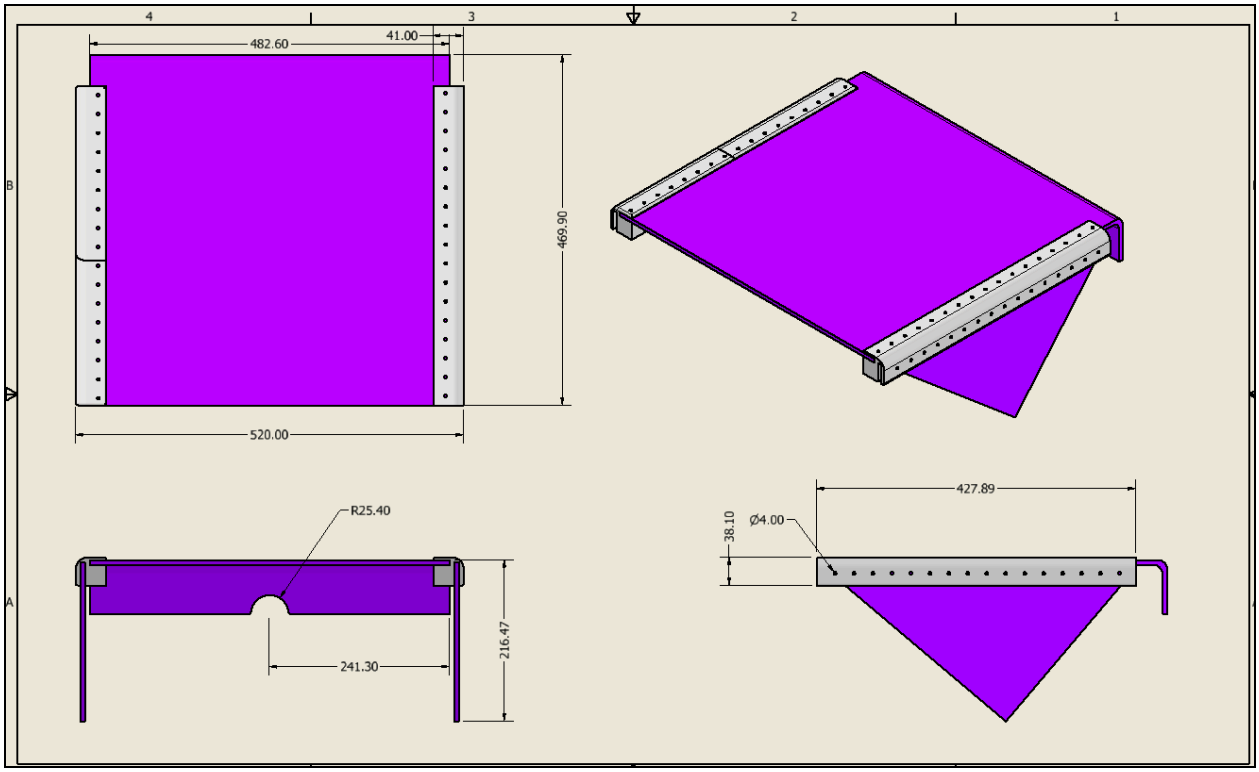


Figure M-4: Laptop Cover

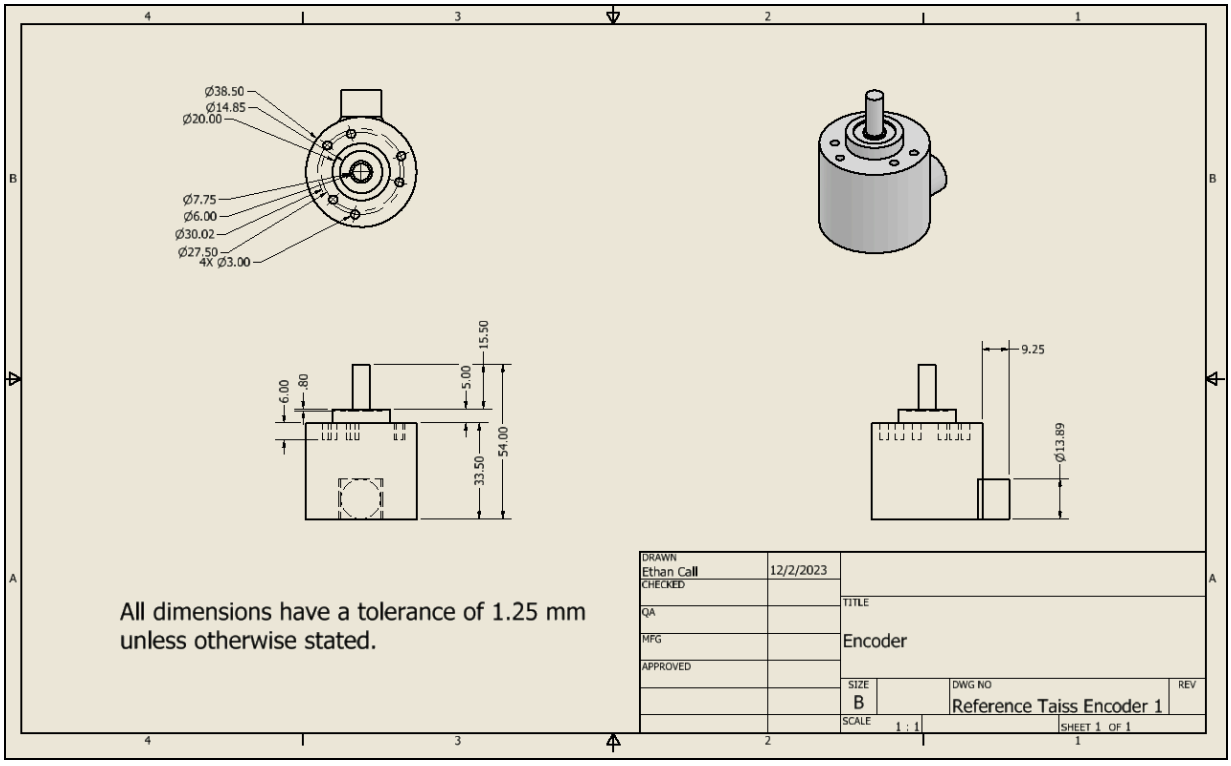


Figure M-5: Taiss Encoder

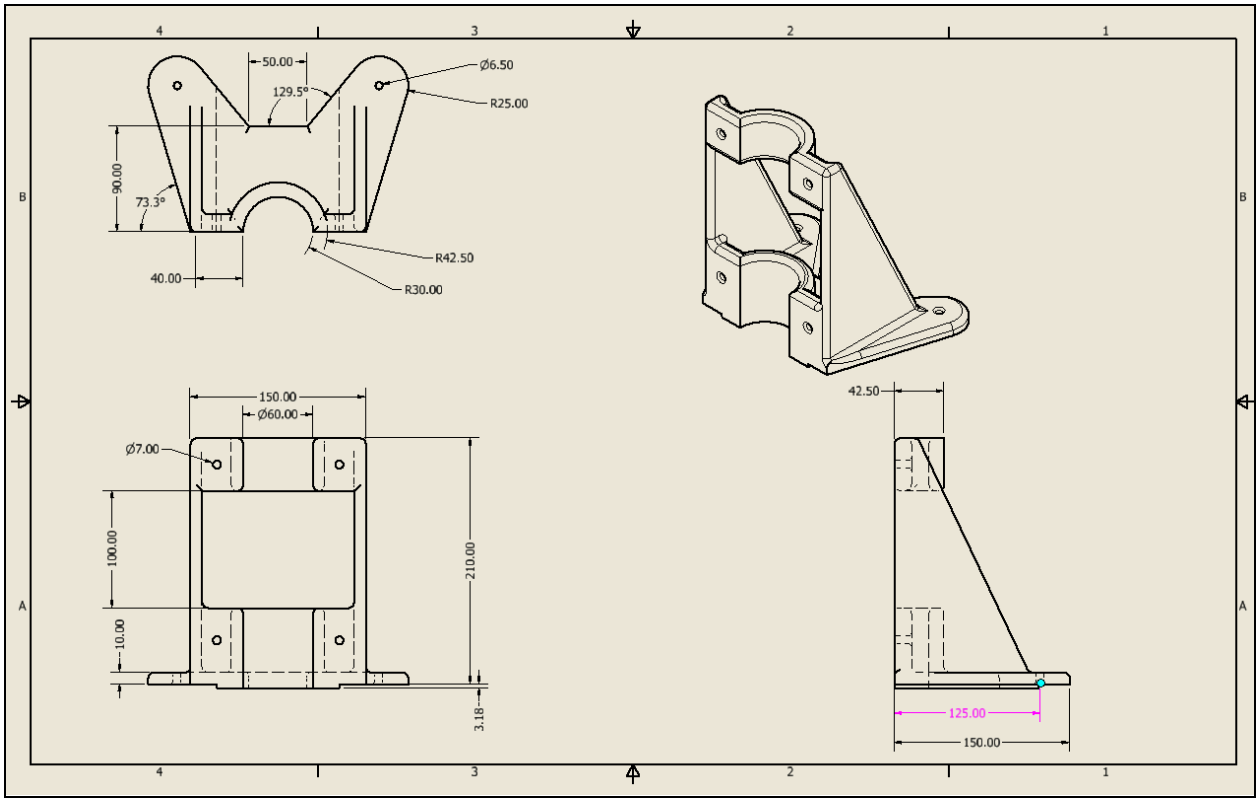


Figure M-6: Light Pole Base

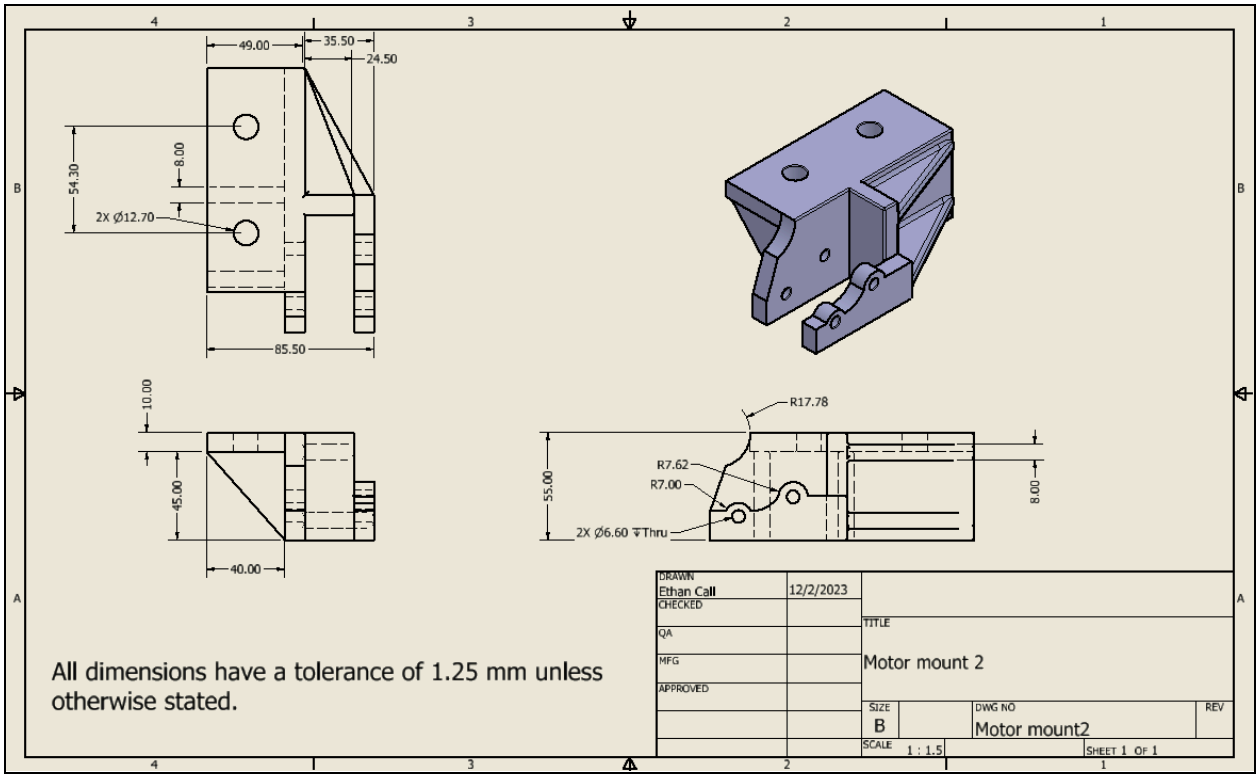


Figure M-7: Motor Mount 2

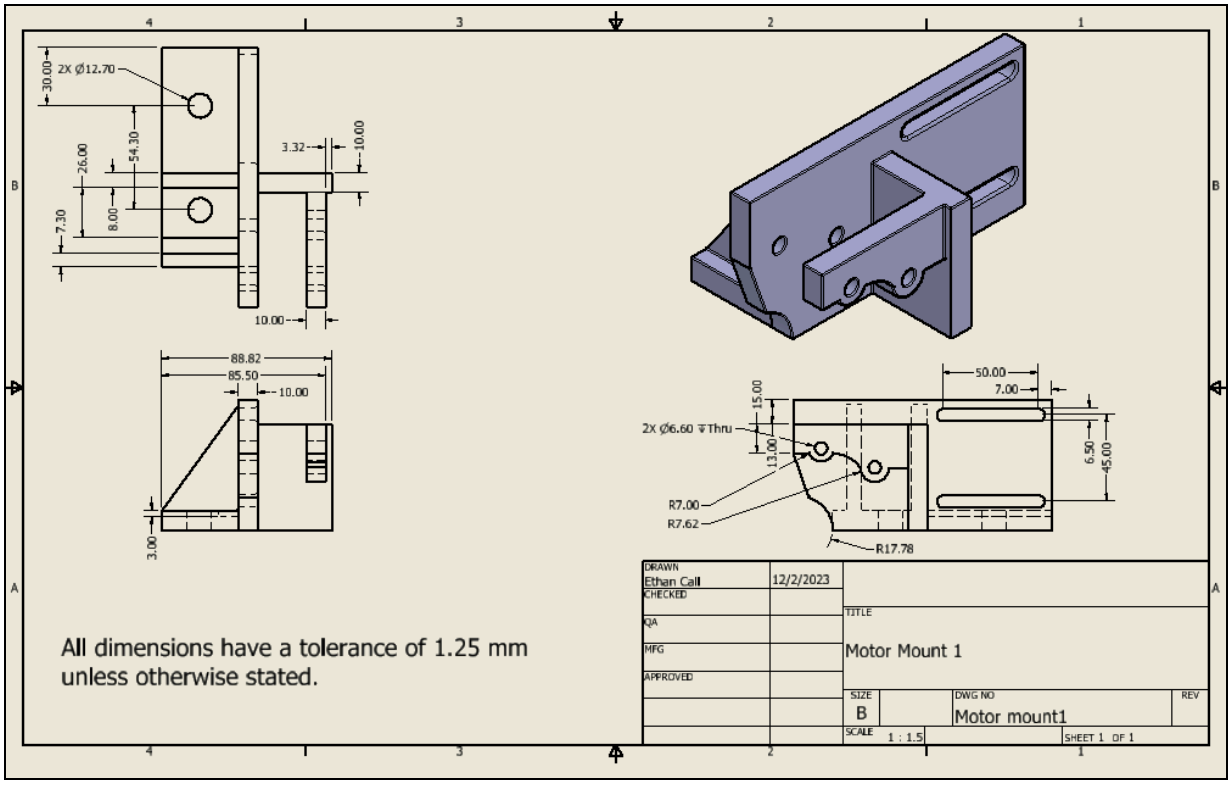


Figure M-8: Motor Mount 1

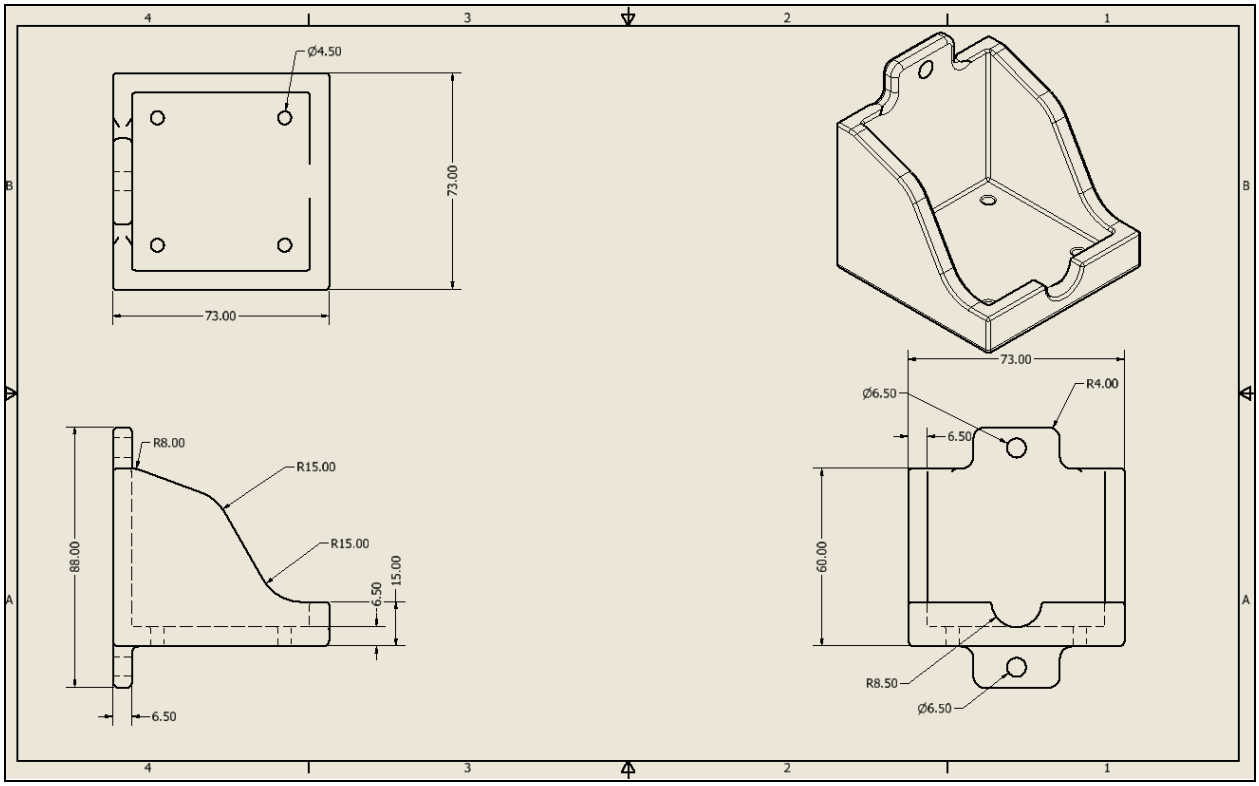


Figure M-9: Lidar Mount

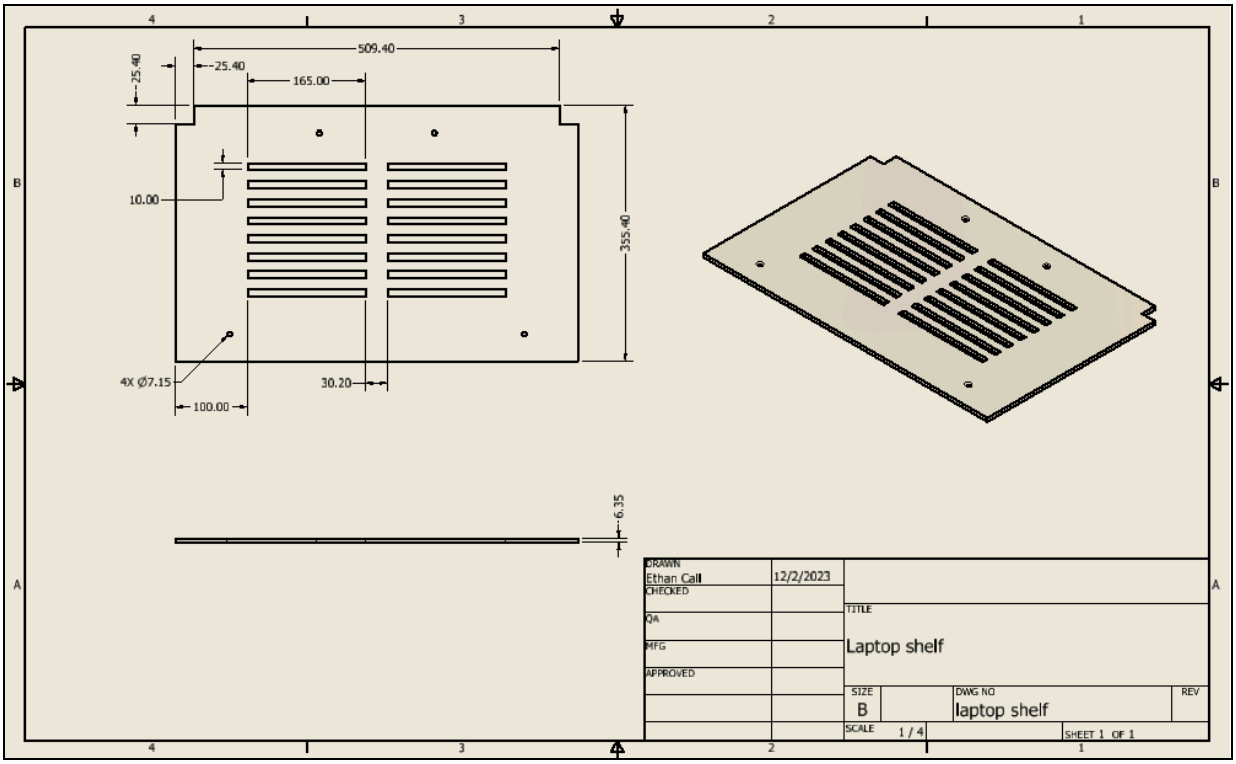


Figure M-10: Laptop Shelf

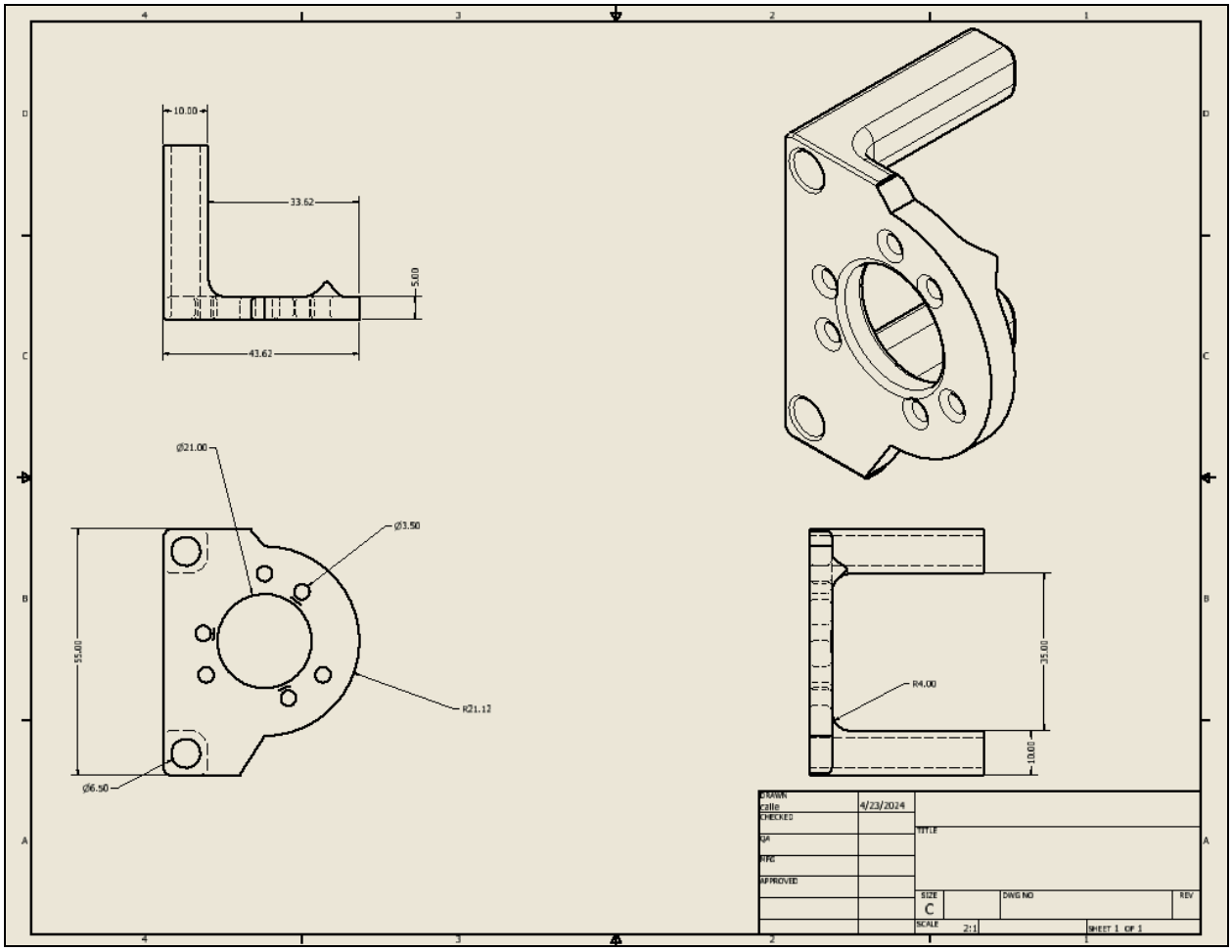


Figure M-11: Encoder Mount

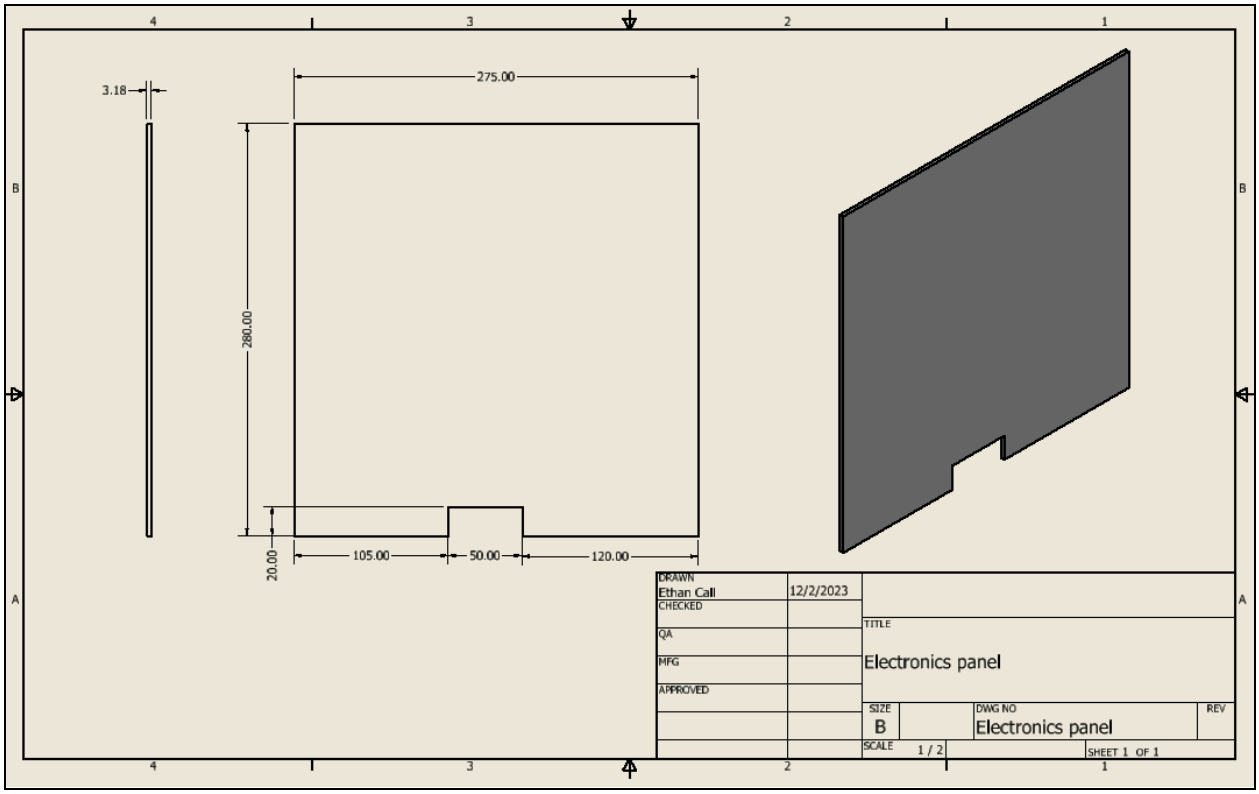


Figure M-12: Electronics Panel

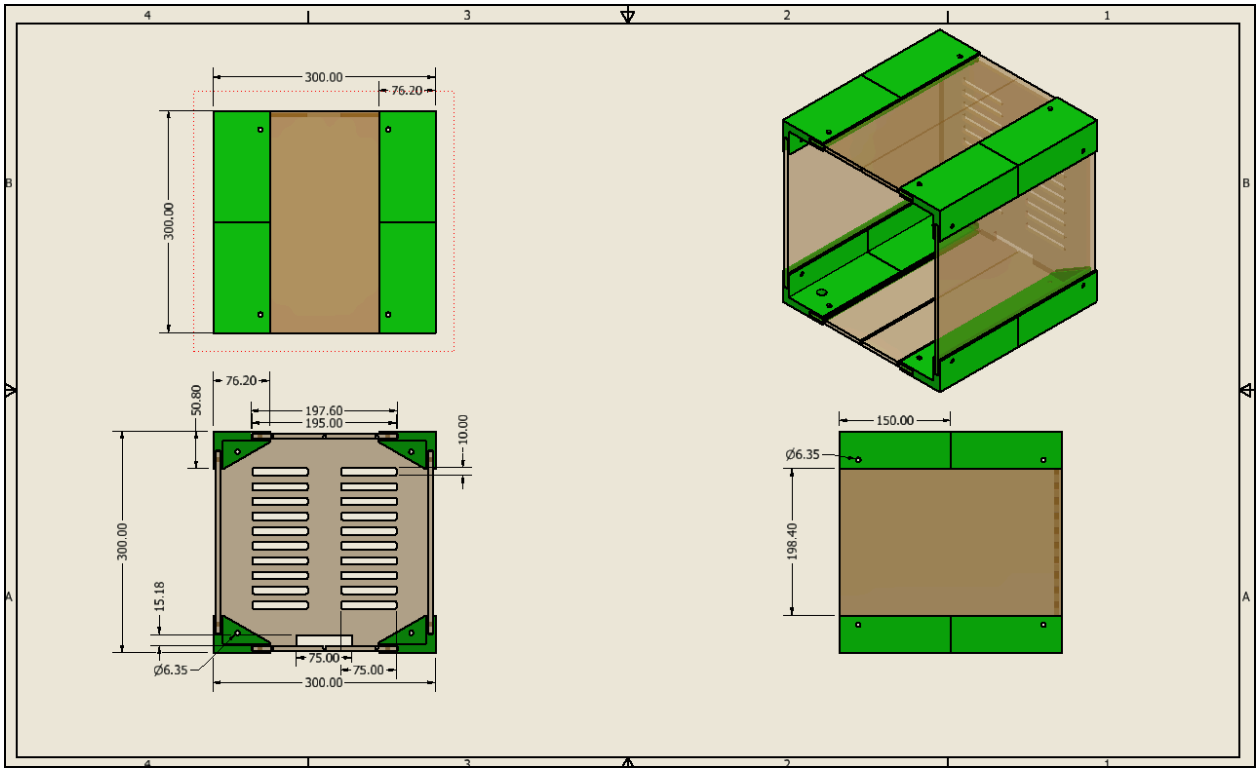


Figure M-13: Electronics Box

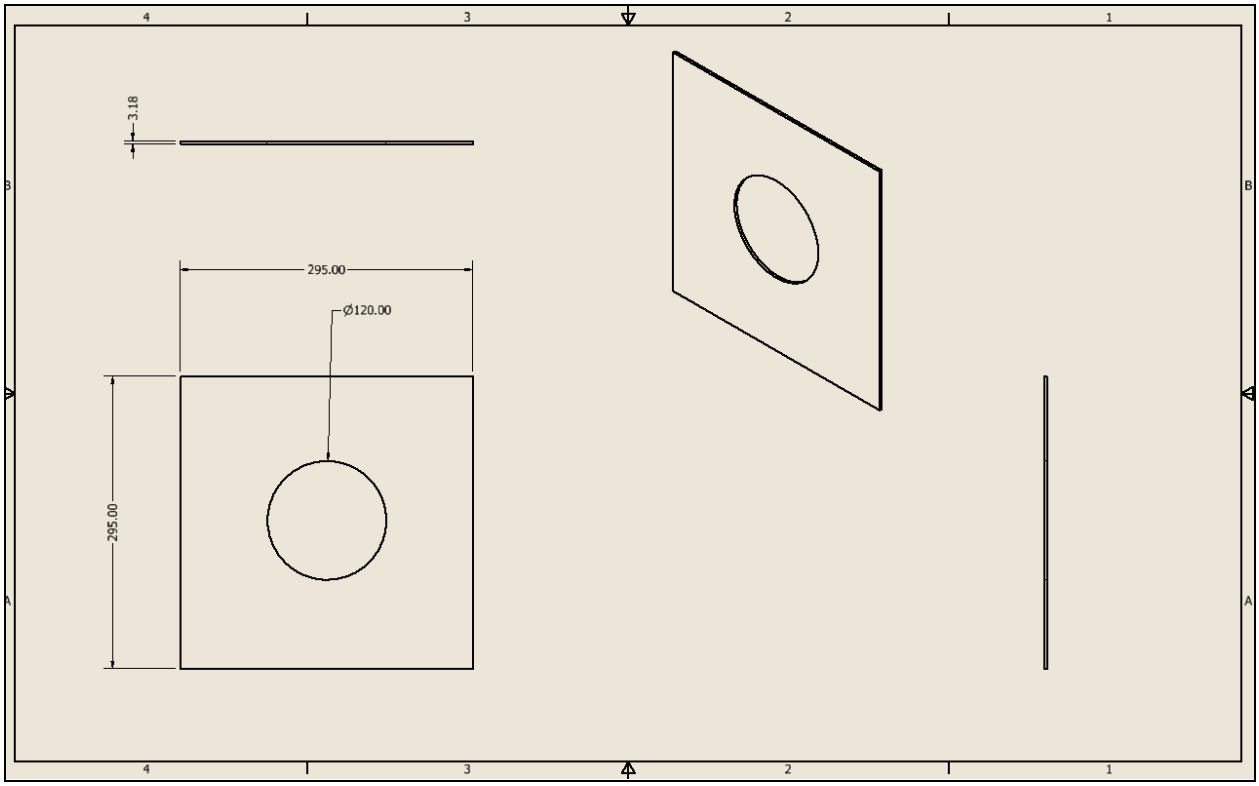


Figure M-14: Electronics Box Lid

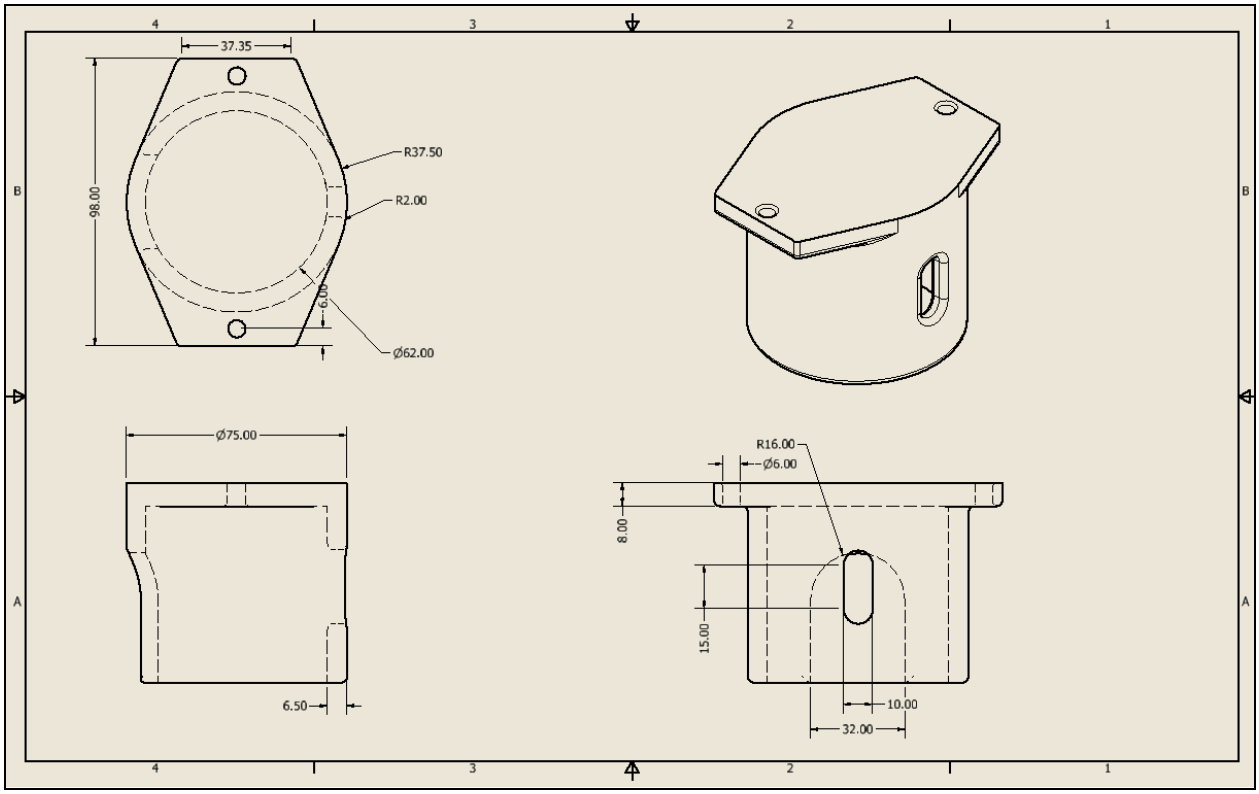


Figure M-15: E-stop/light Mount

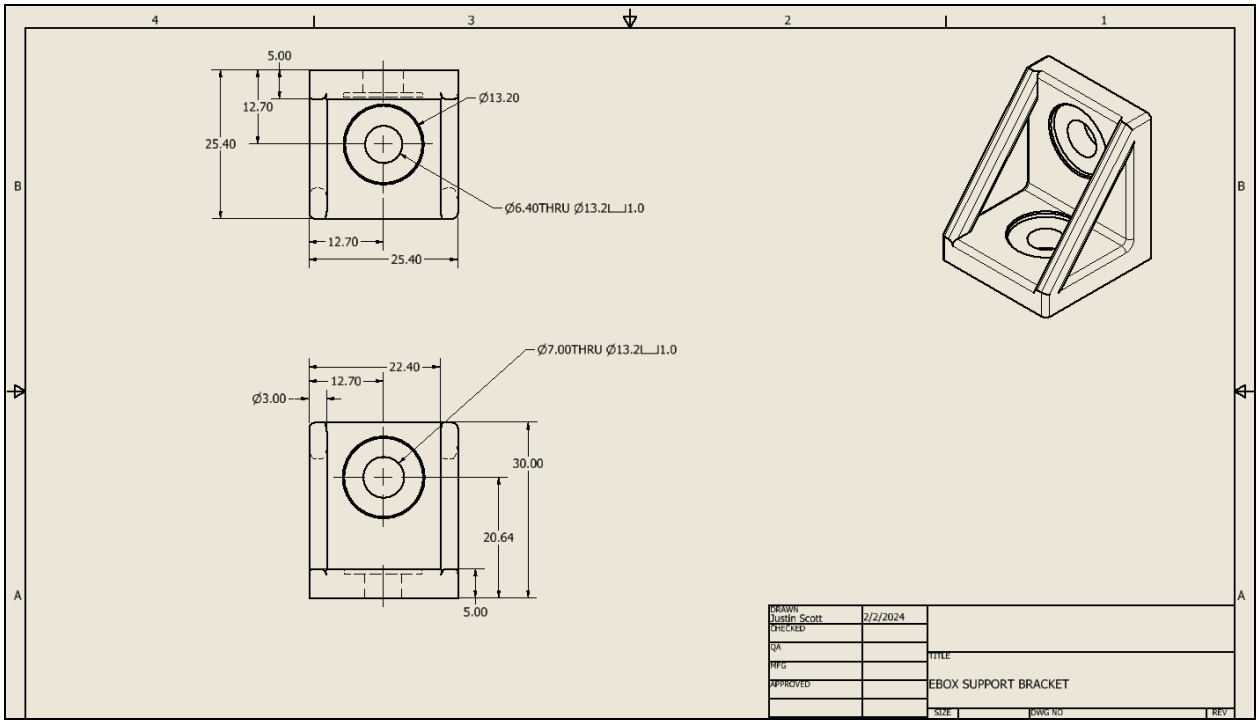


Figure M-16: Ebox Rail Support

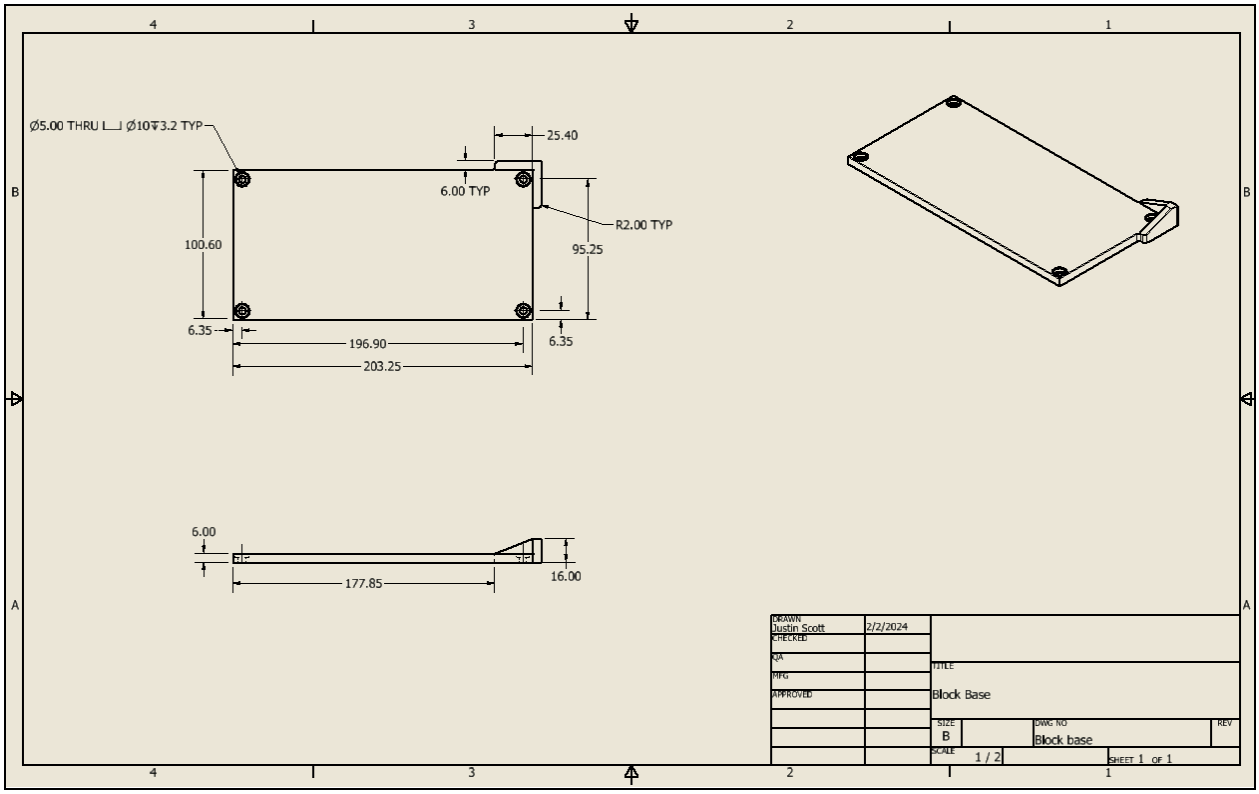


Figure M-17: Cinder Block Base

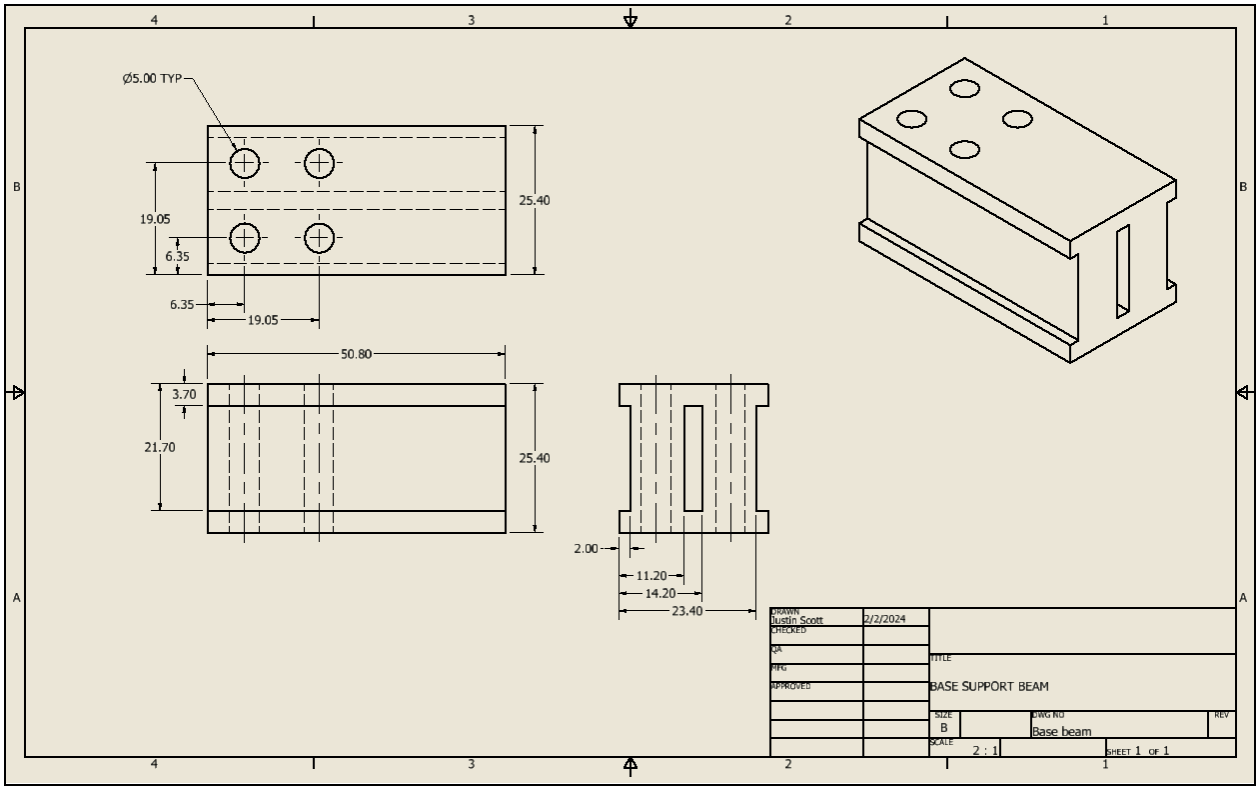


Figure M-18: Cinder Block Rail

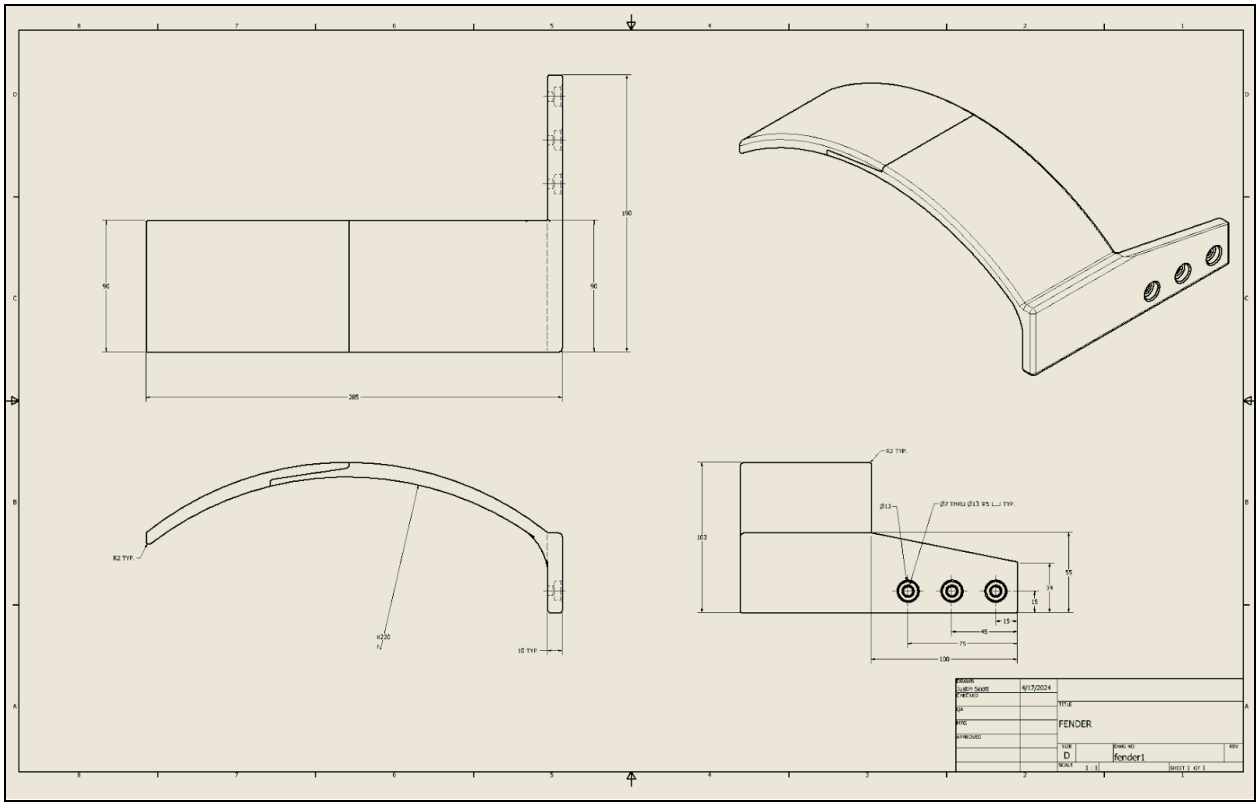


Figure M-18: Fender

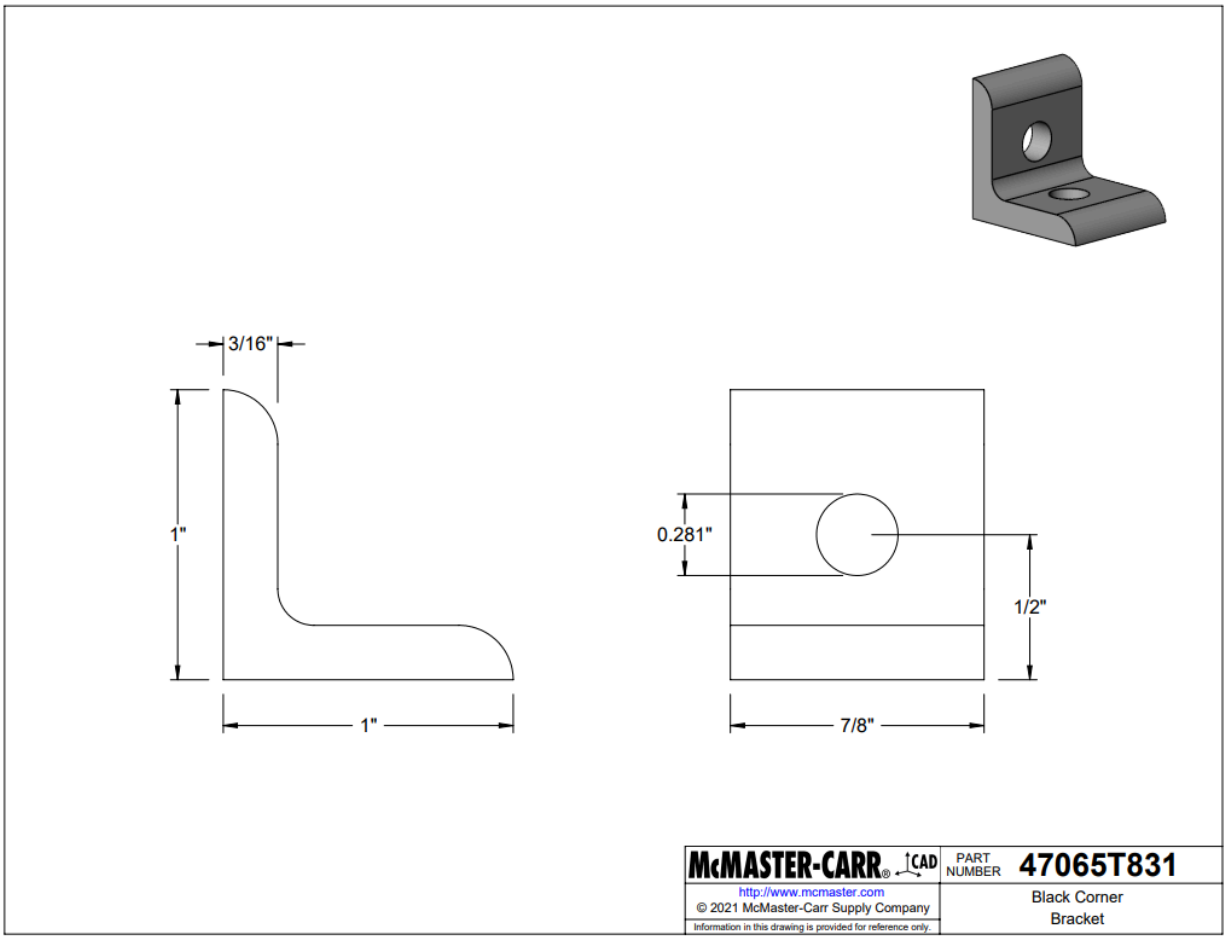


Figure M-19: Black 1” Corner Bracket, Part 47065T831

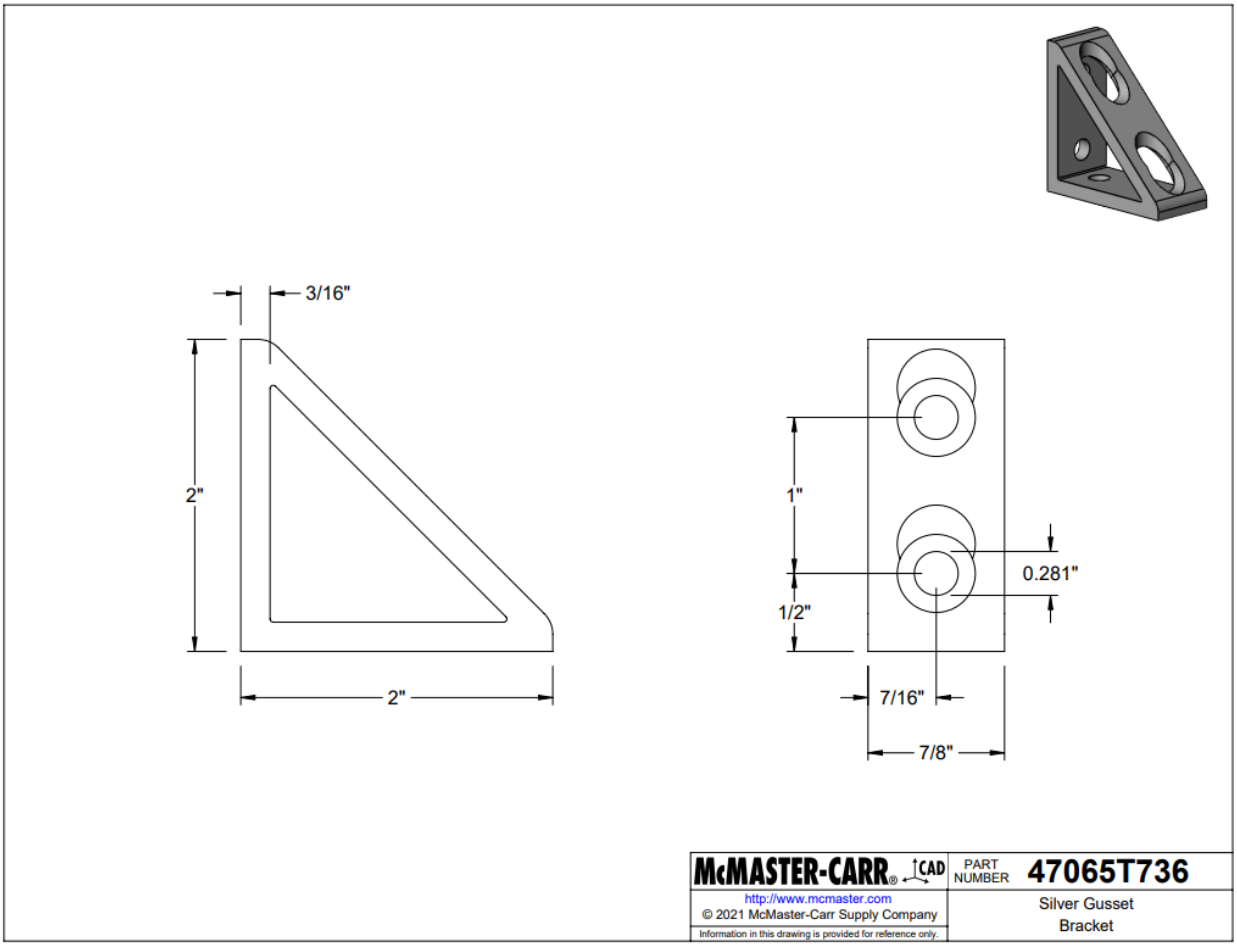


Figure M-20: Silver 1” Gusset Bracket, Part 47065T736

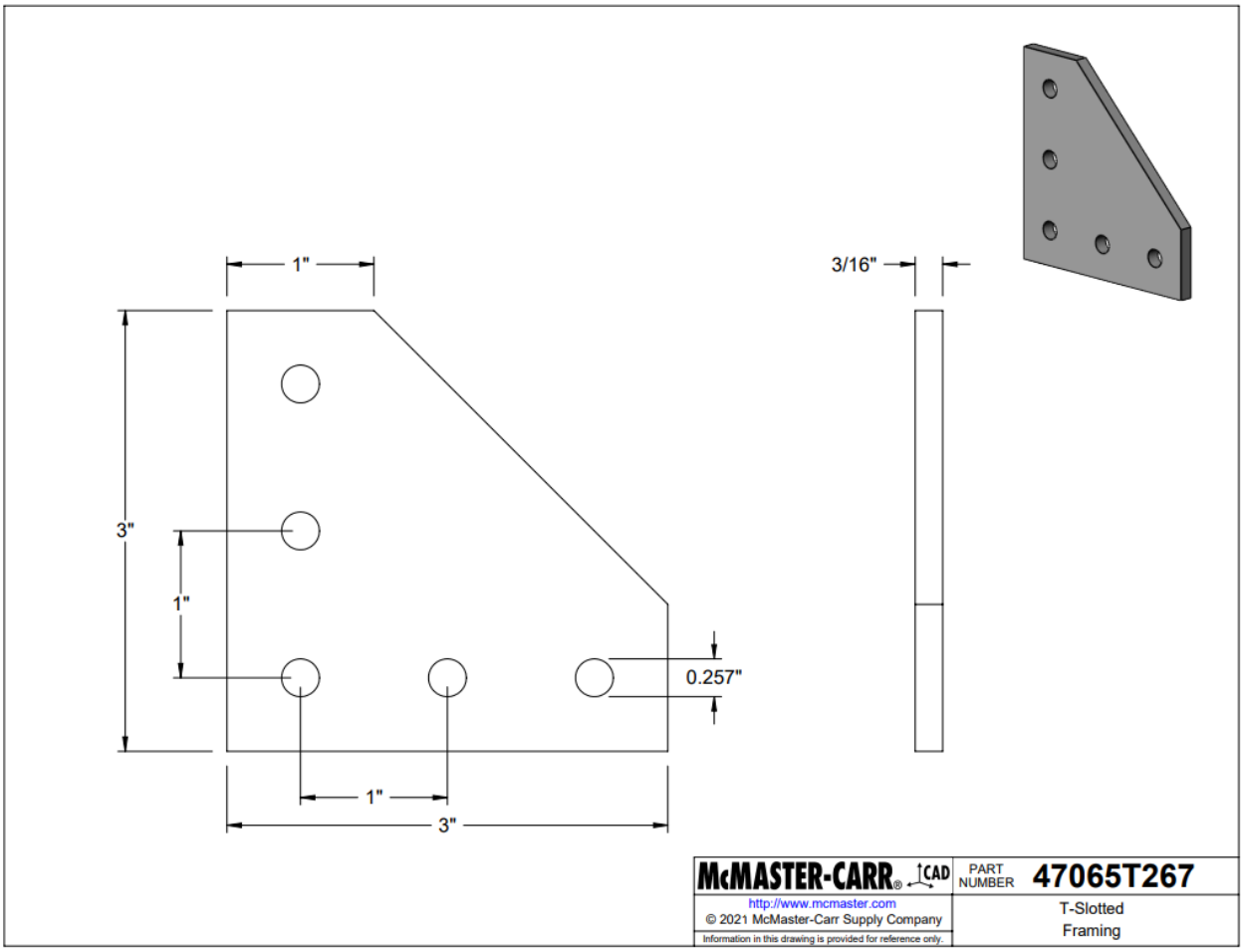


Figure M-21: 1” 90 Degree Outer Bracket, Part 47065T267

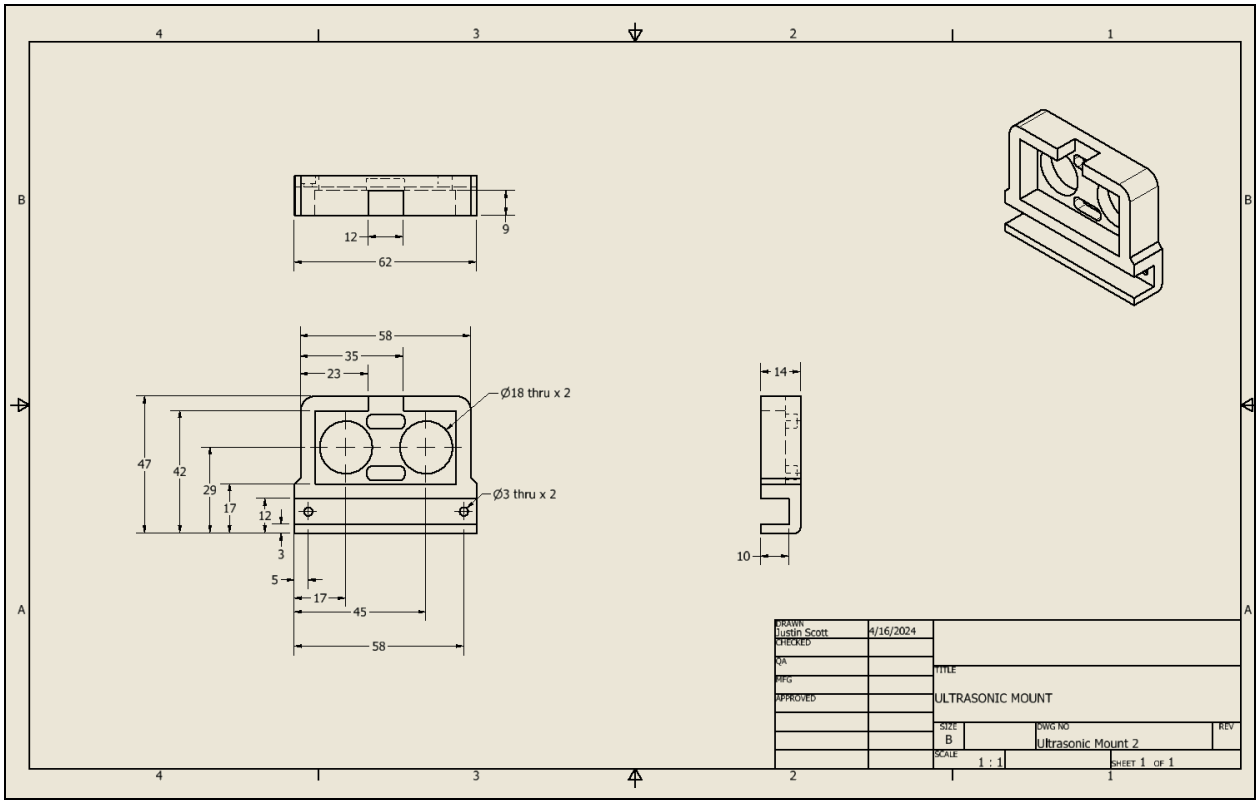


Figure M-22: Ultrasonic Mount

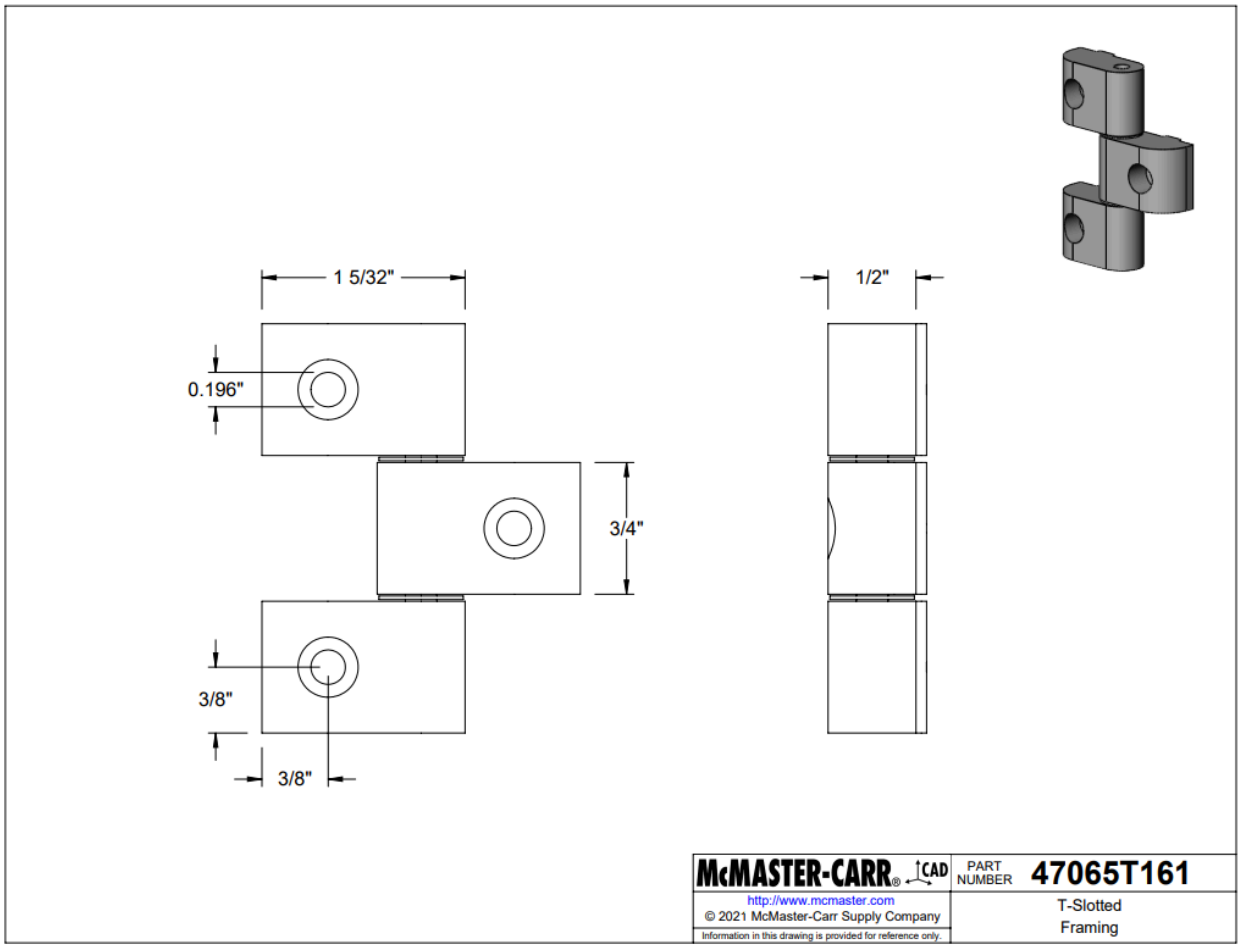


Figure M-23: 1” T-slot Hinge, Part 47065T161

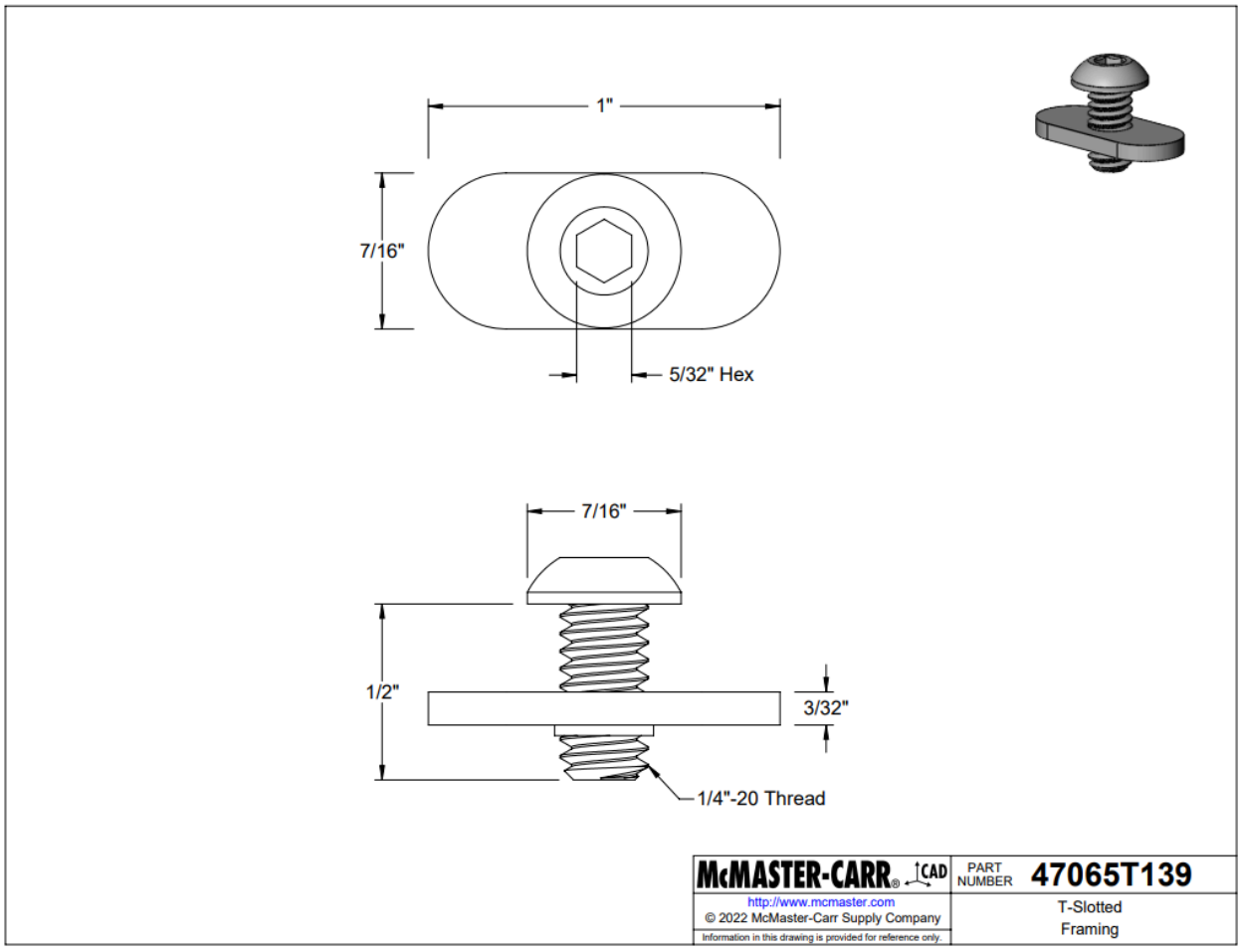


Figure M-24: T-slot Bolt and Washer, Part 47065T139

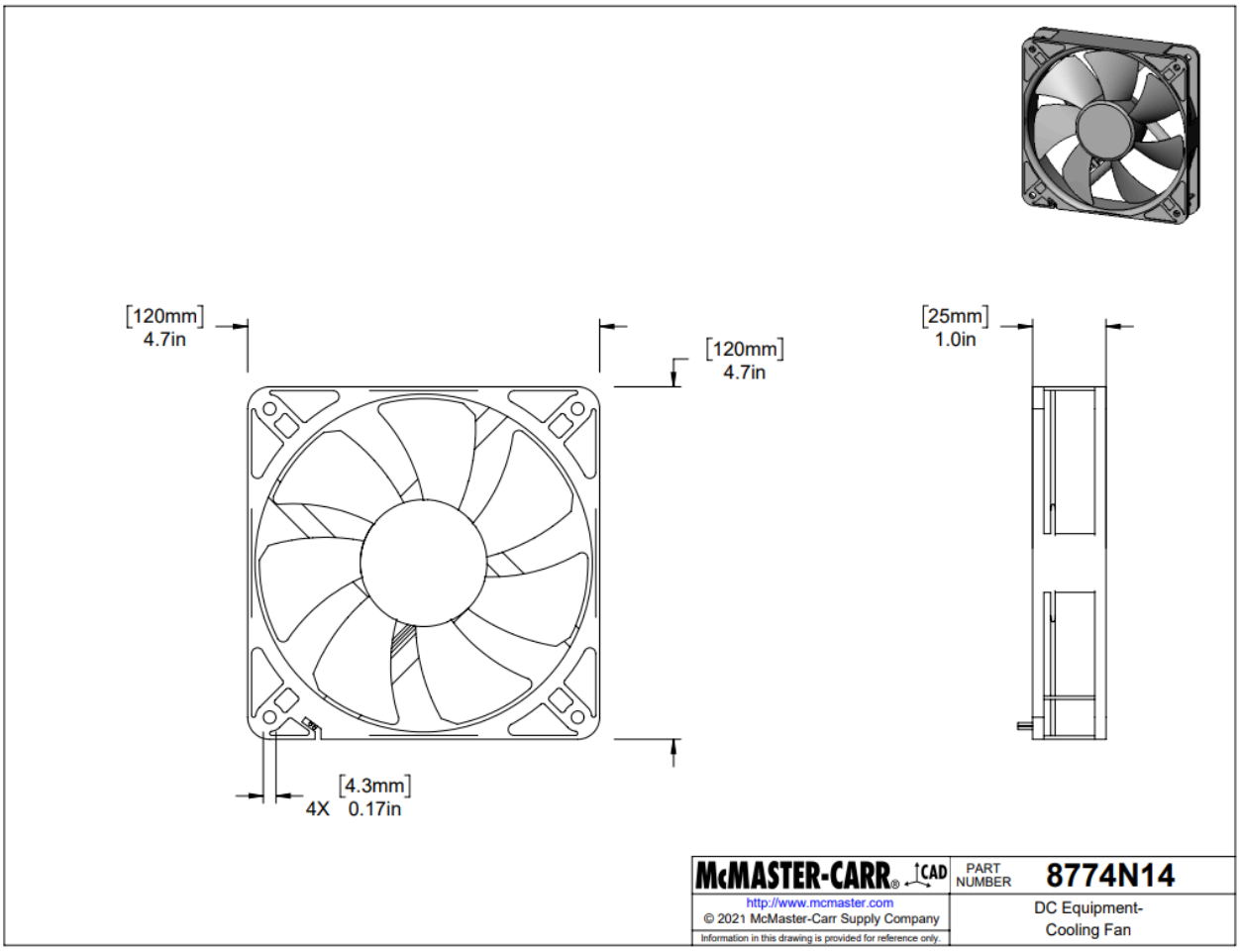


Figure M-25: Cooling Fan, Part 8774N14

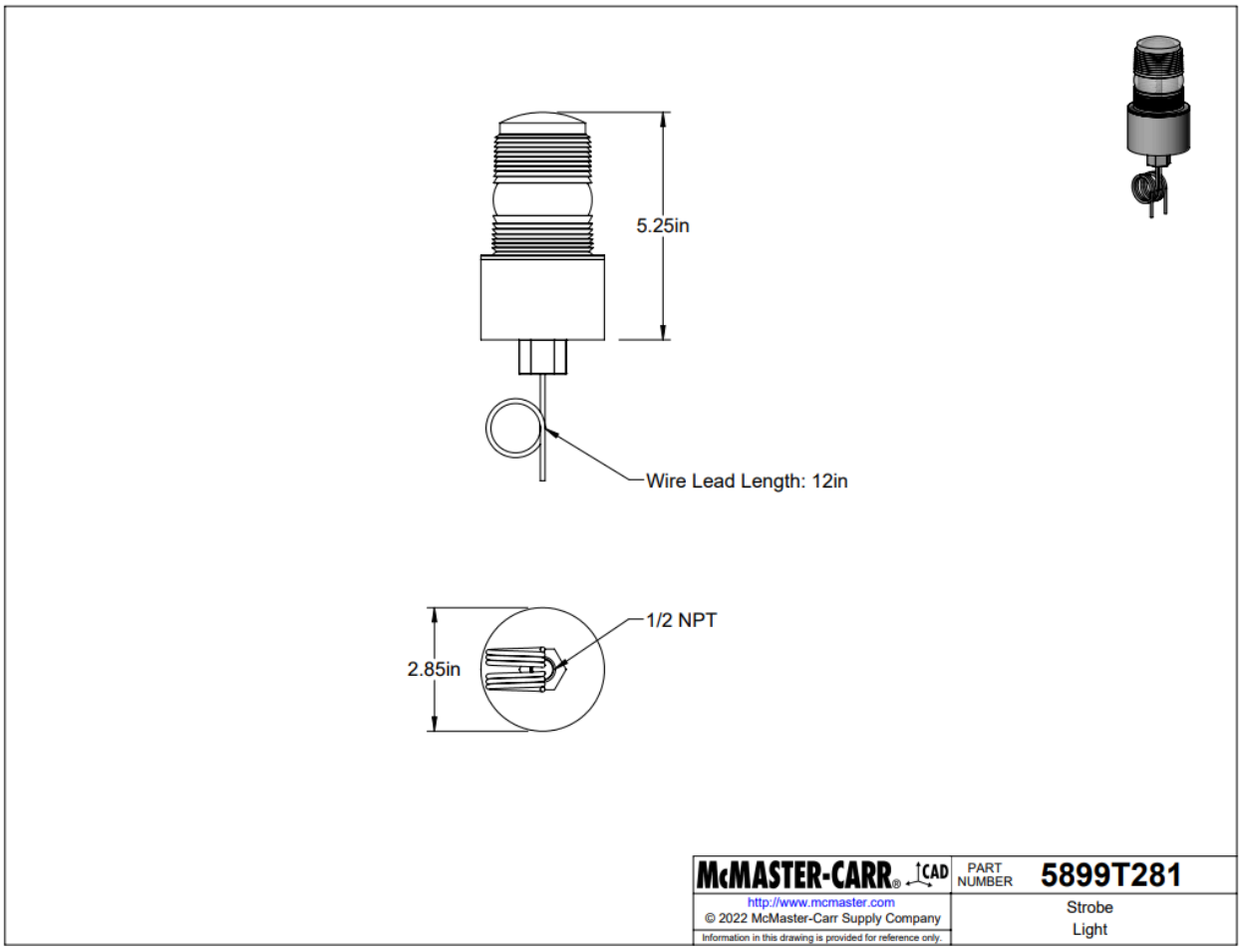


Figure M-26: Strobe Light, Part 5899T281

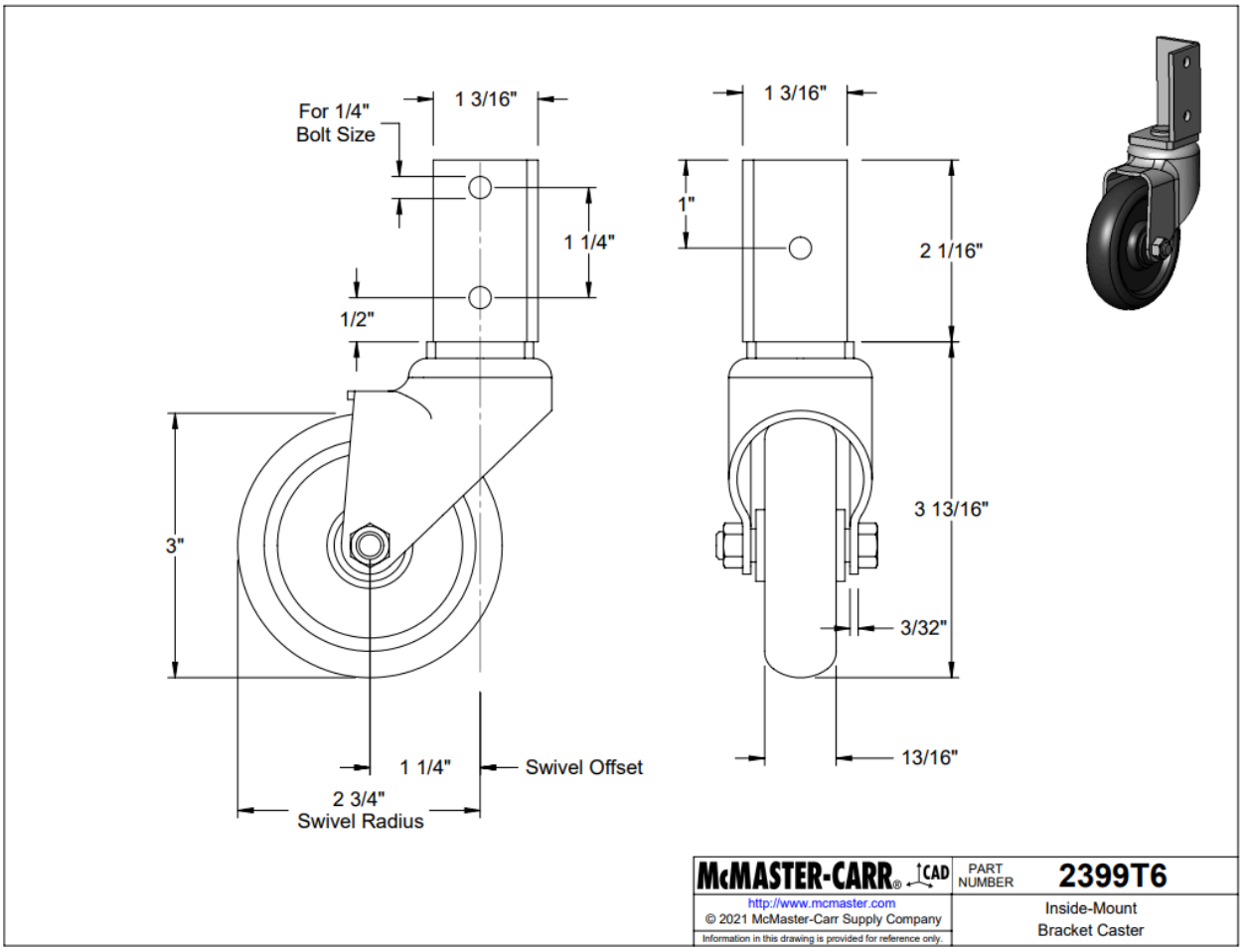


Figure M-27: Caster Wheel, Part 2399T6

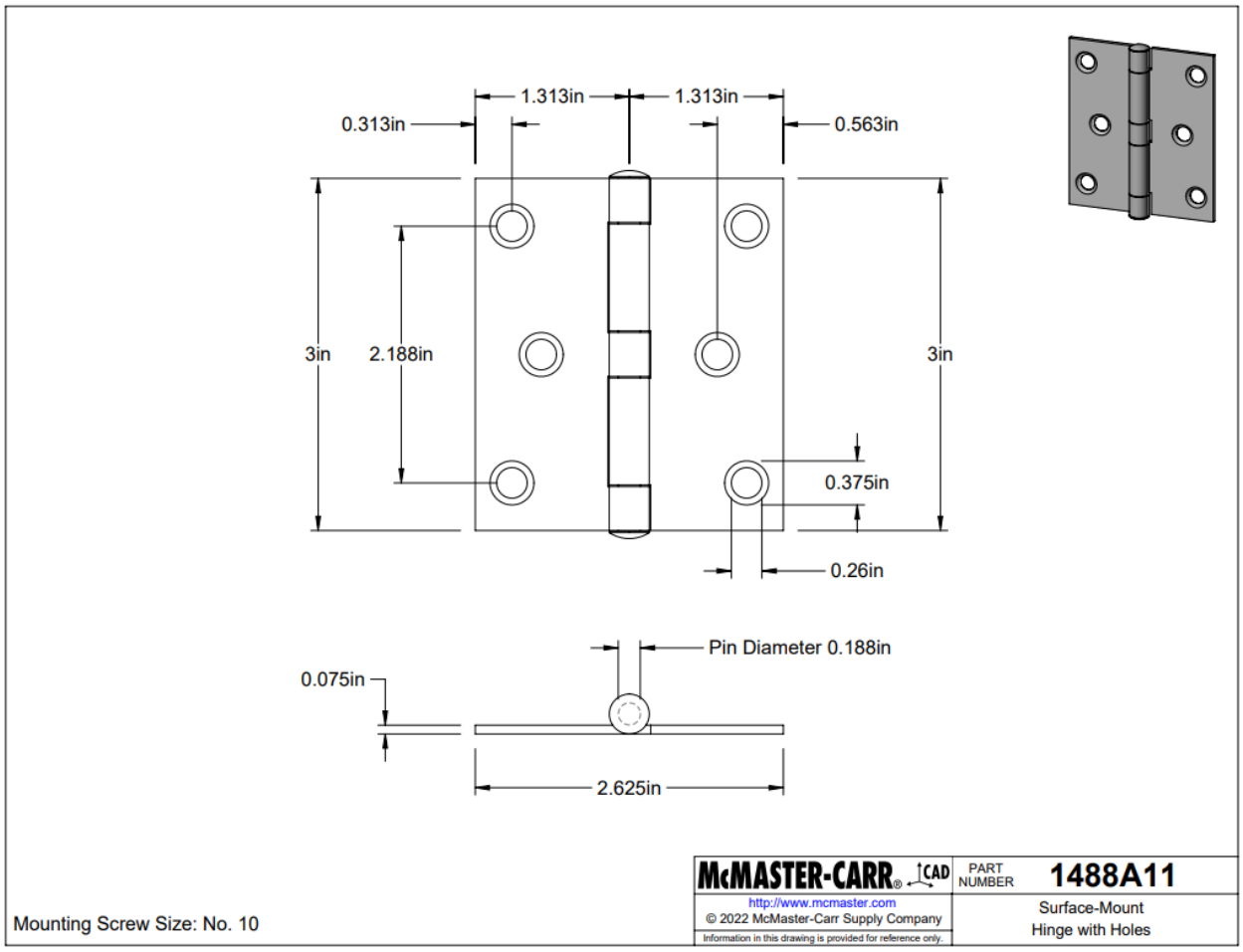


Figure M-28: Hinge for Electronics Box, Part 1488A11

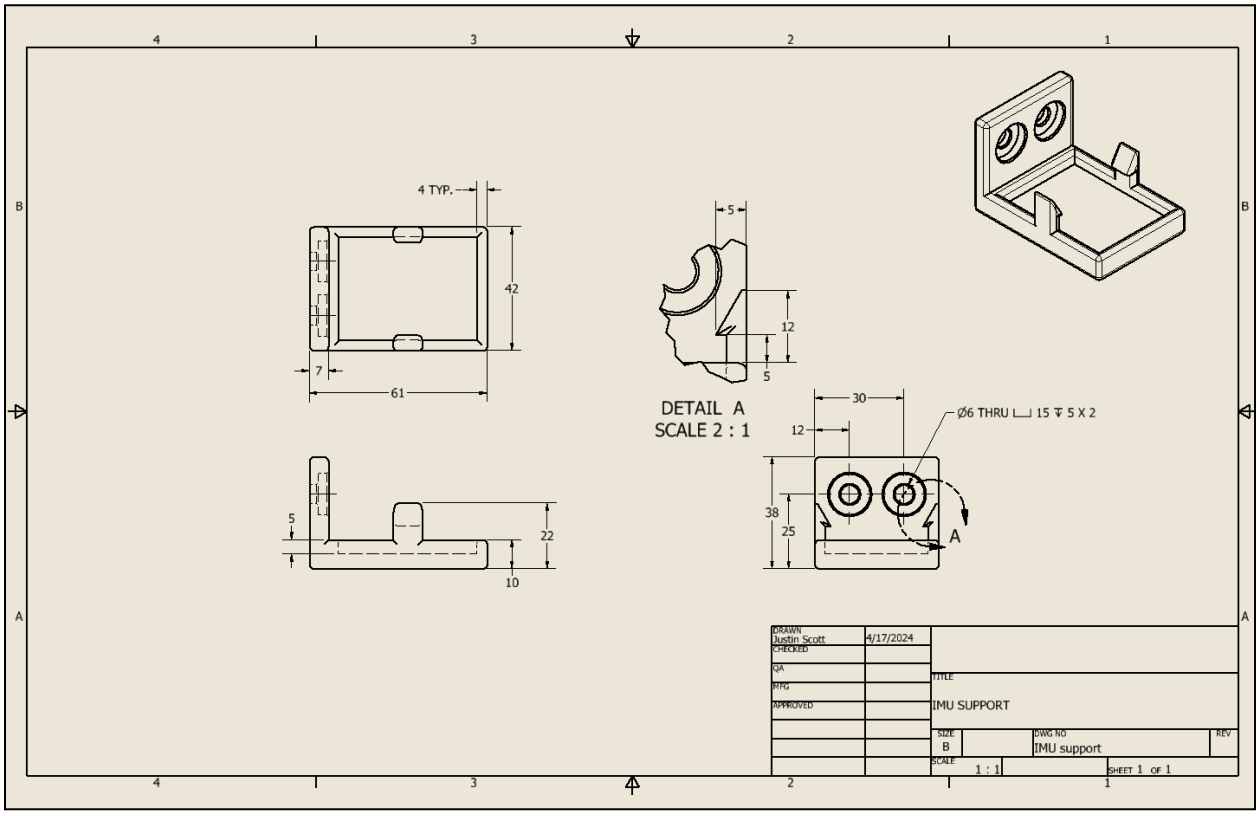


Figure M-29: IMU Support

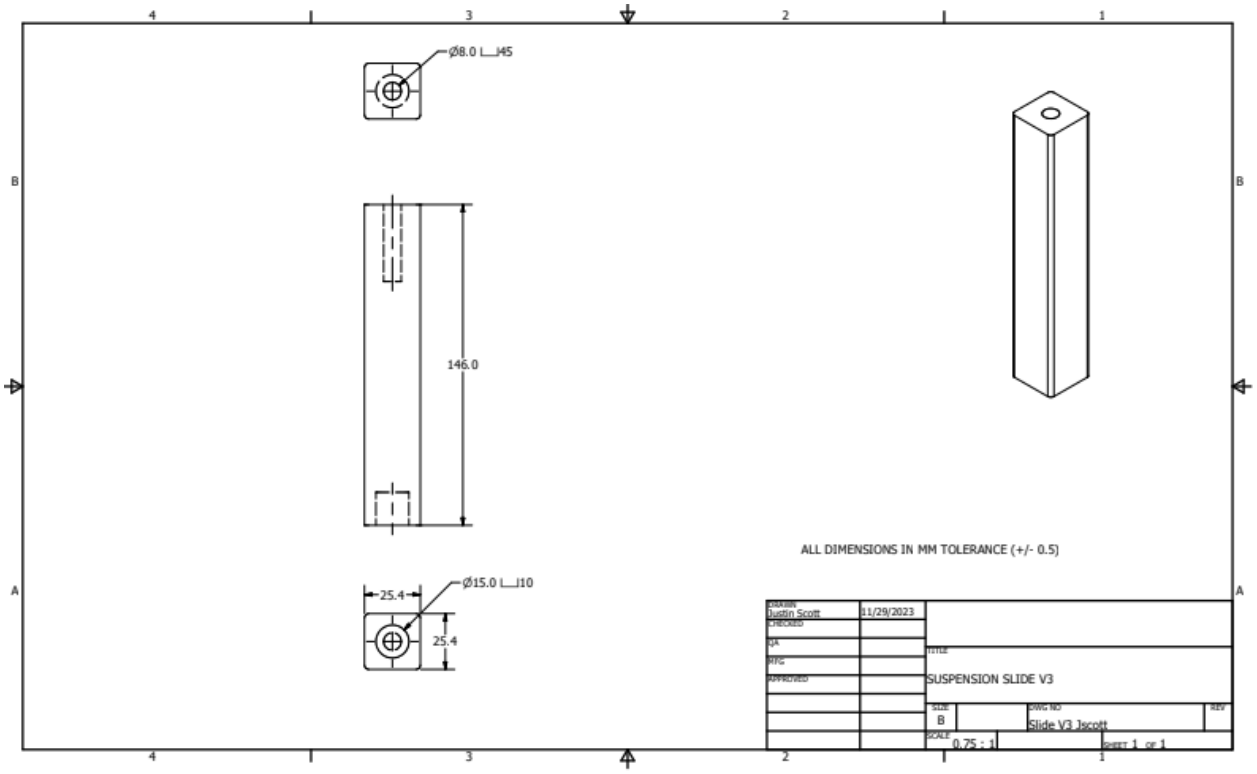


Figure M-30: Suspension Slide

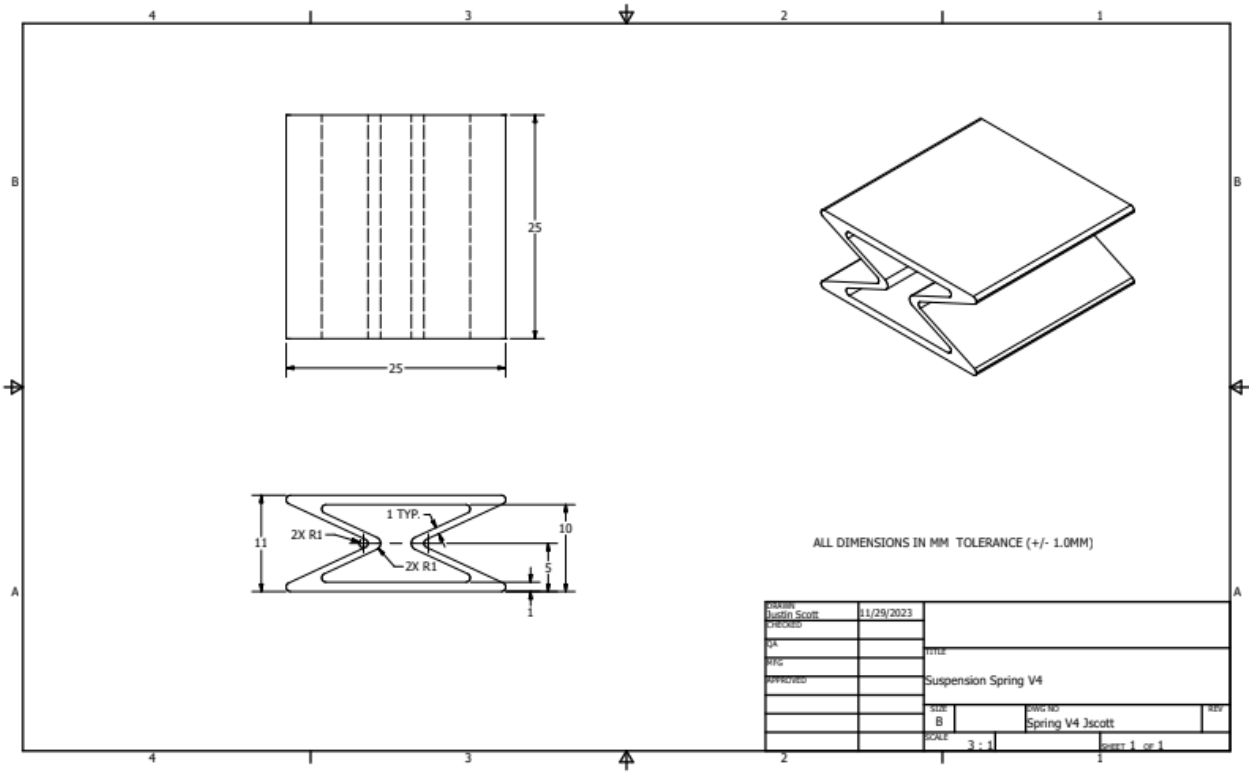
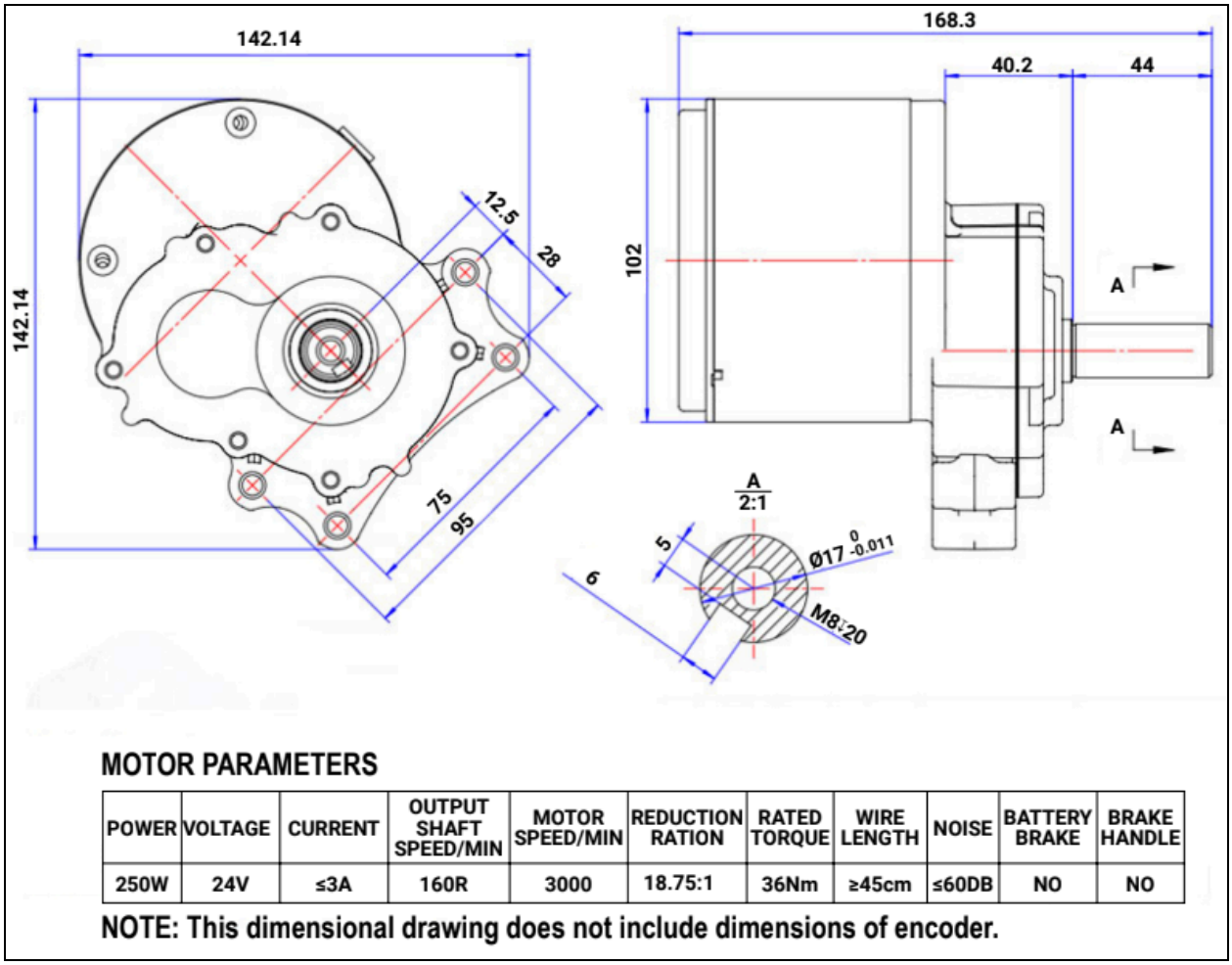


Figure M-31: Suspension Spring



MOTOR PARAMETERS

POWER	VOLTAGE	CURRENT	OUTPUT SHAFT SPEED/MIN	MOTOR SPEED/MIN	REDUCTION RATION	RATED TORQUE	WIRE LENGTH	NOISE	BATTERY BRAKE	BRAKE HANDLE
250W	24V	≤3A	160R	3000	18.75:1	36Nm	≥45cm	≤60DB	NO	NO

NOTE: This dimensional drawing does not include dimensions of encoder.

Figure M-32: Motor Drawings, Model PCS-250 or MY1016Z

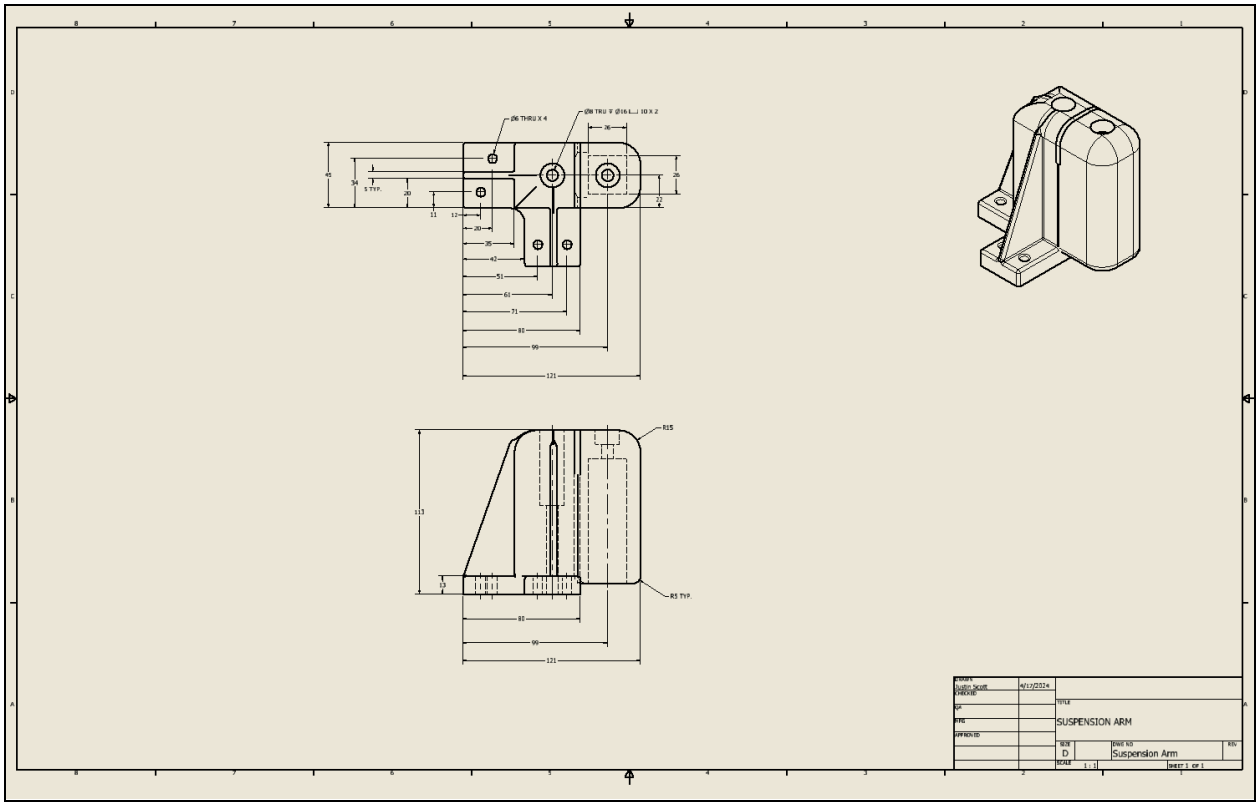


Figure M-33: Suspension Arm

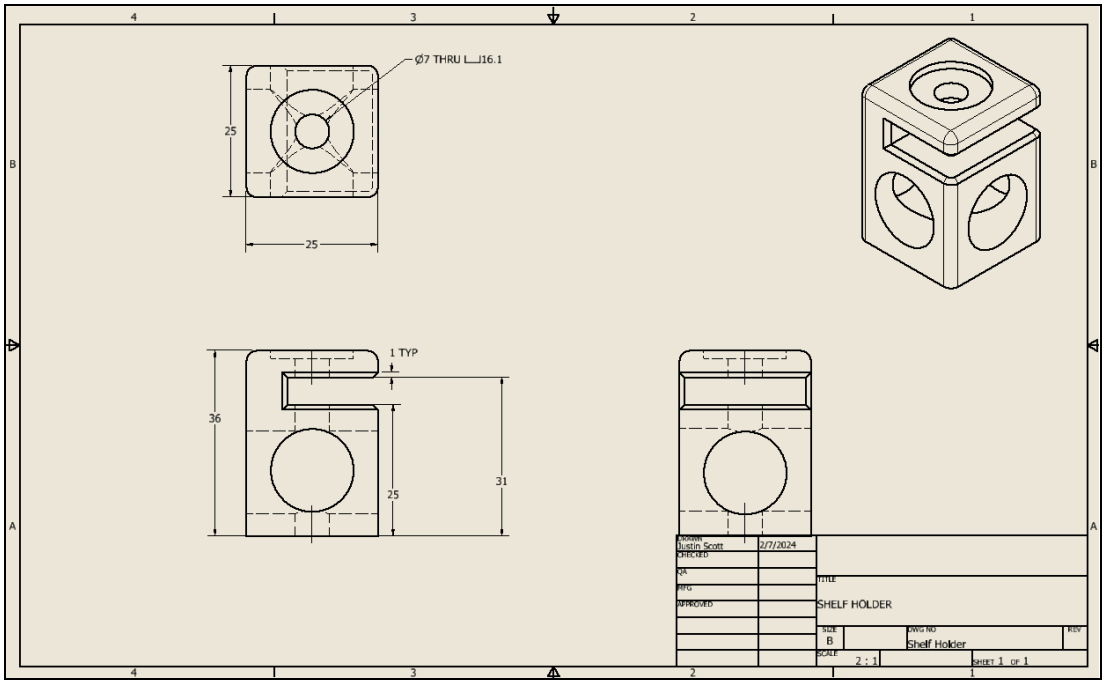


Figure M-34: Laptop Shelf Support

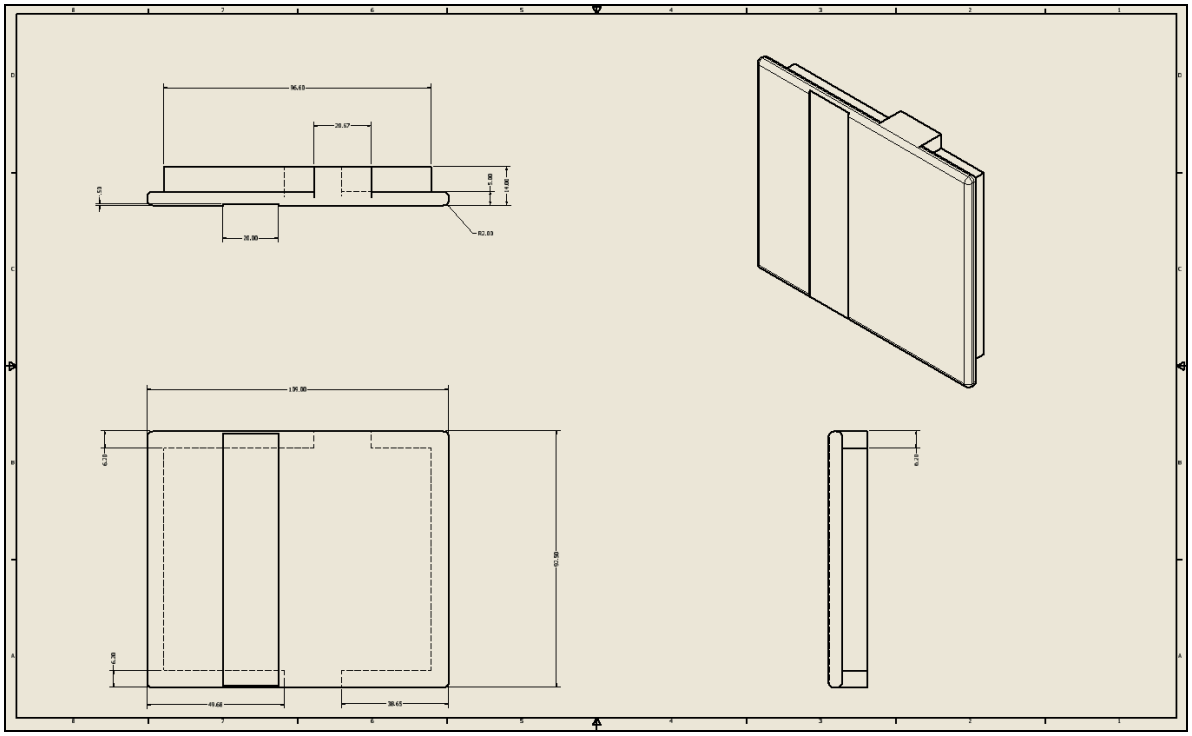


Figure M-35: Motor driver box lid

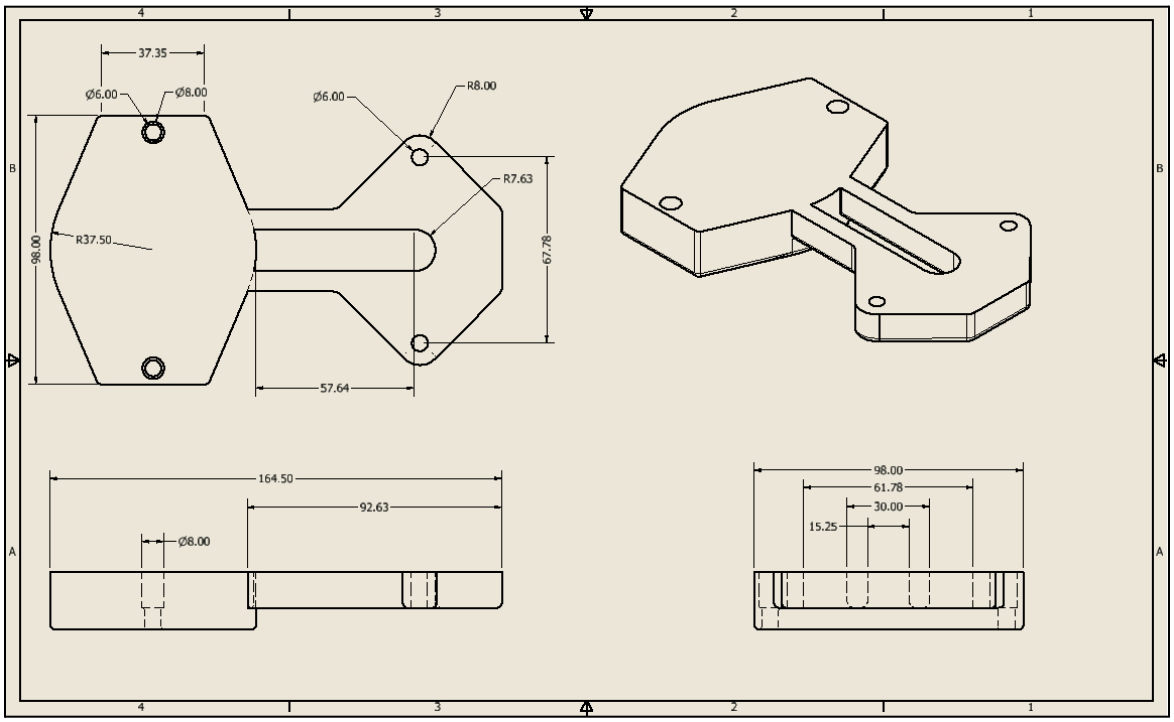


Figure M-36: GPS Mount

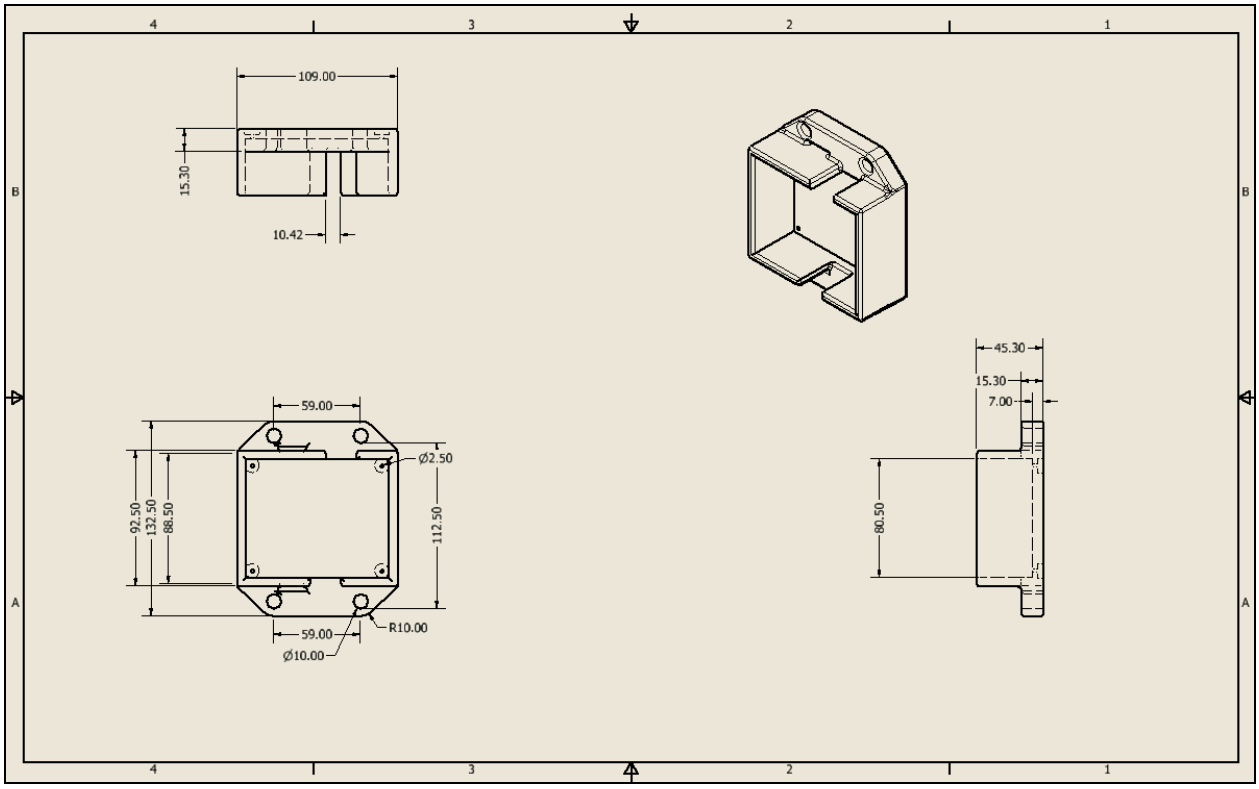


Figure M-37: Motor driver box

Appendix N: Insights

During the design process of the low-fidelity prototype the team made several changes to the original design ideas. Originally the design chosen was that of the rear-wheel drive, zero-turn style. However, fairly early on it was decided to reverse the design and instead make it front-wheel drive. The team believes this would help the navigation team by decreasing the distance between the sensory gear and the drive wheels, and thus prevent pathing and navigation from needing to account for that extra distance (no "predicting the future").

The mechanical team also considered that using undamped casters and wheels may be too jarring on the vehicle over the roughness of a parking lot. This is what inspired the idea to use a suspension system for those wheels on casters. It was considered to simply purchase casters that had a built in suspension, but it was decided to make the design of the vehicle more original and variable, thus the decision to make a custom suspension system. This can be seen in higher detail in Appendix G.

The team created another iteration on the design. It had the electronics box centered in the middle of the vehicle and the cinder block located at the rear as well as a central camera pole. However, the cinder block protruded into the electronics box and the camera pole required extra support in order to keep the camera stable and was impeding the ability of the hood to open properly. It was decided to rearrange the electronics box to the rear and put it into a 12"x12" box that has three panels inside to mount electronics to. The central camera pole was removed and the camera was moved to the front frame as a mount. These changes were made in order to center the mass of the cinder block closer to the main drive wheels and to make the electronics easier to access. The changes also cleaned up the appearance of the robot and also simplified some manufacturing hurdles, eliminating the chance for delays to occur in this space. The cinder block remains accessible for serviceability and verification purposes. The final proposed design and the final actual design can be seen on the following page (**Figure N-1 and N-2**) for comparison.

During the progression from the low-fidelity prototype to the mid-fidelity prototype, several small changes were made to the model in order to ease manufacturability and increase overall functionality.

The first of these changes was to switch from using the imperial system for dimensioning to the Metric system. This improves the compatibility of designs and eases transitions between measurements. Previous to this, a mix of metric and imperial was used that sometimes complicated the constraining of parts in assemblies and building parts to match the others. Therefore, it was determined that standardization was the best method for the team. Due to this change, a ground up redesign of the CAD model was required in order to standardize all measurements to the Metric system. During this rebuild some other small changes were made to the design to optimize the shape and form that simplified the build process for the future. Changes were made to the framing which simplified the structural shape. Changes to the laptop compartment and the cargo area were made that decreased the amount of materials that would be needed and simplified the overall shape of the vehicle.

A change that required dedicating quite some time was optimization of the caster suspension. The original design was found to be flimsy at the joint and was feared to be in danger of failure while turning. A new design was created that deleted the joint in favor of a solid suspension. This model was much stronger and would serve the required purpose, however, it was now a tad too rigid for what was needed, possibly losing the dampening benefits. In order to add the damping effect needed, a detached slide was added with adjustable springs that kept the strength of the solid arm but decreased the rigidity. This new suspension can be viewed more in depth in Appendix G. After these changes, the final iteration of the mid-fidelity design was completed and can be seen in **Figure N-3**.

Figure N-1: Proposed Early Design

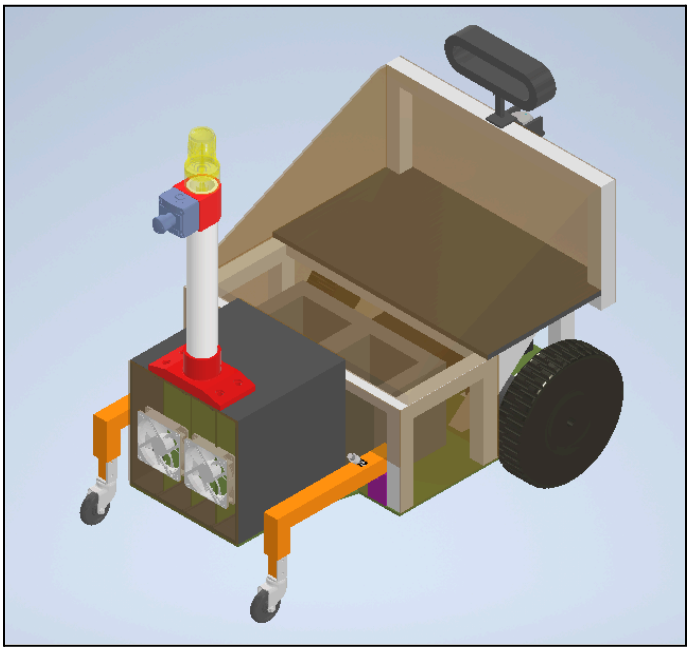


Figure N-2: Final PDR design

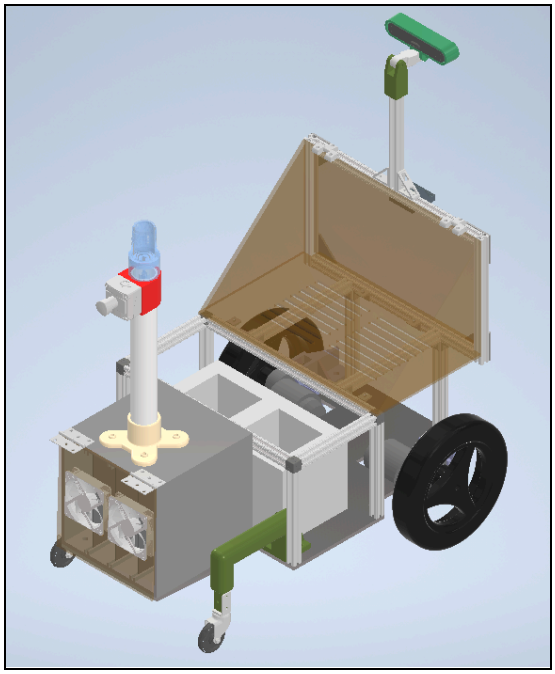


Figure N-3: Final CDR design

Suspension

The suspension design for Rock-E has gone through a number of iterations before a final design was chosen. In the original design, the suspension was going to be included in the rear support of the vehicle only. It was determined that trying to design a suspension for the front of the vehicle was too time consuming and would only yield marginal results. The suspension design would include an L-shaped arm that was supported by a spring and rod. The spring and rod would be made of steel while the rest of the arm was 3D printed from PETG (**Figure N-4**).

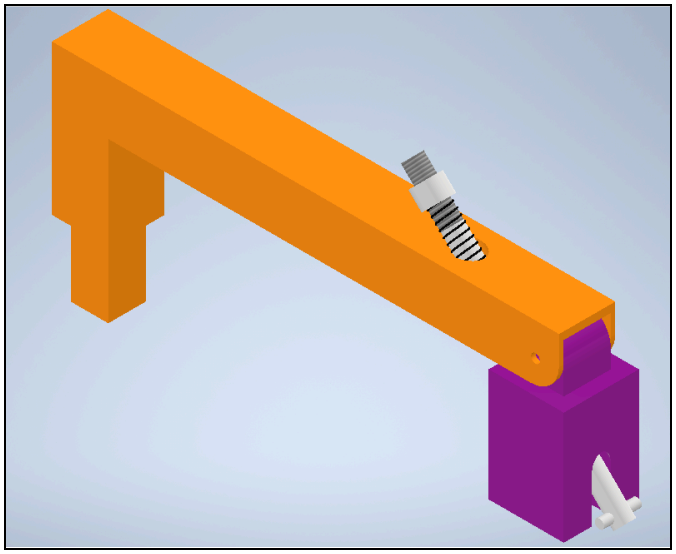


Figure N-4: Original Suspension Design

After the suspension was created, it was tested for strength (**Figure N-5**). It was determined that the suspension design was not going to be adequate for the intended purpose. The design could not withstand the lateral forces the suspension was going to incur.

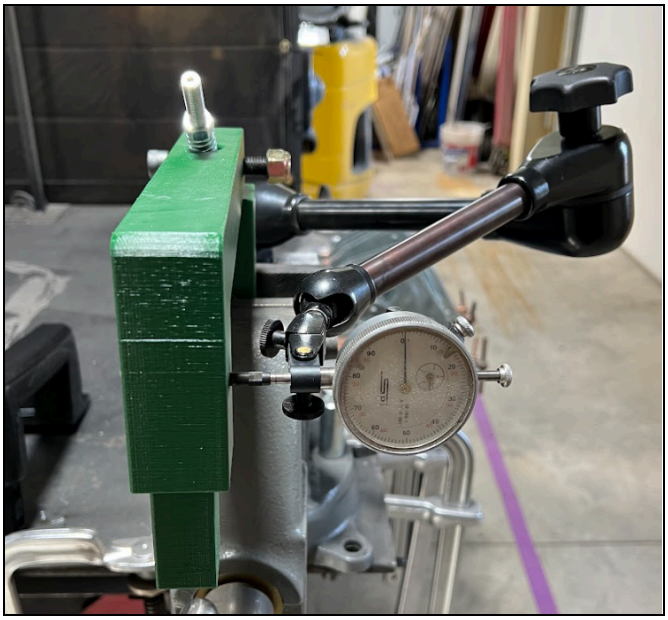


Figure N-5: Lateral Deflection Test of Suspension Arm

At this point the suspension was redesigned as a single, solid, 3D printed arm (**Figure N-6**). The arm was printed from PETG to see if it would be strong enough to support Rock-E. After testing for lateral forces and before installation, it was determined the arm would support the robot. The suspension arm was installed and it did indeed support Rock-E. Unfortunately, once the robot was under power it was determined that the arm was still deflecting too much. One would not be able to control the robot with this design.

Figure N-6: Second Iteration of the Suspension Arm

To minimize the deflection of the arm a third iteration was made which consisted of a two part design. The caster support was turned into a separate slide as seen in **Figure N-7**. This version was made out of PETG.A spring was designed and 3D-printed to install in the arm. The spring would now soak up vibrations from the rear of the robot, just like what was intended for the original design (**Figure N-9**).

Figure N-7: Third Iteration of the Suspension Arm

However, this version of the suspension also ran into concerns. After going over a large bump, the slide part of the suspension broke (**Figure N-10**). The break was consistent with a similar point of high stress seen in the dynamic simulation of an earlier iteration (**Figure H-7**).

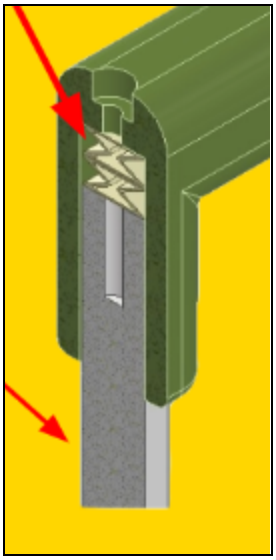


Figure N-10: Spring and Slide Setup

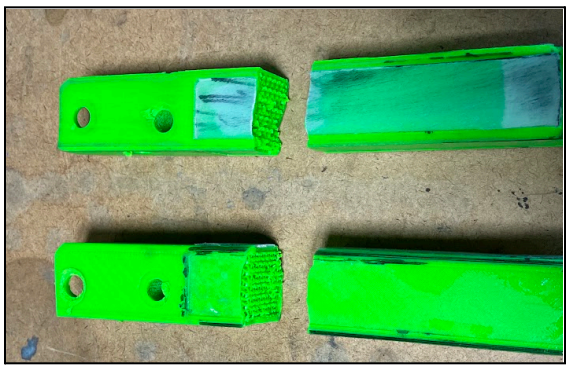


Figure N-10: Broken Suspension slide

The arm was once again redesigned. To reduce the stress, the suspension arm was shortened considerably and it was printed out of carbon-fiber reinforced nylon to increase durability (**Figure N-11**). This design was much stronger and the lateral forces were not deflecting the arm. The springs also helped to reduce vibration.

Figure N-11: Final Suspension Arm Design

Chassis/Frame

Due to a desire to incorporate adjustability to the LIDAR mounting system, changes were necessary to the framing on the front of Rock-E. Limited by a lack of spare aluminum extrusion and bracket pieces, the decision was made to shorten the existing pieces on the front corners to match the length of one remaining piece that would then be added to the middle of the front. This middle piece enabled a custom LIDAR mount to slide vertically to the desired height (rather than its previous static mount on the topmost beam). Thanks to anticipating the need for modularity, this change did not require many other elements to change and was incorporated seamlessly.

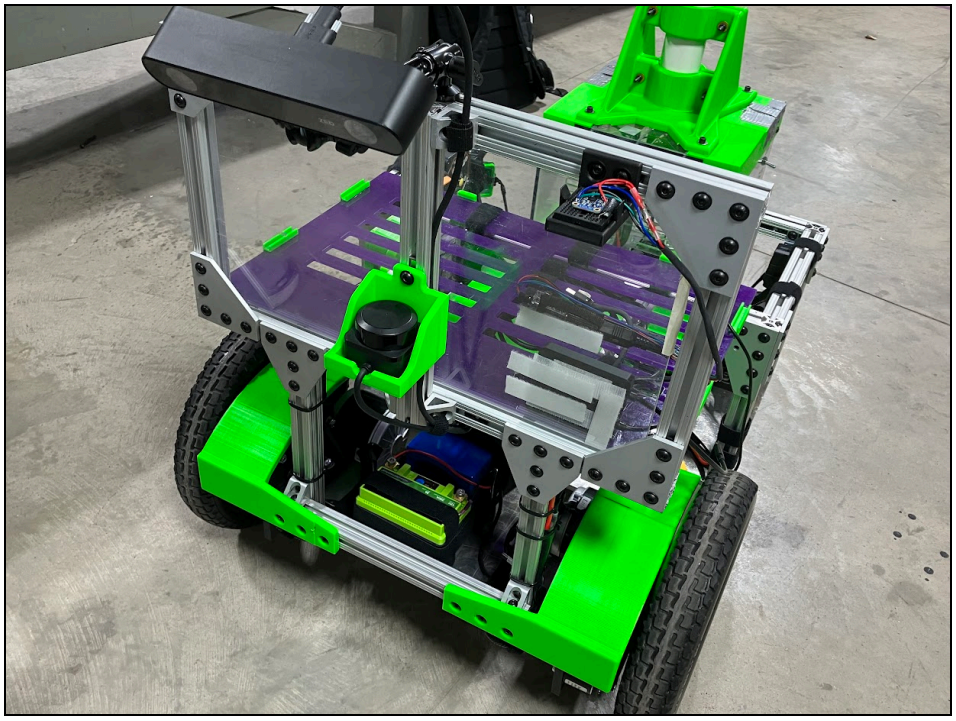


Figure N-11: Revised LiDAR Mount and Front End

Physical construction of the robot showed a need for wiring to be routed under the cinderblock payload. A design utilizing additive manufacturing was added, utilizing tapped holes in the baseplate and bolts going through spacer columns and flat plates. These flat plates initially incorporated their own corners to form a tray for the payload cinderblock to be set upon. When physical testing began, one particular test resulted in the payload shifting forward and breaking off the corners of the plates. As such, it was deemed necessary to reinforce these retention pieces with metal brackets that were recycled from scrap in the lab and sandwiched between the plate and columns. Further physical testing showed that this design held up to the forces of the payload if it shifts.

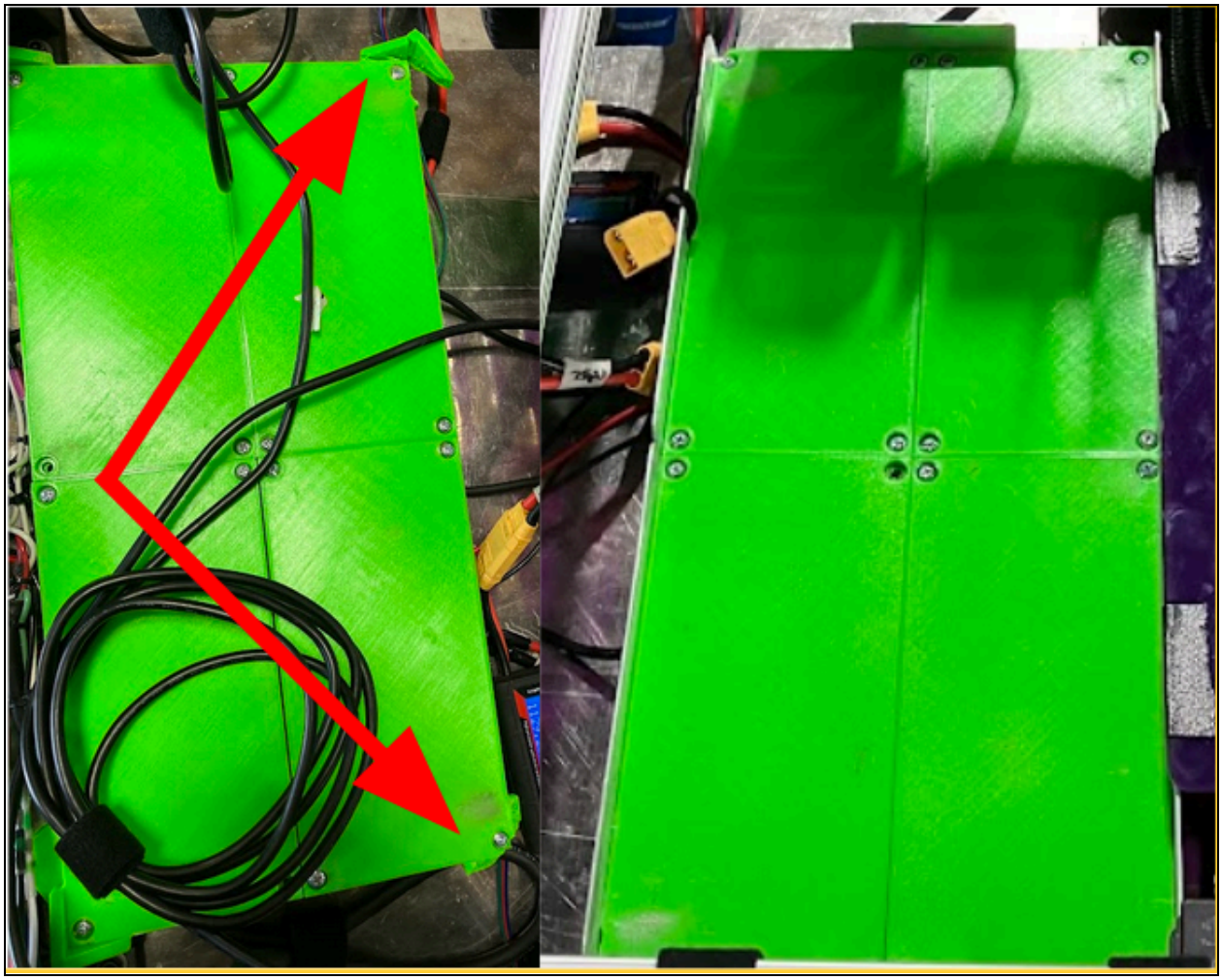


Figure N-12: Initial Design of Payload Shelf after Damage (Left) and Improved Design (Right)

During the fabrication phase, further insight was progressively collected but not implemented regarding manufacturability of the Rock-E design if the design were to pass beyond the prototype stage. Most approaches taken to make Rock-E were appropriate as the vehicle is a one-off prototype that will not be mass produced. However, it should be noted that many changes could be made to subassemblies to enable production if given sufficient funding. For instance, most or all additively manufactured parts could be better made with polymers and molding, such as the LiDAR, motor driver, and ultrasonic sensor mounts. Quite a few subassemblies could better be approached by consolidating down into a bare minimum of parts, such as the

electronics box or payload holder. Hardware used to assemble the prototype was found around the lab, and so there is suboptimal usage and hand-fitting that was required in assemblies such as the motor and light mounts. Like any initial design, design for manufacturing could be applied to generally improve the cost, aesthetics, and ease of assembly. Naturally, the design could always be further improved upon to incorporate different features or a functional payload.

Electronics Box

In the original design of the robot (at the CDR stage) the electronics box was not designed in any great detail. Unsure of the manner of construction that was to be used the design was little more than just a cubic box. The original thought was to merely print the entire box as a whole. However, upon the initiation of construction it was realized that the 3D printer that was being used did not have a sufficiently large build plate. It was therefore decided that the electronics box would be divided into several pieces to be printed separately and assembled, in conjunction with polycarbonate sheets, into a much more dynamic box.

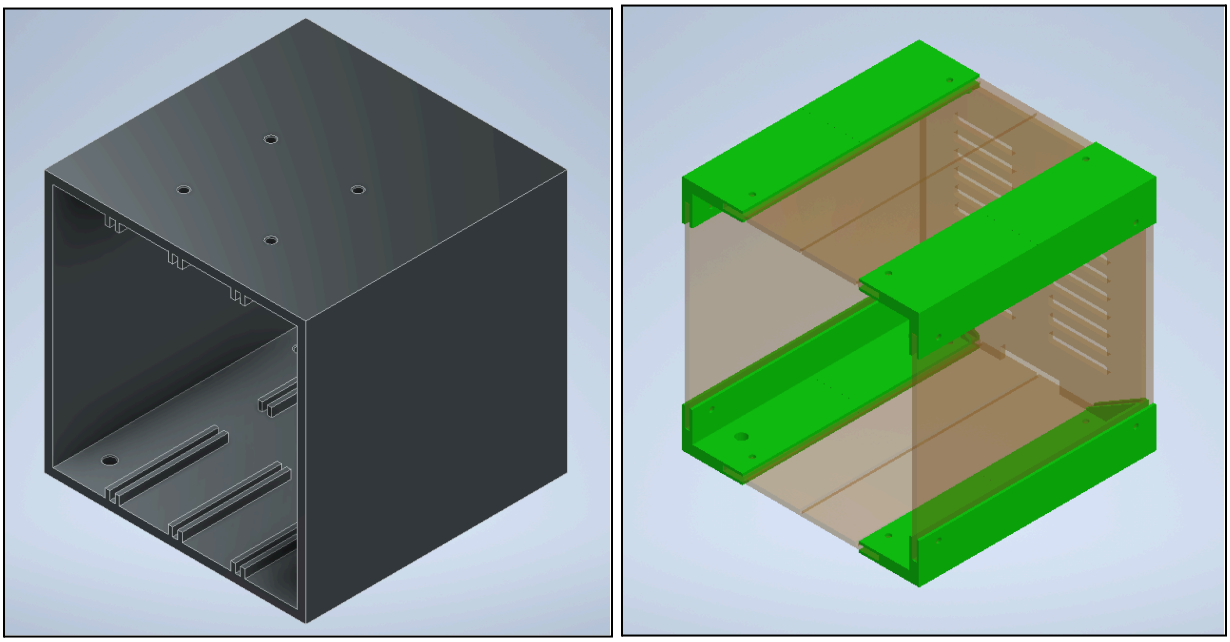


Figure N-13: Original Electronics Box Design vs Final Design

Light Pole and Mount

After the electronics box was redesigned, the mount for the light pole and the height of the light pole was also modified. It was realized that the pole did not need to be quite so tall and so it was shortened. The original mount design (**Figure N-14**) was also not as robust nor stable as was desired.

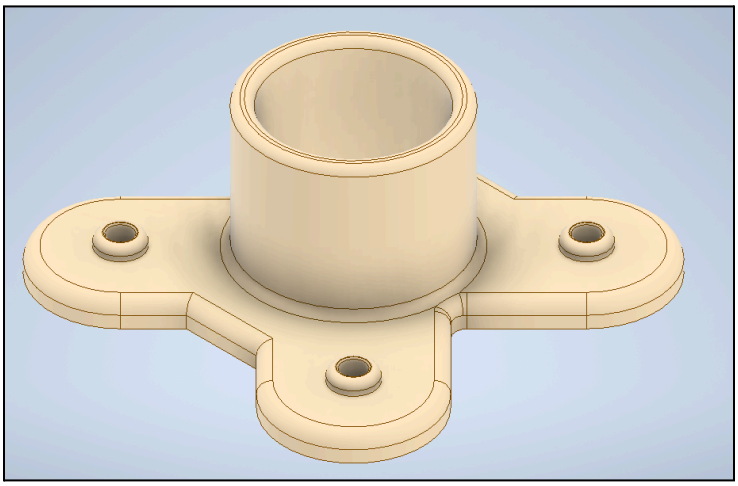


Figure N-14: Original Light Pole Mount

With the new design, two sandwiching mounts with much more structure were used to mount the light pole to the top of the electronics box (**Figure N-15**). Along with those, a new light mount and gps mount were added to the top of the pole giving the pole even more functionality and usability.

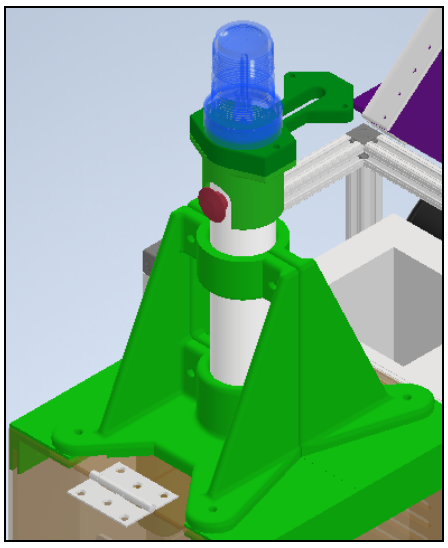


Figure N-15: Final Light Pole Assembly Design

Bumper and Fenders

Following a collision that ended up damaging certain parts of the robot, the decision to add a bumper to the robot was made. The bumper was just an additional bar of square T-slot aluminum that was installed on the front of the robot. As well as the bumper to fenders were also designed and installed in order to better cover the encoders and protect them, in some small way, from the elements (**Figure N-16**).

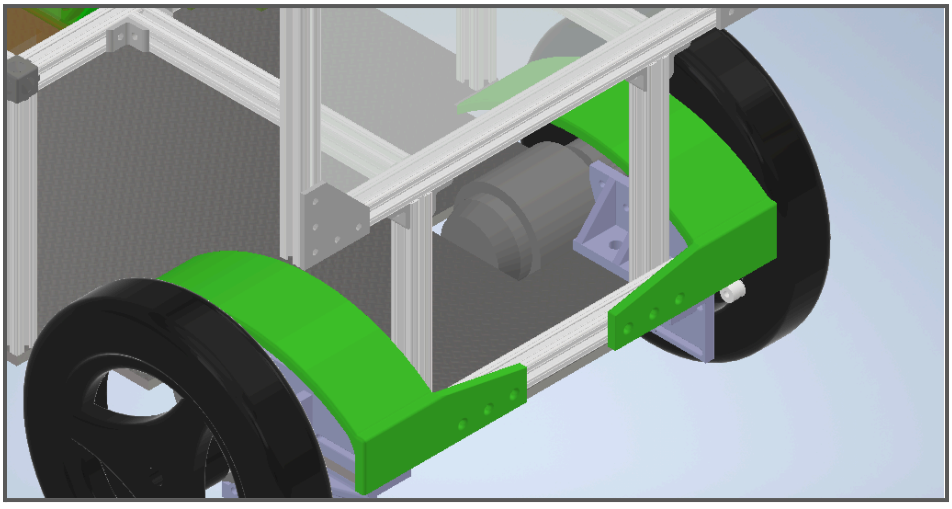


Figure N-16: Fenders mounted to front bumper

IMU Mount

Rock-E needed a suitable place to put the IMU so that it is convenient to calibrate and accessible. The IMU mount consists of a breadboard mounted to a 3D-printed mount, which is then fastened to the front of the chassis. The breadboard serves as a mount for the IMU itself. The IMU has pins soldered to it, but are not connected to anything as its sole purpose is to fix itself to the breadboard. This way, one can remove the IMU and calibrate it and then put it back on the breadboard. The bracket for the IMU mount is also adjustable if need be. A 3D-printed cover was added for the IMU board so in the event that it rains, it is protected.

Battery Change

Rock-E originally had a 20Ah 24V battery on hand from WIU's previous vehicles. This worked fine for a while, but the battery pack became bulged and unsafe. The voltage also was reading lower than rated. To fix these issues, the team ordered a new 10Ah 24V battery from Amazon. This battery actually reads higher than rated, at almost 29V on a full charge. This is fine and does not hurt the system, as it actually makes the vehicle drive faster. The team made sure to test this system given the voltage increase, and there were no noticeable changes other than the speed of the vehicle. This means that the wiring is still safe and the vehicle is still able to perform all of its safety functions when needed.

Laptop Cover

The laptop cover for Rock-E was originally designed to be a single piece of material (**Figure N-17**). The material to be used was ¼ inch thick purple acrylic. After some experimentation with the equipment that was available, it was determined that the "one piece" solution would not be feasible. The acrylic was too thick for the bender and the heat source to ease the bending process was not powerful enough to fully heat the material.

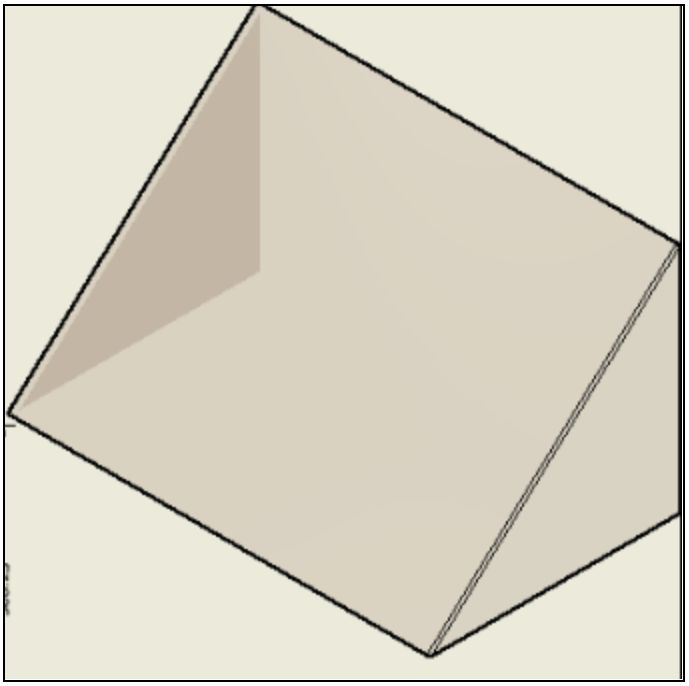


Figure N-17: Original Laptop Cover Design

At this point it was determined the cover would be made out of three pieces of acrylic and 3D-printed sections to connect the pieces. The part was modeled and the printed sections were formed (**Figure N-18**). The printed sections had to be split into two sections each since a single print was too big for the print bed. The printing material used is PETG for strength.

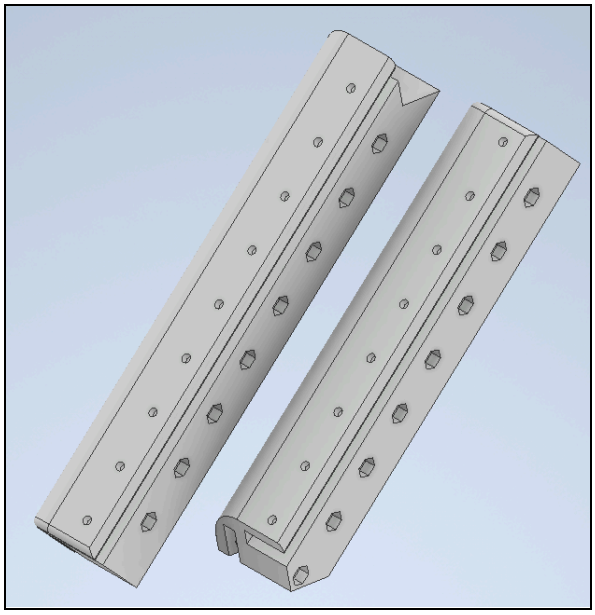


Figure N-18: 3D-Printed Sections of Cover

After the parts were printed, the cover was assembled using 2-inch 8-32 screws with Nyloc nuts to hold everything together. The team was able to bend the top edge of the acrylic to act as a rain shield in case of inclement weather (**Figure N-19**).

Figure N-19: Rain Drip Edge

The original laptop cover design also called for a hinge that would allow anyone operating the navigation equipment access to the laptop between runs. Unfortunately, the hinge locations that were designed would not allow the cover to open far enough to operate the laptop. This was because the camera location had been changed since the original design was created, causing an interference issue.

Another issue brought up by the navigation team was the laptop screen is hard to see in the sunlight. At this time it was decided that the laptop cover would only be used during inclement weather and a sunscreen (**Figure N-20**) would be created to shade the laptop screen during operation on sunny days.

Figure N-20: Sunscreen For Laptop

The laptop cover was one of the most difficult parts of the mechanical build to design. It was a part that was going to require a precise fit. Because of this, and the fact that the cover was believed to not be as important in the overall scheme of the build, its assembly was left until the end of the design process.

If the Rock-E design were to be used again, the laptop cover should be designed and built at the same time as the frame, making sure its fit and finish were correct before assembly of the rest of the robot.

Appendix O: Motor Torque Calculations

In order for Rock-E to have enough power to propel itself around the course, the torque needed at the drive motors needed to be found and used with the mass of the robot to find the power needed to allow Rock-E to travel at a max 5 MPH. The approach was to first assess where the most power would be needed and then find torque based on the force needed at this section of the competition course.

On page ten of the IGVC 2024 rule book [1] it states that the gradient of the course will not exceed 15%. The gradient (or ramp) will be the location where the most power is needed to keep the robot traveling at 5 MPH. Gradient can be converted to degrees using the following equation:

Θ = arctan (0.15) = 8.53 degrees

This angle will be used to help find force. The roll coefficient of the wheels is also needed. This can be found with the following equation:

$$c = 0.005 + \left(\frac{1}{p}\right)\left(0.01 + 0.0095\left(\frac{v}{100}\right)^2\right)$$ [2]

Where:

c = Roll Coefficient

P = Pressure (in Bar)

V = velocity

To solve this a max velocity of 5 MPH (8.04 km/h) and a pressure of 2.06 bar (common in tires) are used. Once the roll coefficient was found one may calculate the force needed using the following formula:

$$F = mg(\sin\theta + c\cos\theta)$$ [3]

Where:

F = Force

M = Mass

c = Roll Coefficient

G = Gravity

Θ = angle

Since the team was unsure of the final mass of the vehicle a range of necessary forces dependent on mass were calculated. Now that the forces are known, one can find torque needed at the wheel:

$$T = rF$$ [4]

Where:

T = Torque

R = Radius of Wheel

F = Force

Now that one has torque and motor rpm (120), one can solve for the power needed to go up the ramp:

$$P = \frac{T\theta}{dt} = T\omega = 2\pi nT = 2\pi\left(\frac{n_{rpm}}{60}\right)T = 0.105 n_{rpm} T$$ [5]

Where:

nrpm = RPM of Motor

T = Torque

The following spreadsheet (Fig. O-1) shows these calculations and can be accessed here:

Rolling Coefficient of Air Filled Tires		
Tire Pressure	2.06	Bar
Velocity	8.04	km/h (max)
Roll Coefficient (c)	0.0099	(dimensionless)

Required Force to Move Robot up the Ramp		
Mass	57	kg
Gravity	9.81	m/s^2
Angle	8.53	Degrees
Force Req'd (F)	88.413	N

Torque Required to Move Robot Up Ramp		
Force	88.413	N
Wheel Diameter	0.305	m
Torque (T)	13.483	Nm

Power Needed for Top Speed		
Motor RPM	120	RPM (max)
Power (P)	170	Watts

Figure O-1: Motor Torque and Power Calculation

Rock-E's mass, including the cinder block, is 57kg. The team used a range of masses to give an idea of the torque (**Figure O-2**) and power (**Figure O-3**) needed to propel Rock-E:

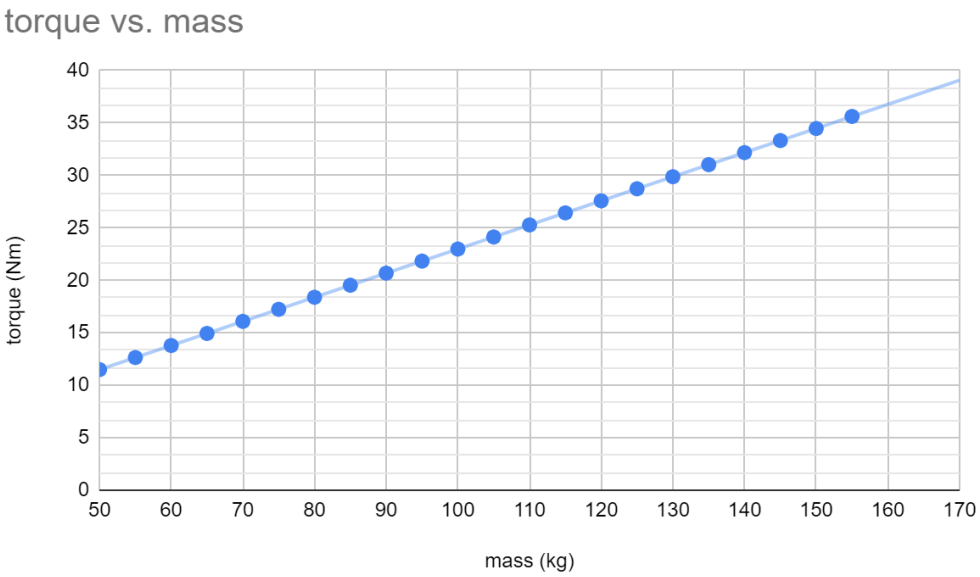


Figure O-2: Torque vs. Mass

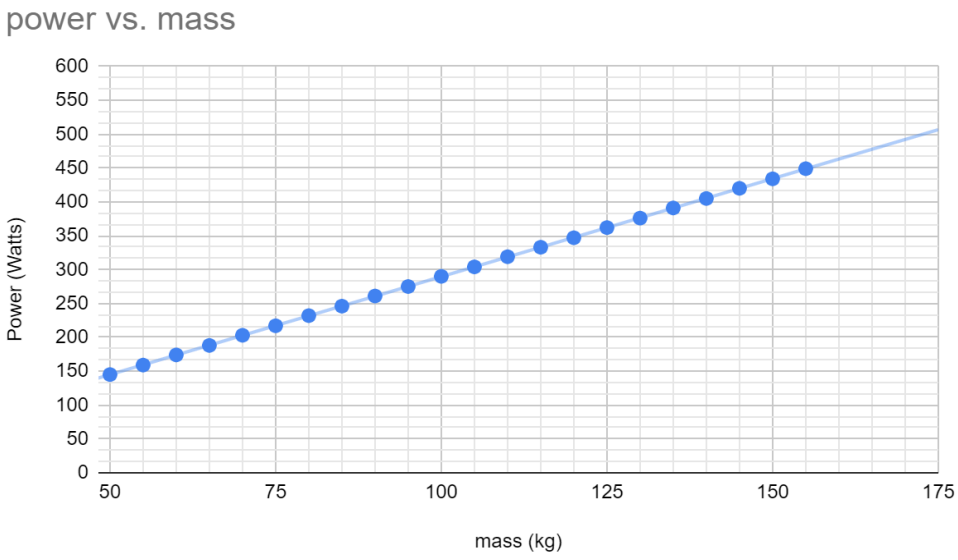


Figure O-3: Power vs. Mass

Appendix P: Component Testing

Motors and Motor Drivers

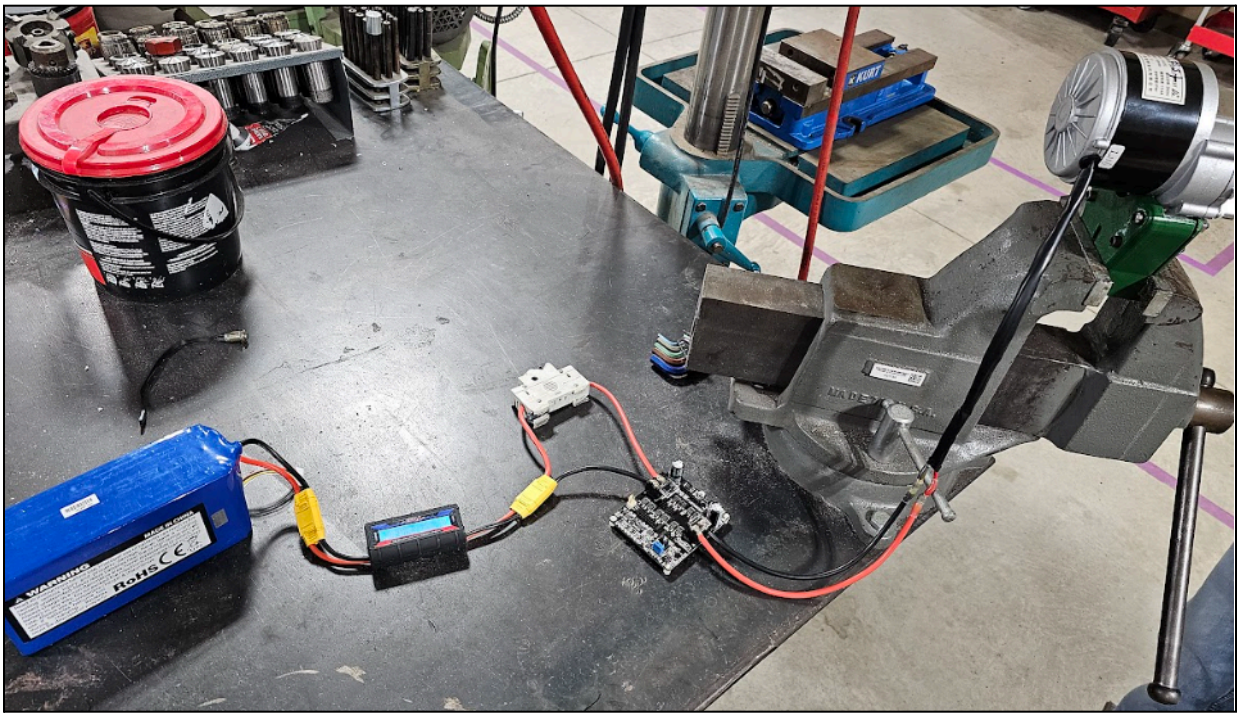


Figure P-1a: Physical Test Circuit

Figure P-2a: Test Circuit Diagram

Shown in **Figure P-1a** is the test circuit for the motor test that was performed in the makerspace on campus. Shown in the figure is a 24V battery connected to a wattmeter that can

show peak voltage, wattage, and current, which is important for this test. After the wattmeter is a 15A in-line fuse for protection over the circuit. Then there is the motor driver, followed by the motor that is mounted to a vise with a custom bracket. **Figure P-2a** shows a simplified diagram of the test circuit.

This motor test was undertaken to verify that the manufacturer's stated specifications are accurate and will thus fulfill the requirements of Rock-E. The Unitemotor MY1016Z motors were selected in part due to their desirable physical capabilities, though suppliers were somewhat spotty when providing specs on their respective websites; for instance, red flags observed during the research phase included that the RPM given on some sites was of the original DC motor's output shaft before it is reduced through the integrated gearbox to the expected 120 rpm. The team wanted to ensure that values stated by the manufacturer and vendor were accurate for all values that are critical to the design. To accomplish this, a test rig was prepared in the makerspace that allowed a verification test to be conducted; this was composed of an additively manufactured motor mount clamped in a vise, the motor connected to the test circuit (**Figure P-2A**), and a load arm with a variable-weight hanger. The watt meter of the test circuit provided a readout of electrical values: max current, max power, and voltage. Weight plates were stacked on a hanger at the end of the load arm and the motor was powered by pressing the directional test button on the motor driver. If the arm successfully traveled to around the vertical position the test was considered successful; the maximum load the test could handle was 30 lbs.

Suspension

From the beginning of the chassis design, the navigation team has requested a suspension system to mitigate vibrations. The mechanical team decided to create a suspension system for the rear trailing arms while allowing the tires on the drive wheels to act as suspension for the front of the vehicle.

It has been observed the competition site is a standard parking lot. The team deduced that since little suspension travel was to occur, very little suspension travel is needed. With this in mind, a trailing arm was created with a suspension component that would allow vertical upwards

movement and downwards movement to a stop . Lateral movement would need to be eliminated as much as possible.

The team decided that the final assembly would be additively manufactured out of nylon layered with carbon fiber as well as with traditional components made out of aluminum. To save on costs, the prototype arm and mount were to be printed out of PETG while the rest of the components were machined out of aluminum and standard bolts (**Figure P-1**).

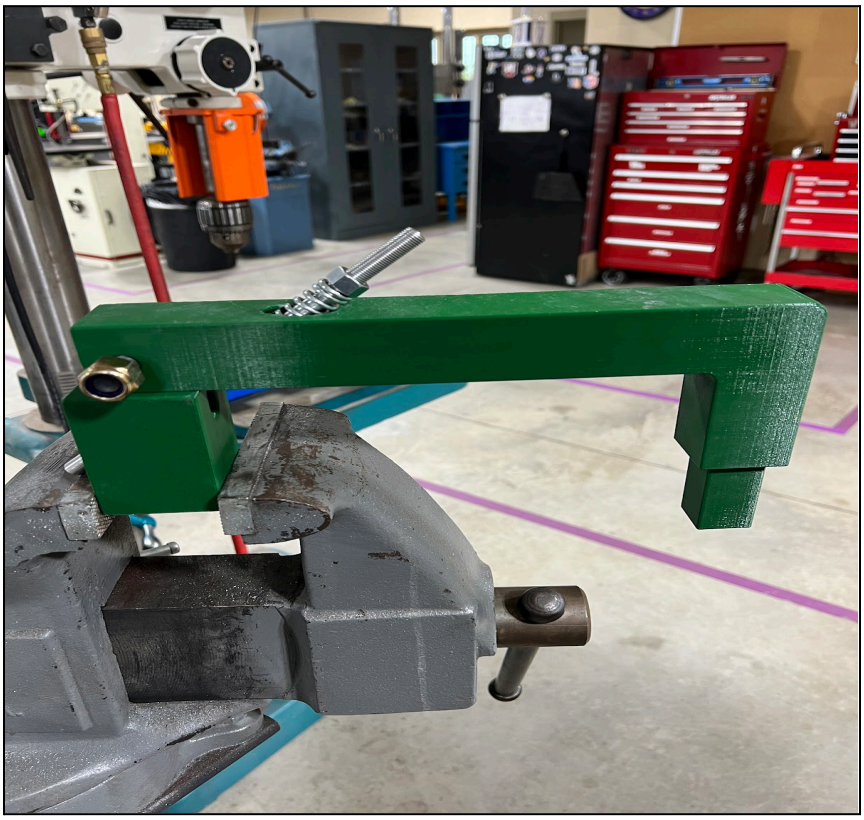


Figure P-1: First Iteration Suspension Design

The first iteration of the trailing arm suspension was tested for vertical and lateral movement. To test the lateral movement of the design, the base was placed in a vise and a long travel indicator was mounted to the side (**Figure P-2**). A force of 3.65 kilograms (8 lbs) was

applied horizontally to the arm. At the end of the test, a lateral movement of 13mm (0.5 inches) was recorded.

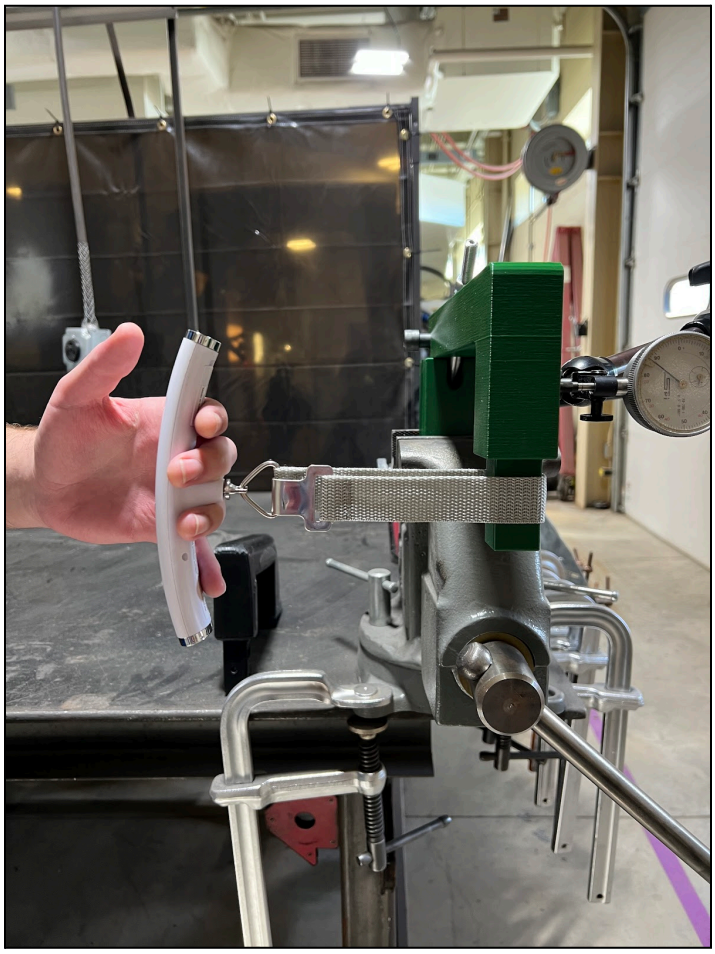


Figure P-2: Suspension Arm Lateral Test 1

The suspension arm was also tested for vertical travel with 3.65 kilograms (8 lbs) of force applied in the vertical direction. This test was performed using a similar setup to the lateral test

except the indicator was placed on top of the arm (**Figure P-3**). At the end of the test, a vertical movement of 7.62mm (0.3 in.) was recorded.

Figure P-3: Suspension Arm Vertical Test 1

The second iteration the suspension arm was redesigned as a single part that would be 3D printed from nylon and carbon fiber (**Figure P-4**). To save on costs, the prototype arm and mount

were to be printed out of PETG. Unlike the first iteration, no components other than the casters and mounting hardware were needed.

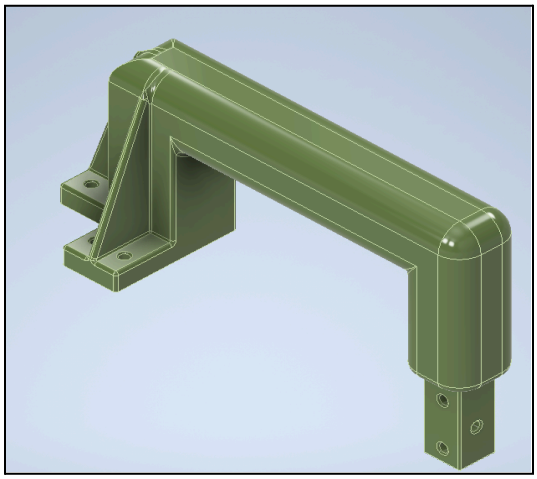


Figure P-4: Second Iteration Suspension Arm Design

After the arm was printed, it was tested for vertical and lateral movement. The parameters used in the first iteration were repeated. Performing the lateral test, 3.65 kilograms (8 lbs) of force were applied in the lateral direction (**Figure P-5**). Lateral movement of 2.54 mm (0.100 in.) was recorded. A vertical force test was also performed (**Figure P-6**). 3.65 kilograms (8 lbs.) of force was applied vertically to the arm. A deflection of 1.9 mm (0.075 in) was recorded.

Figure P-5: Second Iteration Suspension Test: Lateral

Figure P-6: Second Iteration Suspension Test: Vertical

Table P-1: Results from Force Tests of Suspension Designs

Test Iteration	Vertical Movement (mm)	Lateral Movement (mm)
1	13	7.62
2	2.54	1.9

The second iteration of the suspension arm design has four to five times less deflection than the first iteration of the design. Since the final product will be made from nylon and carbon fiber (which is much stiffer than the PETG), the team was satisfied the second iteration design would meet the needs of Rock-E.

With the deflection of the arm reduced, it was ready to integrate a spring system to handle bumps and vibrations actuated by the vehicle's movement. A third iteration of the arm was created using the main body from the second iteration but incorporating a slide on the caster wheel mount (**Figure P-7**). The slide would allow space for a spring to be placed between the arm and the slide. A spring design was developed that could be 3D printed out of PETG (**Figure P-8**). PETG was chosen because of its elastic properties when configured correctly.

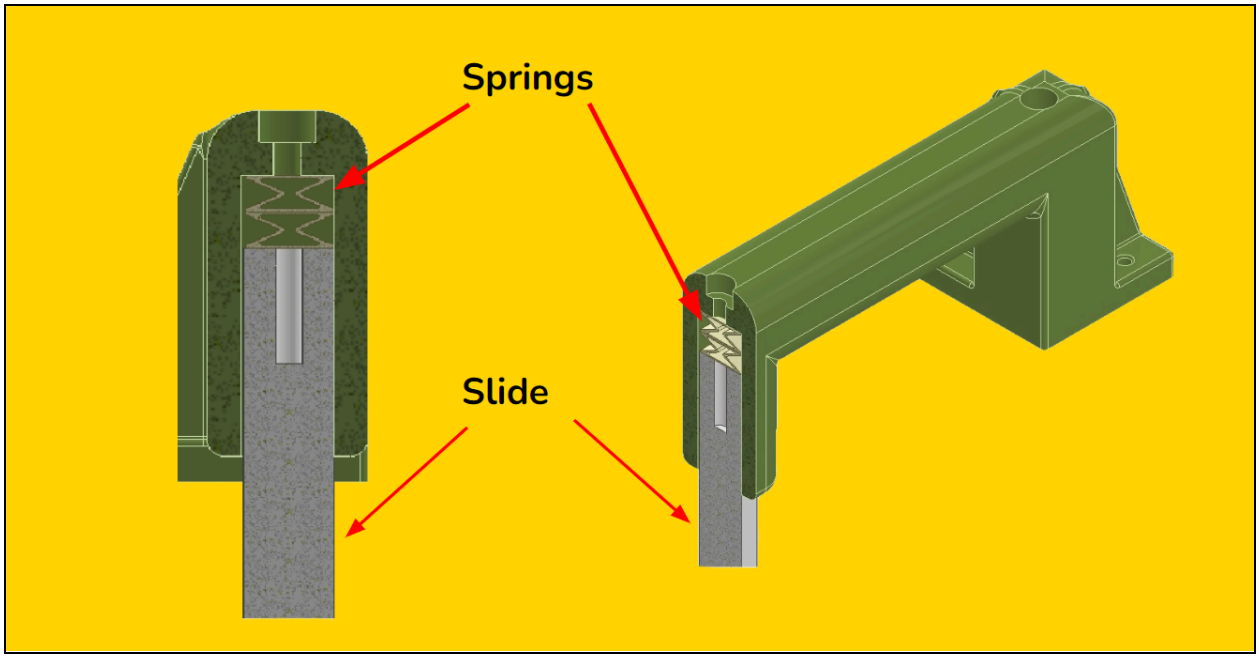


Figure P-7: Third Iteration Suspension Arm with Spring

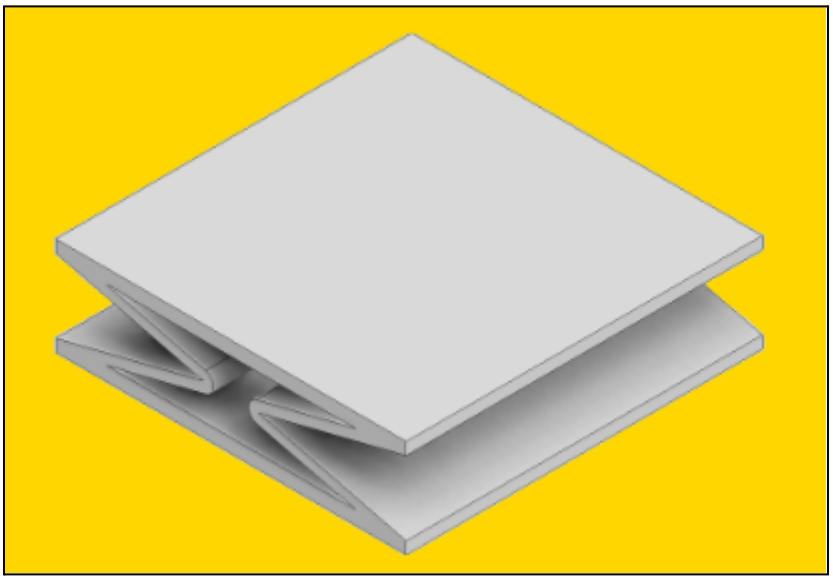


Figure P-8: First Iteration Spring Design

Testing of the current suspension design is underway and the team is seeing promising results so far. The suspension is reacting properly without binding. Further testing is underway to determine the spring rate needed to support the robot and also dampen vibrations coming from the wheels.Spring rate will be adjusted by changing the thickness of the springs in 0.5mm increments and tested on the vehicle. A second spring design has been developed that will require testing as well (**Figure P-9**).

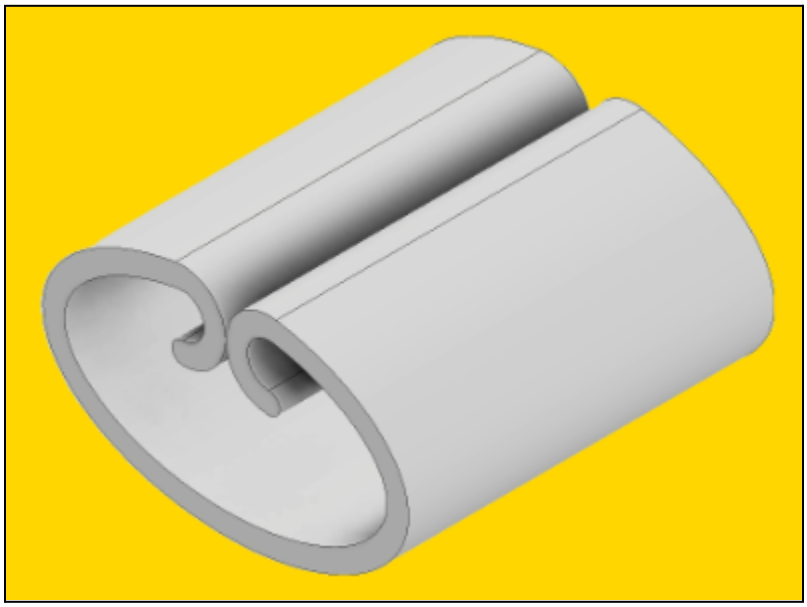


Figure P-9: Second Iteration Spring Design

Appendix Q: Gear Motor Selection

With the information gained from appendix M, it was decided to choose a motor that would meet Rock-E's torque needs, was easy to use, has a built in encoder, and could produce the power needed for Rock-E to travel up the ramp at full speed. Limited to parse through the options available from preferred vendors, the top choice was the model PCS-250 produced by SuperDroid Robots [1]. This is the same make and model of motor that was used by team Ollie last year and more importantly it fit all power needs, though at a somewhat high price [2]. Comparing collected data to the data sheet from the manufacturer, one can see that the PCS-250 motors can produce a torque of 36 Nm per motor and 250 Watts of power, capable of propelling a robot of over 150 kg at a speed of 5 MPH up the ramp, if needed [**Fig. N-1** & appendix K]. While other motors were considered, such as IG-52 class tube-mounted DC motors, it was apparent that most anything less powerful than wheelchair motors may not be capable of propelling Rock-E at the realistically estimated weight class. To use smaller motors a dedicated effort would be needed to minimize weight in nearly all components, namely in batteries, structure, and paneling, while still needing to transport the cinder block payload. With these factors considered, it remained the plan to use wheelchair motors.

MOTOR PARAMETERS

POWER	VOLTAGE	CURRENT	OUTPUT SHAFT SPEED/MIN	MOTOR SPEED/MIN	REDUCTION RATION	RATED TORQUE	WIRE LENGTH	NOISE	BATTERY BRAKE	BRAKE HANDLE
250W	24V	≤3A	160R	3000	18.75:1	36Nm	≥45cm	≤60DB	NO	NO

Figure Q-1: Motor Parameters

Unfortunately, when attempting to purchase the motors, SuperDroid Robots was in the process of discontinuing the model. They will not be making more nor did they have any in stock. With some research, it was found that the motor SuperDroid Robots was using is on Amazon. This motor is a MY1016Z made by Unitemotor [3]. It has the same specs as the PCS-250 motors sold by SuperDroidRobots except there is not an encoder attached. Using the off-the-shelf solution PCS-250 motors visible on Ollie as a reference, a process began to reverse engineer the package that can no longer be directly purchased. To do so, the team would need to

modify the motors, cutting away material on the back plate, then adding an extended tailshaft to reach the US Digital encoders that would be bolted to the plate. The model number of the encoders sold by SuperDroid Robots, visible on Ollie, is E2-500-315-NE-D-G-1. This approach was deemed too complicated and so an externally linked encoder was decided upon. The method to do so was to purchase a Taiss 2-phase Incremental Rotary Encoder and some gears to be connected via belt to the motors' driveshafts. This is a method that needs minimal modification to the motors while adding similar functionality.

Appendix R: Bill of Materials

The following is the bill of materials for Rock-E (**Figure R-1a & b**). The BOM is comprehensive, including parts purchased from outside vendors and parts designed and created by Rock-E's mechanical/electrical team. There are two columns for pricing, one for purchased items and one for items procured from the School of Engineering and Technology prototype shop.

As part of an effort to increase sustainability, team Rock-E is trying to reuse parts and overstock previously purchased by the school. As you can see, significant savings have been achieved by reusing previously purchased parts (**Figure R-2**). An interactive version of the bill of materials can be found here [1].

Part Desciption	Part Number	Website Link	Unit Cost	Quantity	WIU Purchase	No Purchase Cost
Velcro Onewrap 1" X 30'	VEL-30768-AMS	Amazon	$11.27	1	$11.27	
24V 250W Electric Wheelchair motors 120 rpm	MY1016Z	Amazon	$129.99	2	$259.98	
12 inch Wheelchair Replacement Wheels, QTY 2	B0BCP3N25K	Amazon	$147.03	1	$147.03	
Emergency stop switch mechanical	RB-Dfr-527	Robotshop.com	$5.90	1	$5.90	
Basic Robot Power Distribution Module	TE-248-110	SuperDroidRobots	$27.22	1	$27.22	
Molex 2-Pin Friction Lock Connector Set - 5 Pack	TE-252-205	SuperDroidRobots	$6.07	2	$12.14	
Cytron 30A Bi-Directional DC Motor Driver MD30C	B01M4PXFBG	Amazon.com	$38.80	2	$77.60	
3/8" Aluminum sheet (baseplate)	N/A	WIU Shop Stock	$126.00	1		$126.00
Bracket-mount casters, 3" diameter	2399T6	McMaster-Carr	$22.24	2	$44.48	
Relay 24V, 30/40 Amp Waterproof (5pin) pack of 5	B07L2H67BN	Amazon	$25.99	1	$25.99	
18 Gauge Wire Spools	B089CW7YSR	Amazon	$29.49	1		$29.49
10 Gauge Wire	B08HSC5NW5	Amazon	$35.99	1		$35.99
10 Gauge Fuse Holders	B08BWRSPY1	Amazon	$9.99	1	$9.99	
120pcs Automotive Fuse Assortment	B09B7K9PDB	Amazon	$8.99	1	$8.99	
250pcs Heat Shrink Wire Connectors	B07ZPF1TTX	Amazon	$16.95	1	$16.95	
180pcs Solder Seal Connectors	B09BQZLJXG	Amazon	$13.59	1	$13.59	
Ferrite Cores	B07CWCSNW9	Amazon	$9.99	1	$9.99	
GT2 Synchronous Wheel 20&80 Teeth (2PCS Set)	B0C4GD4HV6	Amazon	$16.89	1	$16.89	
Taiss/AB 2 phase Incremental Rotary Encoder 360P/R	B07MWZ4CLT	Amazon	$18.36	2	$36.72	
ZED 2i Stereo Camera with polarizer	RM-STER-006	RobotShop	$569.00	1	$569.00	
SparkFun ZED-F9P GPS Application Board	B087T7BV6K	Amazon	$274.95	1	$274.95	
SparkFun GNSS Accessory Kit	B09ZH8XWGK	Amazon	$84.95	1	$84.95	
Slamtec RPlidar S3 360° Laser Scanner (40 m)	RB-Rpk-29	Robotshop	$549.00	1	$549.00	
Adafruit 9-DOF Absolute Orientation IMU Fusion Breakout	BNO055	Amazon	$32.95	1	$32.95	
Arduino Due	B00A6C3JN2	Amazon	$43.56	2	$87.02	
Dell Mobile Precision Workstation 3480 CTO	GOKZFS4	File Receipt	$1,921.88	1		$1,921.88
Suspension Arm	SUS-001	PDF Link	$20.00	2		$40.00
Suspension spring	SUS-002	PDF Link	$0.63	4		$2.52
suspension slide	SUS-003	PDF Link	$3.00	2		$6.00
Motor Mount (Top)	MM-001	PDF Link	$10.00	2		$20.00
Motor Mount (Bottom)	MM-002	PDF Link	$10.00	2		$20.00
T-slot aluminun extrusion (1" square 2' long)	47065T209	McMaster-Carr	$11.67	12		$140.04
T-slot 3-way corner bracket	47065T278	McMaster-Carr	$12.10	4		$48.40
90 degree corner bracket (1")	47065T236	McMaster-Carr	$7.92	14		$110.88
2 inch PVC pipe (2' long)	150748	K&K hardware	$17.99	1		$17.99
Rail-to-Rail Bi-fold Hinge	47065T161	McMaster-Carr	$34.16	2	$68.32	
Door Leaf Hinges	1488A11	McMaster-Carr	$7.10	2	$14.20	
Pipe Mount Dome Light	5899T28	McMaster-Carr	$102.44	1		$102.44
E-Stop Mount	SAF-001		$20.00	1		$20.00
DC Cooling Fan	8774N14	McMaster-Carr	$17.89	2		$35.78
Gusset Bracket, Silver (2" long)	47065T736	McMaster-Carr	$15.20	2		$30.40
Lidar Mount	LM-001	PDF Link	$10.00	1		$10.00
Acrylic Paneling (96"x48" full sheet)	MP-111	Midland Plastics	$118.00	1	$118.00	
T-Slot Framing Nut W/Buttonhead Bolt (4 pack)	47065T139	McMaster-Carr	$3.05	35		$106.75
Kar-Tech Wireless E-stop	3A548	Kar-Tech	$260.00	1	$260.00	

Figure R-1a: Bill of Materials

Part Desciption	Part Number	Website Link	Unit Cost	Quantity	WIU Purchase	No Purchase Cost
X2 Lithium 12 volt battery	X2P-Li-5	Batteries Plus	$89.99	1		$89.99
Turnigy 22.2 volt 6-cell lithium battery (20 mAH)	9067000388-0	HobbyKing	$201.59	1		$201.59
Button Head Hex Drive Screw - 1/4"-20 Thread, 2-1/2" Long	91255A547	McMaster-Carr	$13.34	1	$13.34	
Button Head Hex Drive Screw - 1/2"-13 Thread, 2" Long	91306A439	McMaster-Carr	$8.88	2	$17.76	
Low-Strength Steel Nylon-Insert Locknut - 1/4"-20	90640A129	McMaster-Carr	$5.34		$5.34	
Low-Strength Steel Nylon-Insert Locknut - 1/2"-13	90640A133	Mcmaster-Carr	$14.60	1	$14.60	
Grade 9 steel Washers - 1/4"	90850A100	McMaster-Carr	$7.20	1	$7.20	
Wheel Spacer	WS-1001	PDF Link	$1.00	2		$2.00
Arduino Due Terminal Shield	HD025	Amazon	$32.00	2	$64.00	
USB A Male to Micro USB B Male Cable	USBA3M0MICROB18(R)	DigiKey	$10.16	3	$30.48	
StarTech USB Hub	HB30C4AIB	DigiKey	$46.10	2	$92.20	
Taiss/AB 2 phase Incremental Rotary Encoder 360P/R	B07MWZ4CLT	Amazon	$18.36	2	$36.72	
2 Pin Circuit Board Connectors for PDM	B01DUC1O68	Amazon	$8.59	1	$8.59	
RobotDigg 340-2GT-6 Endless GT2 Belt	B07P54P5YS	Amazon	$19.99	1	$19.99	
EBOX Rail Support	EB-10	PDF Link	$0.20	2		$0.40
Block Base	BB-10	PDF Link	$0.40	4		$1.60
Base Support Beam	BB-20	PDF Link	$0.30	12		$3.60
100ft 16 Gauge Red & Black Wire	B07W8WVVV2	Amazon	$19.49	1		
260pcs Male & Female Spade Connectors	B07RT4RSGJ	Amazon	$19.99	1		
Shelf Holder	SH-001	PDF Link	$0.25	4	$1.00	
Playstation 4 Controller	B0C8J3R84H	Amazon	$27.99	1	$27.99	
ZED 2i Camera Mount	CMPHD006	Amazon	$37.48	1	$37.48	
Arduino Due Microcontroller	B00A6C3JN2	Amazon	$52.00	1	$52.00	
IMU Support Bracket	SB-001	PDF Link	$1.00	1	$1.00	
2x 12V fans	B07D493BDX	Amazon	$16.25	2	$32.50	
0-24V MOSFET Button	B07GLNCRR4	Amazon	$9.99	1		$9.99
DC to DC Buck Converter	B0BTBVDB4S	Amazon	$9.98	1		$9.98
24V 10000mAh Ebike Battery	B0B5DK4NN8	Amazon	$129.79	1	$129.79	
usb extension cable	B0776WVPMN	Amazon	$6.99	1	$6.99	
Laptop Power Bank	BI-B63	Amazon	$59.98	2	$119.96	
USB Splitter	B0CB3M46Y5	Amazon	$12.99	1	$12.99	
Adafruit BNO055 IMU	B017PEIGIG	Amazon	$34.95	1	$34.95	
UART to USB Bridge	SH-U09C	Amazon	$12.49	1	$12.49	

Figure R-1b: Bill of Materials

Purchase Total	$3,562.48
No Purchase Total	$3,143.71
Gross	$6,706.19

Figure R-2: Purchase Totals for Rock-E Mechanical/Electrical

Appendix S: IGVC Rules

IGVC Rules

Vehicle Configuration

- Design: Must be a ground vehicle propelled by wheels, tracks, pods, etc.
- Length: Between 3 and 7 feet.
- Width: Between 2 and 4 feet.
- Height: Less than 6 feet.
- Propulsion: No internal combustion engines. Must be electric.
- Average Speed: must be at least 1 mph.
- Minimum Speed: 1 mph.
- Maximum Speed: 5 mph.
- Mechanical E-stop location: Must be a red push-to-stop, at least 1 inch diameter, located at the rear center of the vehicle. Must be hardware based and not software controlled.
- Wireless E-stop: Must work at a minimum of 100 feet and must be hardware based and not software controlled. Will be held by the judges during competition.
- Safety Light: Must be an easily visible light source that is on when the vehicle is turned on. Must be flashing when in autonomous mode and solid when out of autonomous mode.
- Payload: Must carry a 20 pound cinder block with dimensions of 16” x 8” x 8”. Must be securely mounted to the vehicle.

Qualification

- Vehicle must follow all Vehicle Configuration rules in order to qualify.
- Lane Following: Vehicle must show that it can detect and follow lanes.
- Obstacle Avoidance: Vehicle must show that it can detect and avoid obstacles.
- Waypoint Navigation: Vehicle must show that it can find a path to a single two meter navigation waypoint by navigating around an obstacle.

Competition Rules

- The competition will take place in the event of light rain or drizzle but not in heavy rain or lightning.
- Each qualified team will have up to two runs (time permitting) in each of three heats.
- Starting order will be based on order of qualification. Teams will set up on-deck in that order. Failure to be on-deck will place you at the end of the order for the run and may forfeit your final (second) run in a heat based on heat time completion.
- No team participant is allowed on the course before the team's first run, and only one student team member is allowed on the course during a run. This shall in no case be the faculty advisor.
- At the designated on-deck time, the competing team will be asked to prepare their vehicle for an attempt. On-deck teams start in the order they arrive in the starting area unless they give way to another team.
- A Starting Official will call teams to the starting line. The Starting Official's direction is final. The Starting Officials may alter the order to enhance the competition flow of entries; e.g. slower vehicles may be grouped together to allow successive running of two vehicles on the course simultaneously.
- A team will have one minute in the starting point to prep the vehicle at the starting line and point out to the Competition Judges the buttons to start and stop the vehicle,
- The Competition Judge will start the vehicle by a one touch motion; i.e. pushing a remote control button, hitting the enter key of a keyboard, a left mouse click, lifting the e-stop up, flipping a toggle switch, etc. The Competition Judge will also carry the E-Stop.
- An attempt will be declared valid when the Competition Judge initiates the start signal at the designated competing time. An attempt will continue until one of the following occurs:
- The vehicle finishes the course.
- The vehicle was E-Stopped by a judge's call.
- The team E-Stops the vehicle.

- The vehicle has not started after one minute after moving to the start line or at the judges' discretion.
- Time for each heat will be strictly observed.
- Tactile sensors will not be allowed.
- Based on the above allowable run times, if the vehicle has not completed the course in the 6 minute time period, the attempt will be ended by a judge's choice E-stop, with no additional penalty for that run.
- Each vehicle must navigate the course by remaining inside the course boundaries and navigating around course obstacles. Crossing internal lines is not allowed and will be judged as an E-Stop end of run with penalty.

Traffic Violations

	Traffic Violations	Ticket Value	E-Stop	Measurement
1	Hold-up Traffic	End of Run	Yes	>60 secs. to 88 ft
2	Leave the Course/Scene	- 10 Feet	Yes	Yes
3	Crash/Obstacle Displacement	- 10 Feet	Yes	Yes
4	Careless Driving	- 5 Feet	No	No
5	Sideswipe/Obstacle Touch	- 5 Feet	No	No
6	Student's Choice E-Stop	- 10 Feet	Yes	Yes
7	Judge's Choice E-Stop	0 Feet	Yes	Yes
8	Blocking Traffic	- 5 Feet	Yes	Yes
9	Loss of Payload	0 Feet	Yes	Yes
10	Too slow, did not average 1 mph	Disqualified	No	No

Figure S-1: Traffic Violations Chart

- Hold-up traffic: Must maintain 1 mph, there will be a speed check at 44/88 foot mark of the course, will result in end of run with time recorded
- Leave the scene\course: All portions of the vehicle cross the boundary. The overall distance will be measured from the starting line to the furthest point where the final part of the vehicle crossed the boundary edge.

- Crash: The overall distance will be measured from the starting line to the collision point with the obstacle.
- Careless Driving: Crossing the boundary while at least some part of the vehicle remains in bounds.
- Student E-Stop: Student e-stop is used if the team feels that there may be damage caused to their vehicle or they know that it is stuck and want to end their time.
- Judge E-Stop: The overall distance will be measured from the starting line to the front of the vehicle or where the final/furthest remaining part of the vehicle has stopped or crossed the boundary outside edge.
- Obstacle Displacement: Defined as displacing permanently the obstacle from its original position.
- Slightly rocking/Tilting an obstacle with no permanent displacement is not considered obstacle displacement. An obstacle that rocks or tilts significantly but with no displacement will still be considered an end of run. Judges' calls are final.
- Blocking Traffic: Vehicles stopping on course for over one minute will be E-Stopped and measured.
- Loss of Payload: If the payload falls off the vehicle the run will be ended.
- Too Slow: If the vehicle does not maintain 1 mph minimum average speed limit throughout the course this run is disqualified.

Appendix T: Wiring Best Practices

Electrical Standards

Wiring Color

Red	24V
Black	Ground
Yellow	12V
White	Tx
Orange	5V
Blue	Rx
Green	Signal

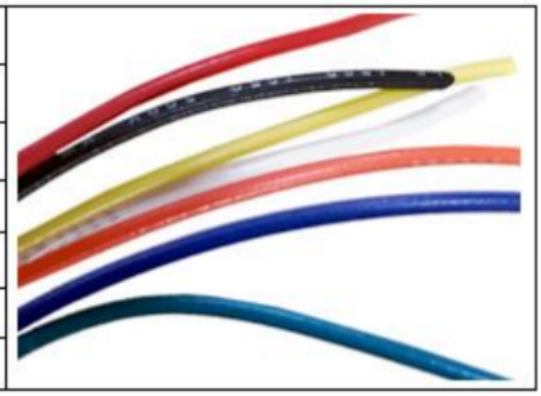


Figure T-1: Wiring Color Chart

Choosing the correct wire color is important for the vehicle as it allows the team to quickly identify each system. This makes the system neat and less confusing. The figure shown is an example of what a common wiring color scheme for a vehicle like Rock-E would look like [1].

Noise Mitigation

Running a system with tons of wires can cause a lot of interference in the system. To prevent this, there are several methods one can use [1].

- Shielded cable is a great way to reduce noise. They can be used for the motor leads and data wires.
- Ferrite beads can help reduce unwanted noise. These can be added to each of the motor leads.
- Twisting power and ground wires together can help reduce noise.
- Make signal wires the same length to avoid any inconsistencies.
- Completely separate power wires and signal wires to avoid noise.

Fuses

Using fuses is absolutely necessary when designing the electrical system. Fuses are put in place to protect the components and to prevent meltdowns. For the system, fuses are put in place where there is high current draw, like between the motors and motor drivers, and between the PDM and 24V battery. Fuses are also put in places where there are very important components that run Rock-E's system, like between the 12V battery and PDM [1].

Appendix U: Standards Referenced

Wire Gauge

To keep Rock-E safe, the correct gauge wire must be chosen. Incorporating the incorrect gauge can cause meltdowns as the wires may not be rated for the current that is being drawn. For the vehicle, Rock-E will be utilizing the American Wiring Gauge (AWG) standard *ASTM B258-18*[1].

Nuts and Bolts

Rock-E will be utilizing the SAE J429 standard for hardware in the vehicle[2]. The SAE J429 standard has specific requirements for bolts, screws, studs, and U-bolts up to 1-1/2" in diameter. Bolts meeting SAE J429 standards have radial lines engraved on the bolt head.

Extruded Aluminum

Rock-E will be utilizing the ASME SB-221-21 standard for the extruded aluminum sections used in the chassis design[3]. This is the standard specifications for Aluminum and Aluminum-Alloy Extruded Bars, Rods, Wire, Profiles, and Tubes. Rock-E will specifically follow the standard for extruded profiles.

IP67 Rating

Many of the consumer-grade electrical components Rock-E is utilizing have an IP67 rating. This is an electrical protection rating found in the ANSI-IEC_60529-2020 standard for consumer electronics[4]. This standard states that a device can withstand water hose and jet sprays along with being able to sustain momentary water immersion in depths of 150mm to 1000mm for up to 30 minutes.

SI Units of Measure

All units of measurement used with Rock-E will utilize SI or the International System of Unit Measurement [5], also known as the International Bureau of Weights and Measures (BIPM). BIPM was set up by the Metre Convention signed in Paris on 20 May 1875 that included seventeen International states during the final session of the diplomatic Conference of the Metre. Even though the United States of America was one of the original seventeen states that signed the convention, the metric system is still not fully adopted in the US.

Appendix V: Final Navigation Prototype

Since the presentation of the Low-Fidelity and Mid-Fidelity Prototypes in the previous reports, lots of progress has been made. The new and improved design makes up the Final Navigation Prototype which is the combination of all subsystems into the ultimate navigation system.

Final Navigation Prototype Considerations and Final Decision

A variety of different methods were considered to fulfill the requirements of the IGVC. The rules state that the autonomous vehicle needs to detect and stay between lines, avoid potholes and obstacles, and navigate to specific coordinates. Due to the versatility of the considered components, decisions had to be made about the purpose of each part. For example, there are some components that fall under the category of a Camera or LiDAR sensor that can carry out obstacle detection, line detection, and pothole detection. Additionally, some form of mapping can be accomplished by a Camera, LiDAR, or GPS sensor. Because of this, it was important from the start to determine the role of each component within the larger network.

There were a large number of ideas considered before reaching a final decision. It was ruled out early on to use either a Camera or LiDAR sensor for all detection purposes. This decision was made in the spirit of learning, the overall goal of the senior project, as implementing more sensors into the system would increase the amount of knowledge learned. Additionally, the considerations of using a Camera or LiDAR for three-dimensional mapping was vetoed due to their complexity. Simplicity and functionality is key, something that could be achieved without the complex mapping of these sensors. After many ideas, it was finally decided to use a Camera for line and pothole detection, a combination of one LiDAR and three ultrasonic sensors to detect obstacles, and a GPS, IMU and Encoder for mapping and localization

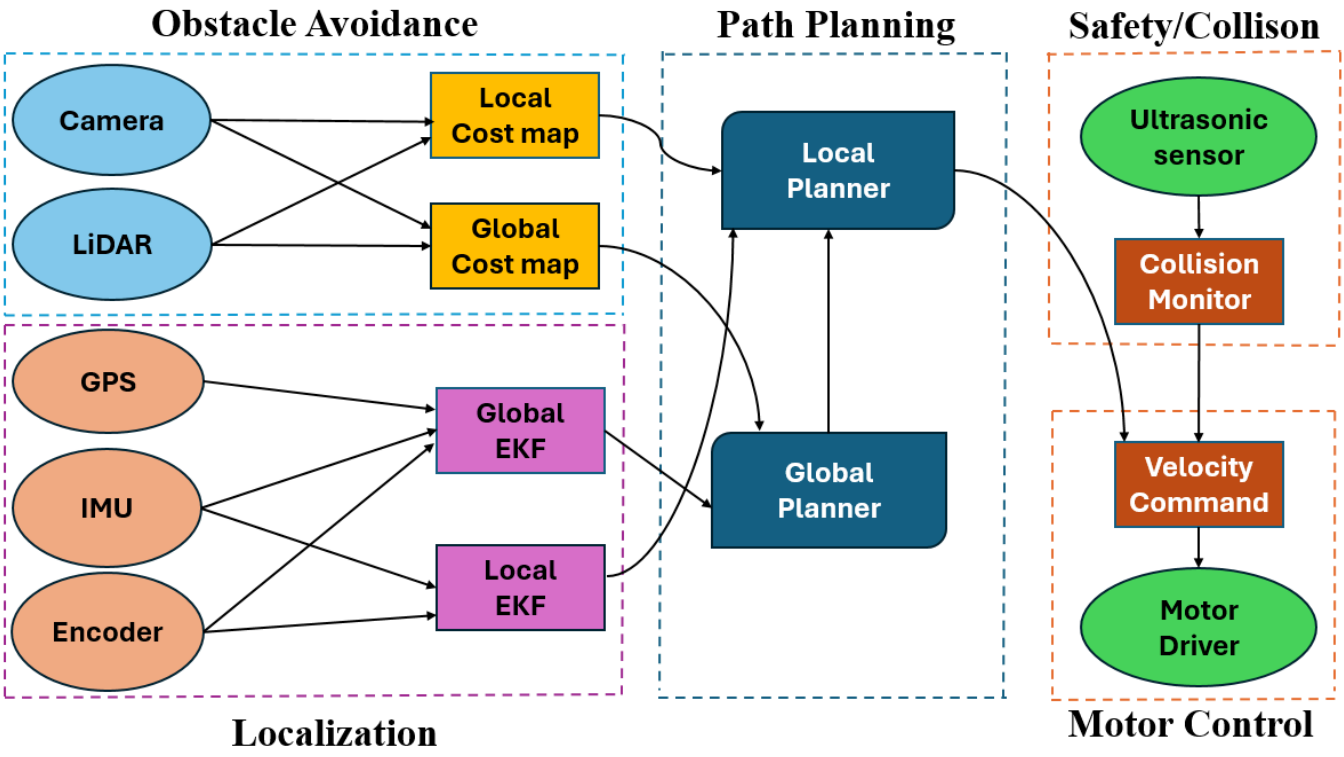


Figure V-1: Final Navigation Prototype

Final Navigation Prototype Description

The Final Navigation Prototype makes use of all the individual systems that will be mentioned later on in this report. Pictured above in **Figure V-1** is a diagram of all of these isolated systems and how they are interconnected. At this time, all of the individual systems and the combined network are functioning. The Central Processor is a Dell Precision 3480 running the Humble Hawksbill distribution of ROS 2. In order to communicate effectively, the LiDAR, GPS, IMU, and Camera components are connected directly by USB to the computer. These provide their respective data to the Central Processor so that it can process the information. The Camera and LiDAR provide the data to form the Local and Global Costmaps, which are used to perform the Obstacle Avoidance of the vehicle. Since data stability is critical for robot control, the Extended Kalman Filter is used to eliminate inaccuracies. The Encoder, IMU, and GPS data is combined in the filter to provide the Local and Global Localization of the robot. Using the

costamps and localization, the Global and Local Planners generate paths which are used to guide the robot to its destination. Two Arduino Due boards act as the Microcontollers of the system and are also directly connected to the computer through the USB hub. These Microcontrollers take a major load off of the Central Processor and provide the required signals for motor control. Three Ultrasonic Sensors and two Encoders are connected to the boards. These sensors all provide their respective data to the Microcontrollers which then pass this information to the Central Processor. On top of this, there are both Motor drivers and a Safety light also communicating through the Arduino Due. After the Central Processor interprets the data, it will send a command to the Microcontroller in charge of motor control which will then send the corresponding Pulse Width Modulation signal to the Motor Drivers. Based on the received signal, these Motor Drivers will send power to the motors which will rotate the wheels and move the vehicle. Furthermore, when the robot moves backward, the Collision system will detect if there are obstacles behind the vehicle, and the Center Processor will change the velocity command to avoid it. Because of this, the Central Processor and Microcontrollers will have constant two-way communication. At its current state, Rock-E can successfully avoid obstacles, stay within the boundary lines, and reach its navigation waypoint.

Appendix W: Ethics

When approaching the topic of ethics in the engineering field, there is always a question of which is more important: moral code or professional efficiency and accuracy. Engineers greatly impact the quality of life for all people because their decision-making can bring out the greater good in society or damage it. The Code of Ethics for Engineering is the moral code that all engineers should abide by because it contains the guidelines for achieving goals in public safety, personal integrity, and knowledgeable contribution.

According to the Institution of Engineering and Technology, there are ten specific rules that engineers must submit. These rules can be summed up into three categories. First, engineers should put public safety as the paramount goal. Safety involves striving for ethical design and practices; disclosing endangered factors without delay; avoiding injury to others in every way by malicious action; and undertaking tasks in which they are properly trained and qualified. Second, engineers must be people of integrity. Excellence in integrity implies that engineers must be able to reject bribery, state truthful data, accept criticism, correct errors, credit properly in technical work, and treat all people fairly without bias. Third, engineers should contribute to the development of the field. This requires engineers to avoid conflicts, assist others in professional work, and improve their understanding of conventional and emerging technologies in society.

All engineering designs are based on theories and data analysis, which may compose unforeseen possibilities. Putting public safety first allows engineers to consider more before a project is performed. For example, a push E-Stop switch is used to stop the vehicle. It is made of plastic to isolate electrical power. It is located in a place that can be reached easily and quickly. The quality of integrity allows engineers to accept criticism more easily which can correct possible mistakes. Following the code of ethics will help an engineer to make the proper decision when dealing with complex issues under pressure. Hence, the Engineering code of ethics should always come first because it is fundamental to a safe and comfortable workplace and society.

Appendix X: Central Processor

In the initial stages of selecting a central processor for the Autonomous Ground Vehicle project, various options were weighed, exploring standard laptops and compact alternatives like NUCs (Next Unit of Computing). In the transition into the Critical Design phase, the imperative for heightened computational power became increasingly evident, prompting a thorough reassessment of the choices available for processors. After meticulous consideration, the preference tilted towards a high-performance solution, ultimately settling on the Dell Precision 3480. This decision was spurred by the laptop's comprehensive specifications, which not only met but exceeded the intricate demands of the autonomous ground vehicle project. The Dell Precision 3480 emerged as an ideal choice, providing the computational prowess necessary for navigating complex algorithms and handling vast datasets seamlessly, aligning perfectly with the evolving needs of the project.

The selection of the Dell Precision 3480 as the central processor was a strategic move, considering its robust features. The 13th Gen Intel Core i7 processor ensures powerful and efficient computational capabilities, capable of handling the complex algorithms and vast datasets integral to autonomous navigation. The inclusion of the NVIDIA RTX A500 graphics ensures not only smooth rendering of graphics but also enhances parallel processing capabilities, which is crucial for real-time decision-making in the vehicle's navigation. The substantial 64GB RAM ensures that the laptop can efficiently manage and process large datasets without experiencing bottlenecks. Additionally, the 1TB storage capacity provides ample space for storing and retrieving data generated during the autonomous vehicle's operation. The 14" LCD display proves beneficial not just for testing but also plays a crucial role in realizing a more compact design for the robot/vehicle. The aspiration for a smaller size aligns with the objective of seamlessly integrating the display onto the vehicle, satisfying both computational needs and mechanical team specifications. This guarantees an enhanced user experience and convenience, especially throughout testing and debugging phases.

We extend our heartfelt gratitude to LeClaire Manufacturing and Caddis for their generous sponsorship of $1,250.00 towards our senior design project. Their invaluable support has enabled our team to acquire the cutting-edge Dell Precision 3480 laptop as the central processor for our autonomous vehicle project. This remarkable contribution not only enhances the overall processing power but also elevates the graphics capabilities, significantly impacting the efficiency of our project. LeClair Manufacturing and Caddis have played a pivotal role in propelling our senior design project to new heights, demonstrating their commitment to fostering innovation and education. We are profoundly thankful for their belief in our project and the impact it will have on advancing technology and autonomous vehicle systems.

Figure X-1: Dell Precision 3480

Robot Operating System (ROS)

In the pursuit of building an Autonomous Ground Vehicle for the Intelligent Ground Vehicle Competition (IGVC), there was careful consideration of the choice of the Robot Operating System (ROS) as the backbone of the project. After a thorough evaluation, the team decided to go with ROS 2 over its predecessor, ROS. ROS 2 presents numerous advantages that align well with the demands of the project. The team has opted for ROS 2 Humble Hawksbill for the project. ROS 2 is chosen for this project due to its superior real-time capabilities and robust middleware infrastructure, ensuring prompt data processing for safe navigation. Its enhanced security and modularity provide a reliable foundation for seamlessly integrating crucial sensors like cameras, LiDAR, GPS, and IMU, essential for navigation and perception tasks in the autonomous vehicle.

Utilizing ROS 2 in the project is instrumental for fostering efficient communication and coordination among vital components like cameras, LiDAR, GPS, and IMU sensors crucial for the vehicle's success in the IGVC competition. ROS 2 facilitates seamless data flow between these devices, allowing the cameras to capture visual input, LiDAR for precise obstacle detection, GPS for accurate localization, and IMU for orientation and motion sensing. The flexibility of ROS 2 permits us to establish well-defined interfaces and data exchange formats, ensuring harmonious collaboration among these sensors for autonomous navigation. Leveraging ROS 2 unleashes the collective potential of these sensor systems, enabling successful navigation through complex environments and excellence in the IGVC competition. Additionally, the availability of ROS 2 packages for GPS, LiDAR, IMU, and cameras streamlines integration, saving valuable time and effort, allowing us to focus on core project aspects like navigation algorithms and decision-making, ensuring a smoother and more efficient development process.

Figure X-2: ROS 2 Humble Hawksbill

Standard for Computer

Making sure the Dell Precision 3480 is up to the necessary standards for the Autonomous Ground Vehicle project is a big deal, and following ISO 9001 standards is key to that. This means Dell is serious about keeping things top-notch in how they make and perform with their products. These standards cover everything - from the 13th Gen Intel Core i7 processor to the NVIDIA RTX A500 graphics, 64GB RAM, 1TB storage, and the 14" LCD display. Each part has to meet specific quality checks through lots of testing and validation. ISO 9001 isn't just about the bits and pieces; it also looks at how the laptop comes together and works as a whole. This matters a lot for making sure the autonomous navigation system is reliable and safe. These standards go beyond the components and dig into how the laptop gets made and controlled for quality. This keeps everything in check, making sure it's consistent, traceable, and reliable, adding up to an overall dependable Dell Precision 3480. In short, ISO 9001 standards make sure the central processor lives up to recognized quality benchmarks, giving confidence in how well the Autonomous Ground Vehicle project will run.

Appendix Y: Microcontroller

Component Research and Choice

Although a laptop will be implemented as the central processor, it is more efficient to use a separate controller to communicate with outer devices with the ROS 2 software. The microcontroller not only works as the medium but also takes some load off of the central processor. Furthermore, the computer is not capable of using Pulse Width Modulation (PWM), a common method of controlling DC motors. Due to these reasons, it was decided to implement a microcontroller of some sort into the design. Originally, a number of microcontrollers that were popular, effective, and had the ability to use Pulse Width Modulation were considered. The three main considerations were an Arduino, Raspberry Pi, or ProductivityOpen microcontroller. For a long time, the ProductivityOpen microcontroller was the most favored one as it was an industrial, and more robust version of the traditional Arduino or Raspberry Pi boards. However, the final decision was made to use one of the Arduino boards: the Arduino Due, which is shown in **Figure Y-1**. It is compatible with micro-ROS, a ROS 2 package that can be downloaded onto microcontrollers to allow for easy communication. The Due has 12 PWM pins available, which makes it capable of controlling four motors. It also has 96 KB of SRAM memory providing adequate processing power to the project. **Figure Y-2** shows part of the Arduino Due's schematic while the full diagram can be viewed at this link.

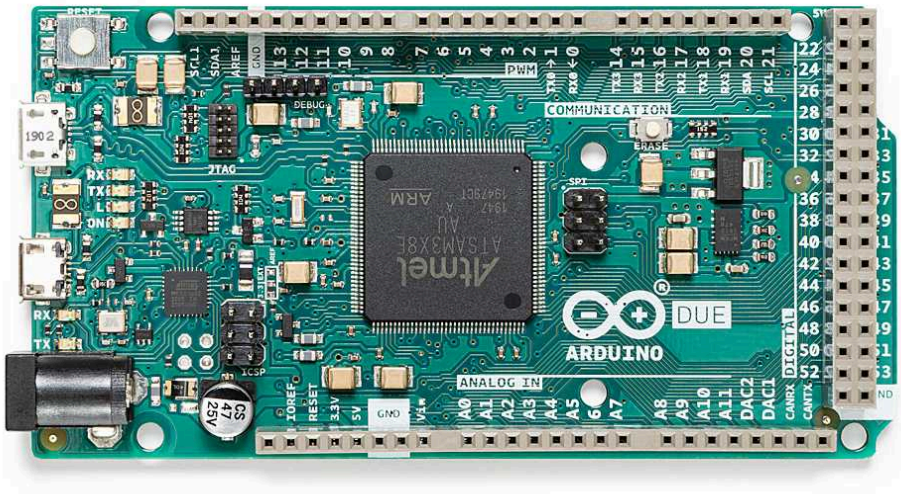


Figure Y-1: Arduino Due Board

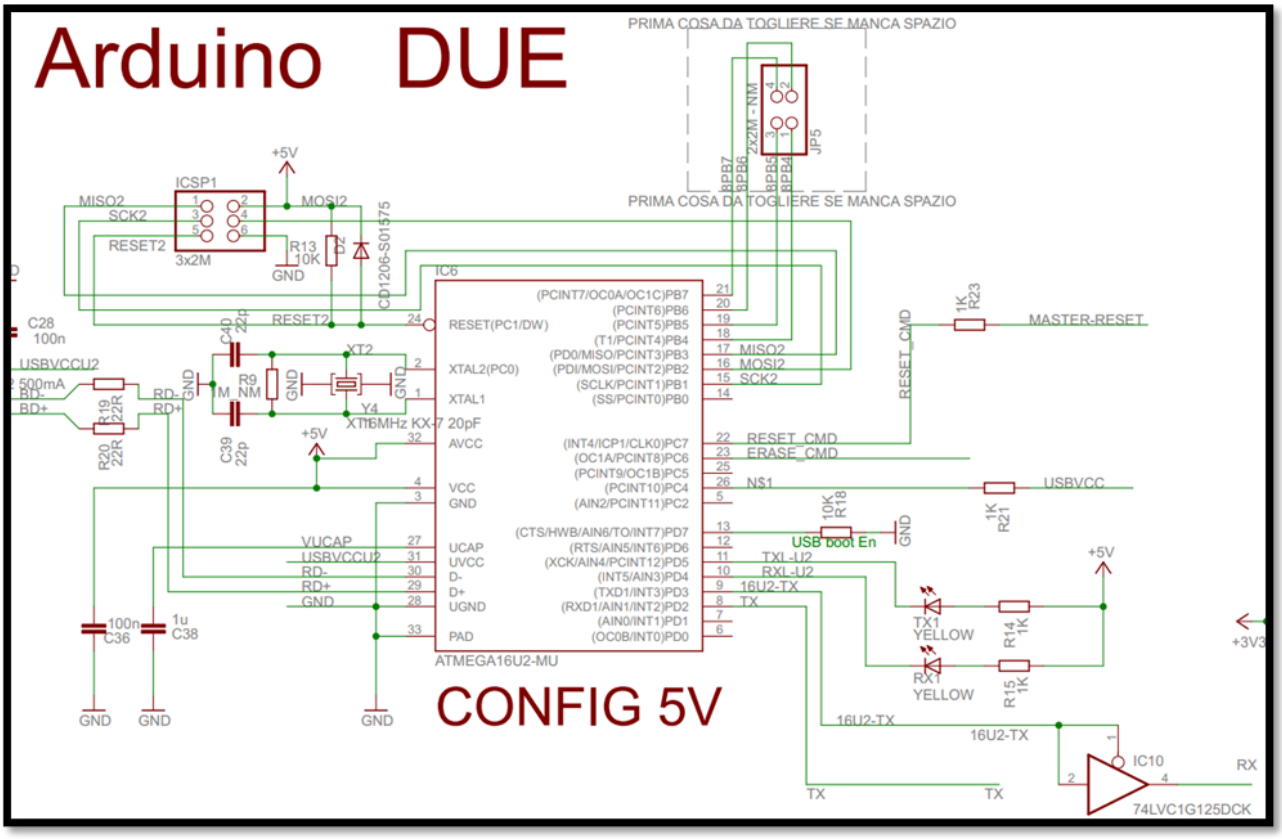


Figure Y-2: Arduino Due Schematic

ROS 2 and Arduino Communication

There are two methods to communicate the Arduino microcontroller with the ROS 2 system. One is the ros2arduino bridge, XRCE-DDS, and the other one is the micro-ROS Agent. After comparing these two methods, the XRCE bridge was first chosen because it had less complicated download procedures. Several problems were encountered during the implementation process. First, the Linux Operating System could not find the bridge due to incorrect setup code. **Figure Y-3** shows the fault issued on the screen and **Figure Y-4** is the incorrect setup code. In order to do it correctly, the "&& git checkout V1.3.0," in the second line must be taken out of the command. Second, Linux could not detect the USB port connection. A special command is needed to permit the use of the port. Third, the ros2arduino only works for boards that have at least 32 KB of SRAM memory. At first, the Arduino Uno was used and an error message indicated the data was too large to be compiled since it has only 2 KB of memory. That is also one of the reasons that the Arduino Due board was selected.

```
caiyun@caiyun-VirtualBox:~$ cd Micro-XRCE-DDS-Agent && git checkout v1.3.0
HEAD is now at 559952b Merge pull request #152 from eProsima/v1.3.0
caiyun@caiyun-VirtualBox:~/Micro-XRCE-DDS-Agent$ MicroXRCEAgent serial --dev /de
v/ttyACM0 -b 115200
MicroXRCEAgent: command not found
```

Figure Y-3: XRCE "command not found" Fault

```
$ git clone https://github.com/eProsima/Micro-XRCE-DDS-Agent.git
$ cd Micro-XRCE-DDS-Agent && git checkout v1.3.0
$ mkdir build && cd build
$ source /opt/ros/dashing/setup.bash # to share libraries with ros2
$ cmake ..
$ make
$ sudo make install
$ sudo ldconfig /usr/local/lib/
```

Figure Y-4: XRCE Incorrect Setup Code

After troubleshooting, only one of the four laptops was able to use the bridge successfully which was not convenient for the project's progress. Furthermore, in the second semester of senior design, the only computer that was able to use the XRCE bridge lost its communication. As a result, the only choice is to use the Mircor_ros Agent. The download process was similar to that of the XRCE bridge. Follow this link step by step to download the micro-ROS Agent. Yet, for the Arduino Due to using the Agent, a SAM patch must be put inside the Arduino IDE's preference file. The patch can only be done on Arduino IDE 1.8.19 or less. With permission to use the DUE board on the computer, a ZIP library is also needed to get access to the example codes specially written for the Mircor_ROS. Modified the original Publisher/Subscriber codes, the communication is then well established to receive and transfer data. **Figure Y-5** shows the architecture of the ROS 2 stack and an Arduino Microcontroller. The first mentioned communication method, XRCE bridge, is a type of Middleware, on the other hand, the second, chosen method for the project is the "micro-ROS Arduino" in the category of RTOS stands for "Real Time Operating System". **Figure Y-6** is an example of data being called and published on the terminal using the micro-ROS Agent Publisher code.

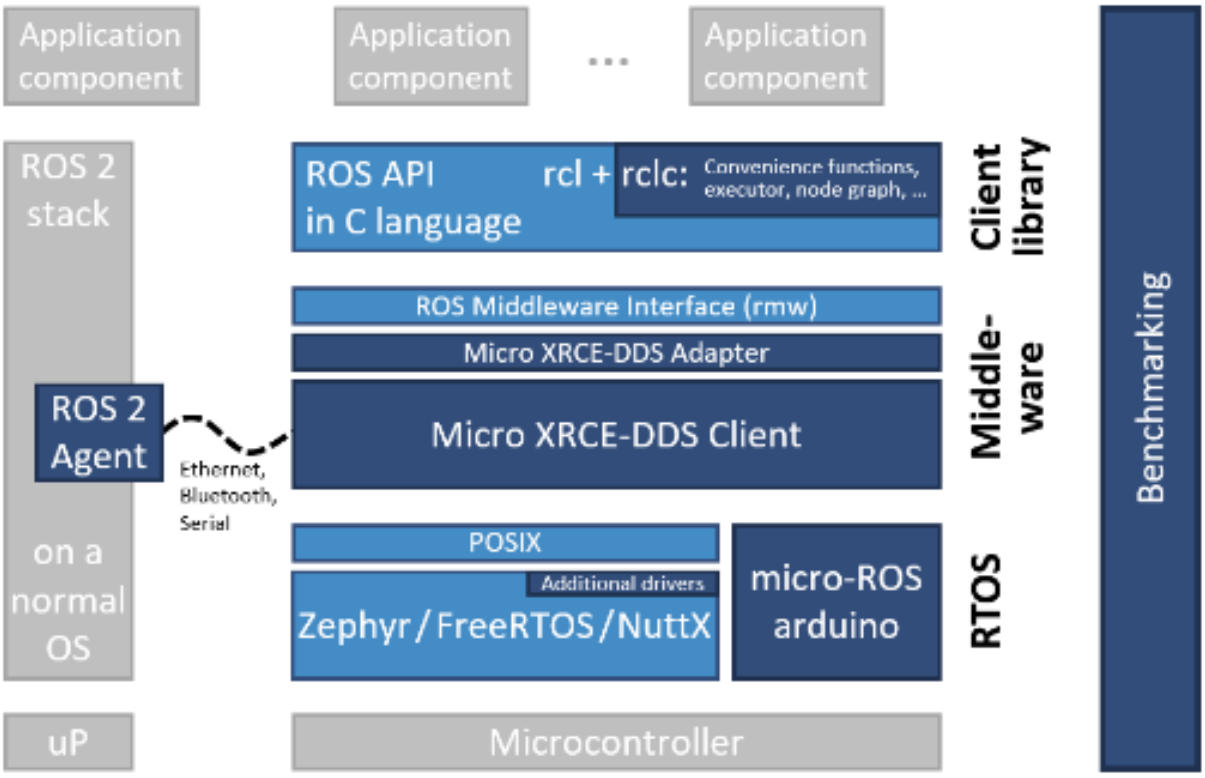


Figure Y-5: Mirco-ros and ROS 2 Architecture

```
caiyun@caiyun-VirtualBox:~/uros_ws$ ros2 run micro_ros_agent micro_ros_agent serial --dev /d
ev/ttyACM0
[1701196833.911463] info     | TermiosAgentLinux.cpp | init                     | running...
             | fd: 3
[1701196833.911882] info     | Root.cpp              | set_verbose_level        | logger setup
          | verbose_level: 4
[1701196833.935243] info     | Root.cpp              | create_client            | create
          | client_key: 0x0793D525, session_id: 0x81
[1701196833.935375] info     | SessionManager.hpp    | establish_session        | session estab
lished    | client_key: 0x0793D525, address: 0
caiyun@caiyun-VirtualBox:~$ ros2 topic echo /micro_ros_arduino_node_publisher
data: 219
---
data: 220
---
data: 221
---
data: 222
---
data: 223
---
data: 224
```

Figure Y-6: Mirco-ros Published Data

Devices Layout and Current State

At the planning stage, the devices that are considered to communicate with ROS2 through microcontrollers are one IMU, two motor drivers, three Ultrasound sensors, one safety light, and two encoders. Since there are quite a few devices, the team determined that using two Arduino boards would improve the data processing speed, which is critical to the performance of the robot. Later on, IMU is able to directly connect to the computer. Hence, the final application plan becomes one board for motor drivers and encoder, and another board for the light and Ultrasound sensors. Both Arduino boards plug into a USB hub and then connect to the laptop. **Figure Y-7** below shows the connection of microcontrollers and their connected devices. At current state, both Arduino boards function correctly with good performance connecting to the central processing unit.

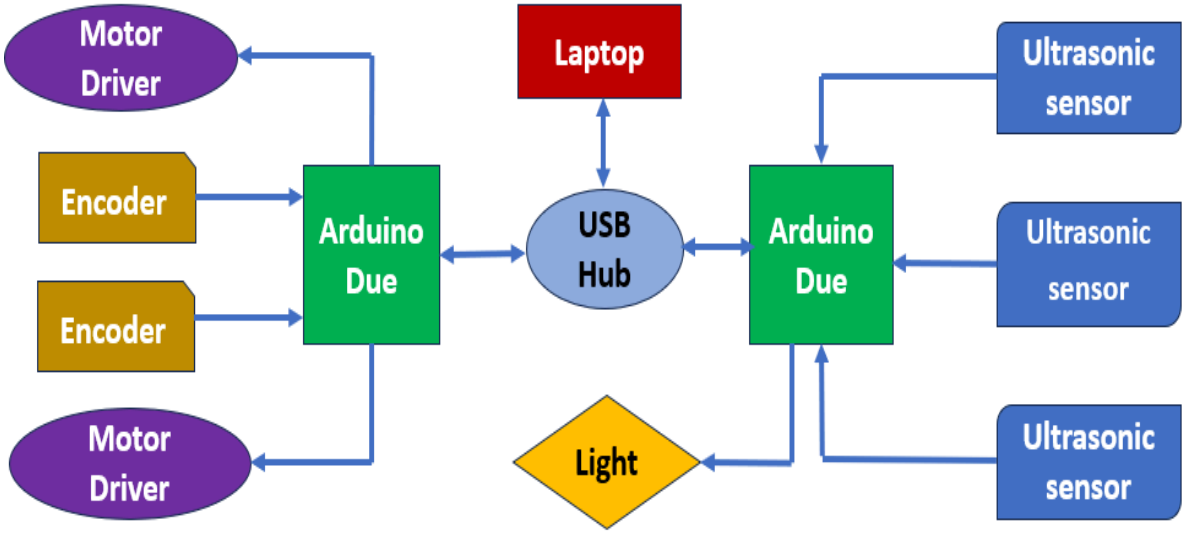


Figure Y-7: Microcontrollers and Devices

Microcontroller Standard

Microcontrollers are defined as "an integrated circuit that contains a microprocessor along with memory and associated circuits and that controls some or all of the functions of an electric device." The basic components of a microcontroller include CPU, memory, I/O ports, Serial Ports, Timers, ADC, DAC, Interpret Control, and Special Functioning Block. The categories of microcontrollers are separated by their memory, architecture, and instruction sets. For example, divided by architecture, there are ARM, 8051, AVR, and MSP types. Since microcontrollers can be used for various applications, such as unmanned aircraft and vessels, it is essential to avoid illicit use. As stated on the government website, the Director of National Intelligence, any usages that are critical to public safety should have reasonable explanations or be reported.

Appendix Z: Ultrasonic Sensors

Ultrasonic Sensors

The team decided to use Ultrasonic Sensors as proximity sensors, mostly as a failsafe when maneuvering around obstacles, or backing up. The issue with Ultrasonic Sensors is that they have a limited range when compared to the LiDAR, and there is also much less accuracy. To choose the Ultrasonic Sensors, it was required to find a sensor that was compatible with Arduino, and as the Ultrasonic Sensors were not a priority, one with a lot of readily available information.

Chosen Component

The Ultrasonic Sensor that best fit this description was the HC-SR04. These sensors have a range of 1 ft to 13 ft and are also cheap and readily available. There is also a large number of open-source code and libraries that can be used with these Ultrasonic Sensors, which made the coding and integration of the sensors much easier.

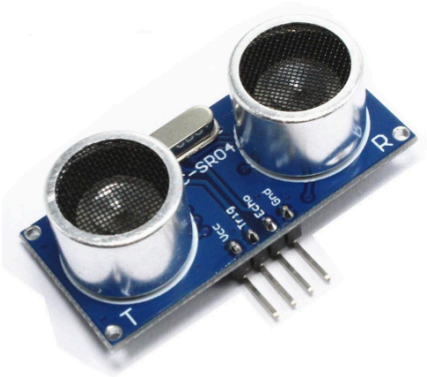


Figure Z-1: HC-SR04 Ultrasonic Sensor

Ultrasonic Sensors Current State

Three Ultrasonic Sensors were placed on the robot to cover the back half of the vehicle. The sensor placement and covered range can be seen in **Figure Z-2**. These sensors were connected to the second Arduino Due of the vehicle which takes the current distance readings

from each sensor and sends the corresponding values to the main computer. Although the sensors have no input into the path planning algorithm, they are used as a last-resort safety measure for the vehicle. Using the Collision Monitor package of Nav2, a "keep-out" zone was designed for the vehicle and is shown in **Figures Z-3 and Z-4**. When the Ultrasonic Sensors detect an obstacle in this range, the Collision Monitor will override the velocity commands being sent to the motors and prevent the robot from moving backwards. However, it still has the ability to move forward and turn when an obstacle is detected so that the vehicle can move out of range of the detected obstacle. A video demonstrating this function can be seen here

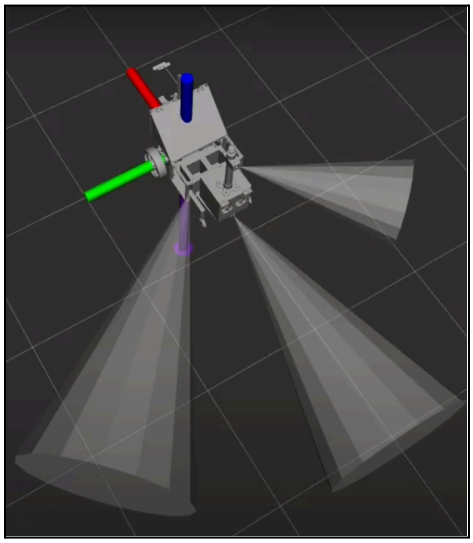


Figure Z-2: Ultrasonic Sensor Placement and Field of View

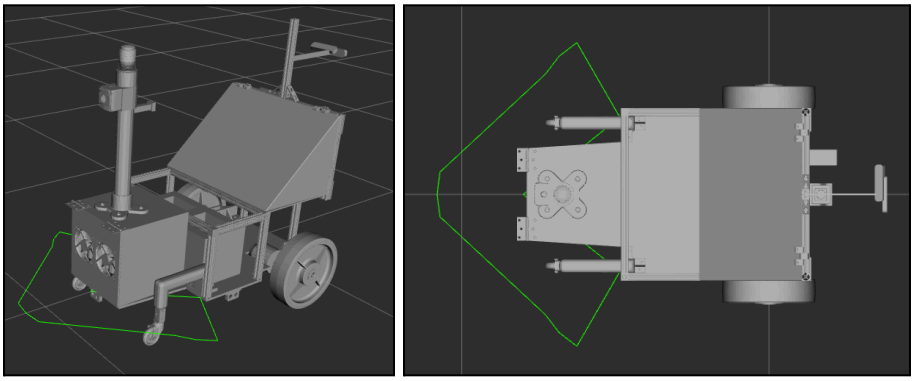


Figure Z-3 and Z-4: Keep-Out Zone for Collision Monitor

Appendix AA: LiDAR

For the autonomous vehicle to function properly, it is necessary to have a means to detect obstacles. For the IGVC competition, there will be obstacles along the course which the vehicle will need to navigate around. There are many means by which obstacle detection can be accomplished, but for the scope of this project, options are limited by budget constraints and the processing power that the sensor might require.

System Research

In order to find the best way to accomplish the obstacle detection, the team first looked at the prior competitions to get a better idea of where to start. Most of the teams went with a LiDAR as their method of obstacle detection. The most recent winning team used only ultrasonic sensors. Certain teams used a depth detection camera instead of a separate sensor subsystem as well.

Chosen Sensors

After research into the components, it was decided to utilize a combination of a LiDAR and ultrasonic sensors. The LiDAR is the primary source of obstacle detection, and needs a field of view of at least 120 degrees, but no more than 180 degrees. Three ultrasonic sensors were also utilized, one mounted at the back of the vehicle, and two on the sides of the robot. As the accuracy of ultrasonic sensors is quite limited, these sensors will be used simply as proximity sensors, and will just be used as a back up to the LiDAR, or in case the vehicle is reversing.

LiDAR

The initial thought for the LiDAR was to have a 3D LiDAR, and have that be the sole means of obstacle detection and line detection as well. 3D LiDARs have incredible capabilities and are often used in autonomous vehicles in use today. However, many of the capabilities possessed by these 3D LiDARs fall outside the scope of this project. The other drawbacks associated with 3D LiDARs are that they generate a lot of data that would cause the processing times to increase greatly, and the costs would definitely be outside of the specified budget.

The next option for LiDARs is the 2D LiDARs. While there are many options for 2D LiDARs, there are certain qualifications that these LiDARs need to meet. As the competition takes place outdoors, the LiDAR needs to be able to detect obstacles without too much interference from sunlight. As ROS 2 is the system used, LiDAR needs to be compatible with it to ensure smooth communication. The LiDAR needs to have an adequate field of view between 120 and 180 degrees.

Table AA-1: LiDAR Consideration

Unit	Compatibility	Range/ FOV	Cost	Ranking	Product
Velarray M1600	Linux	30m, 180°	$650.00	4	Link
SICK TiM P LiDAR	Linux/ ROS 2	25m, 270°	$2,856.00	3	Link
Slamtec RPlidar A3	Linux/ROS 2	25m, 360°	$509.15	2	Link
Slamtec RPlidar S3	Linux/ROS 2	40m, 360°	$549.00	1	Link

The research done showed that the LiDAR that most fit this description was the Rplidar series. The winning team of the IGVC competition in 2022 used the Rplidar A2. The two frontrunners in the Rplidar series were the Rplidar A3 and the Rplidar S3. In the end, the team has decided to go with the Rplidar S3.

The reason for this is because the Rplidar S3 has a higher sampling rate than the Rplidar A3. The Rplidar A3 has a sampling rate of 16k samples per second while the Rplidar S3 has a sampling frequency of 32k samples per second. Moreover, the Rplidar S3 has an improved resistance to daylight interference and is therefore more suitable for outdoor applications, particularly when compared to the Rplidar A3, or any LiDARs in the A series.

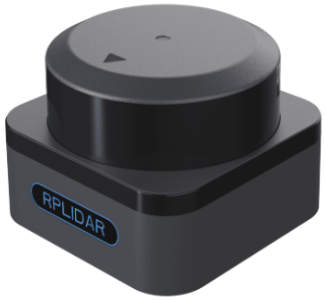


Figure AA-2: RPlidar S3 LiDAR

Past Progress

The RPlidar S3 was delivered and tested. While as of now, the LiDAR is still not connected to Linux or ROS 2, the product has been tested and point cloud data has been generated of the obstacles in a room. This was done by using the proprietary software of Slamtec, Robostudio. This software takes data from the LiDAR and creates a two dimensional map of the surroundings from data collected by the LiDAR.

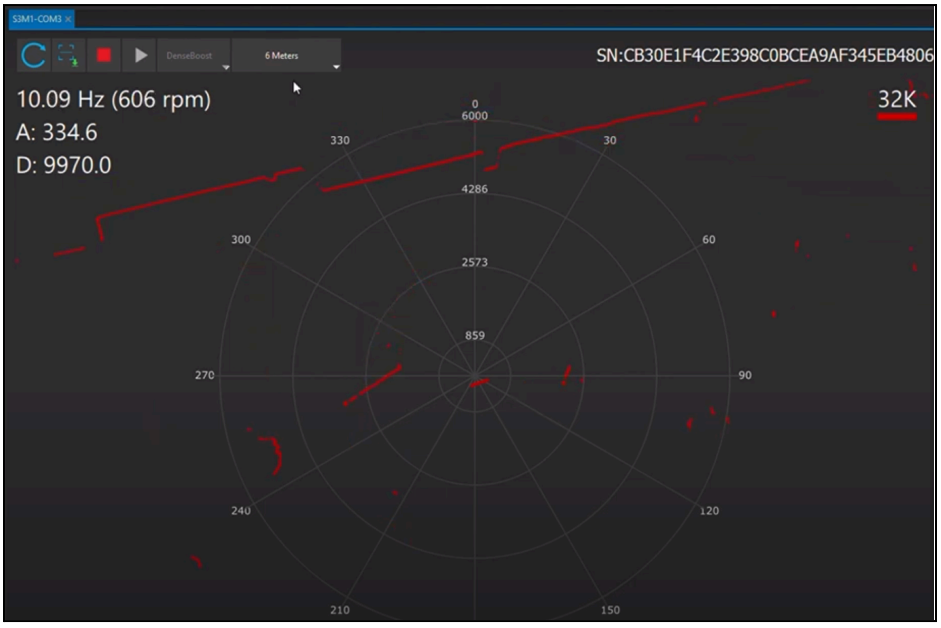


Figure AA-3: Point Cloud Data Generated from RoboStudio

After the LiDAR was tested, and point cloud data on the RoboStudio app was found to be accurate, the next step was to get the LiDAR publishing data to ROS 2. To do this, a driver found on GitHub was cloned and used. This driver extracted data from the LiDAR, specifically ranges or distances from obstacles along with angles. The scan data was extracted and appeared as seen in **Figure AA-4**.

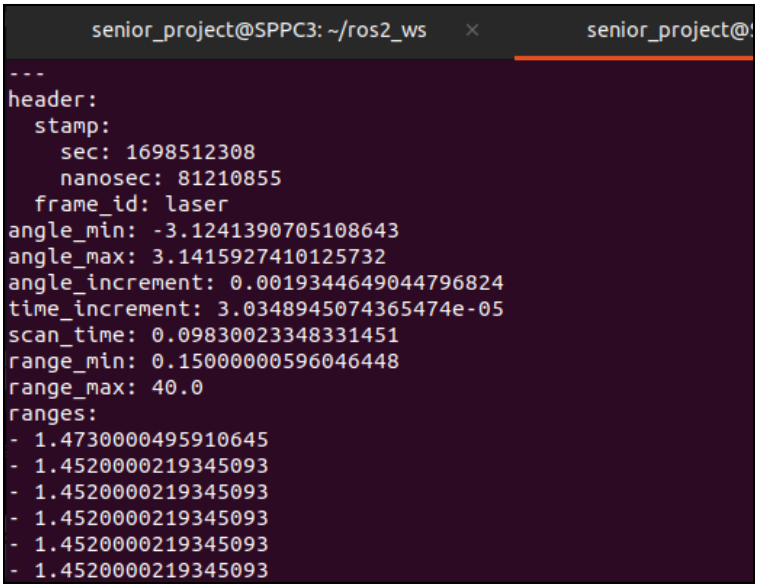


Figure AA-4: Scan Data Extracted from LiDAR

There are many different ways to code for obstacle detection and avoidance, but the method chosen was to convert the incoming scan data into a point cloud. The point cloud topic, 'PointCloud2' used on ROS 2 visualizes the scan data as points on a grid. This appears as seen in **Figure AA-5**.

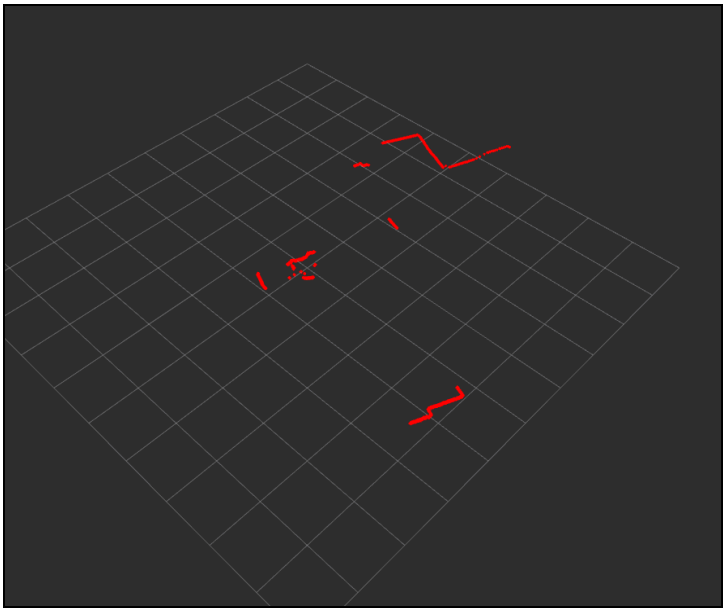


Figure AA-5: Point Cloud Data Visualized on RVIZ2

These points are defined by x, y, and z coordinates, giving the precise location of the obstacle. As the RPlidar S3 is a 2D LiDAR, the z coordinate is always zero. An obstacle avoidance code was built from these points. This was done by limiting the LiDAR's field of view to 120 degrees with a 3D printed cover and further dividing that range into two zones, the left and the right. If an obstacle is detected in the left zone as determined by the coordinates, the code prints a message to turn right and vice versa. The video of the LiDAR detecting obstacles and displaying messages can be viewed here. These messages will be sent to the motor controls once the LiDAR is fully integrated with the other components.

LiDAR Current State

The LiDAR publishes its data in the form of the ROS 2 form of Laserscan. This output is taken and merged with the data taken from the camera as described in Appendix AD. This merged data is then utilized by the robot to create costmaps and global maps of the obstacles, using which the robot can create paths to its required destination and waypoints without collisions with the obstacles. More specifics on this obstacle avoidance method can be found in Appendix AI.

LiDAR Standards

It is essential for the chosen LiDAR to meet safety standards. The RPlidar S3 uses a low powered laser as its light source. It meets the Class 1, IEC-60825 Laser Safety Standard which makes it safe to human eyes. The LiDAR also complies with 21 CFR 1040.10 and 1040.11, the specifications of which can be found in the RPlidar S3 datasheet. The raw point cloud must be compliant with LAS v1.2 or v1.3 and Point Record Format 1,3,4, or 5, according to the US Geological Survey LiDAR Guidelines and Base Specifications draft.

Appendix AB: GPS

Waypoint navigation is a necessary method of operation at the IGVC. Because GPS coordinates will be provided throughout the course, the robot will need a GPS module to navigate to these points. This unit needs to be accurate within the scope of the competition, while also being cost-effective to fit into the budget.

Previous Team Research

Research began by looking at the GPS units that were implemented in the past competitions because these modules would likely be sufficiently accurate with a reasonable cost. Although there were several different GPS units put to use by the different teams in the competition, the most popular units by far were the ublox GPS modules. On top of this, both WIU teams from last year used one of these ublox GPS modules.

Continued Research

Further research on the Internet was conducted after looking at the teams from last year. A wide range of cost and accuracy were considered early on. Unsurprisingly, the ublox GPS modules were extremely popular online as well, due to their wide use in similar projects. Despite this, some cheaper options and some expensive options with more of an industrial purpose were examined before narrowing down the modules to those shown in Table AB-1 below.

Table AB-1: GPS Module Considerations

Unit	Cost	Accuracy	Product	Rank
Goouu Tech GT-U7 Module	$14.99	2.5 m	Link	5
EMLID Reach M2	$649	7 mm	Link	4
Ublox C94-M8P Application Board	$399.99	2.5 m	Link	3
Ublox C009-F9P Application Board	$301.79	1 cm	Link	2
SparkFun ZED-F9P Application Board	$274.95	1 cm	Link	1 ✅

SparkFun ZED-F9P Application Board

The final decision was made to purchase the SparkFun ZED-F9P Application Board. This GPS module was chosen for a number of reasons; it was available for delivery on Amazon in less than a week, it has a high accuracy in comparison to similar receivers, the board allows for easy integration, it is reasonably priced, and there is lots of documentation for this unit. All of this contributed to the decision to implement the SparkFun ZED-F9P GPS module into the autonomous vehicle. The receiver is pictured below in **Figure AB-2**, and a detailed schematic can be viewed here.

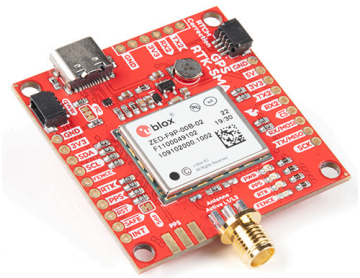


Figure AB-2: SparkFun ZED-F9P Application Board

GPS Current State

In order to use the chosen ublox GPS unit with ROS2, a driver for this module was implemented. The developers of this driver have published the free-to-download code on GitHub which is compatible with several ublox GPS modules including the ZED-F9P. This package takes care of the serialization, message retrieval, and a lot more of the finer details associated with booting and using the ublox GPS modules. It then publishes the data to a number of nodes and even allows for easy configuration of the baud rate, device path, and other settings. In the first semester, the GPS data was combined with the BNO055 IMU data for localization. At this time, the GPS coordinate data has been fused with the wheel encoder and BNO055 IMU data in the Extended Kalman Filter to produce an accurate position of the robot in space.

GPS Standard

Because this project will obviously be taking place in the United States, it is important to be aware of the GPS (Global Positioning System) standards for GNSS. There are a couple of things that go into this standard for satellite navigation. One of which is the message form all GPS receivers must output. The SparkFun ZED-F9P follows the NMEA 0183 and ISO 6709 specifications for communication between electronics, which are used worldwide in numerous navigation applications. The NMEA protocol includes specifications on the serial configuration, message structure, sentence format, and is supported by certain software. ISO 6709 is an international standard for how coordinates should be represented using longitude, latitude, and altitude. In addition to these specifications, users of the United States provided GPS Standard Positioning Service are guaranteed a certain performance standard. This performance standard is detailed extensively in this document.

Appendix AC: IMU

The GPS is an important unit of the vehicle, but it cannot function properly without an IMU. IMU is short for Inertial Measurement Unit. It is typically a solid-state device that contains an array of sensors that can measure motion. This unit ensures the central processor recognizes the vehicle's heading direction and the angle away from the designed goal point. The IMU is integrated with three main components: an accelerometer, a gyrometer, and a magnetometer. The accelerometer measures the linear acceleration, the gyrometer calculates the rotation, and the magnetometer acts like a compass.

Research on Components

Once knowing that IMU is a required component, research was carried out on other winning teams' choices in 2023. Only one team specified their IMU unit which is built in with a GPS with the price of $349.99. Furthermore, after deciding to use ROS 2 as the integration control, one IMU specially made for the ROS is found on the Internet at an affordable price. However, there are no quality reviews on the product. This item is out of consideration for the quality concern. Since there is no need to buy an expensive, super-precise unit, the choice is narrowed down to the Adafruit BNO055. Table AC-1 shows the IMU units being considered with the choice ranking.

Table AC-1: IMU Researched Models

Brand	Model	Advantage	Cost	Ranking
Cube Orange	Hex	GPS built-in	$349.99	4
ZED	2i	Camera built-in	$549.00	3
ROS IMU	T1-901	ROS friendly	$34.44	2
Adafruit	BNO055	Arduino friendly	$33.7	1 ✅

Component Choice and Application

The Adafruit BNO055 IMU was chosen due to its compatibility with Arduino and low cost at around $ 33 on Amazon. It also has good reviews and is listed as one of the top 10 student project IMUs. The BNO055 has an I2C connection directly connecting to the Arduino's analog pins. And its size is considerably small. The Adafruit library allows the IMU to use the Arduino IDE software conveniently. IDE provides part of the code that can extract data from the IMU and do some advanced programming; the unit can provide a vehicle bearing angle in the range of -180 to 180 degrees. **Figure AC-2** shows the IMU, and **Figure AC-3** demonstrates the electrical diagram of the unit.

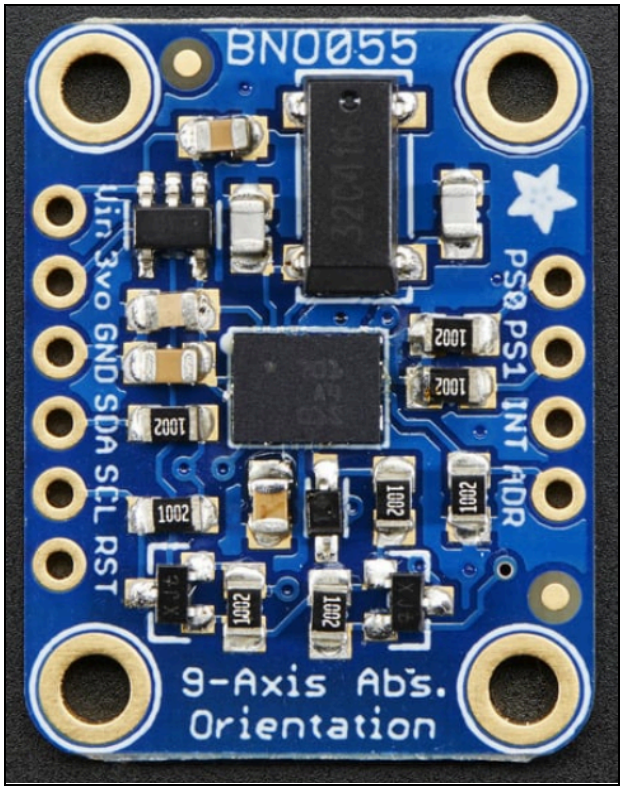


Figure AC-2: Adafruit 9-DOF IMU

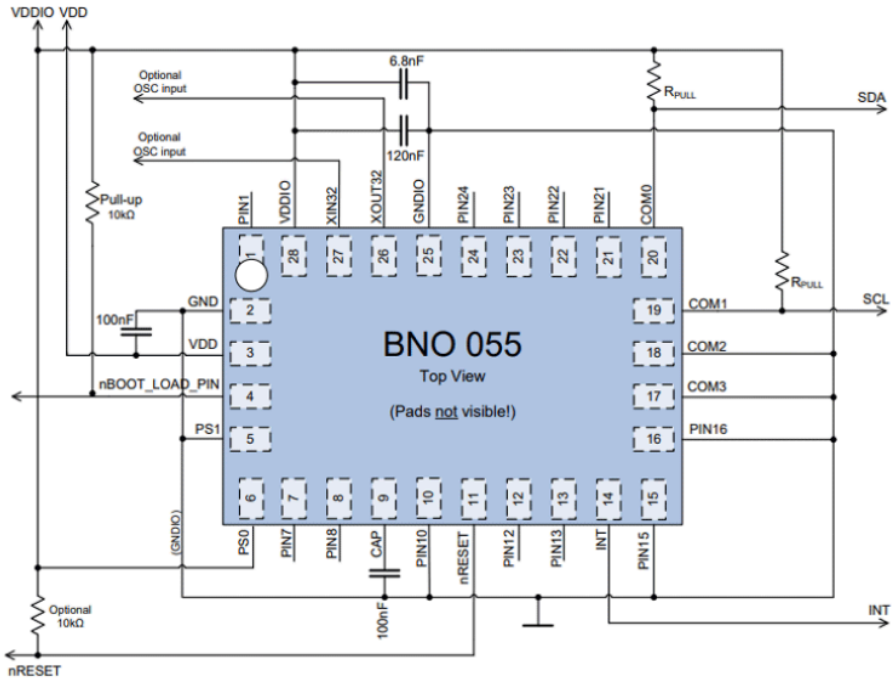


Figure AC-3: IMU Schematic

IMU Current State

In the first semester, the IMU was connected to an Arduino Due which relayed the sensor's information to the main computer using micro-ROS. However, in the second semester, a USB to UART adapter was used to directly connect the IMU to the laptop. The pin layout along with the adapter is shown in **Figure AC-4** below. This decision was made to reduce the processing power of the Arduino Due, to make the IMU data stream more efficient, and allowed a ROS2 driver to be implemented. The driver used for this project can be found here, and its main functions include fetching the required data from the IMU sensor and converting it into a Imu ROS2 message type. Although the data conversion is not necessary, it makes the data compatible to be used with the Extended Kalman Filter.

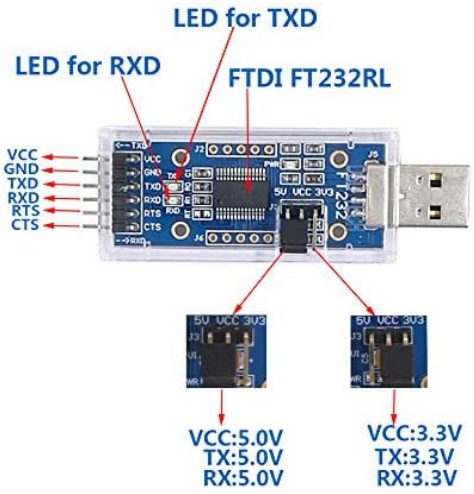


Figure AC-4: UART to USB Adapter Pinout

Standard of IMU

Two inertial systems terminology standards are developed by the IEEE/AESS Gyro and Accelerometer Panel by the IEEE Standard Boards as IEEE Std 1559 and IEEE Std 528-2001. According to these documents, an inertial sensor is defined as "a position, attitude, or motion sensor whose references are completely internal, except possibly for initialization." and all inertial systems require calibration and initialization. All inertial measurement devices have errors that grow with time which means the bias will drift during the operation through time. This drift is called the In-Run Bias Stability (IRBS). IMUs are separated into several performance grades based on the specifications of the accelerometer and gyro: Consumer/Automotive Grade, Industrial Grade, Tactical Grade, and Navigation Grade Specific. For example, IMU is industrial grade when the gyro is operated less than 10 degrees/hour IRBS and qualified as "Tactical Grade" when the gyroscopes are operated between 0.5 degrees/hour and 5 degrees/ hour. The highest grade is the "Navigation Grade" which requires less than 0.1 degree/hour. The BNO 055 has an IRBS of 20 degree/hour which belongs to the "Consumer Grade". Specific information can be found on grouper.ieee.org.

Appendix AD: Camera

The camera subsystem is an indispensable element for the Autonomous Ground Vehicle, as mandated by the Intelligent Ground Vehicle Competition (IGVC) requirements. In the context of the IGVC, the camera system serves as the "eyes" of the autonomous vehicle, enabling it to perceive and interpret its environment effectively. The camera subsystem must maintain precision within the competition's requirements while remaining budget-friendly to align with financial constraints.

System Research

In the search for the right camera system, the team looked at what cameras past teams used in the IGVC competitions. Research was also conducted to understand exactly what this project needed. This process led us to find four different camera options. Each was carefully looked into and the information was compiled in a table to help make a smart choice that matches the project's goals and budget.

Table AD-1: Camera Consideration

Unit	Compatibility	Resolution	Cost	Product
ZED X	Linux/ROS 2	2560x1080	$599.00	link
Intel RealSense D455i	Linux/ROS 2	2560x1080	$419.00	link
Intel RealSense D435	Linux/ROS 2	2560x1080	$314.00	link
ZED 2i Stereo Camera	Linux/ROS 2	4416x1242	$499.00	link

ZED 2i Stereo Camera

The team has chosen the ZED 2i stereo camera with a polarizer as the ideal camera subsystem for the project. This decision was based on a careful evaluation of various camera options and their alignment with the specific requirements of the Autonomous Ground Vehicle. The ZED 2i stands out as an exceptional choice, primarily due to its effectiveness in line and pothole detection, which are critical aspects of the project. The stereo vision capabilities provided by this camera not only enhance the vehicle's depth perception but also enable it to accurately identify and navigate around lines and potholes on the competition course. The addition of a polarizer further optimizes the camera's performance in various lighting conditions, ensuring that the vehicle can maintain a robust and reliable performance throughout the IGVC competition.

Figure AC-2: ZED 2i Stereo Camera

Camera current state

The camera system has undergone significant development since its acquisition during the Preliminary Design Review (PDR). Initially, the camera's alignment with project specifications was confirmed using the ZED SDK and ZED Explorer software, ensuring its capability to capture visual data effectively for the autonomous vehicle.

Following this, coding became a crucial aspect of optimizing the camera's functionality. Firstly, Hough Lines were considered for straight line detection, aiding the vehicle in following predefined paths. Subsequently, efforts were directed towards Canny edge detection, a code identifying curves along the vehicle's path, showcasing proficiency in handling complex vision algorithms. Additionally, during the development process, a challenge emerged as the video feed exhibited considerable noise as shown in **Figure AD-3**, potentially impacting the accuracy of the vision algorithms. To address this, a strategic decision was made to incorporate a masking function into the code. This masking function proved instrumental in mitigating unwanted noise, providing a more refined and accurate output. **Figure AD-3** illustrates the effectiveness of the mask in eliminating undesirable elements from the video feed. This approach not only enhances the overall performance of the camera subsystem but also underscores the adaptability and problem-solving capabilities employed in refining the system's functionality.

Figure AD-3: Unmasked image (Left), Masked Image (Right)

Another milestone involved the development of codes for circle detection, representing potholes on the path as shown in **Figure AC-4**. This functionality is crucial for the vehicle to identify and navigate around potential obstacles. The standardized 2 feet size of the detected potholes ensures accurate recognition, enabling the vehicle to make informed decisions in real-time and enhancing its capability to traverse the designated course with precision.

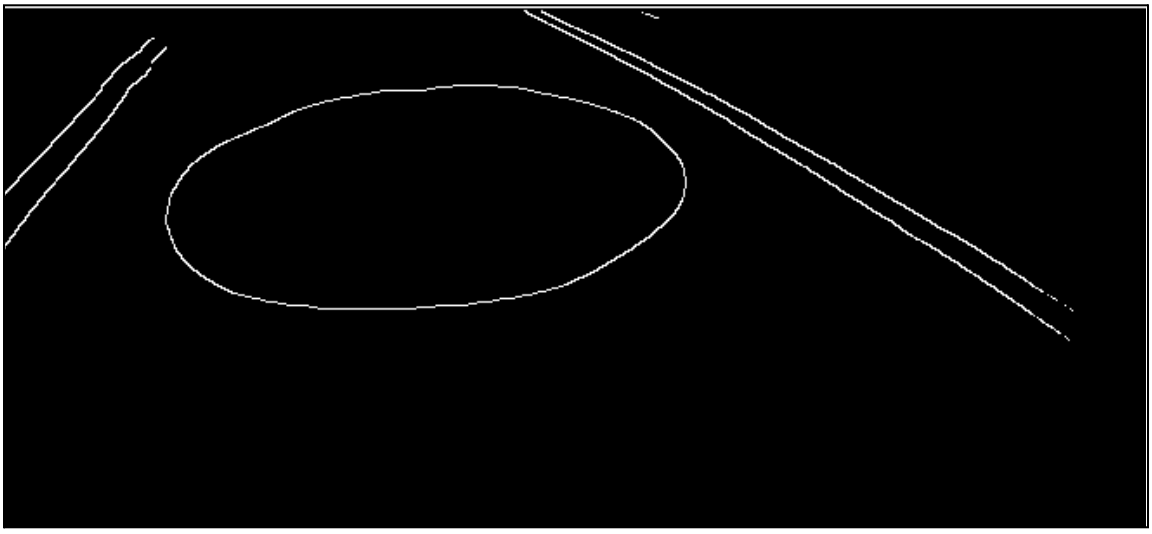


Figure AC-4: Pothole Detection

Focus then shifted towards enhancing the vehicle's navigation safety by extending the codes to include strategies for avoiding both lines and potholes. White lines were established as boundaries, and circles were coded as objects, providing the robot with the necessary parameters for effective navigation between lines and avoidance of potholes. Although the codes worked fine on ROS 2, a challenge arose with the ZED2i requiring a graphics pack not available on the laptop provided by the school. To address this, a new laptop has been ordered, expected to arrive before the semester concludes, ensuring access to the full capabilities of the ZED2i. While awaiting the new laptop, efforts were dedicated to creating a standalone subsystem. Coding focused on displaying avoidance information directly on the video feed. The avoidance video, featuring clear instructions based on left and right line detections, demonstrates a creative approach allowing the camera subsystem to operate autonomously and contribute to the vehicle's

navigation. The avoidance code gives output: if the camera detects the left line, the instruction prompts a turn to the right as shown in **Figure AC-5**. And if the camera detects the right line, the instruction prompts a turn to the left, as shown in **Figure AC-6**.

Figure AC-5: Turn Right Instruction Triggered by Left Line Detection

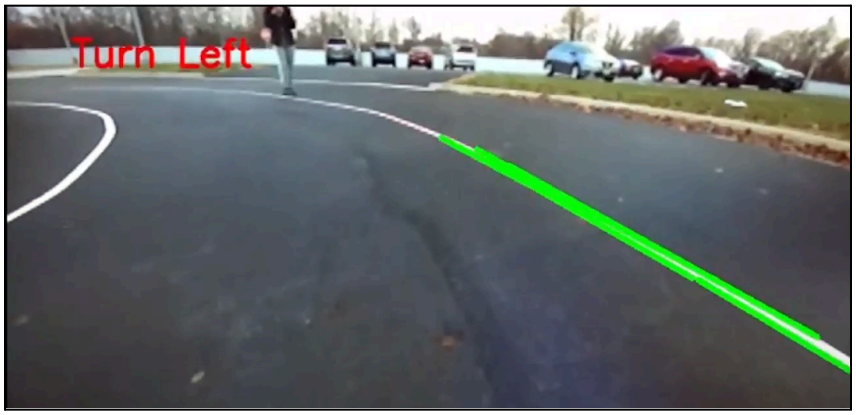


Figure AC-6: Turn Left Instruction Triggered by Right Line Detection

For a comprehensive understanding and a visual demonstration of the standalone subsystem, showcasing the system in action and its significant role in autonomous navigation. The camera subsystem has made significant progress from camera acquisition to the development of sophisticated vision algorithms. Proficiency in coding for line and pothole detection sets the stage for effective navigation and obstacle avoidance.

With the successful implementation of the decision-making process, the team's focus naturally shifted towards the integration aspect. The team recognized the potential to optimize decision-making by leveraging the capabilities of the ZED Wrapper. This decision led to the development of a custom code, '/tanaji', which served as a pivotal component in integration efforts. Through this code, the team established a direct connection to the point cloud data provided by the ZED Wrapper and tailored the output to suit the team's specific needs. The filtering process implemented within '/tanaji' played a crucial role in refining the data output. By isolating white data points, generating Hough lines, and detecting circles, the team was able to extract valuable information necessary for effective navigation. However, despite these significant advancements, challenges persisted, particularly in dealing with noise inherent within the data. To overcome this hurdle, the team introduced a speckle filter into the system. This filter proved instrumental in reducing noise by identifying and removing data points that did not exhibit clustering characteristics. By applying this filter, the team ensured that only coherent groups of points representing lines and circles remained, thus enhancing the accuracy and reliability of the navigation system.

The meticulous approach to addressing these challenges underscores the team's unwavering commitment to refining the safety and efficiency of vehicle navigation. By integrating advanced algorithms and techniques, the team continually strives to enhance the capabilities of the system and overcome potential obstacles more effectively. Ultimately, the overarching goal remains unchanged: to ensure seamless navigation and obstacle avoidance, thereby bolstering the overall safety and reliability of autonomous vehicles in real-world scenarios.

Standard for Camera

The ZED 2i stereo camera with a polarizing lens is primarily designed for capturing high-quality images and videos. When it comes to assessing its performance and capabilities, several ISO standards can be relevant. ISO 12233 is suitable for measuring resolution, which helps evaluate the camera's ability to capture detailed images. ISO 15739 is essential for noise measurements, indicating how well the camera handles image noise, an important aspect for

image clarity. ISO 12232 can be used for sensitivity measurements, which assess the camera's ability to work well in different lighting conditions. ISO 19093 is particularly relevant for low light measurements, which are essential for evaluating the camera's performance in challenging lighting conditions. Each of these standards serves a specific purpose in evaluating the ZED 2i stereo camera's capabilities and performance, making it a versatile tool for various applications.

Appendix AE: Motor Control

To control the motors efficiently, the Cytron MD30C Motor Driver is used. Like most motor drivers, the MD30C does not have memory itself, so the Arduino Due is implemented to gain accurate control of the motor through the drivers. Furthermore, it is practical to separate the motor and control power because the signals can interfere with each other due to electromagnetic induction. The Cytron driver has three control pins: GND (ground), PWM, and DIR. The PWM and DIR pins are connected to Due's two digital pins, and GND is connected to the ground pin. **Figure AE-1** shows the connection between the Arduino Due and the driver.

Figure AE-1: Arduino Due with Cytron Motor Driver

Basic Motor Function Setup

In the Arduino IDE, there is a Cytron motor driver library that can be utilized easily. However, through careful design processing, the idea of using two motor drivers instead of one came into consideration. With each driver for each wheel, the motion of the robot would be controlled more precisely. Furthermore, with two motor encoder feedback data being processed, it is wiser to control motor motion individually. Since the project calls for two single drivers individually, it is inconvenient to use the library. Because of this, a different program was written for motor control, which allows one Arduino controller to run two drivers at the same time. Since each motor needs two PWM pins, four PWM pins are initialized at the beginning of the program. And in the Setup section, these four pins are set as Output. Unlike the Cytorn library, which utilizes the positive and negative signs as direction control, the new code uses the DIR pin. When set 1 is the forward direction and number 0 is for the backward direction. The ultimate vehicle running command comes from the central processor using the data from other devices. Hence, to simplify the motor calling process, six individual motor functions: start, stop, forward, backward, turn-left, and turn-right, are created using the Void command. **Figure AE-2** shows part of the motor code.

When doing the code testing, a problem was encountered. One of the motors only ran in one direction. After troubleshooting, it was determined the direction pin was not working correctly. In addition, the direction pin (DIR) cannot be written as “analogWrite”, instead, “digitalWrite” must be used to change motor direction. Now, the motor control system can function properly as a stand-alone system.

```
int pwmpin = 9;      // PWM pin speed for motor 1
int pwmpin2 =10;    //direction pin motor 1
int pwmpin3 =11;   // PWM pin speed for motor 2
int pwmpin4 =12; // motor 2 direction

pinMode(pwmpin,OUTPUT);// set pin 9,10,11,12 as output pins
pinMode(pwmpin2,OUTPUT);
pinMode(pwmpin3,OUTPUT);
pinMode(pwmpin4,OUTPUT);

void forward() {
  // run forward slow at 50
   analogWrite(pwmpin,50);// NOT use negative valude to change direction
   digitalWrite(pwmpin2,1);//always reverse o or 1 when use analogWrite..
   analogWrite(pwmpin3,50);//  NOT use negative valude to change direction
   digitalWrite(pwmpin4,1);
   delay (2000);
```

Figure AE-2: Part of Motor Code

At this point, all tests were done only through Arduino IDE. However, for automation, the command for motor control must be sent by the ROS2 platform. This requires communication between the Arduino IDE and ROS2 software.

Motor Control through ROS2

It has been decided that this project utilizes Navigation 2 as the main automation platform, which means the central processor will combine all sensor data. After integration, it will develop a path plan to move the robot by sending a "twist" type velocity command to the motors. **Figure AE-3** below demonstrates the motor control data flow out of the Controller Server, as a Twist data type node, "cmd_vel-nav", and then to the Velocity Smoother, Collision Monitor (when using remote control), to Twist_to_motor node, and finally, to the Arduino Due board to move the robot. For the robot to move efficiently and smoothly, the NAV2 Velocity Smoother code combined with encoder feedback is implemented in the motor control function. **Figure AE-4** describes the velocity data flow in the ROS2

Within the Velocity Smoother parameters, the robot's acceleration and deceleration time, along with maximum and minimum speed will be limited to suitable values. **Figure AE-5** shows the section of smoother parameter setting.

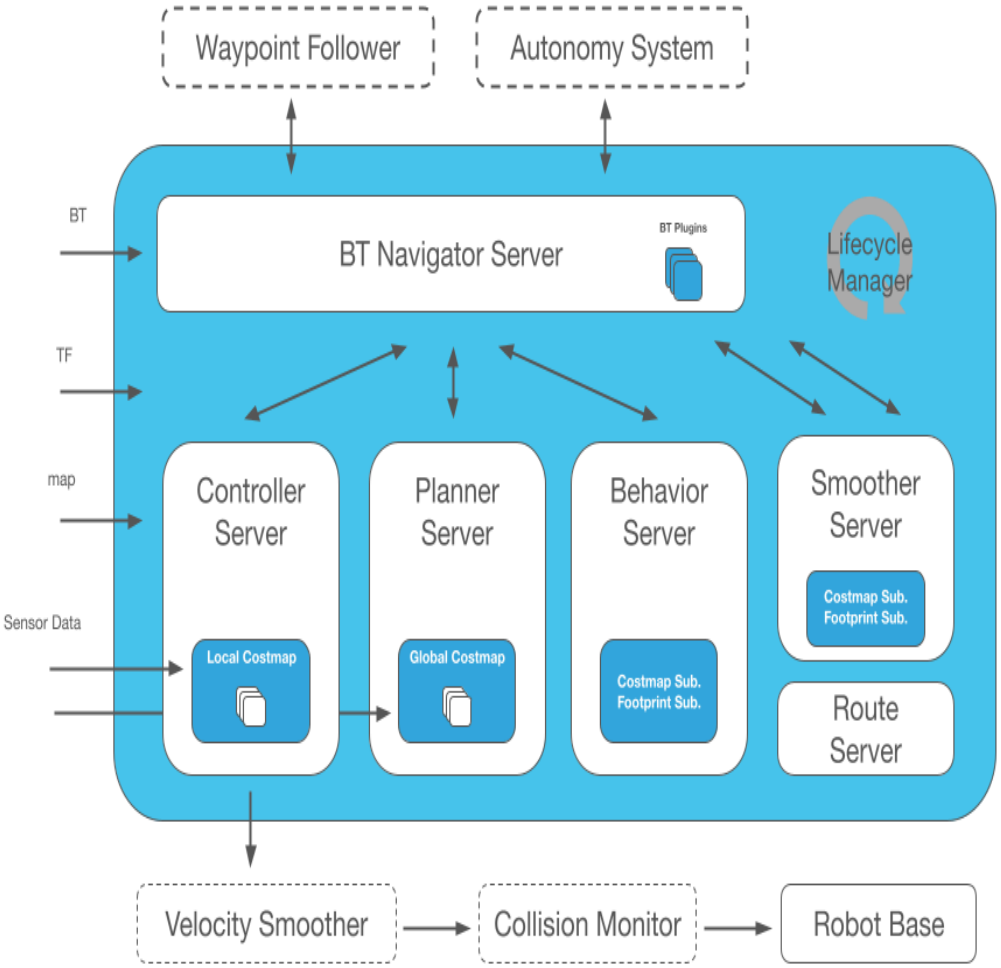


Figure AE-3: Nav2 Motor Control Topic cmd_vel

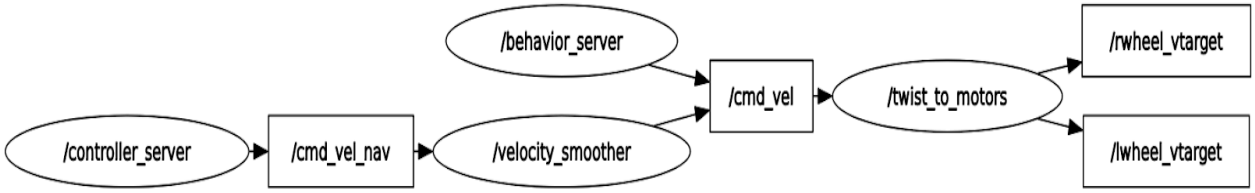


Figure AE-4: Motor Control Data Flow

```
velocity_smoother:
  ros__parameters:
    use_sim_time: false #was True
    smoothing_frequency: 20.0
    scale_velocities: False
    feedback: "OPEN_LOOP"
    max_velocity: [0.26, 0.0, 1.0]
    min_velocity: [-0.26, 0.0, -1.0]
    max_accel: [2.5, 0.0, 3.2]
    max_decel: [-2.5, 0.0, -3.2]
    odom_topic: "odom"
    odom_duration: 0.1
    deadband_velocity: [0.0, 0.0, 0.0]
    velocity_timeout: 100000.0
```

Figure AE-5: Nav2 Velocity Smoother Parameters

After subscribing to the smothered velocity values, the "Twist to Motor " code performs mathematical calculations then publishes the targeted left and right velocity to each motor drive separately . The equation curves of the left wheel and right wheel target velocity are obtained by experiment. **Figure AE-6** and **Figure AE-7** show the two equations.

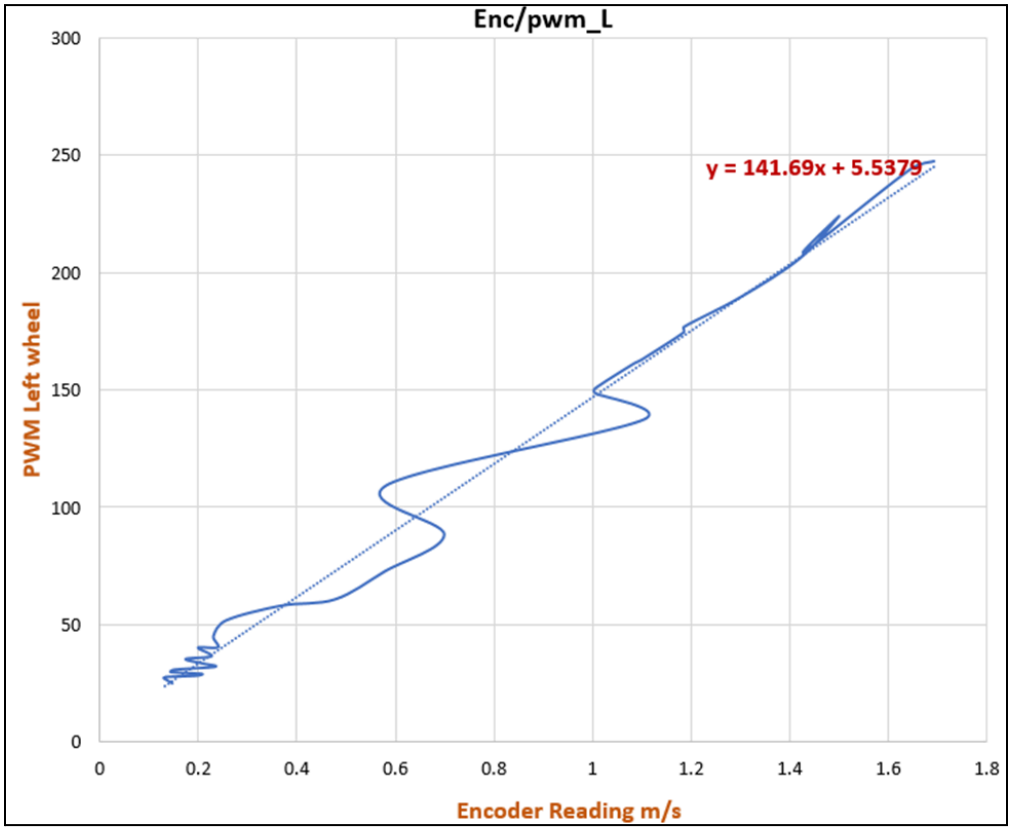


Figure AE-6: Left Wheel Velocity Curve

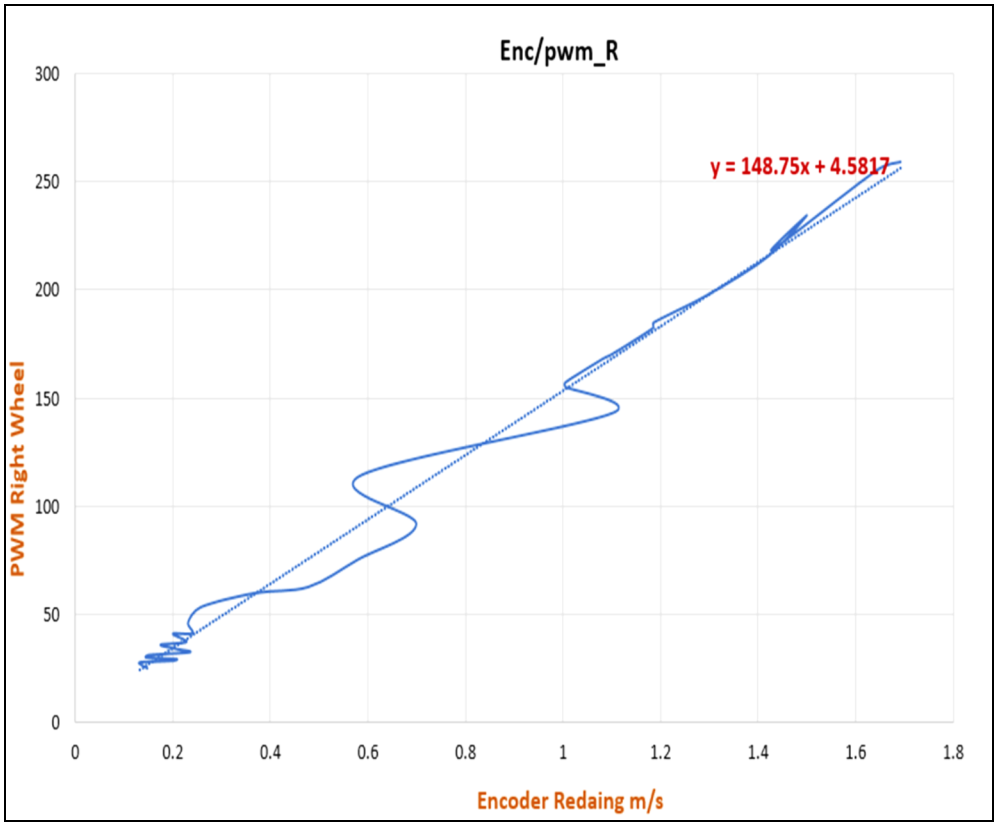


Figure AE-7: Right Wheel Velocity Curve

Motor Control Testing

There are a few problems occurring while doing motor testing through the ROS2 platform. One is that the robot always has a two-second delay for all direction commands. The problem is fixed by deleting the 100-millisecond delay in the original Arduino IDE loop function. The second problem is, that once in a while, the robot would lose control and keep on going with the last velocity command received by the Arduino. And the only way to stop it is to use the Emergency Stop. Multiple observations have indicated two different causes: lost communication with the laptop and a PWM value greater than 255. To fix this problem, two more section codes are added. One is to detect the communication status; once the robot loses communication, the motors will stop. Another set of code limits the PWM between 0 to 255. **Figure AE-8** demonstrates part of the PWM limiting code. For the current state, the robot motor motion is able to function properly through ROS2 Navigation stacks.

```
void move_left_wheel(int pwm) {
  if (pwm < 0) {
    digitalWrite(PWMpin2, LOW);  // direction backward
  } else {
    digitalWrite(PWMpin2, HIGH);  // direction forward
  }
  if (abs(pwm) >= 255) {
    analogWrite(PWMpin1, 255);
  } else {
    analogWrite(PWMpin1, abs(pwm));  // speed at pwm1
  }
}
```

Figure AE-8: Arduino code for limiting PWM

Appendix AF: Remote Control

Although the ultimate goal of this project was to create an autonomous vehicle, the robot still has the ability to be controlled remotely. Remote control allows the robot to be moved with less hassle and was very helpful during the testing phase of the motors. Due to the Bluetooth compatibility of the main computer, almost any controller can be connected to the computer and be used for controlling the robot. A PlaySation 4 controller was purchased and used for the remote control, which is pictured in **Figure AF-1** below. Once the controller has been connected, code is run to transform the joystick input into a ROS2 Twist message. A Twist message contains values for the linear and angular velocity in the x, y, and z directions. The code used for the joystick conversion can be found here. The controller is also configured so that the left bumper button acts as an "enable" button and the right bumper button acts as a "turbo" button. In order to send a velocity command, the "enable" button has to be pressed. This is an extra safety measure to ensure that velocity commands aren't accidentally sent to the robot. The "turbo" button simply increases the speed of the vehicle when pressed. Additionally, a launch file was constructed for conveniently setting the robot to remote control mode. This launch file simultaneously runs the controller program, smooths the velocity commands, and sends a signal to turn on the safety light. A video showing the functionality of the remote control can be seen here.

Figure AF-1: PlayStation 4 Controller

Appendix AG: Unified Robot Description Format (URDF)

The URDF is a three-dimensional representation of the vehicle that can be visualized in RViz2. Although it is unnecessary to display the robot in the simulation software, it looks much nicer and has a practical purpose as well. Each sensor on the vehicle is located in a different spot and at varying orientations. The URDF describes the placement and direction of each sensor, which is necessary to use the data effectively.

URDF Fusion Export

In order to avoid building a URDF from scratch, the robot's three-dimensional CAD model was used. There are several different GitHub packages that provide the ability to export the CAD model directly into a URDF. The package used can be found here, and is compatible with Autodesk Fusion. Autodesk Inventor was used to design the robot initially, so the .stl file was transferred to Fusion which could be used to export the CAD file as a URDF. The final product can be seen in **Figure AG-1**.

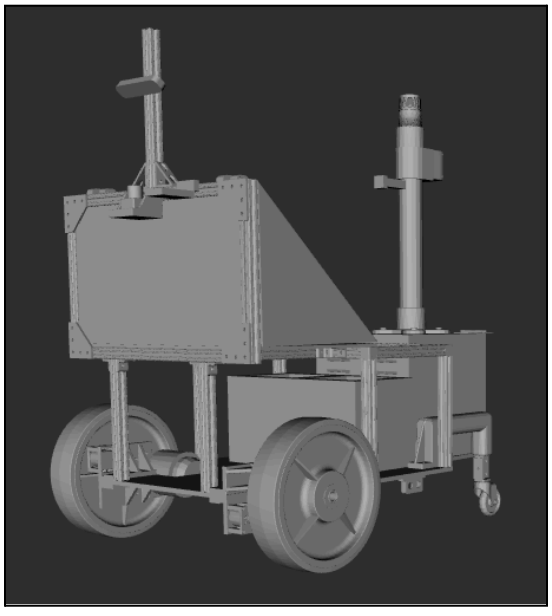


Figure AG-1: URDF Model

Transforms and Frames

In order to use the data from each sensor, transforms and frames are required. All of the sensors could be producing highly accurate information, but the navigation system cannot use any of it effectively if it doesn't know where each set of data is coming from. Frames are basically a point and orientation used to represent the location of an important body or sensor. The URDF has a single frame called the "base_link" which is a point located directly between the two wheels. This frame is oriented so that the vehicle travels in the positive x-direction when moving forward and so the positive z-axis points upwards through the robot. The "base_link" is very important because every other frame is described with reference to the "base_link" in one way or another. This is done using transforms, which can be static or dynamic. For example, the "laser" frame represents the location and orientation of the LiDAR. The URDF stores the distance and difference in orientation between the "laser" frame and "base_link" frame as a static transform. The URDF holds these static transforms for every sensor being used. This allows the navigation system to know the context of each sensor and make educated decisions. All of the transforms and frames produced by the URDF can be seen in **Figure AG-2**.

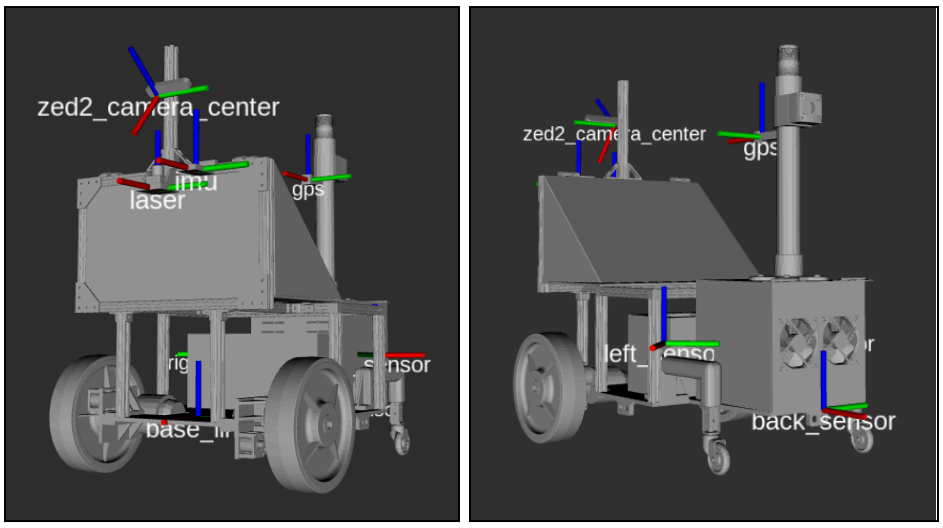


Figure AG-2: URDF Sensor Frames

Appendix AH: Localization

A vital function of the robot is the ability to tell where it is in space. The vehicle needs to know where it is in reference to obstacles and lines in order to avoid them, while also being able to tell its global position and heading to navigate to GPS waypoints.

Wheel Encoders Data

As mentioned earlier, the robot uses two Taiss E38S6-360-24G rotary encoders which are connected to the wheels by a belt. These encoders measure the amount of rotation with ticks so that every revolution of the encoder will output 360 ticks. In order to use this information effectively, the encoder data has to be manipulated. Both of the wheel encoders are connected to the first Arduino Due which reads the ticks of each encoder and sends the information to the main computer. Code was developed to receive this data and transform it into an Odometry ROS2 data type. An Odometry message, simply put, describes the position, orientation, and velocity of the vehicle. It is necessary to convert the encoder data to this message type so it is compatible with the Extended Kalman Filter. Using the width of the wheels and the number of ticks accumulated in a meter, the position, orientation, linear velocity, and angular velocity of the vehicle is calculated. A video showing the manipulated encoder data and its output can be seen here, and the data pathing can be seen in **Figure AH-1** below. Encoder Video Need to add pic of rqt_graph also

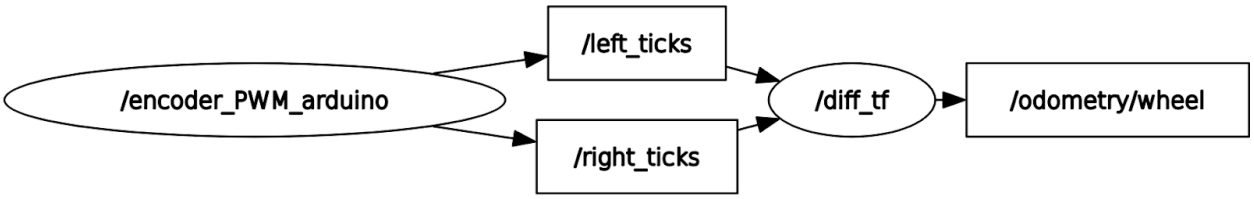


Figure AH-1: Encoder Data Pathing

Extended Kalman Filter

In order to localize itself as accurately as possible, the robot fuses data from a number of sensors. This technique is made possible with an Extended Kalman Filter. There are several different types of Kalman Filters, but due to the unpredictable nature of the system, an Extended Kalman Filter was the best choice as it can handle nonlinear systems. A ROS2 package, which can be found here, was downloaded and is used as the Extended Kalman Filter for the system. In most cases, encoder and IMU data is sufficient to effectively localize a robot in its environment, but GPS waypoint navigation is necessary for the IGVC. GPS data can be very inaccurate, and is subject to discrete jumps, so two instances of the Extended Kalman Filter are run simultaneously. The first filter takes data from the BNO055 IMU and wheel encoders. This filter is accurate in the short term, and produces continuous data that is used for the Local Planner and for short term navigation. The second filter fuses all of the data, which includes the GPS. In order to be used with the other data, the GPS data is first converted into Universal Transverse Mercator coordinates. The traditional Geographic coordinates are based on a spherical, three-dimensional system, and are subject to distortion. However, UTM coordinates are two-dimensional and fix most distortion, making them more suitable for the system. The converted GPS data is then combined with an absolute orientation provided by the Adafruit IMU. This orientation has to be based in the ENU frame, which means the IMU has to read 0 degrees when facing East, and 90 degrees when facing North. The data from the wheel encoders is then fused to produce a combined output that is used in the Global Planner for waypoint navigation. A detailed graph showing the data mapping can be seen in **Figure AH-2** below.

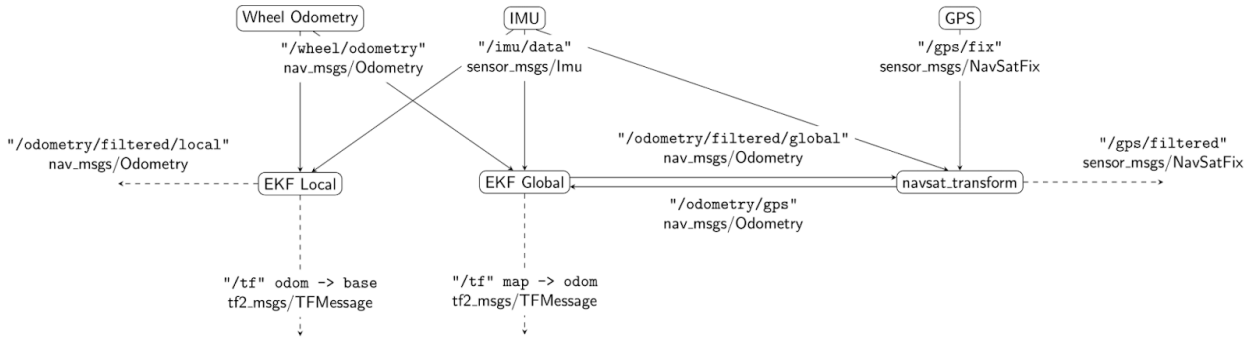


Figure AH-2: Dual Extended Kalman Filter Chart

Waypoint Navigation

With the Extended Kalman Filter outputting a global position, orientation, and velocity, it is possible to navigate to GPS waypoints. The navigation system can take multiple waypoints and move to each point one at a time. These waypoints have a longitude, latitude, and even a yaw that represents the orientation of the robot at that point. The desired waypoints are written in a .yaml file and are passed to the Nav2 Waypoint Follower when the GPS waypoint code is run. Much like the GPS data in the filter, these points are transformed into UTM coordinates to be used by the system. The robot will then move until it has reached the current goal pose. It will then move to the next point and continue until the last waypoint has been achieved. An example .yaml file of waypoints is shown below in **Figure AH-3**.

```yaml
waypoints:
- latitude: 41.513420032486444
  longitude: -90.49105243191832
  yaw: 3.92699
- latitude: 41.513420032486444
  longitude: -90.49112283990048
  yaw: 3.1415
- latitude: 41.513420032486444
  longitude: -90.49105243191832
  yaw: 0.785398
- latitude: 41.51348580884368
  longitude: -90.49104304418735
  yaw: 1.57079
```

Figure AH-3: Waypoints .yaml File

Appendix AI: Obstacle Avoidance

In the pursuit of advancing autonomous systems, the integration of multiple sensor modalities has become imperative to achieve robust and reliable perception capabilities. Among these modalities, LiDAR, camera, and ultrasonic sensors play pivotal roles in providing comprehensive environmental awareness.

The integration process begins with the camera, which captures visual information and outputs it in the form of a 3D point cloud. This data representation offers rich spatial information about the scene, including object shapes and their positions in three-dimensional space. However, for seamless integration with LiDAR data, which also provides 3D spatial information, the camera's output needs to be transformed into a compatible format. This involves converting the 3D point cloud data from the camera into a 2D scan, aligning it with the data format commonly used by LiDAR sensors. This transformation not only simplifies the integration process but also ensures that the data from both sensors can be easily fused and utilized together for perception tasks. Once aligned, the camera and LiDAR data work in tandem to provide a more comprehensive understanding of the environment. LiDAR sensors excel in capturing precise distance measurements to objects, while cameras offer rich color and texture information. By combining these modalities, the perception system gains the ability to accurately detect and classify objects in the environment, such as vehicles, pedestrians, and obstacles, with improved accuracy and reliability.

However, even with the combined capabilities of LiDAR and camera sensors, there are limitations and scenarios where additional sensor modalities are necessary to ensure robustness and safety. This is where ultrasonic sensors come into play. Ultrasonic sensors complement LiDAR and camera systems by providing close-range distance measurements, particularly useful for detecting objects in the immediate vicinity of the vehicle, such as curbs, walls, or backing up the vehicle.

Moreover, ultrasonic sensors serve as a fail-safe mechanism, offering redundancy in the perception system. In the event of a failure or limitation in either the LiDAR or camera sensors,

ultrasonic sensors can provide essential distance measurements to prevent collisions, thereby enhancing the overall safety of the autonomous system.

In order to plan a path for the robot, it needs to be aware of the obstacles surrounding it and how close it can safely get to it without coming too close to hitting it. These criteria can be met with the help of two packages, SLAM, and Nav2.

SLAM

SLAM, or Simultaneous Localization and Mapping is an integral part of navigating around obstacles. In the context of ROS2, the SLAM package creates a map of its surroundings, in this case, based off of the input received by the devices that detect obstacles, here, a LiDAR and camera in the form of the input topic Laserscan.

The localization aspect of the SLAM package involves estimating the location and pose of the robot in real-time as it moves through the environment. ROS2 utilizes various localization algorithms and packages to perform this task, including popular methods like Adaptive Monte Carlo Localization (AMCL), and the Extended Kalman Filter (EKF). The former method, AMCL, is extremely useful when it comes to noting the position and orientation of the robot on a predefined map. The EKF is more useful for real time navigating and localization.

Mapping refers to the construction of a permanent representation of the robot's environment. In ROS2, mapping algorithms process sensor data, which in this case is Laserscan from both the LiDAR and the camera, to create a map that captures the layout and features of the surroundings. When an obstacle is detected, it is registered and added to the permanent map of the surroundings of the robot. An example of a map created by the SLAM package can be seen in **Figure AI-1**, which depicts data taken from the integrated sensor data.

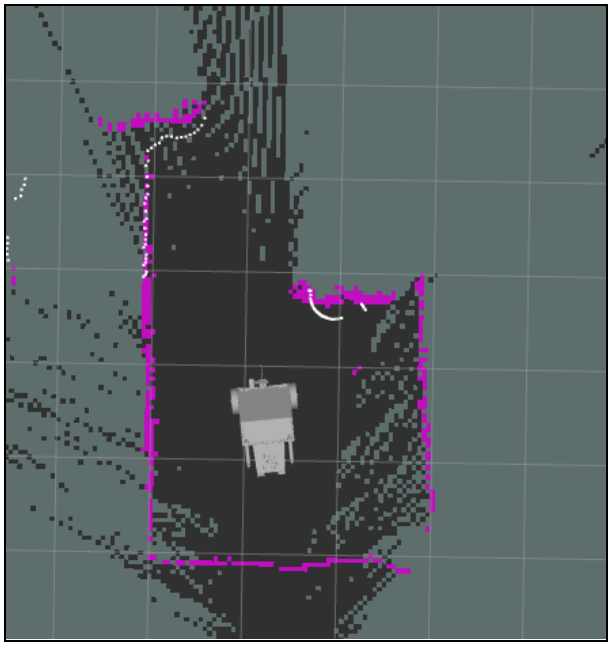


Figure AI-1: Map Created Using SLAM

Nav2 Costmap 2D

While the SLAM toolbox focuses on creating a map of all the obstacles with respect to the current position and orientation of the robot, the costmaps, taken from the Nav2 costmaps focus on how to navigate around those obstacles. This is done by adding a layer of cost around each obstacle to serve as an indication of how close the robot can safely approach the obstacle. Similar to the SLAM toolbox, the Nav2 costmaps take in data from the sensors, and utilize the localization from other devices such as the IMU and GPS, and the odometry data from the wheel encoders to create a moving zone of cost around the obstacles that are detected by the sensors, all the while creating a static map that captures the zones of cost and saves them.

The costmaps are structures in three layers. These layers are the static layer, inflation layer, and the obstacle layer. The static layer detects permanent or stationary objects. These obstacles are extracted from static maps and are unchanging. The obstacle layer represents dynamic obstacles which are detected by the sensors during the runtime of the robot and is constantly updating. The final layer is the inflation layer which adds a buffer zone, or a zone of

cost around the obstacles to account for the robot size, often based on the base footprint of the robot to ensure safe navigation.

Local Costmaps

There are two types of costmaps, the local and global costmaps. Local costmaps represent the immediate surroundings of the robot, and are used for short term obstacle avoidance and local path planning. The local costmap updates far more frequently, often at a rate equal to the rate at which the sensor data is published. This allows for the constant adjustment of the path in real time based on the immediate obstacles detected by the robot. **Figure AI-2** shows how the local costmap detects obstacles in the vicinity and adds cost around them.

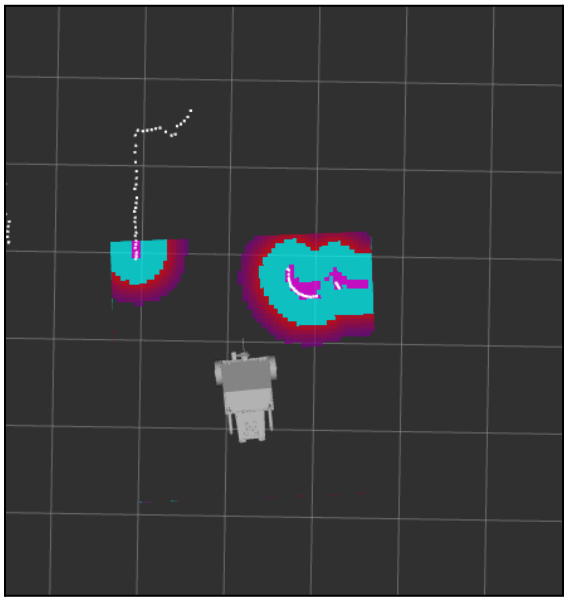


Figure AI-2: Local Costmap Created Using Nav2

Global Costmaps

The global costmap is a map of the entire known environment surrounding the robot, specifically the static obstacles, taken from a static map, here produced by the SLAM toolbox. Unlike the map produced by the SLAM toolbox, it contains zones of cost around all the obstacles present in the static map. The global costmaps only update when there is an update in the static

map that it is based on, which is far less than the local costmap. It does play an integral role in the long term, high level path planning, when navigating from the current position to another position on the map. This is better for obstacle detection and avoidance over a longer distance as compared to the local costmaps and is used for global planning. **Figure AI-3** shows an image of a global costmap with data from the LiDAR and camera, surrounding the robot.

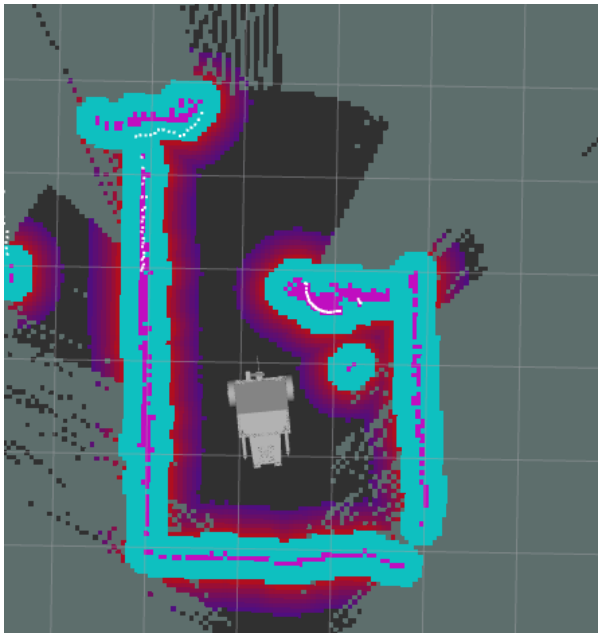


Figure AI-3: G;obal Costmap Created Using Nav2

Appendix AJ: Path Planning

Path Planning is arguably the most important part of the navigation system. Even if every other system is working to perfection, the robot would not be able to navigate autonomously without this ability. The Nav2 package provides a simple way to give the vehicle the capacity to make effective obstacle avoidance and waypoint navigation decisions. In short, the path planning of the robot is broken down into a global and local plan to allow for more precise GPS wayfinding in the long term and much safer obstacle navigation in the short term.

Nav2

Nav2 is a highly detailed, open-source package for ROS2 that is professionally maintained. This package provides a number of navigation abilities and options to common, mobile robots. This set of libraries is the standard for many industrial robots, so it was an easy choice to implement this into the navigation system. Nav2 is used for a number of different reasons on the robot; for example, the robot's velocity commands are being smoothed by the Nav2 Velocity Smother, the Nav2 Waypoint Follower is being used to navigate from point to point, a number of fallback actions are contained within the Nav2 Behavior Server, local and global costmaps are generated by Nav2 Costmap 2D, and the local and global plan is produced by the Nav2 Controller and Planner Servers. This package is extremely useful and its documentation can be found here.

Figure AJ-1: Nav2 Logo

Global Planner

The global plan represents the overall path the robot needs to follow in order to reach its ultimate goal, which in this case, is each GPS waypoint. This path is generated with information from a bunch of different areas. The global costmap, global localization, and each waypoint are all required to produce a global plan. Nav2 allows for custom algorithms to be developed and implemented into the code, but it was decided to use the Smac 2D Planner for this project. This provided planner is Grid Based and uses the A* algorithm to produce a global path depending on the goal, detected obstacles, and its current location. Additionally, the Smac Planner was chosen because it can be used for robots that do not have a circular footprint. Due to the robot being more rectangular in shape, the popular NavFn Planner was ruled out since it is used for circular robots. The Global Planner saves all of the cost data so that the system knows the location of all detected obstacles and lines in the past and present. This way it can make more educated decisions on how to proceed to the next waypoint. An image showing the global costmap and the global plan produced can be seen in **Figure AJ-2** below.

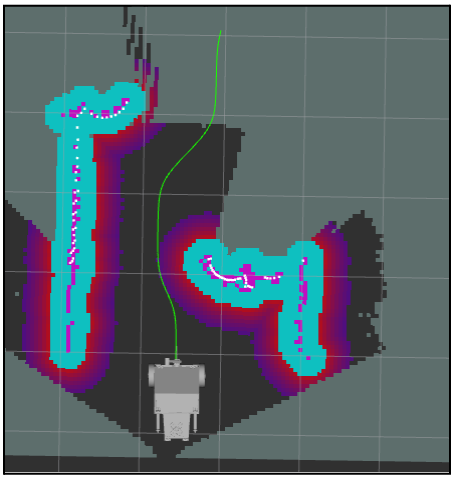


Figure AJ-2: Global Plan

Local Planner

While the Global Planner holds the ultimate goal of the robot, the Local Planner is used for short term navigation. Much like the Global Planner, the local path needs the local costmap, local localization, and short term goal to produce a path. The main goal of the local path is to avoid nearby obstacles and stay within the lines while staying as close as possible to the global plan. The Nav2 Controller Server is used to produce this path, and it was decided to use the default DWB Controller from the package. This controller takes in the current local plan, outputs the required Twist velocity commands to proceed, and continuously checks to see if the goal has been reached. Unlike the Global Planner, the Local Planner only sees obstacles in its immediate area from the local costmap. This reduces the computation power required and allows for more efficient paths to be generated. In addition to this, the Local Planner receives data for the robot footprint to produce a much more accurate path and to avoid collisions. The robot footprint used and the local plan produced from the local costmap is pictured in **Figures AJ-3 and AJ-4** below.

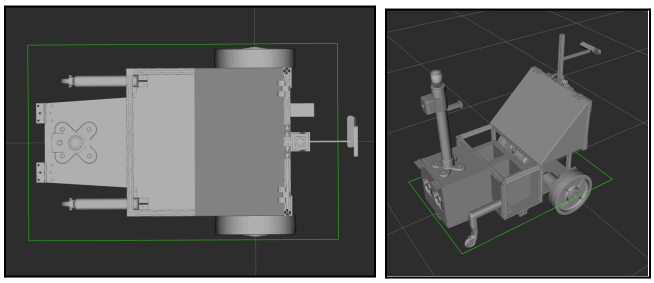


Figure AJ-3: Robot Footprint

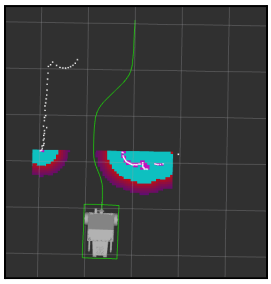


Figure AJ-4: Local Plan

Appendix AK: Results

At this time, Rock-E the robot has complete, autonomous function. The mechanical design of the robot has been optimized and the electrical wiring has been complete and properly protected. The robot has the ability to travel to waypoints, avoid obstacles, and navigate between detected lines autonomously. Although the GPS waypoint navigation is functioning, it is still being optimized for accuracy and function. The functioning robot can be seen below in **Figure AK-1**, Autonomous Rock-E (LiDAR), Indoor Rock-E and Final Product .

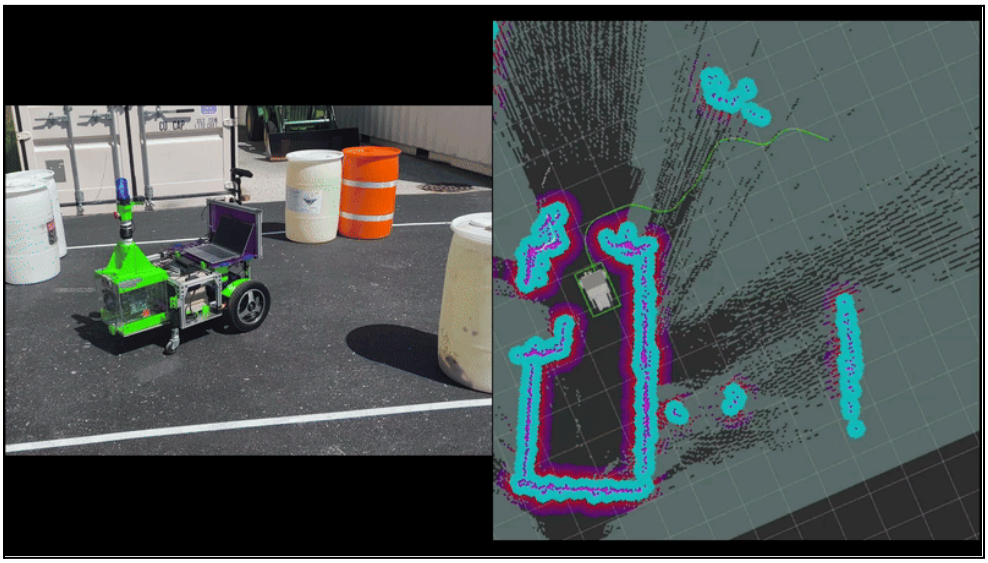


Figure AK-1: Autonomous Rock-E

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